MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Pharmacology

PHARMACOLOGY

complete study manual

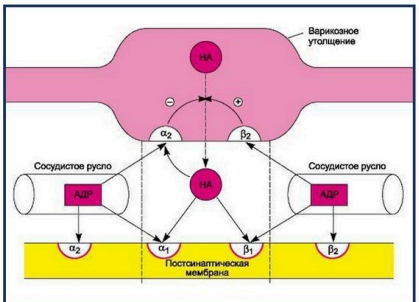
A study manual for practical classes



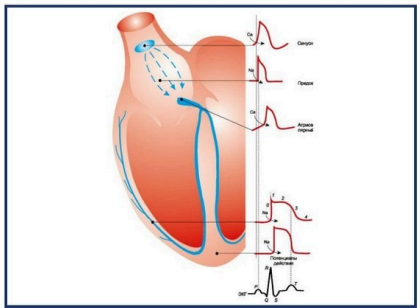
Dosage forms



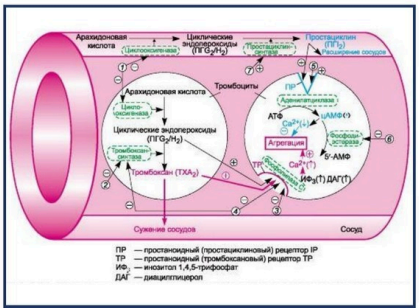
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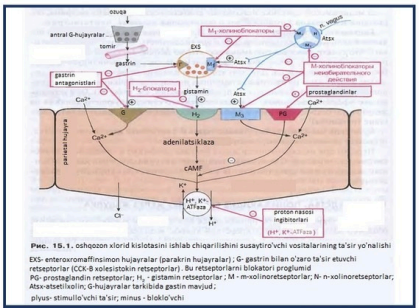
Adrenergic synapse



Cardiac conduction system



Platelet aggregation



Hydrochloric acid secretion

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Pharmacology: Complete Study Manual. A study manual for practical classes. — Tashkent, 2026.

This manual is based on the 18 practical class topics of the calendar-thematic plan of the course “Pharmacology” approved for the 2026–2027 academic year. The book consists of two parts: Part I covers the 9 topics of the 5th semester (prescription writing, general pharmacology, drugs acting on the peripheral and central nervous systems and on the respiratory system), and Part II covers the 9 topics of the 6th semester (drugs acting on the cardiovascular, digestive, urinary and blood systems, hormonal and anti-inflammatory preparations, antiseptics and chemotherapeutic agents).

At the end of each chapter there are algorithms for practical skills (prescription writing), review questions, test questions and clinical cases. The answers to the tests are given at the end of the book.

The manual is based on D.A. Kharkevich's textbook “Pharmacology” (2010), S.S. Azizova's textbook “Pharmacology” (2006), the manual “Fundamentals of Prescription Writing” of the TMA Department of Pharmacology (2020), the works of V.V. Maisky, B.G. Katzung, Kaplan USMLE, First Aid, the Color Atlas of Pharmacology and other textbooks, as well as modern clinical guidelines. The figures are taken from these sources, and the source is given under each of them. The figures on the cover are taken from the teaching material of the TMA Department of Pharmacology.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before applying the clinical information in practice, especially doses, it is recommended to compare it with the main textbooks and the current official prescribing information.

The manual is intended for 3rd-year students of the faculties of general medicine of medical higher education institutions.

This English edition is a translation of the Uzbek original. Drug names are given as International Nonproprietary Names (INN), with the name used in the original in brackets where it differs.

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Preface

Pharmacology is the bridge between theoretical medicine and clinical practice. Whatever disease a doctor deals with, the main tool in his or her hands remains the drug. A correctly chosen drug cures the patient, while a wrongly chosen drug or a wrong dose causes harm. That is why a thorough knowledge of pharmacology is essential for every future doctor.

This manual follows the calendar-thematic plan of the Department of Pharmacology of Tashkent State Medical University approved for the 2026–2027 academic year. The book consists of two parts:

- **Part I (5th semester)** — prescription writing, general pharmacology, drugs acting on the peripheral and central nervous systems and on the respiratory system (Topics 1–9);
- **Part II (6th semester)** — drugs acting on the cardiovascular, digestive, urinary and blood systems, hormonal preparations, anti-inflammatory, antiseptic and chemotherapeutic agents (Topics 10–18).

Each topic has the same structure:

1. **Learning objectives** — what the student should know and be able to do.
2. **Theoretical part** — classification of drug groups, mechanism of action, pharmacological effects, uses, adverse effects and contraindications.
3. **Clinical part** — drug choice, interactions, poisonings and their management.
4. **Practical skills** — algorithms for writing prescriptions and prescriptions for the main drugs.
5. **Review questions, 15 test questions and 3 clinical cases** (with answers).
6. **Quick review table** — for revision before an exam.

The colored boxes in the text mean the following:

REMEMBER
Yellow box — information that must be memorized (classification, mechanism, key numbers).

CLINICAL SIGNIFICANCE
Blue box — the clinical significance of the information and its use in practice.

MEMORY AID
Green box — a technique that makes memorizing easier (mnemonic, association).

CAUTION
Orange box — dangerous mistakes, differences between textbooks, outdated or changed information.

GRAY BOX — SAMPLE PRESCRIPTION
Rp.: The Latin part
D.S. The signatura (in the patient's language).

The manual is based mainly on D.A. Kharkevich's textbook “Pharmacology”, S.S. Azizova's Uzbek-language textbook “Pharmacology”, the department's teaching manual “Fundamentals of Prescription Writing”, and Katzung, Kaplan, First Aid and other international sources. Where textbooks differ or information has been updated, this is pointed out separately.

Doses are given for educational purposes. In practice, always rely on the official prescribing information of the preparation and the current clinical protocols.

PART I. PRACTICAL CLASSES OF THE 5TH SEMESTER

Prescription writing. General pharmacology. Neurotropic drugs. Drugs acting on the respiratory system

TOPIC 1

The importance of prescription writing in training general practitioners. The prescription and its structure. Solid and soft dosage forms and the rules for prescribing them

Learning objectives

A general practitioner (GP) writes dozens of prescriptions every day. A single error in a prescription (a misplaced decimal point, a tenfold dose overshoot, a mix-up of dosage forms) can endanger the patient's life. That is why prescription writing is the most practical branch of pharmacology.

The student should know:

- the concepts of prescription writing, drug substance, medicinal product, drug preparation and dosage form;
- the essence of the State Pharmacopoeia, lists A and B, and officinal and magistral drugs;
- the types of doses and the rules for writing doses in a prescription;
- the parts of a prescription and the procedure for filling in a prescription form;
- the abbreviations used in writing prescriptions in Latin and the genitive case;
- the classification, advantages and disadvantages of solid and soft dosage forms.

The student should be able to:

- write prescriptions for powders, tablets, dragées, capsules, granules, ointments, pastes, creams, gels, liniments, suppositories and plasters;
- calculate the amount of a substance from a percentage concentration;
- find and correct errors in a written prescription.

1. Prescription writing and its basic concepts

Prescription writing (recipe writing) (Lat. recipere — to take) — the branch of pharmacology that studies the rules for prescribing, preparing and dispensing medicines to the patient. It is divided into two parts:

- **general prescription writing** — dosage forms and the general rules for prescribing them;
- **special prescription writing** — prescribing specific drug groups (studied in each subsequent topic). **A prescription** is the physician's **written instruction** to the pharmacist (dispensing chemist) to prepare or dispense a medicinal product in a certain form and dose, and on how to use it. A prescription is simultaneously a medical, legal and financial document.

Table 1.1. *Significance of the prescription*

Aspect	Content
Medical	Determines which drug the patient takes, how much and how
Legal	The physician who wrote the prescription is legally responsible for the drug's effect; it may serve as evidence in court
Financial and accounting	The basis for keeping records of substance consumption and of free and subsidized drugs in the pharmacy
Statistical	A source for conclusions about drug consumption, morbidity and drug effectiveness

Key terms

- **Drug substance (active substance)** — an individual chemical compound or natural substance that has pharmacological activity and is approved for medical use (e.g., paracetamol).
- **Medicinal product** — a substance or mixture of substances approved for the prevention, diagnosis and treatment of disease.
- **Drug preparation** — a ready-to-use medicinal product brought into a particular dosage form (e.g., “Paracetamol 500 mg tablets”).
- **Dosage form** — a state of the medicinal product convenient for use (tablet, solution, ointment, etc.).
- **Medicinal plant raw material** — plants or their parts containing biologically active substances.
- **Galenic preparations** — preparations obtained by processing medicinal plant raw material and containing ballast substances (infusion, decoction, tincture, extract). **Neogalenic preparations** — preparations maximally purified from ballast substances that can be given parenterally.

REMEMBER

Drug substance → medicinal product → drug preparation. The same drug substance may be produced in different dosage forms: for example, diclofenac is available as tablets, a solution for injection, a gel, suppositories and eye drops.

State Pharmacopoeia

Pharmacopoeia (Gr. pharmacon — drug, poieo — I make) — a collection of state standards **having the force of law** that set the quality requirements for medicines, the methods of preparing, storing and analyzing them, the maximum doses and the lists. Compliance with the pharmacopoeia is mandatory for all pharmaceutical companies and pharmacies. There is also the WHO **International Pharmacopoeia**, which is advisory in nature.

In the pharmacopoeia, drugs are classified by their degree of danger:

- **List A (Venena — poisonous)** — poisonous drugs that are prescribed, stored and dosed with great care. They are stored separately in a lockable cabinet.
- **List B (Heroica — potent)** — drugs that may cause complications when used without medical supervision.
- In addition, **narcotic drugs, psychotropic substances and precursors** are controlled by separate legislation. **Officinal drugs** (Lat. officina — pharmacy) — drugs whose composition is set by the pharmacopoeia or a regulatory document and which are produced ready-made in a factory. **Magistral drugs** (Lat. magister — master) — drugs prepared individually in the pharmacy according to a physician's prescription. There are also magistral formulations named after their authors: Votchal's drops, Pavlov's mixture, Vishnevsky's liniment.

Names of medicines

Every drug has three kinds of names:

Table 1.2. *Types of drug names*

Type of name	Description	Example
Chemical name	Reflects the chemical structure of the substance; rarely used in practice	2-(acetyloxy)benzoic acid

Type of name	Description	Example
International nonproprietary name (INN)	A single generic name assigned by WHO	Acetylsalicylic acid
Trade (proprietary) name	A commercial name given by the manufacturer	Aspirin, Thrombo ASS, Cardiomagnyl

CLINICAL SIGNIFICANCE

In modern practice, it is recommended to write prescriptions by the **international nonproprietary name (INN)**. This allows the patient to choose a cheaper generic preparation with the same active substance and prevents the same substance from being prescribed twice under two different trade names (dose duplication).

MEMORY AID

Common INN stems help to recognize the drug group: **-olol** — beta-blockers (metoprolol), **-pril** — ACE inhibitors (enalapril), **-sartan** — angiotensin receptor blockers (losartan), **-statin** — statins (atorvastatin), **-prazole** — proton pump inhibitors (omeprazole), **-floxacin** — fluoroquinolones (ciprofloxacin), **-mab** — monoclonal antibodies (rituximab).

2. Doses and writing units of measurement in a prescription

Dose — the amount of a drug given to the body once or over a certain period that produces the desired effect.

Table 1.3. *Types of doses*

Dose	Latin name	Definition
Single	Dosis pro dosi	The amount for one intake
Daily	Dosis pro die	The total amount taken in one day
Course dose	Dosis pro cursu	The amount over the whole course of treatment
Minimum (threshold) dose	Dosis minima	The smallest amount that just begins to produce an effect
Average therapeutic dose	Dosis therapeutica	The amount that produces the desired effect in most patients
Maximum dose	Dosis maxima	The largest amount considered safe (given in the pharmacopoeia for list A and B drugs)
Toxic dose	Dosis toxica	The amount that causes signs of poisoning
Lethal dose	Dosis letalis	The amount that causes death
Loading dose	—	A large initial dose to rapidly achieve the required blood concentration
Maintenance dose	—	Subsequent doses to maintain the concentration achieved

Therapeutic range (therapeutic width) — the interval between the minimum therapeutic and the minimum toxic doses. The wider it is, the safer the drug (this concept is discussed in detail in topic 3).

Writing amounts in a prescription

In a prescription, the unit of mass is the **gram** and the unit of volume is the **milliliter**. The name of the unit is not written, only the number. Grams are written as whole numbers, and fractions of a gram as decimals.

Table 1.4. *Writing mass and volume in a prescription*

Amount	How it is written in the prescription
1 gram	1,0
0.5 gram (500 mg)	0,5
0.1 gram (100 mg)	0,1
0.01 gram (10 mg)	0,01
0.001 gram (1 mg)	0,001
0.0001 gram (100 mcg)	0,0001
10 ml	10 ml
5 drops	gtts. V (in Roman numerals)
500,000 units of action	500,000 U (ED)

REMEMBER

There must be at least one digit after the decimal point in a prescription: **1.0** (not “1”), **0.5** (not “.5”). Shifting the decimal point by one place changes the dose 10-fold! Drugs dosed in units of action (U): some antibiotics (benzylpenicillin, nystatin), heparin, insulin, and certain hormones and enzymes.

Volume of spoons and drops

Table 1.5. *Household dosing units*

Unit	Volume
1 teaspoon	5 ml
1 dessert spoon	10 ml
1 tablespoon	15 ml
1 ml of aqueous solution	20 drops
1 ml of alcoholic solution	30–50 drops (depending on alcohol concentration)
1 ml of oily solution	30 drops
1 ml of ether	60 drops

Doses for children and the elderly

Traditional prescription-writing manuals give simplified formulas for calculating a child's dose from the adult dose:

- by age: **child's dose = adult dose × child's age / 24**;
- by weight: **child's dose = adult dose × child's weight / 70**. Elderly patients (over 60), especially for drugs acting on the CNS, cardiac glycosides and diuretics, are prescribed **1/2–3/4 of the adult dose**.

CAUTION

The formulas above are for rough estimates only. In modern pediatrics, doses for each drug are set in **mg/kg** or **mg/m²** (body surface area), based on the package insert and pediatric reference books. For most drugs, calculating a child's dose from age alone can be dangerous.

3. The prescription and its structure

A prescription is written on a special **prescription form** clearly and legibly with a ballpoint pen or ink. Corrections are not allowed. The size of an ordinary form is 105 × 148 mm.

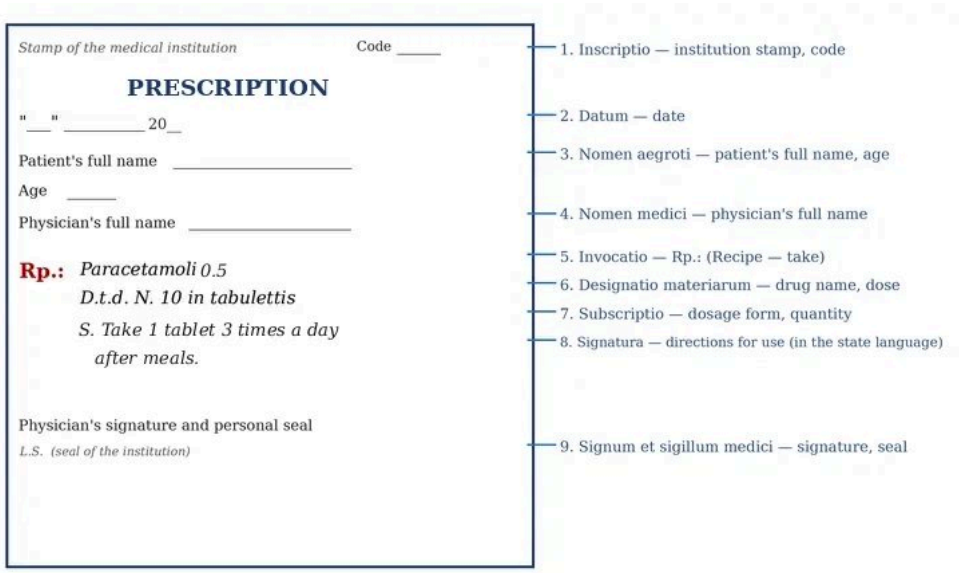


Figure 1.1. *The prescription form and the parts of a prescription (compiler's diagram, based on the TMA manual "Fundamentals of prescription writing")*

Parts of a prescription

1. **Inscriptio** — the stamp and code of the medical institution.
2. **Datum** — the date the prescription was written.
3. **Nomen aegroti** — the patient's surname, first name, patronymic and age.
4. **Nomen medici** — the physician's surname, first name and patronymic.
5. **Invocatio** — the physician's address to the pharmacist: **Rp.:** (Recipe — take).
6. **Designatio materiarum** — the names and amounts of the drug substances (in Latin, in the genitive case).
7. **Subscriptio** — instructions to the pharmacist about the dosage form and the amount to be dispensed (M.f., D.t.d. N.).
8. **Signatura** — the method of use for the patient (after S. or D.S., in the state language).
9. **Signum et sigillum medici** — the physician's signature and personal seal.

MEMORY AID

A prescription is read "from top to bottom": **Stamp → Date → Patient → Physician → Rp. → Drug → Instructions → Signatura → Physician's signature**. Initial letters of the Latin names: **I-D-N-N-I-D-S-S-S**. Only **parts 5, 6 and 7** (Rp., drug names, instructions to the pharmacist) are written in Latin; the rest is in the state language.

General rules for writing prescriptions

- Each drug substance is written **on a new line and with a capital letter**.
- Drug substances are written **in the genitive case** (Genetivus), because it means “Take [of what?] ...”: Rp.: Analgini 0.5 — “Take 0.5 of analgin”.
- Order of the ingredients in compound prescriptions: **main drug (basis) → auxiliary drug (adjuvans) → taste and smell corrector (corrigens) → form-giving substance (constituens)**.
- If several substances are prescribed in equal amounts, the amount is placed opposite the last one, preceded by **ana** (**āā**) — “in equal amounts”.
- If two drugs are written on one form, they are separated by the # sign (except list A and narcotic drugs, which are written on separate forms).
- If the drug must be dispensed urgently, **Cito!** (quickly) or **Statim!** (immediately) is written at the top of the form.
- If the prescription is to be repeated, **Repetatur!** (let it be repeated) is written.
- If the dose of a drug substance exceeds the maximum dose, the physician writes the amount **in words**, puts an exclamation mark and confirms it with a signature.

> **CAUTION**
> Vague expressions such as “internally” or “as directed” must not be written in the **signatura**. The amount, the number of times a day, the timing in relation to meals and the duration must be stated clearly: “Take 1 tablet 3 times a day after meals for 5 days”.

Types of prescription forms

- **Ordinary forms** — for most drugs; certified with the institution's “For prescriptions” seal.
- **Subsidized and free prescriptions** — the form is stamped “free” or “subsidized” and signed by the physician and the head of the institution.
- **Special forms** (serially numbered, strictly accounted for) — for narcotic and psychotropic substances and certain potent drugs. They are certified with the institution's official seal with the coat of arms and are kept in a safe by the chief physician.

> **CAUTION**
> **Electronic prescription.** In Uzbekistan, the “Electronic prescription” system was launched in December 2025 in Tashkent and a number of pilot districts and is planned to be rolled out nationwide by the end of 2026. The physician confirms the prescription electronically, a QR code is assigned automatically and the prescription is sent to the pharmacy automatically; it was first made mandatory for antibiotics and hormonal preparations. A separate procedure remains for narcotic and potent drugs. Because these rules keep changing, the current orders of the Ministry of Health should be checked. The skill of composing the Latin part of a prescription correctly remains the foundation in the electronic system as well.

Abbreviations used in prescriptions

Abbreviation rule: a word is abbreviated **at a consonant** (extr. — extractum, ung. — unguentum). Names of drug substances, plant names and chemical elements are **not abbreviated** (exceptions: aq. — aqua, sp. — spiritus, ac. — acidum).

Table 1.6. *Abbreviations most commonly used in prescription writing*

Abbreviation	Full form	Translation
Rp.:	Recipe	Take

Abbreviation	Full form	Translation
M.f.	Misce, fiat (fiant)	Mix to make
D.S.	Da. Signa.	Give. Label
D.t.d. N.	Da (dentur) tales doses numero	Give such doses in the number of ...
S.	Signa	Label
ana (āā)	ana partes aequales	In equal amounts
q.s.	quantum satis	As much as needed
ad	ad	up to ...
in tab.	in tabulettis	In tablets
in caps.	in capsulis	In capsules
in amp.	in ampullis	In ampoules
in ch. cer.	in charta cerata	In waxed paper
in vitr. nigr.	in vitro nigro	In a dark glass bottle
steril.!	sterilisetur!	Sterilize!
supp.	suppositorium	Suppository
tab.	tabuletta	Tablet
ung.	unguentum	Ointment
sol.	solutio	Solution
tinct.	tinctura	Tincture
extr.	extractum	Extract
inf.	infusum	Infusion
dec.	decoctum	Decoction
gtt.	gutta (guttae)	Drop (drops)
p.o.	per os	By mouth
i/v, i/m, s/c	intra venam, intra musculum, sub cutem	Intravenously, intramuscularly, subcutaneously

Declensions of Latin nouns

Since drug substances are written in the genitive case in a prescription, the main declension endings must be known.

Table 1.7. *Main declensions of Latin nouns (singular)*

Declension	Nominative	Genitive	Examples
I	-a	-ae	Tinctura → Tincturae; Tabuletta → Tabulettae; Aqua → Aquae
II	-us, -um	-i	Analginum → Analgini; Oleum → Olei; Acidum → Acidi
III	various (-o, -is, -as, -x, -en, etc.)	-is	Solutio → Solutionis; Sulfas → Sulfatis; Radix → Radicis
IV	-us, -u	-us	Spiritus → Spiritus; Fructus → Fructus
V	-es	-ei	Species → Specierum (plural)

MEMORY AID

Anions of salts belong to the 3rd declension: **-as** → **-atis** (Natrii sulfas → Natrii sulfatis), **-is** → **-itis** (Natrii nitris → Natrii nitritis), **-idum** → **-idi** (Natrii chloridum → Natrii chloridi, 2nd declension). The cation (metal) is always in the genitive: Natrii, Kalii, Calcii, Magnii, Zinci.

4. Classification of dosage forms

Dosage form — a convenient state given to a medicinal product for use. The dosage form determines the rate of absorption, bioavailability, onset and duration of action of the drug.

Table 1.8. *Classification of dosage forms*

Criterion	Types
State of aggregation	Solid, soft, liquid, gaseous
Route of administration	Enteral (oral, sublingual, rectal), parenteral (injection), topical (skin, mucous membranes), inhalation
Dosing	Dosed (tablet, capsule, suppository) and non-dosed (ointment, solution, powder)
Preparation	Officinal (in a factory) and magistral (in a pharmacy, ex tempore)
Composition	Simple (one active substance) and compound (two or more substances)

This topic covers **solid** and **soft** dosage forms; liquid dosage forms are studied in topic 2.

5. Soft dosage forms (Formae medicamentorum molles)

Soft dosage forms include **ointments, pastes, creams, gels, liniments, suppositories and plasters**. They are widely used in dermatology, ophthalmology, proctology, gynecology and surgery; some (nitroglycerin ointment, hormonal patches) are used for a **resorptive** (systemic) effect.

Any soft dosage form consists of two parts:

- **the active substance (basis)** — has the therapeutic effect;
- **the form-giving substance, the base (constituens)** — forms the dosage form and holds the active substance.

5.1. Ointments (Unguenta)

Ointment (Unguentum, i n) — a non-dosed dosage form of soft consistency intended for application to the skin and mucous membranes. The amount of active substance is usually **up to 25%** (if more — a paste).

Table 1.9. *Ointment bases (constituens) and their properties*

Base	Property	Use
Petrolatum (Vaselinum)	Indifferent, does not spoil, not absorbed through the skin	If no base is specified in the prescription, petrolatum is used; for a superficial effect
Lanolin (Lanolinum)	Obtained from sheep's wool, well absorbed through the skin, absorbs up to 150% water	For a deep and resorptive effect; moist wounds
Petrolatum + lanolin (in equal amounts)	Combines the advantages of both	Eye ointments, many magistral ointments
Lard (Adeps suillus)	Well absorbed but quickly turns rancid	Rarely used now

Base	Property	Use
Polyethylene glycol bases	Water-soluble, washable	Modern officinal ointments

Ointments are prescribed in **two** ways:

Abbreviated form — if the ointment is officinal or has a single active substance with a petrolatum base. After Rp.: the name of the dosage form in the genitive (Unguenti), the substance name, its concentration and the total amount are written.

PRESCRIPTION 1.1. TETRACYCLINE EYE OINTMENT
Rp.: Unguenti Tetracyclini ophthalmici 1% — 10.0
D.S. Apply inside the lower eyelid 3 times a day.

PRESCRIPTION 1.2. OFFICINAL ZINC OINTMENT
Rp.: Unguenti Zinci 10% — 30.0
D.S. Apply to the affected skin area 2 times a day.

Extended form — for magistral ointments, or if the base is not petrolatum, or if there are two or more active substances. Each substance is written on a separate line, the base is brought up to the total mass with "**ad**", and then **M.f. unguentum** is written.

PRESCRIPTION 1.3. 1% NEOMYCIN OINTMENT ON A LANOLIN BASE, 50.0
Rp.: Neomycini sulfatis 0.5
Lanolini ad 50.0
M.f. unguentum
D.S. Apply to the injured area 2 times a day.

Calculation: 100.0 of ointment contains 1.0 of neomycin; 50.0 of ointment — x; x = 50 × 1 / 100 = **0.5**.

REMEMBER
Formula for calculating the amount of a substance from a percentage concentration: **amount of substance (g) = total mass × percentage / 100**. This formula is the same for ointments, pastes, liniments and solutions.

CLINICAL SIGNIFICANCE
Eye ointments are prepared sterile and prescribed in amounts of 5.0–10.0, skin ointments in amounts of 20.0–100.0. The word ophthalmicum is added to the name of an eye ointment.

5.2. Pastes (Pastae)

Paste (Pasta, ae f) — a thick ointment containing **25% to 65%** powdered substances (zinc oxide, talc, starch, white clay). Thanks to the large amount of powder, pastes have an **adsorbing, drying and anti-inflammatory** effect and stay on the skin surface for a long time. They are used for weeping dermatitis, eczema and diaper rash.

If the total powdered substances in the prescription are less than 25%, an indifferent powder (starch, talc) is added to obtain the required consistency.

PRESCRIPTION 1.4. OFFICINAL ZINC PASTE
Rp.: Pastae Zinci 40.0
D.S. Apply to the affected skin area 2 times a day.

PRESCRIPTION 1.5. MAGISTRAL PASTE (5% ANESTHESIN AND 25% ZINC OXIDE, 50.0)
Rp.: Anaesthesini 2.5
Zinci oxydi 12.5
Vaselini ad 50.0
M.f. pasta
D.S. Apply to the itching skin area.

5.3. Creams (Cremores) and gels (Gela)

Cream — a soft dosage form that is more liquid than an ointment and contains water (emulsion base). It is well absorbed into the skin, cosmetically convenient (not shiny) and convenient for use in skin folds and on the face. **Gel** — a viscous dosage form based on gelling agents such as gelatin, agar-agar and carbomer; it is quickly absorbed by the skin and has a cooling effect. Both are officinal only and are therefore written in the **abbreviated** form.

PRESCRIPTION 1.6. HYDROCORTISONE CREAM
Rp.: Cremoris Hydrocortisoni 1% — 15.0
D.S. Apply a thin layer to the affected skin area 2 times a day.

PRESCRIPTION 1.7. DICLOFENAC GEL
Rp.: Geli Diclofenaci 1% — 50.0
D.S. Apply to the painful joint area 3 times a day.

CAUTION
Different manuals use different spellings for cream and gel, such as “Crem”, “Cremoris”, “Gel”, “Geli”. According to Latin grammar, Cremor, oris m → genitive **Cremoris**, Gelum, i n → **Geli**. Some Uzbek manuals also give the undeclined form “Gel Troxevasin”. Follow your department's requirements.

5.4. Liniments (Linimenta)

Liniment (Linimentum, i n) — a dosage form in the form of a **liquid ointment** that melts at body temperature. Vegetable oils (sunflower, cottonseed, castor, peach oil) are used as the base. It is shaken before use. It is used for joint and muscle pain, burns and wounds. Officinal liniments are written in the abbreviated form, magistral liniments in the extended form.

PRESCRIPTION 1.8. OFFICINAL SYNTHOMYCIN LINIMENT
Rp.: Linimenti Synthomycini 10% — 25.0
D.S. Apply to the wound surface under a dressing.

PRESCRIPTION 1.9. MAGISTRAL LINIMENT (5% TURPENTINE, 10% METHYL SALICYLATE, 75.0)
Rp.: Olei Terebinthinae 3.75
Methylii salicylatis 7.5
Olei Helianthi ad 75.0
M.f. linimentum
D.S. Rub into the painful joint.

CLINICAL SIGNIFICANCE

Vishnevsky's liniment (balsamic liniment) — a classic magistral formulation of xeroform, birch tar and castor oil. It is still used to treat purulent wounds and abscesses.

5.5. Suppositories (Suppositoria)

Suppository (Suppositorium, i n) — a dosed dosage form that is solid at room temperature and **melts at body temperature**. It is intended for insertion into body cavities.

Table 1.10. *Types of suppositories*

Type	Shape	Mass
Rectal suppositories (suppositoria rectalia)	Cone or cylinder with a pointed end	1.1–4.0 g (usually 3.0 for adults)
Vaginal suppositories (suppositoria vaginalia)	Ball (globuli), egg-shaped (ovula), flat (pessaria)	1.5–6.0 g (usually 4.0)
Sticks (bacilli)	Cylinder with a pointed end	0.5–1.0 g; for the urethra, cervical canal and fistula tracts

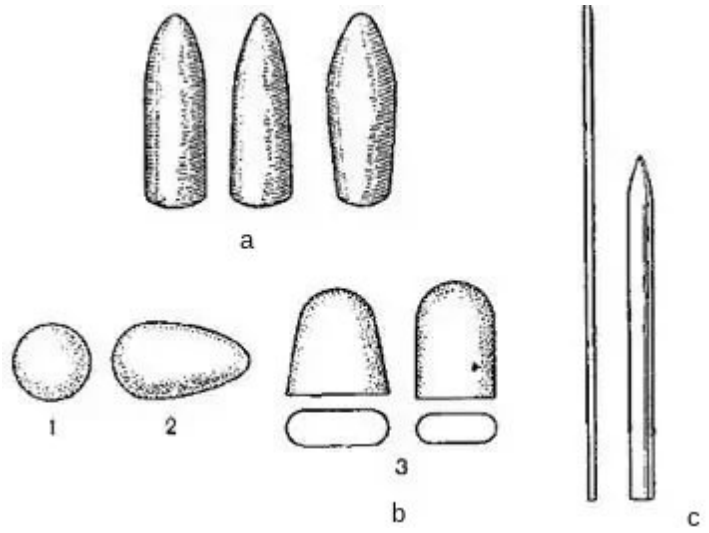


Figure 1.2. *Shapes of suppositories: a — rectal; b — vaginal (1 — globule, 2 — ovule, 3 — pessary); c — sticks (TMA "Fundamentals of prescription writing", 2020)*

Bases: **cocoa butter** (Oleum Cacao) — the most commonly used base, melts at 30–34 °C; gelatin-glycerin mass; polyethylene oxide bases; butyrol.

CLINICAL SIGNIFICANCE

Rectal suppositories are used for a **local** (hemorrhoids, proctitis) and **resorptive** effect. A drug absorbed from the lower part of the rectum enters the inferior vena cava directly through the inferior and middle rectal veins, bypassing the liver. The drug is therefore protected from partial breakdown in the liver. This route is convenient in vomiting patients, young children (paracetamol suppositories) and unconscious patients.

Officinal suppositories are prescribed in the abbreviated form: Rp.: Suppositorium cum (with …) + the substance name **in the ablative** + dose.

PRESCRIPTION 1.10. OFFICINAL PARACETAMOL SUPPOSITORIES (FOR A CHILD)
Rp.: Suppositorium cum Paracetamolo 0.25
D.t.d. N. 10
S. Insert 1 suppository into the rectum when the temperature is above 38.5 °C (not more than 4 times a day).

Compound officinal suppositories with a trade name: Rp.: Suppositoria “Anusolum” N. 10.

Magistral suppositories are written in the extended form: the dose of the active substance for one suppository, the amount of base, **M.f. suppositorium rectale (vaginale)**, then **D.t.d. N.** and the signatura.

PRESCRIPTION 1.11. RECTAL SUPPOSITORIES WITH ANESTHESIN (MAGISTRAL)
Rp.: Anaesthesini 0.3
Olei Cacao 3.0
M.f. suppositorium rectale
D.t.d. N. 10 in charta cerata
S. Insert 1 suppository into the rectum when in pain.

REMEMBER
If the amount of base is not specified in a suppository prescription, a rectal suppository is made from 3.0 and a vaginal suppository from 4.0. If the exact amount of base is not known, **q.s.** (quantum satis — as much as needed) is written: Olei Cacao q.s. ut fiat suppositorium.

5.6. Plasters (Emplastra) and transdermal therapeutic systems

Plaster (Emplastrum, i n) — a plastic mass that softens at body temperature and sticks to the skin, or a fabric coated with such a mass. Plasters may consist of the base only (adhesive plaster — for securing a dressing) or contain a drug (pepper and corn plasters).

Transdermal therapeutic systems (TTS) — multilayer patches that release the drug through the skin into the blood at a constant, controlled rate. Examples: nitroglycerin, fentanyl, nicotine, estradiol, rivastigmine and scopolamine patches. They bypass the liver and keep the blood concentration steady.

PRESCRIPTION 1.12. ADHESIVE PLASTER
Rp.: Emplastri adhaesivi 3 cm × 500 cm
D.S. For securing a dressing.

PRESCRIPTION 1.13. FENTANYL TRANSDERMAL SYSTEM
Rp.: Systematis therapeutici transdermalis Fentanyli 25 mcg/hour
D.t.d. N. 5
S. Apply one patch to the skin of the chest or shoulder every 72 hours.

CAUTION
Fentanyl is a narcotic analgesic. It is prescribed on a **special form** and is strictly accounted for. Cutting or heating the patch (heating pad, sauna) is forbidden: when heated, absorption rises sharply and respiratory depression may occur.

6. Solid dosage forms (Formae medicamentorum durae)

Solid dosage forms include **powders, granules, tablets, dragées, capsules, pills, caramels, lozenges (troches), eye films** and medicinal herbal species.

Table 1.11. *Advantages and disadvantages of solid dosage forms*

Advantages	Disadvantages
Accurate dosing	Relatively slow onset of action
Convenient to store, transport and use	Unsuitable for unconscious or vomiting patients and for children who cannot swallow
Physically and chemically stable, long shelf life	May irritate the gastrointestinal mucosa
Resistant to microbial contamination	Absorption depends on food, gastric pH and intestinal motility
An unpleasant taste and smell can be masked with a coating	Some drugs are broken down in the liver during first pass

6.1. Powders (Pulveres)

Powder (Pulvis, eris m) — a free-flowing solid dosage form. Types:

- **by composition:** simple (simplices) and compound (compositi);
- **by dosing:** divided into doses (divisi) and undivided (indivisi);
- **by fineness:** coarse (grossi), fine (subtiles), very fine (subtilissimi).



Figure 1.3. *Powder (TMA "Fundamentals of prescription writing", 2020)*

Undivided powders are prescribed for dusting (5.0–100.0) or for substances taken orally in large doses where an exact dose is not important. Undivided powders for oral use are measured with a teaspoon.

PRESCRIPTION 1.14. SIMPLE UNDIVIDED POWDER FOR DUSTING
Rp.: Pulveris Streptocidi subtilissimi 20.0
D.S. Dust onto the wound surface.

PRESCRIPTION 1.15. COMPOUND UNDIVIDED POWDER FOR ORAL USE
Rp.: Magnesii oxydi 40.0
Natrii hydrocarbonatis 10.0
M.f. pulvis
D.S. Take 1/3 teaspoon 3 times a day 1 hour after meals.

Divided powders — each dose is wrapped in a separate paper. The optimal mass of one powder is **0.1–1.0 g** (usually 0.3–0.5 g). If the dose of a substance is less than 0.1, an **indifferent filler** — sugar (Saccharum), glucose or sodium bicarbonate — is added so that the mass of one powder reaches at least 0.3.

PRESCRIPTION 1.16. CODEINE PHOSPHATE POWDERS (WITH A FILLER)
Rp.: Codeini phosphatis 0.015
Sacchari 0.3
M.f. pulvis
D.t.d. N. 12
S. Take 1 powder 3 times a day when coughing.

Powders of hygroscopic, volatile or odorous substances are wrapped in **waxed** (in charta cerata) or **paraffined** (in charta paraffinata) paper: D.t.d. N. 12 in charta cerata.

CAUTION
Substances that cannot be prescribed as powders: strongly **hygroscopic** substances (calcium chloride, potassium acetate), substances that **liquefy** when mixed together (eutectic mixtures, e.g., camphor with menthol), and **easily decomposing** and **explosive** combinations (potassium permanganate with organic substances).

6.2. Granules (Granula)

Granules — small uniform grains 0.2–3 mm in size. Substances with an unpleasant taste, irritants and antibiotics for children are often produced as granules; a suspension is prepared from them. All granules are officinal and are therefore written in the abbreviated form.

PRESCRIPTION 1.17. ACETYLCYSTEINE GRANULES
Rp.: Granulorum Acetylcysteini 0.2
D.t.d. N. 20
S. Dissolve 1 sachet of granules in 1/2 glass of water and take 3 times a day after meals.

6.3. Tablets (Tabulettae)

Tablet (Tabuletta, ae f) — a dosed solid dosage form made in a factory by compressing drug and auxiliary substances. All tablets are **officinal**. Their mass is 0.05–1.0 g; if the diameter is more than 9 mm, a score line is made to ease splitting. Auxiliary substances (starch, talc, lactose, magnesium stearate) are **not written** in the prescription.



Figure 1.4. *Tablets and capsules (TMA "Fundamentals of prescription writing", 2020)*

Types of tablets:

- **by route of administration:** oral; sublingual (sublinguales, nitroglycerin); buccal; chewable; water-soluble (solubiles, effervescent); vaginal tablets; tablets for preparing solutions;
- **by coating:** uncoated and coated (obductae); enteric-coated; film-coated;
- **by drug release:** immediate and extended-release (retard, SR, CR, XL, MR);

- **by dose:** forte (increased dose), mite (reduced dose). Tablets can be prescribed in **two ways**.

First method (without the name of the dosage form, with D.t.d.):

PRESCRIPTION 1.18. PARACETAMOL TABLETS (METHOD 1)
Rp.: Paracetamoli 0.5
D.t.d. N. 20 in tabulettis
S. Take 1 tablet when the temperature rises (not more than 4 times a day, at least 4 hours apart).

Second method (begins with the name of the dosage form in the accusative case — Tabulettas):

PRESCRIPTION 1.19. PARACETAMOL TABLETS (METHOD 2)
Rp.: Tabulettas Paracetamoli 0.5 N. 20
D.S. Take 1 tablet when the temperature rises.

Tablets of **compound composition** are prescribed by trade name; the dose is not indicated, and the name is in quotation marks in the nominative case:

PRESCRIPTION 1.20. COMPOUND TABLET WITH A TRADE NAME
Rp.: Tabulettas “Co-trimoxazolum 480” N. 20
D.S. Take 2 tablets 2 times a day after meals for 7 days.

Coated tablets: Rp.: Tabulettas Omeprazoli obductas 0.02 N. 14.

CLINICAL SIGNIFICANCE
Retard, enteric-coated and film-coated tablets **must not be split, chewed or crushed**: if the coating is damaged, the whole dose may be absorbed at once and cause a toxic effect (nifedipine retard, morphine retard), or the active substance may be destroyed by gastric acid (omeprazole, pancreatin). The patient should be reminded of this in the signatura: “swallow without chewing”.

MEMORY AID
D.t.d. N. ... in tabulettis — “**D**a **t**ales **d**oses, **N**umero ...” (give such doses, in number ...). The dose of one tablet is written after Rp., and their number after D.t.d.

6.4. Dragées (Dragee)

Dragée — a spherical dosed dosage form made in a factory by coating sugar granules layer by layer with drug and auxiliary substances (dragée coating). Its mass is 0.1–0.5 g. The name dragée is not declined in Latin.

PRESCRIPTION 1.21. DRAGÉE
Rp.: Dragee Drotaverini 0.04
D.t.d. N. 20
S. Take 1 dragée 3 times a day when in pain.

6.5. Capsules (Capsulae)

Capsule — a shell that holds powdered, granular, pasty or liquid drugs. The capsule masks the taste and smell of the drug, protects it from light and air, and protects the gastric mucosa from the irritant effect of the drug.

- **Gelatin capsules** (capsulae gelatinosae): hard (with a cap — operculatae) and soft (elastic — elasticae; for oily solutions).

- **Starch capsules** (capsulae amylaceae, cachets) — hardly used now.
- **Enteric (glutoid) capsules** — dissolve only in the intestine.



Figure 1.5. *A hard gelatin capsule filled with granules (TMA "Fundamentals of prescription writing", 2020)*



Figure 1.6. *A micropellet capsule and types of capped capsules (TMA "Fundamentals of prescription writing", 2020)*

PRESCRIPTION 1.22. AMOXICILLIN CAPSULES

Rp.: Amoxicillini 0.5
D.t.d. N. 21 in capsulis
S. Take 1 capsule 3 times a day every 8 hours for 7 days.

PRESCRIPTION 1.23. OMEPRAZOLE CAPSULES (METHOD 2)

Rp.: Capsulas Omeprazoli 0.02 N. 28
D.S. Take 1 capsule 2 times a day 30 minutes before breakfast and dinner.

6.6. Pills (Pilulae) and other solid forms

Pill (Pilula, ae f) — a spherical dosage form weighing 0.1–0.5 g; it is prepared in the pharmacy from a special mass. If its mass is more than 0.5, it is a **bolus**. Today tablets and capsules have almost completely replaced them.

PRESCRIPTION 1.24. MAGISTRAL PILLS (0.001 STRYCHNINE NITRATE IN EACH)

Rp.: Strychnini nitratis 0.03
Massae pilularum q.s. ut fiant pilulae N. 30
D.S. Take 1 pill 2 times a day.

In pills, the amount of substance is written for the total number: 0.001 × 30 = 0.03.



Figure 1.7. *Pills (TMA "Fundamentals of prescription writing", 2020)*

- **Caramels** and **lozenges (troches, pastilles)** are sucked in diseases of the mouth and throat: Rp.: Trochiscos "Strepsils" N. 24 / D.S. Dissolve 1 lozenge in the mouth every 2–3 hours.
- **Eye films** (Membranulae ophthalmicae) — sterile polymer films that dissolve in tear fluid: Rp.: Membranulas ophthalmicas cum Pilocarpini hydrochlorido N. 30 / D.S. Place 1 film behind the lower eyelid once a day.
- **Medicinal herbal species** (Species) — a mixture of crushed plant parts; infusions or decoctions are prepared from them at home.

7. Clinical part: choosing the right dosage form

The effectiveness and safety of the same drug substance depend strongly on the dosage form. The GP chooses the dosage form taking the following into account:

Table 1.12. *Choosing the dosage form according to the clinical situation*

Situation	Appropriate form	Reason
Angina attack	Nitroglycerin sublingual tablet or spray	Acts in 1–2 minutes, bypasses the liver
Vomiting child, fever	Paracetamol rectal suppository	Oral intake impossible, absorption adequate
Child under 3 years	Suspension, syrup, mixture prepared from granules	Cannot swallow tablets; easy to adjust the dose to weight
Chronic pain, oncology	Fentanyl TTS, morphine retard	Steady concentration, fewer doses
Local skin inflammation	Ointment, cream, gel	High local concentration, few systemic adverse effects
Weeping dermatitis	Paste, powder (dusting)	Drying and adsorption
Dry, flaking skin	Ointment (petrolatum or lanolin base)	Occlusion, moisturizing
The patient forgets to take the drug many times a day	Retard tablet, TTS	Reducing the number of doses

CLINICAL SIGNIFICANCE

Adherence (compliance). Patients adhere much better to preparations taken once a day than to those taken 3–4 times a day. That is why, in chronic diseases (arterial hypertension, diabetes mellitus), extended-release forms are chosen whenever possible.

CAUTION

Common prescription errors: (1) a missing or misplaced decimal point in the dose; (2) writing the substance name in the nominative case (“Analginum 0.5” instead of “Analgini 0.5”); (3) omitting the dosage form or the number in the D.t.d. part; (4) writing the signatura in Latin or vaguely; (5) prescribing a list A or narcotic drug on an ordinary form; (6) no confirmation mark when the maximum dose is exceeded.

Practical skills

Skill 1. Algorithm for prescribing a dosed solid dosage form (tablet, capsule, dragée)

1. Fill in the top part of the form: date, patient's full name and age, physician's full name.
2. Write the **Rp.:** sign.
3. Write the name of the drug substance **with a capital letter, in the genitive case** and indicate its **single** dose in grams (Paracetamoli 0.5).
4. On a new line write **D.t.d. N.** and the number of doses, then add the dosage form with the preposition in in the ablative case (in tabulettis, in capsulis; the form is not written for dragées).
5. On a new line, after **S.**, write the method of use in the state language: amount, number of doses, timing in relation to meals, duration.
6. Sign and apply your personal seal.
7. Compare the dose with the maximum dose; adjust it to age for children and the elderly.

Skill 2. Algorithm for writing a prescription for a magistral ointment (paste, liniment)

1. Determine the total mass and the concentration of each active substance.
2. Calculate the amount of each active substance: **mass × percentage / 100**.
3. After Rp.: write each active substance on a separate line.
4. Write the base with **“ad”** up to the total mass (Vaselini ad 50.0).
5. Write **M.f. unguentum** (pasta, linimentum).
6. After **D.S.** write the signatura.
7. For a paste, check that the total of powdered substances is not less than 25%.

Skill 3. Algorithm for writing a prescription for a magistral suppository

1. After Rp.: write the dose of the active substance **for one suppository**.
2. Write the base (Olei Cacao) in the amount for one suppository or with **q.s.**
3. Write **M.f. suppositorium rectale (vaginale)**.
4. Write **D.t.d. N.** and the number of suppositories.
5. In the signatura, state into which cavity, how many times and when it is to be inserted.

Review questions

1. What is prescription writing and into which parts is it divided?
2. Explain the medical, legal and financial significance of a prescription.
3. What is the difference between a drug substance, a medicinal product, a drug preparation and a dosage form?
4. What is the State Pharmacopoeia? Which drugs belong to lists A and B?

5. Explain the difference between officinal and magistral drugs with examples.
6. Give examples of chemical, international nonproprietary and trade names of drugs.
7. What types of doses do you know? What is the therapeutic range?
8. How are mass, volume, drops and units of action written in a prescription?
9. List the parts of a prescription. Which of them are written in Latin?
10. What do the abbreviations Cito!, Statim!, Repetatur!, ana, q.s. and ad mean?
11. What types of prescription forms are there?
12. What is the difference between an ointment and a paste? What requirements are placed on ointment bases?
13. When are the abbreviated and extended forms of prescribing ointments used?
14. What are the types and bases of suppositories, and what is their resorptive effect?
15. Describe the classification of powders and the rules for prescribing them.
16. List the types of tablets. Why must retard tablets not be split?
17. What types of capsules are there and what are their advantages?
18. What are the advantages of transdermal therapeutic systems?

Test questions

1. Which part of the prescription contains the physician's instructions to the pharmacist about the dosage form?
    A) Subscriptio
    B) Designatio materiarum
    C) Invocatio
    D) Signatura
    E) Inscriptio

2. How is 10 mg of a substance written in a prescription?
    A) 0,1
    B) 0,01
    C) 1,0
    D) 10,0
    E) 0,001

3. Translation of the abbreviation “D.t.d. N. 10”:
    A) Give, label
    B) As much as needed
    C) Mix to make a powder
    D) Give 10 such doses
    E) Let it be repeated 10 times

4. If no ointment base is specified in the prescription, which base is used?
    A) Cocoa butter
    B) Lard
    C) Lanolin
    D) Petrolatum
    E) Sunflower oil

5. The main difference between a paste and an ointment:
   A) It contains more than 25% powdered substances
   B) It is a dosed dosage form
   C) It melts at body temperature
   D) It is used only in the eye
   E) It is prepared only on a lanolin base

6. How much active substance is taken to prepare 40.0 g of a 5% ointment?
   A) 4,0
   B) 1,0
   C) 0,5
   D) 2,0
   E) 5,0

7. The most commonly used base for rectal suppositories:
   A) Petrolatum
   B) Paraffin
   C) Lanolin
   D) Cocoa butter
   E) Starch

8. What is done when a substance with a dose of less than 0.1 is prescribed as a powder?
   A) An indifferent filler (e.g., sugar) is added
   B) It is prescribed only in capsules
   C) It is wrapped in waxed paper
   D) Prescribing it as a powder is forbidden
   E) The dose is increased to 0.1

9. Which dosage form must not be split or chewed?
   A) Ordinary scored tablet
   B) Water-soluble tablet
   C) Chewable tablet
   D) Granules
   E) Extended-release (retard) tablet

10. Which parts of the prescription are written in Latin?
    A) Invocatio, Designatio materiarum, Subscriptio
    B) All parts
    C) Only the signatura
    D) Inscriptio and Datum
    E) Nomen aegroti and Nomen medici

11. Meaning of the abbreviation “Ana” (āā):
    A) Let it be repeated
    B) In equal amounts
    C) Mix
    D) up to ...
    E) As much as needed

12. The main advantage of a transdermal therapeutic system:
    A) It has only a local effect
    B) The drug bypasses the liver and a steady blood concentration is maintained
    C) It does not cause allergic reactions
    D) The effect begins within a few seconds
    E) Its effectiveness increases safely when heated

13. What is written on the prescription if the drug must be prepared urgently?
    A) Repetatur!
    B) Da!
    C) Cito!
    D) Misce!
    E) Sterilisetur!

14. When prescribing officinal suppositories, in which case is the name of the active substance written after “cum”?
    A) Genitive
    B) Ablative case
    C) Accusative case
    D) Nominative
    E) Dative case

15. Which substance is not recommended to be prescribed as a powder?
    A) Talc
    B) Magnesium oxide
    C) Streptocide
    D) Calcium chloride
    E) Zinc oxide

Clinical cases

Case 1

A GP found that a 4-year-old child (weight 16 kg) has a temperature of 39 °C during an acute respiratory infection. The child is vomiting and refuses to take medicine by mouth. The physician wants to prescribe paracetamol.

Questions: 1) Which dosage form is most appropriate? 2) Calculate the single dose (the single dose of paracetamol is 10–15 mg/kg). 3) Write a prescription.

Answer: 1) Because the child is vomiting, a rectal suppository is appropriate: the drug partly bypasses the liver and is absorbed even when oral intake is impossible. 2) 16 kg × 15 mg/kg = 240 mg ≈ 0.25 g. 3) Prescription: ANSWER TO CASE 1 Rp.: Suppositorium cum Paracetamolo 0.25 D.t.d. N. 10 S. Insert 1 suppository into the rectum when the temperature is above 38.5 °C; not more than 4 times a day, at least 6 hours apart.

Case 2

A student wrote the following prescription: Rp.: Neomycinum sulfas 1% 50 / Vaselinum / D.S. for application.

Questions: Find the errors in the prescription and write the correct version.

Answer: Errors: 1) the substance name is in the nominative case — it should be in the genitive (Neomycini sulfatis); 2) the mass should be “50.0”, not “50”; 3) if the base is petrolatum and there is a single active substance, the abbreviated form is sufficient; in the extended form, the amount of active substance should be calculated in grams (0.5) and “ad” and “M.f. unguentum” written; 4) the signatura is vague — where and how many times to apply is not stated. Correct version: ANSWER TO CASE 2 Rp.: Unguenti Neomycini sulfatis 1% — 50.0 D.S. Apply a thin layer to the infected wound surface 2 times a day.

Case 3

A 68-year-old patient needs a nonsteroidal anti-inflammatory drug for knee joint pain caused by gonarthrosis. The history includes a gastric ulcer and arterial hypertension.

Questions: 1) Which dosage form is safer and why? 2) Write a prescription.

Answer: 1) In an elderly patient with a history of gastric ulcer, a systemic NSAID (tablet) increases the risk of gastrointestinal bleeding and of a rise in blood pressure. A topical gel creates a sufficient concentration in the joint area, very little is absorbed into the blood, and the risk of systemic adverse effects is much lower. 2) Prescription: ANSWER TO CASE 3 Rp.: Geli Diclofenaci 1% — 50.0 D.S. Rub a thin layer into the knee joint area 3–4 times a day; do not use for more than 14 days.

Quick review table

Table 1.13. *Prescribing solid and soft dosage forms — summary*

Dosage form	Latin (genitive)	Officinal / magistral	Prescription formula
Tablet	Tabulettae	Officinal only	Rp.: Moddai 0.5 / D.t.d. N. 20 in tabulettis / S. ... or Rp.: Tabulettas Moddai 0.5 N. 20 / D.S. ...
Dragée	Dragee (not declined)	Officinal only	Rp.: Dragee Moddai 0.05 / D.t.d. N. 20 / S. ...
Capsule	Capsulae	Mainly officinal	Rp.: Moddai 0.25 / D.t.d. N. 20 in capsulis / S. ...
Powder (divided)	Pulveris	Both	Rp.: Moddai 0.3 / D.t.d. N. 10 / S. ... (if the dose is less than 0.1, a filler + M.f. pulvis)
Powder (undivided)	Pulveris	Both	Rp.: Pulveris Moddai 20.0 / D.S. ...
Granules	Granulorum	Officinal	Rp.: Granulorum Moddai 100.0 / D.S. ...
Ointment	Unguenti	Both	Abbreviated: Rp.: Unguenti Moddai 1% — 30.0 / D.S.; extended: substance / base ad / M.f. unguentum / D.S.
Paste	Pastae	Both	Rp.: Pastae Zinci 40.0 / D.S. or extended + M.f. pasta
Cream	Cremoris	Officinal	Rp.: Cremoris Moddai 1% — 15.0 / D.S. ...
Gel	Geli	Officinal	Rp.: Geli Moddai 1% — 50.0 / D.S. ...
Liniment	Linimenti	Both	Rp.: Linimenti Moddai 10% — 25.0 / D.S. or extended + M.f. linimentum
Suppository	Suppositorii	Both	Officinal: Rp.: Suppositorium cum Moddao 0.25 / D.t.d. N. 10 / S.; magistral: substance / Olei Cacao 3.0 / M.f. suppositorium / D.t.d. N. / S.
Plaster	Emplastri	Officinal	Rp.: Emplastri ... / D.S. ...

TOPIC 2

Liquid dosage forms for enteral use and the rules for prescribing them. Liquid dosage forms for parenteral and topical use and the rules for prescribing them

Learning objectives

Liquid dosage forms are used most often in pediatrics, geriatrics and emergency care. A solution is absorbed quickly and its dose is easy to adjust to the patient's weight. However, prescriptions for solutions involve a lot of calculation: concentration, total volume and number of doses depend on one another. A calculation error leads to the patient receiving a dose several times too high.

The student should know:

- the classification of liquid dosage forms and their advantages and disadvantages;
- the types of solutions (aqueous, alcoholic, oily; for oral, topical and injection use) and the ways of expressing concentration;
- the properties of infusions, decoctions, tinctures, extracts, mixtures, emulsions, suspensions and mucilages;
- the requirements for solutions for injection and eye drops;
- the basic concepts of gaseous dosage forms (aerosols, inhalers).

The student should be able to:

- write prescriptions for solutions in the extended, abbreviated and percentage forms;
- calculate the total amount of substance and the volume from the single dose and the number of doses;
- perform calculations for preparing and diluting solutions of different concentrations;
- write prescriptions for drugs in ampoules and vials, including a separate prescription for the solvent.

1. Classification of liquid dosage forms

Liquid dosage forms (Formae medicamentorum fluidae) — dosage forms in which the drug substance is dissolved, suspended or emulsified in a liquid.

Table 2.1. *Liquid dosage forms*

Dosage form	Latin (nominative / genitive)	Brief description
Solution	Solutio / Solutionis	A clear liquid in which the substance is completely dissolved in the solvent
Infusion	Infusum / Infusi	An aqueous extract from the delicate parts of a plant
Decoction	Decoctum / Decocti	An aqueous extract from the coarse parts of a plant
Tincture	Tinctura / Tincturae	An alcoholic extract from a plant (officinal)
Liquid extract	Extractum fluidum / Extracti fluidi	A concentrated extract (usually 1:1)
Mixture	Mixtura (not written in the prescription)	A liquid mixture of three or more ingredients
Emulsion	Emulsum / Emulsi	A suspension of oil droplets in water

Dosage form	Latin (nominative / genitive)	Brief description
Suspension	Suspensio / Suspensionis	A suspension of solid particles in a liquid
Mucilage	Mucilago / Mucilaginis	A viscous colloidal solution; reduces irritant effects
Syrup	Sirupus / Sirupi	A thick sugar solution; a corrigent or a dosage form
Neogalenic preparations	Praeparata neogalenica	Plant preparations purified from ballast substances
Injection solutions	Formae pro injectionibus	Sterile, pyrogen-free solution, suspension or emulsion

Table 2.2. *Advantages and disadvantages of liquid dosage forms*

Advantages	Disadvantages
Rapid absorption, rapid onset of action	Unstable, spoil quickly (especially infusions and decoctions — 3–4 days)
Convenient for children and patients who cannot swallow	Dosing is less accurate than with tablets (spoons, drops)
Easy to adjust the dose to weight	Taste and smell are harder to mask
Irritant substances can be given diluted	Inconvenient to transport and store, prone to microbial contamination
Can be given parenterally (injection solutions)	Often require refrigerated storage

2. Solutions (Solutiones)

Solution (Solutio, onis f; abbreviated Sol.) — a clear liquid dosage form obtained by dissolving one or more substances in a solvent. A solution consists of two parts: the **solute** and the **solvent**.

Solvents: distilled water (Aqua destillata), purified water (Aqua purificata), ethyl alcohol (Spiritus aethylicus), glycerin (Glycerinum), vegetable oils (peach, almond, olive, sunflower oil), liquid paraffin.

> **REMEMBER**
> If no solvent is specified in the prescription, the solution is prepared in **distilled water**. For an alcoholic solution the word spirituosae is added, for an oily solution oleosae: Solutionis Iodi spirituosae 5%, Solutionis Camphorae oleosae 10%.

2.1. Ways of expressing the concentration of a solution

Table 2.3. *Ways of expressing concentration*

Method	Meaning	Example
Percentage (%)	Grams of substance in 100 ml of solution	2% — 2.0 g in 100 ml
Ratio	The ratio of the substance to the volume of solution	1:5000 — 1.0 g in 5000 ml
Mass – volume	Total amount of substance and volume of solution	3.6 — 180 ml
Mass/volume (mg/ml)	Milligrams in 1 ml	40 mg/ml
Units of action	U in 1 ml	5000 U in 1 ml (heparin)

> **MEMORY AID**
> Converting a percentage into a ratio: **1% = 1:100**, **0.1% = 1:1000**, **0.02% = 1:5000** (furacilin), **0.01% = 1:10,000**. The second number of the ratio = 100 / percentage.

2.2. Solutions for oral use

Solutions for oral use are dosed with a spoon or in drops. Solutions taken by the spoon are usually prescribed **3 times a day for 4 days — 12 doses in total** (the solution does not keep long).

Calculation algorithm:

1. Number of doses: times a day × days (e.g., 3 × 4 = 12).
2. Amount of substance: single dose × number of doses.
3. Volume of solution: spoon volume × number of doses (tablespoon: 15 × 12 = 180 ml; dessert spoon: 10 × 12 = 120 ml; teaspoon: 5 × 12 = 60 ml). Example. Write a prescription for a sodium bromide solution with a single dose of 0.3, to be taken 1 tablespoon 3 times a day.

Calculation: 0.3 × 12 = **3.6**; 15 ml × 12 = **180 ml**; percentage: 3.6 × 100 / 180 = **2%**.

Solutions can be prescribed in **three forms**:

PRESCRIPTION 2.1. EXTENDED FORM
Rp.: Natrii bromidi 3.6
Aquae destillatae ad 180 ml
M.f. solutio
D.S. Take 1 tablespoon 3 times a day.

PRESCRIPTION 2.2. ABBREVIATED FORM (MASS – VOLUME)
Rp.: Solutionis Natrii bromidi 3.6 — 180 ml
D.S. Take 1 tablespoon 3 times a day.

PRESCRIPTION 2.3. ABBREVIATED PERCENTAGE FORM
Rp.: Solutionis Natrii bromidi 2% — 180 ml
D.S. Take 1 tablespoon 3 times a day.

REMEMBER
The three forms express the same drug. The extended form is used for magistral solutions; the abbreviated forms are used more often. In **officinal solutions** (Cordiamine, Validol, dilute hydrochloric acid) the concentration is **not indicated** — only the name and the amount are written: Rp.: Validoli 5 ml.

2.3. Drops for oral use

Potent and low-dose substances are given in drops. The single dose is usually **5–20 drops**; the prescription is written for **20 doses**. 1 ml of aqueous solution = 20 drops.

Example. Write a prescription for an atropine sulfate solution with a single dose of 0.0005, to be taken 10 drops at a time.

Calculation: 0.0005 × 20 = **0.01**; 10 drops × 20 = 200 drops = **10 ml**; percentage: 0.01 × 100 / 10 = **0.1%**.

PRESCRIPTION 2.4. DROPS (PERCENTAGE FORM)
Rp.: Solutionis Atropini sulfatis 0.1% — 10 ml
D.S. Take 10 drops 2 times a day before meals.

3. Liquid forms prepared from medicinal plant raw material

3.1. Infusions and decoctions (Infusa et Decocta)

Infusion — prepared from the delicate parts of a plant (leaf — folium, flower — flos, herb — herba): the raw material is heated with boiling water in a water bath for **15 minutes**, then cooled for **45 minutes** and strained.

Decoction — prepared from coarse parts (bark — cortex, root — radix, rhizome — rhizoma): heated in a water bath for **30 minutes** and strained **while hot**. Exception: bearberry (Uva ursi) leaf is thick, so a decoction is made from it; conversely, an infusion is made from roots containing essential oils (valerian).

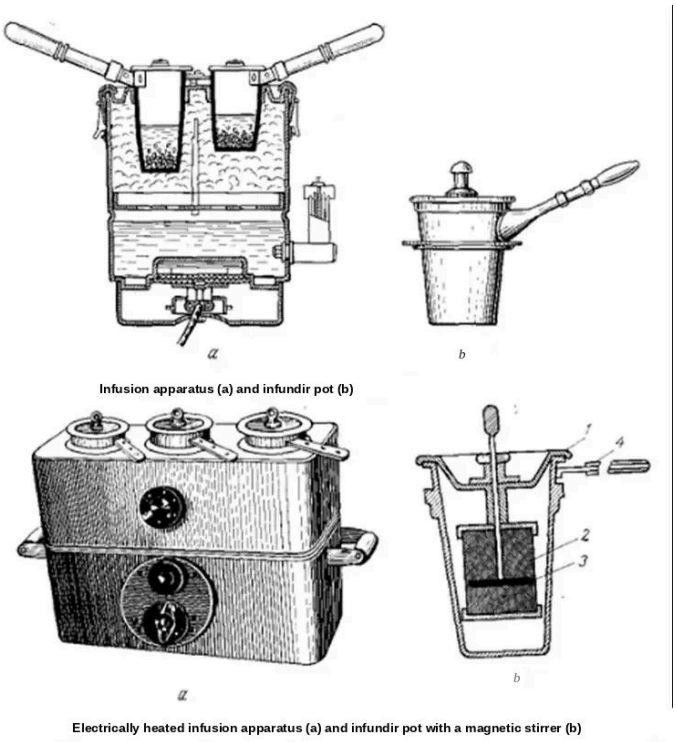


Figure 2.1. *Equipment for preparing an infusion: an infundir pot and a water bath (TMA "Fundamentals of prescription writing", 2020)*

Table 2.4. *Standard ratios for infusions and decoctions (if not specified in the prescription)*

Raw material	Ratio	Examples
Ordinary (mild) plants	1:10	Chamomile flower, peppermint leaf, oak bark, bearberry leaf
Potent plants	1:30	Spring adonis herb, valerian root, lily of the valley herb
Poisonous plants (containing potent substances)	1:300	Foxglove leaf, thermopsis herb, ipecacuanha root

CAUTION

The standard ratios are used only when the amount of raw material is not given in the prescription (S.S. Azizova, department manual). When the amount of raw material is calculated from the single dose (e.g., thermopsis herb 0.6 — 180 ml), the ratio follows from the prescription itself. Infusions of foxglove, thermopsis and ipecacuanha are hardly ever prescribed in current practice — they have been replaced by standardized tablets and syrups.

Infusions and decoctions are prescribed **only in the abbreviated form**: the dosage form (Infusi, Decocti), the plant part with a small letter, the plant name with a capital letter, the mass of raw material and the volume of the finished drug.

PRESCRIPTION 2.5. INFUSION (SPRING ADONIS HERB, SINGLE DOSE 0.5)
Rp.: Infusi herbae Adonidis vernalis 6.0 — 180 ml
D.S. Take 1 tablespoon 3 times a day.

Calculation: 0.5 × 12 = 6.0; 15 × 12 = 180 ml; ratio 6:180 = 1:30.

PRESCRIPTION 2.6. DECOCTION (OAK BARK, 1:10)
Rp.: Decocti corticis Quercus 20.0 — 200 ml
D.S. For rinsing the mouth 4–5 times a day.

3.2. Tinctures and extracts

Tincture (Tinctura) — a clear, colored extract obtained from plant raw material with 40% or 70% ethyl alcohol. Because alcohol is a preservative, tinctures keep for a long time. All tinctures are **officinal**; they are dosed in drops.

PRESCRIPTION 2.7. TINCTURE
Rp.: Tincturae Valerianae 30 ml
D.S. Take 20–25 drops 3 times a day.

Several tinctures can be prescribed as a mixture; in that case **M.D.S.** is used:

PRESCRIPTION 2.8. MIXTURE OF TINCTURES
Rp.: Tincturae Valerianae Tincturae Leonuri ana 15 ml
M.D.S. Take 30 drops 3 times a day.

Extracts (Extracta) — concentrated extracts: liquid (fluida, 1:1), thick (spissa, moisture up to 25%) and dry (sicca, moisture up to 5%). Liquid extracts are given in drops; thick and dry extracts are given as part of tablets, capsules and suppositories.

PRESCRIPTION 2.9. LIQUID EXTRACT
Rp.: Extracti Crataegi fluidi 25 ml
D.S. Take 20–30 drops 3 times a day before meals.

Neogalenic preparations — standardized plant preparations maximally purified from ballast substances; some are suitable for injection. Examples: Corglycon, strophanthin, Adonisid.

4. Mixtures, emulsions, suspensions, mucilages

4.1. Mixture (Mixtura)

Mixture — a liquid mixture of **3 or more** ingredients consisting of dissolved solids and various liquid dosage forms (solution, infusion, tincture, syrup). Mixtures that form a precipitate are shaken before use (mixturae agitandae). The word “Mixtura” is not written in the prescription; **M.D.S.** is placed at the end.

PRESCRIPTION 2.10. MIXTURE (ADONIS INFUSION + SODIUM BROMIDE + CODEINE)
Rp.: Infusi herbae Adonidis vernalis 6.0 — 180 ml Natrii bromidi 3.6
Codeini phosphatis 0.12
M.D.S. Take 1 tablespoon 3 times a day.

Corrigents — substances that improve the taste and smell of a drug: sugar syrups (Sirupus simplex, Sirupus Rubi idaei — raspberry syrup), aromatic waters (Aqua Menthae piperitae — peppermint water). **Mucilages** are added to irritant drugs: starch mucilage (Mucilago Amyli), marshmallow root mucilage (Mucilago radicis Althaeae). Syrups and mucilages are usually added in an amount of **1/3 of** the total volume.

4.2. Emulsions (Emulsa)

Emulsion (Emulsum, i n) — a milky dosage form consisting of two immiscible liquids (water and oil). Stability is provided by **emulsifiers**: gelatose, gum arabic, egg yolk, Tweens. The classic ratio for preparing an oil emulsion: **oil : emulsifier : water = 2 : 1 : 17**; if the amount of oil is not specified, the emulsion is prepared at a ratio of 1:10.

PRESCRIPTION 2.11. OIL EMULSION (ABBREVIATED FORM)
Rp.: Emulsi olei Ricini 20.0 — 200 ml
D.S. Take all at once (as a laxative).

CLINICAL SIGNIFICANCE
In modern practice, the most important emulsions are **emulsions for intravenous use**: propofol (for general anesthesia) and fat emulsions for parenteral nutrition. Oily solutions normally cannot be given intravenously, but special emulsions with very fine particles (less than 1 μm) are an exception.

4.3. Suspensions (Suspensiones)

Suspension (Suspensio, onis f) — solid drug particles that are insoluble in the liquid, held in suspension. It is used orally, topically and intramuscularly. **It must not be given intravenously.** The signatura must include "**Shake before use!**".

PRESCRIPTION 2.12. ANTIBIOTIC SUSPENSION FOR A CHILD
Rp.: Suspensionis Amoxicillini 250 mg/5 ml — 100 ml
D.S. Shake before use. Take 5 ml 3 times a day for 7 days.

CLINICAL SIGNIFICANCE
Antibiotics for children are often produced as **powders or granules for oral suspension**. Boiled and cooled water is poured into the bottle up to the mark and shaken. The finished suspension is kept in the refrigerator for 7–14 days. Parents should be taught to use a dosing syringe or spoon: the volume of kitchen spoons is not standard.

5. Liquid dosage forms for topical use

Solutions for topical use are used for wiping the skin, washing wounds, rinsing, compresses (lotions), douching and enemas, and also as eye, ear and nose drops.

Table 2.5. *Amounts of solutions for topical use*

Purpose	Usual amount
Eye drops	5–10 ml

Purpose	Usual amount
Ear and nose drops	10–20 ml
Gargling, compresses	100–200 ml
Douching, washing wounds	500–1000 ml
Application to the skin (alcoholic solutions)	10–50 ml

PRESCRIPTION 2.13. NITROFURAL (FURACILIN) SOLUTION (RATIO FORM)
Rp.: Solutionis Furacilini 1:5000 — 200 ml
D.S. Gargle 3–4 times a day.

PRESCRIPTION 2.14. ALCOHOLIC SOLUTION
Rp.: Solutionis Viridis nitentis spirituosae 1% — 10 ml
D.S. Apply to the skin around the wound.

PRESCRIPTION 2.15. OFFICINAL SOLUTION (CONCENTRATION NOT WRITTEN)
Rp.: Solutionis Hydrogenii peroxydi diluti 50 ml
D.S. For washing the wound.

Prescribing a concentrated solution. When a large volume is needed (gargling, douching), the patient is prescribed a concentrated solution with instructions for diluting it at home.

Example: the throat needs to be gargled 10 times with a 0.02% potassium permanganate solution (200 ml each time). For one gargle: 200 × 0.02 / 100 = 0.04; for 10 times: **0.4**. Let each dose be contained in 1 tablespoon (15 ml) of concentrated solution: 15 × 10 = **150 ml**.

PRESCRIPTION 2.16. CONCENTRATED SOLUTION
Rp.: Kalii permanganatis 0.4
Aquae destillatae ad 150 ml
M.f. solutio
D.S. Dilute 1 tablespoon of the solution in 1 glass of warm water and gargle.

5.1. Eye drops (Guttae ophthalmicae)

Requirements for eye drops: they must be **sterile**, **free of mechanical particles**, as far as possible **isotonic** (osmotic pressure equal to 0.9% NaCl) and close to the pH of tear fluid, and chemically stable. Eye drops are prescribed in 5–10 ml, enough for **20 applications**; 1–2 drops are instilled into each eye at a time.



Figure 2.2. *Eye drops (TMA "Fundamentals of prescription writing", 2020)*

PRESCRIPTION 2.17. EYE DROPS

Rp.: Solutionis Pilocarpini hydrochloridi 1% — 10 ml
D.S. Instill 1–2 drops into each eye 3 times a day.

CLINICAL SIGNIFICANCE

Pressing on the lacrimal duct area (the inner corner of the eye) with a finger for 1–2 minutes after instilling eye drops reduces drainage of the drug into the nasopharynx and its absorption into the blood. This is especially important with beta-blocker drops (timolol) to prevent bronchospasm and bradycardia.

5.2. Enemas and mucilages

Therapeutic (medicated) enemas are given after a cleansing enema; the single volume does not exceed **50–100 ml**. Irritant substances are given mixed with a mucilage. The prescription is written only in the extended form.

PRESCRIPTION 2.18. THERAPEUTIC ENEMA

Rp.: Chlorali hydrati 1.0
Mucilaginis Amyli Aquae destillatae ana 25 ml
M.D.S. For one enema. Introduce warm into the rectum after a cleansing enema.

6. Dosage forms for injection (Formae medicamentorum pro injectionibus)

For injection, aqueous and oily solutions, suspensions, emulsions and sterile powders to be dissolved before use are used.



Figure 2.3. *Dosage forms for injection: ampoules, vials, infusion solutions (TMA "Fundamentals of prescription writing", 2020)*

Table 2.6. *Requirements for injection solutions*

Requirement	Content
Sterility	No microorganisms; preparation under aseptic conditions or sterilization in an autoclave
Pyrogen-free	No fever-causing substances (bacterial endotoxins); especially important for large-volume infusions
No mechanical particles	Clarity; prevention of the risk of embolism
Stability	Does not decompose during storage; stabilizers may be added
Isotonicity	Especially for large volumes given intravenously and into tissues (0.9% NaCl, 5% glucose)
Closeness to physiological pH	Less pain and phlebitis

Solvents: water for injection (Aqua pro injectionibus), isotonic sodium chloride solution (Solutio Natrii chloridi isotonica 0.9%), 5% glucose solution, sterile peach or almond oil (intramuscular only).

Table 2.7. *Parenteral routes and volumes*

Route	Single volume	What must not be given
Intradermal (i/d)	0.1–0.2 ml	— (allergy tests, BCG)
Subcutaneous (s/c)	1–2 ml	Irritant and hypertonic solutions, alcoholic solutions
Intramuscular (i/m)	1–10 ml	Calcium chloride, irritant solutions, large volumes
Intravenous (i/v)	1–500 ml and more	Oily solutions, suspensions, ordinary emulsions, air bubbles
Into the spinal canal (intrathecal)	1–10 ml	Solutions with preservatives

CAUTION

Oily solutions, suspensions and ordinary emulsions must not be given intravenously — risk of fat and particle embolism. **Alcoholic solutions are not given subcutaneously** — tissue necrosis. **Calcium chloride is given only intravenously** (necrosis when given intramuscularly or subcutaneously). **A concentrated potassium chloride solution is not given as an intravenous push** — only diluted, slowly by drip (risk of cardiac arrest).

6.1. Prescribing injection solutions

Solutions in ampoules (divided into doses):

PRESCRIPTION 2.19. SOLUTION IN AMPOULES (PERCENTAGE FORM)

Rp.: Solutionis Diclofenaci natrii 2.5% — 3 ml
D.t.d. N. 5 in ampullis
S. Inject 3 ml deep intramuscularly once a day.

PRESCRIPTION 2.20. SOLUTION IN AMPOULES (MASS – VOLUME FORM)

Rp.: Solutionis Dexamethasoni 0.004 — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml intravenously or intramuscularly 1–2 times a day.

Infusion solutions in bottles (not divided into doses) — **only in the percentage form**:

PRESCRIPTION 2.21. INFUSION SOLUTION

Rp.: Solutionis Natrii chloridi isotonicae 0.9% — 400 ml
D.S. Give by intravenous drip at a rate of 40–60 drops per minute.

Oily solutions are given only intramuscularly or subcutaneously, **warmed to body temperature**:

PRESCRIPTION 2.22. OILY SOLUTION IN AMPOULES

Rp.: Solutionis Progesteroni oleosae 2.5% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml intramuscularly (warm to body temperature first).

Sterile powders. Drugs that are unstable in solution (most antibiotics) are produced as dry powder in vials and dissolved before use (ex tempore). In this case, **a separate prescription is written for the solvent**, and the two prescriptions are separated by the # sign.

PRESCRIPTION 2.23. STERILE POWDER IN A VIAL AND ITS SOLVENT

Rp.: Ceftriaxoni 1.0
D.t.d. N. 10
S. Dissolve the powder in the vial in 3.5 ml of 1% lidocaine solution and inject intramuscularly once a day.
#
Rp.: Solutionis Lidocaini hydrochloridi 1% — 3.5 ml
D.t.d. N. 10 in ampullis
S. For preparing the ceftriaxone solution.

CAUTION

Ceftriaxone diluted with lidocaine is given **only intramuscularly**; for intravenous use it is dissolved in water for injection. Lidocaine is not used in patients allergic to lidocaine or with impaired cardiac conduction. Giving ceftriaxone in the same line with calcium-containing solutions (Ringer's solution), or simultaneously with calcium solutions in newborns, is forbidden (precipitate formation in the lungs and kidneys).

Officinal and neogalenic preparations in ampoules are written by name only: Rp.: Corglyconi 0.06% — 1 ml / D.t.d. N. 10 in ampullis / S. ...

7. Gaseous dosage forms: aerosols and inhalers

Aerosol (Aerosolum, i n) — a dosage form in which the drug substance is dispersed as fine particles or droplets in a gaseous medium (air or a propellant). It is produced in pressurized canisters. **Spray** — a propellant-free form sprayed with a mechanical pump.

Table 2.8. *Types of inhalation devices*

Device	Working principle	Advantage / disadvantage
Metered-dose aerosol inhaler (MDI)	Under propellant pressure, an exact dose is released with each press	Cheap and widespread; requires coordinating the press with inhalation (a spacer helps)
Dry powder inhaler (DPI: Turbuhaler, Diskhaler, capsule-based)	The patient's deep inhalation carries the powder into the lungs	No coordination needed; a strong inhalation is required (unsuitable in a severe attack and in young children)
“Soft mist” inhaler (Respimat)	The energy of a spring creates a slow aerosol cloud	More drug reaches the lungs; a low-velocity cloud
Nebulizer (compressor, ultrasonic, mesh)	Turns a solution into a fine aerosol	Used at any age and in severe conditions; bulky, requires electricity

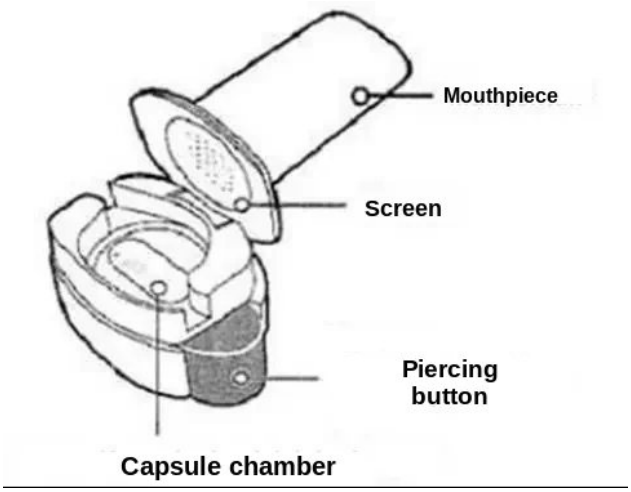


Figure 2.4. *Structure of a capsule-based dry powder inhaler (TMA “Fundamentals of prescription writing”, 2020)*

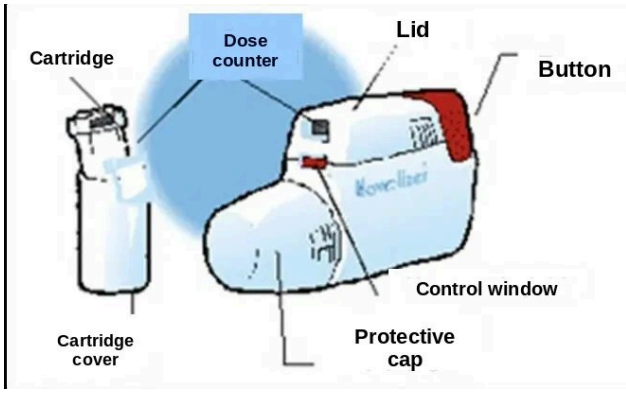


Figure 2.5. *Structure of a metered-dose aerosol inhaler (TMA “Fundamentals of prescription writing”, 2020)*

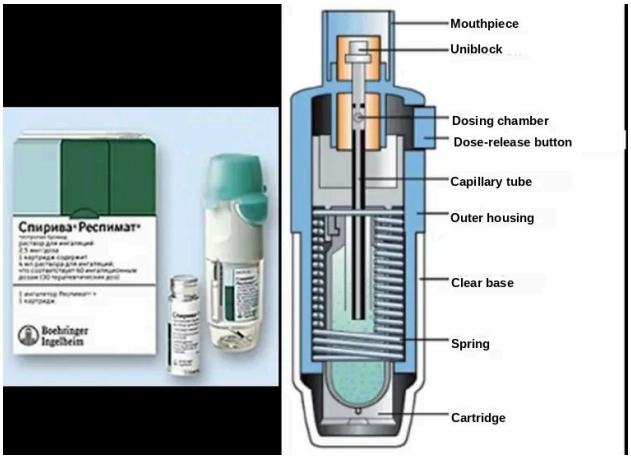


Figure 2.6. *"Respimat" soft mist inhaler (TMA "Fundamentals of prescription writing", 2020)*



Figure 2.7. *Compressor nebulizer (TMA "Fundamentals of prescription writing", 2020)*

REMEMBER

Rules for using a metered-dose aerosol inhaler: 1) remove the cap and shake; 2) breathe out fully; 3) seal the lips tightly around the mouthpiece; 4) press the canister once while starting a slow, deep breath; 5) hold the breath for **5–10 seconds**; 6) breathe out slowly. Rinsing the mouth after a glucocorticoid inhaler is mandatory (to prevent oral candidiasis).

PRESCRIPTION 2.24. METERED-DOSE AEROSOL

Rp.: Aerosoli Salbutamoli 0.0001/dosi — 200 dosi

D.S. Take 1–2 inhalations during an attack of breathlessness; not more than 8 inhalations a day.

PRESCRIPTION 2.25. SOLUTION FOR A NEBULIZER

Rp.: Suspensionis Budesonidi pro inhalationibus 0.25 mg/ml — 2 ml

D.t.d. N. 20

S. Inhale 2 ml through a nebulizer 2 times a day.

8. Dose calculation and preparing solutions of different concentrations

This section corresponds to the 5th-semester laboratory class ("Dose calculation. Preparing solutions of different concentrations").

8.1. Basic formulas

Table 2.9. *Basic calculation formulas for working with solutions*

Quantity sought	Formula
Mass of substance (g)	m = V × C% / 100
Percentage concentration	C% = m × 100 / V
From mg/ml to percentage	C% = (mg/ml) / 10
From percentage to mg/ml	mg/ml = C% × 10
Dilution	$C_1 \times V_1 = C_2 \times V_2$
Substance per ml (ampoule)	mg/ml = C% × 10
Required volume (ml)	V = required dose (mg) / concentration (mg/ml)
Infusion rate (drops/min)	drops/min = volume (ml) × 20 / time (min)

MEMORY AID

“Multiply the percentage by ten — you get mg/ml”: 1% = 10 mg/ml, 2% = 20 mg/ml, 0.1% = 1 mg/ml, 10% = 100 mg/ml, 40% glucose = 400 mg/ml.

8.2. Sample calculations

Example 1. An ampoule contains 2 ml of 2% lidocaine solution. How much lidocaine is in the ampoule?

2% = 20 mg/ml; 20 × 2 = **40 mg**.

Example 2. A patient needs 4 mg of dexamethasone. The ampoule contains a 4 mg/ml solution. How many ml are drawn up?

V = 4 / 4 = **1 ml**.

Example 3. 200 ml of a 5% solution must be prepared from a 40% glucose solution. How much of the 40% solution is taken?

$C_1 \times V_1 = C_2 \times V_2 \rightarrow 40 \times V_1 = 5 \times 200 \rightarrow V_1 =$ **25 ml**; the remaining 175 ml of water for injection is added.

Example 4. A 1000 ml infusion must be given over 8 hours. The drip set gives 1 ml = 20 drops. What is the rate?

1000 × 20 / 480 ≈ **42 drops/minute**.

Example 5. A child weighing 12 kg is prescribed amoxicillin 25 mg/kg 2 times a day. The suspension is 250 mg/5 ml. How many ml are given at a time?

Dose: 12 × 25 = 300 mg; 250 mg — 5 ml, 300 mg — x; x = 300 × 5 / 250 = **6 ml**.

Example 6. How many tablets are needed for 500 ml of a 0.02% nitrofural (furacilin) solution (1 tablet = 20 mg)?

m = 500 × 0.02 / 100 = 0.1 g = 100 mg; 100 / 20 = **5 tablets**.

CAUTION

The most common error in calculations is **mixing up units** (mg and g, mcg and mg). Before starting a calculation, convert all quantities to one unit. For “high-alert” drugs such as insulin, heparin and potassium chloride, it is recommended that a second staff member independently check the calculation.

9. Clinical part: choosing the liquid dosage form and the route of administration

Table 2.10. *Comparison of routes of administration*

Route	Onset of action	Bioavailability	Main indications
Oral (solution)	15–30 minutes	Variable (first pass through the liver)	Planned treatment, children
Sublingual	1–3 minutes	High (bypasses the liver)	Angina attack, hypertensive crisis
Intramuscular	10–20 minutes (aqueous solution)	High	When oral intake is impossible; depot preparations
Subcutaneous	15–30 minutes	High	Insulin, heparins, vaccines
Intravenous	A few seconds	100%	Emergencies, precise titration, large volumes
Inhalation	1–5 minutes	Local (high in the lungs)	Bronchial asthma, COPD

CLINICAL SIGNIFICANCE

In emergencies (anaphylactic shock, status epilepticus, severe hypoglycemia), the **intravenous** or intramuscular route is chosen. In anaphylaxis, giving epinephrine (adrenaline) **intramuscularly** (into the anterolateral thigh) is the first choice: it acts quickly and carries no risk of the arrhythmias seen with intravenous use.

Practical skills

Skill 1. Algorithm for prescribing a solution for oral use

1. Determine the single dose (g) and the method of intake (spoon or drops).
2. Set the number of doses: usually 12 for a spoon, 20 for drops.
3. Total amount of substance = single dose × number of doses.
4. Total volume = spoon volume × 12 (or number of drops × 20 / 20).
5. If necessary, calculate the percentage: substance × 100 / volume.
6. Write the prescription in the chosen form and state the type of spoon and the number of doses in the signatura.

Skill 2. Algorithm for prescribing an injection

1. Determine whether the drug comes in an ampoule (dosed) or a bottle (non-dosed).
2. For a solution in ampoules: Solutionis ... % — ... ml / D.t.d. N. ... in ampullis.
3. For an infusion solution in a bottle: percentage form only, D.S.
4. For a dry powder: a prescription for the powder (D.t.d. N.), then # and a separate prescription for the solvent.
5. State clearly in the signatura the route (intravenously, intramuscularly, subcutaneously), the rate and the number of injections.
6. Check that no oily solution or suspension is prescribed intravenously.

Skill 3. Preparing the required concentration from a concentrated solution

1. Write down the volume and concentration of the required solution (C_2, V_2).
2. Write down the concentration of the available solution (C1).
3. Find the required volume from the formula $V_1 = C_2 \times V_2 / C_1$.
4. Add solvent in a volume of $V_2 - V_1$.
5. Check the result with a reverse calculation.

Review questions

1. List the liquid dosage forms. What are their advantages and disadvantages?
2. Into which types are solutions divided by solvent and by use?
3. How is the concentration of a solution expressed? What percentage is 1:5000?
4. How are the three forms of a prescription for an oral solution written?
5. How many doses are prescribed for solutions dosed in spoons and in drops, and why?
6. How are infusions and decoctions prepared? What is the difference between them?
7. What standard ratios exist for infusions and decoctions?
8. What is the difference between tinctures and extracts? How are they prescribed?

9. What is a mixture? Why are corrigents and mucilages added?
10. What is the difference between emulsions and suspensions? Can they be given intravenously?
11. What requirements are placed on eye drops?
12. What requirements are placed on injection solutions? What does pyrogen-free mean?
13. Which solutions must not be given intravenously, and which must not be given subcutaneously?
14. How are a dry powder in a vial and its solvent prescribed?
15. What types of inhalers are there? How is a metered-dose aerosol inhaler used?
16. How many mg of substance are in 1 ml of a 10% solution? Write the dilution formula.

Test questions

1. How many doses is an oral solution dosed by the tablespoon usually prescribed for, and what is its total volume?
   A) 12 doses, 60 ml
   B) 12 doses, 180 ml
   C) 20 doses, 300 ml
   D) 10 doses, 100 ml
   E) 6 doses, 90 ml

2. What percentage is a 1:5000 nitrofural (furacilin) solution?
   A) 0,002%
   B) 0,02%
   C) 0,5%
   D) 0,2%
   E) 5%

3. Which plant part is used to prepare an infusion?
   A) Bark
   B) Root
   C) Leaf and flower
   D) Rhizome
   E) Seed coat

4. The main difference between preparing a decoction and an infusion:
   A) It is prepared in cold water
   B) It is prepared with alcohol
   C) It is heated in a water bath for 30 minutes and strained while hot
   D) It is dosed only in drops
   E) It is only officinal

5. Find the incorrect statement about tinctures:
   A) They keep for a long time
   B) They are an officinal dosage form
   C) They are alcoholic extracts
   D) They are dosed in drops
   E) They are prepared ex tempore in the pharmacy

6. Dosage form that must not be given intravenously:
   A) 10% calcium chloride solution
   B) Oily solution
   C) Isotonic sodium chloride solution
   D) Aqueous solution in an ampoule
   E) 5% glucose solution

7. By which route is calcium chloride solution given?
   A) Intradermally
   B) Subcutaneous
   C) By any route
   D) Intramuscular
   E) Only intravenously

8. How is a solution prepared when the solvent is not specified in the prescription?
   A) In peach oil
   B) In glycerin
   C) In 70% alcohol
   D) In isotonic solution
   E) In distilled water

9. How much lidocaine is in 1 ml of a 2% lidocaine solution?
   A) 2 mg
   B) 20 mg
   C) 2 g
   D) 0.2 mg
   E) 200 mg

10. What does it mean that an injection solution is pyrogen-free?
    A) Its pH equals 7.4
    B) It contains no fever-causing substances
    C) It contains no microorganisms
    D) It is isotonic
    E) It contains no mechanical particles

11. What is done when prescribing a dry antibiotic powder in a vial?
    A) Separate prescriptions for the powder and the solvent are written, separated by the # sign
    B) M.f. solutio is written
    C) Sterilisetur! is written
    D) Only the powder is written
    E) The powder and the solvent are written on one line

12. What must always be stated in the signatura of a prescription for a suspension?
    A) Take warm
    B) Shake before use
    C) Store in a cool place
    D) Store in the dark
    E) Dispense sterile

13. How much of a 40% glucose solution is taken to prepare 200 ml of a 10% solution?
    A) 100 ml
    B) 20 ml
    C) 50 ml
    D) 40 ml
    E) 80 ml

14. The main disadvantage of a dry powder inhaler:
    A) It contains a propellant
    B) Pressing and inhaling must be coordinated
    C) A strong inhalation is required
    D) It is used only in hospital
    E) It requires a power supply

15. How many doses is a prescription for oral drops usually written for?
    A) 12
    B) 40
    C) 20
    D) 10
    E) 5

Clinical cases

Case 1

A 2-year-old child (weight 12 kg) presents with acute otitis media. The pediatrician wants to prescribe amoxicillin at a daily dose of 90 mg/kg, divided into 2 doses, for 10 days. The pharmacy has powder for preparing a 400 mg/5 ml suspension (70 ml of finished suspension).

Questions: 1) Calculate the single dose in mg and ml. 2) How many bottles are needed for 10 days? 3) Write a prescription.

Answer: 1) Daily dose 12 × 90 = 1080 mg; single dose — 540 mg; 400 mg — 5 ml, 540 mg — x; x = 540 × 5 / 400 ≈ 6.75 ml (in practice 6.8 ml). 2) Daily volume ≈ 13.5 ml; over 10 days ≈ 135 ml → 2 bottles (2 × 70 ml). 3) Prescription: ANSWER TO CASE 1 Rp.: Pulveris pro suspensione Amoxicillini 400 mg/5 ml — 70 ml D.t.d. N. 2 S. Add cold boiled water to the bottle up to the mark and shake. Give 6.8 ml 2 times a day (every 12 hours) for 10 days with a dosing syringe. Shake before use. Store in the refrigerator.

Case 2

A nurse was about to give 10 ml of a 7.5% potassium chloride solution (in an ampoule) as an intravenous push to a patient with hypokalemia.

Questions: 1) What is the nurse's error and what is the danger? 2) State the correct method of administration.

Answer: 1) An intravenous push of concentrated potassium chloride sharply raises the blood potassium concentration and creates a risk of cardiac arrest (asystole, ventricular fibrillation). 2) Potassium chloride must always be diluted (e.g., in 400–500 ml of 0.9% NaCl or 5% glucose solution) and given slowly by drip (into a peripheral vein usually not more than 10 mmol per hour), under ECG and blood potassium monitoring. 10 ml of the 7.5% solution contains ≈ 750 mg KCl ≈ 10 mmol of potassium.

Case 3

A patient needs a 1:10,000 potassium permanganate solution for washing a wound. It is used at 1 liter a day for 10 days. A concentrated solution should be obtained from the pharmacy, with the daily dose in 1 tablespoon.

Questions: 1) Calculate the total amount of substance and the volume of the concentrated solution. 2) Write a prescription.

Answer: 1) 1:10,000 → 0.1 g in 1 liter; for 10 days 1.0 g. If the daily dose is 1 tablespoon (15 ml), the total volume is 15 × 10 = 150 ml. 2) Prescription: ANSWER TO CASE 3 Rp.: Kalii permanganatis 1.0 Aquae destillatae ad 150 ml M.f. solutio D.S. Dilute 1 tablespoon of the solution in 1 liter of warm boiled water and use once a day for washing the wound.

Quick review table

Table 2.11. *Prescribing liquid dosage forms — summary*

Dosage form	Prescription formula	Note
Solution (for oral use)	Rp.: Sol. Moddai 2% — 180 ml / D.S. 1 tablespoon ...	12 doses; three forms
Drops	Rp.: Sol. Moddai 0.1% — 10 ml / D.S. 10 drops ...	20 doses; 1 ml = 20 drops
Officinal solution	Rp.: Validoli 5 ml / D.S. ...	Concentration not written
Alcoholic / oily solution	Rp.: Sol. Moddai spirituosae (oleosae) 1% — 10 ml / D.S.	Type of solvent in words
Infusion / decoction	Rp.: Infusi (Decocti) herbae (corticis) ... 6.0 — 180 ml / D.S.	Abbreviated only; 1:10, 1:30, 1:300
Tincture	Rp.: Tincturae Valerianae 30 ml / D.S. 20 drops ...	Officinal only; M.D.S. in a mixture
Liquid extract	Rp.: Extracti ... fluidi 25 ml / D.S.	1:1
Mixture	All ingredients in sequence / M.D.S.	The word “Mixtura” is not written
Emulsion	Rp.: Emulsi olei ... 20.0 — 200 ml / D.S.	2:1:17
Suspension	Rp.: Susp. Moddai ... — 100 ml / D.S. Shake!	Not for intravenous use
Eye drops	Rp.: Sol. Moddai 1% — 10 ml / D.S. 1–2 drops ...	Sterile, isotonic
Ampoule	Rp.: Sol. Moddai 2% — 2 ml / D.t.d. N. 10 in ampullis / S.	Route stated in the signatura
Infusion solution	Rp.: Sol. Natrii chloridi isotonicae 0.9% — 400 ml / D.S.	Percentage form only
Dry powder + solvent	Powder / D.t.d. N. / S. # Solvent / D.t.d. N. in ampullis / S.	Two prescriptions
Aerosol	Rp.: Aerosoli Moddai 0.0001/dosi — 200 dosi / D.S.	Teach the technique of use

TOPIC 3

General pharmacology. Pharmacokinetics and pharmacodynamics of drug substances. Drugs acting on the afferent nervous system

Learning objectives

General pharmacology is the “grammar” of all pharmacology. When studying each drug group, we keep returning to the same questions: how the drug is absorbed, where it is distributed, how it is broken down and excreted (pharmacokinetics); which target it acts on and what effect it produces (pharmacodynamics). A student who knows these concepts can understand even a new drug logically.

The student should know:

- the tasks and branches of pharmacology;
- the routes of drug administration, the mechanisms of absorption, bioavailability and the first-pass effect;
- the principles of distribution, biological barriers, biotransformation and excretion;
- the main pharmacokinetic parameters (Cmax, AUC, Vd, Cl, t½, Css);
- the types of drug action, drug targets, the types of receptors, and the concepts of agonist and antagonist;
- the dose–effect relationship, the therapeutic index, phenomena on repeated use and drug interactions;
- the classification of drugs acting on the afferent nervous system, and the mechanism and use of local anesthetics.

The student should be able to:

- calculate the time to reach steady-state concentration from the half-life;
- calculate the maximum dose of local anesthetics according to the patient's weight;
- write prescriptions for local anesthetics, astringents, coating agents, adsorbents and irritants.

1. Pharmacology as a science and its branches

Pharmacology (Gr. pharmacon — drug, logos — study) — the science that studies the interaction of chemical compounds with the living organism. Its main task is to search for new effective and safe medicines and to justify the rational use of existing ones.

Table 3.1. *Main branches of pharmacology*

Branch	What it studies
Pharmacokinetics	What the body does to the drug: absorption, distribution, biotransformation, excretion
Pharmacodynamics	What the drug does to the body: site of action, mechanism, effects
Pharmacogenetics	The influence of hereditary factors on drug action
Pharmacoepidemiology	The use and effects of drugs in the population
Toxicology	The toxic effects of substances and the treatment of poisoning
Clinical pharmacology	The effects of drugs in the patient's body; rational pharmacotherapy

MEMORY AID

The four stages of pharmacokinetics — **ADME**: **A**bsorption, **D**istribution, **M**etabolism (biotransformation), **E**limination (excretion). Pharmacokinetics is “the drug's journey”, pharmacodynamics is “the drug's work”.

2. Pharmacokinetics

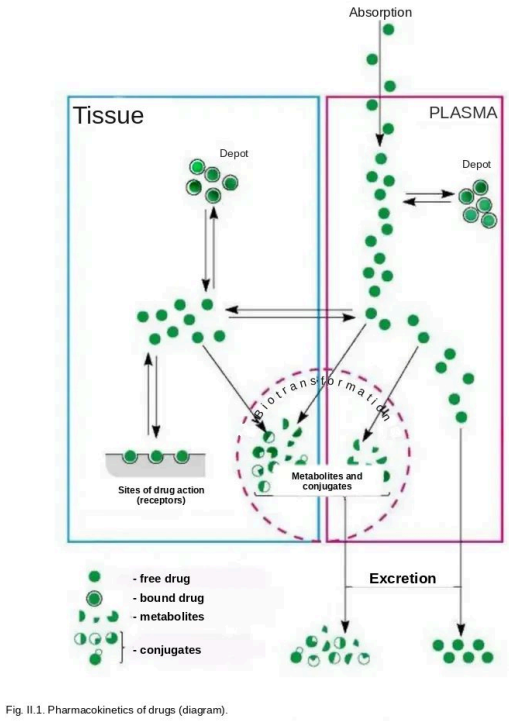


Fig. II.1. Pharmacokinetics of drugs (diagram).

Figure 3.1. *General scheme of drug pharmacokinetics: absorption, distribution to plasma and tissues, depot, metabolism and excretion (teaching material of the TMA Department of Pharmacology)*

2.1. Routes of drug administration

Table 3.2. *Routes of drug administration*

Group	Route	Advantage	Disadvantage
Enteral	Oral (per os)	Simple, safe, cheap	Slow; first-pass effect; influence of gastric juice and food
Enteral	Sublingual, buccal	Rapid action, bypasses the liver	Only low-dose, lipophilic drugs
Enteral	Rectal	In vomiting, in children; partly bypasses the liver	Unpredictable absorption, inconvenient
Parenteral	Intravenous	Immediate action, exact dose, 100% bioavailability	Risk of infection, embolism, anaphylaxis; cannot be reversed
Parenteral	Intramuscular	Rapid; depot preparations	Pain, abscess; dangerous in coagulation disorders
Parenteral	Subcutaneous	Steady absorption (insulin, heparin)	Small volume; irritant substances unsuitable
Parenteral	Intrathecal, intra-articular, intraosseous	Directly to the target	Special skill needed

Group	Route	Advantage	Disadvantage
Other	Inhalation	Rapid and local action in the lungs	The device must be learned
Other	Transdermal, topical (skin, mucous membranes)	Local action or steady systemic absorption	Only lipophilic, potent drugs

2.2. Mechanisms of absorption

Absorption — the passage of a drug from the site of administration into the blood. Main mechanisms:

1. **Passive diffusion** — along the concentration gradient, without energy expenditure, through the lipid membrane. The most common mechanism. Lipophilic and **non-ionized** substances pass well.
2. **Filtration** (through aqueous pores) — water-soluble substances of small molecular weight (up to 100–200 Da).
3. **Active transport** — with the help of carrier proteins, with energy (ATP) expenditure, against the gradient; it can be saturated and competitively inhibited (levodopa, iron, 5-fluorouracil).
4. **Facilitated diffusion** — with a carrier but along the gradient, without energy (glucose).
5. **Pinocytosis (endocytosis)** — engulfment of large molecules by the membrane (vitamin B_{12} — intrinsic factor complex). **Ionization and pH.** Most drugs are weak acids or bases. Their non-ionized (lipophilic) fraction crosses the membrane. The degree of ionization depends on the pH of the medium and the pKa of the substance (**Henderson–Hasselbalch equation**):

- weak **acids** (acetylsalicylic acid, barbiturates) are poorly ionized in an **acidic** medium and are well absorbed;
- weak **bases** (alkaloids: morphine, atropine) are poorly ionized in an **alkaline** medium.

> **REMEMBER**
> **“Like is absorbed in like, trapped in the opposite”**: an acid is non-ionized (absorbed) in an acidic medium and ionized (trapped and excreted) in an alkaline medium. By this rule, alkalinizing the urine (sodium bicarbonate) in poisoning speeds up the excretion of **weak acids** (salicylates, phenobarbital).

2.3. Bioavailability and the first-pass effect

Bioavailability (F) — the fraction (%) of the administered dose that reaches the **systemic circulation** unchanged. With intravenous administration, F = 100%. With oral intake, F depends on: the dosage form, completeness of absorption, and metabolism in the intestinal wall and liver.

First-pass effect (presystemic elimination) — a drug absorbed from the intestine first reaches the liver through the portal vein, and part of it is broken down before reaching the systemic circulation. Drugs with a strong first-pass effect: nitroglycerin (given sublingually), propranolol, morphine, lidocaine (unsuitable for oral use), verapamil.

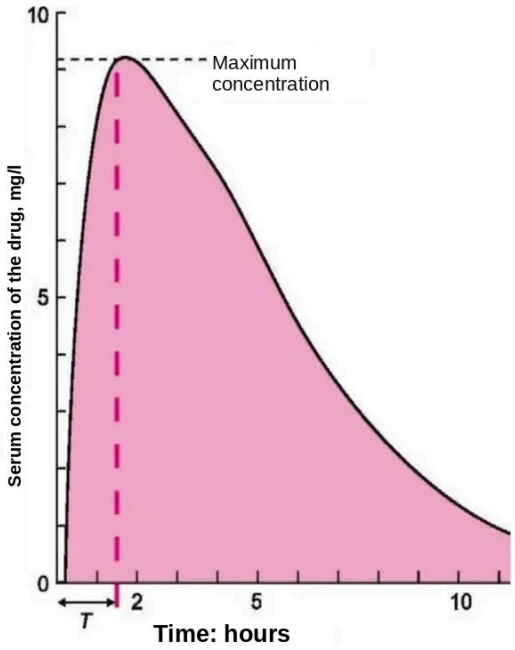


Fig. II.4. Determination of the bioavailability of a substance after enteral administration (after Kalant et al.).

Figure 3.2. *Serum concentration curve of an orally administered drug: Cmax — maximum concentration, T — time to reach it; the area under the curve (AUC) reflects bioavailability (teaching material of the TMA Department of Pharmacology)*

CLINICAL SIGNIFICANCE

Bioequivalence — two preparations with the same active substance (original and generic) at the same dose produce the same Cmax, Tmax and AUC in the blood. Generic preparations must prove their bioequivalence to be registered.

2.4. Distribution

After absorption, the drug is distributed to the tissues through the blood. Factors affecting distribution:

- **blood flow** (the brain, heart, kidneys and liver are saturated quickly; adipose tissue slowly);
- **lipophilicity** and molecular size;
- **binding to plasma proteins** (acidic drugs — to albumin, basic drugs — to alpha-1-acid glycoprotein). Only the **free (unbound) fraction** is active, is metabolized and is filtered;
- **biological barriers**: the blood–brain barrier (BBB), the placental barrier, the blood–ocular and blood–testis barriers.

CAUTION

Displacement from protein binding. Drugs that are more than 90% protein-bound (warfarin, sulfonylureas, phenytoin), when given together with another strongly bound drug (NSAIDs, sulfonamides, valproate), may have a sharp rise in their free fraction: risk of bleeding or hypoglycemia. In newborns, sulfonamides and ceftriaxone can displace bilirubin from albumin and cause **bilirubin encephalopathy**.

The blood–brain barrier — tight junctions of the endothelium of brain capillaries and the P-glycoprotein pump. Lipophilic substances (diazepam, thiopental) cross well; quaternary ammonium compounds (neostigmine, metocinium) and many antibiotics cross poorly. In inflammation (meningitis), the permeability of the barrier increases — benzylpenicillin reaches the cerebrospinal fluid.

The placental barrier — lipophilic, small-molecule drugs cross easily. Therefore, during pregnancy any drug is regarded as potentially dangerous.

Volume of distribution (Vd) — a notional volume that relates the amount of drug in the body to its plasma concentration: **Vd = D / C_0**. Small Vd (5–15 l) — the drug is mainly in the blood (heparin, warfarin); large Vd (hundreds of liters) — the drug accumulates in the tissues (digoxin, amiodarone, chloroquine). In poisoning with drugs of large Vd, hemodialysis is ineffective.

Depot formation — accumulation of a drug in a tissue: thiopental — in adipose tissue, tetracyclines — in bone and teeth, iodine — in the thyroid gland, amiodarone — in fat and the lungs.

2.5. Biotransformation (metabolism)

Biotransformation takes place mainly in the **liver** (also in the intestinal wall, lungs, kidneys and blood). Its aim is to make a lipophilic substance **hydrophilic** (water-soluble) and so make its excretion easier.

Table 3.3. *The two phases of biotransformation*

Phase	Reactions	Main enzymes	Result
Phase I (metabolic transformation)	Oxidation, reduction, hydrolysis	Cytochrome P450 (CYP3A4, CYP2D6, CYP2C9, CYP2C19, CYP1A2), esterases	A functional group appears; activity may decrease, be retained or increase
Phase II (conjugation)	Conjugation with glucuronic acid, sulfate, glycine, glutathione; acetylation, methylation	UDP-glucuronosyltransferase, N-acetyltransferase, etc.	A hydrophilic, usually inactive conjugate is formed

REMEMBER

Prodrug — a drug that is inactive itself and is converted into an active substance in the body: enalapril → enalaprilat, clopidogrel → active thiol metabolite, levodopa → dopamine, codeine → morphine (CYP2D6), prednisone → prednisolone. Some metabolites are toxic: paracetamol → N-acetyl-p-benzoquinone imine (NAPQI) — hepatotoxic.

Table 3.4. *Inducers and inhibitors of microsomal enzymes (clinically important)*

Inducers (reduce the effect of other drugs)	Inhibitors (enhance the effect of other drugs, toxicity)
Rifampicin	Macrolides (clarithromycin, erythromycin)
Phenobarbital and other barbiturates	Azole antifungals (ketoconazole, fluconazole, itraconazole)
Carbamazepine, phenytoin	Cimetidine
Chronic alcohol use	Amiodarone
Tobacco smoking (CYP1A2)	Fluoroquinolones (ciprofloxacin — CYP1A2)
St John's wort	Grapefruit juice (intestinal CYP3A4)
Glucocorticoids (some CYPs)	HIV protease inhibitors (ritonavir)

MEMORY AID

Inducers: **“Rifa, Pheno, Carba and Pheny drank, smoked and ate St John's wort”** — rifampicin, phenobarbital, carbamazepine, phenytoin, alcohol, smoking, St John's wort. Inhibitors: **“MAKC CIMA GRAPE”** — **MA**crolides, **K**etoconazole (azoles), **C**iprofloxacin, **CIM**etidine, **A**miodarone, **GRAPE**fruit.

CLINICAL SIGNIFICANCE

Rifampicin speeds up the metabolism of oral contraceptives and can lead to **unintended pregnancy**. When clarithromycin is given together with simvastatin, the statin concentration rises and there is a risk of **rhabdomyolysis**. Ciprofloxacin can raise the theophylline concentration and lead to seizures.

2.6. Excretion

Drugs are excreted mainly through the **kidneys**, and also with the **bile** through the intestine, through the lungs (volatile substances — ether, halothane), sweat, saliva and **breast milk**.

Renal excretion is the sum of three processes:

1. **Glomerular filtration** — small-molecule substances not bound to protein;
2. **Tubular secretion** (in the proximal tubule, active) — separate carriers for acids (penicillins, diuretics) and bases. Probenecid competitively inhibits penicillin secretion;
3. **Tubular reabsorption** (passive) — lipophilic, non-ionized substances are reabsorbed. Reabsorption can be controlled by changing the urine pH.

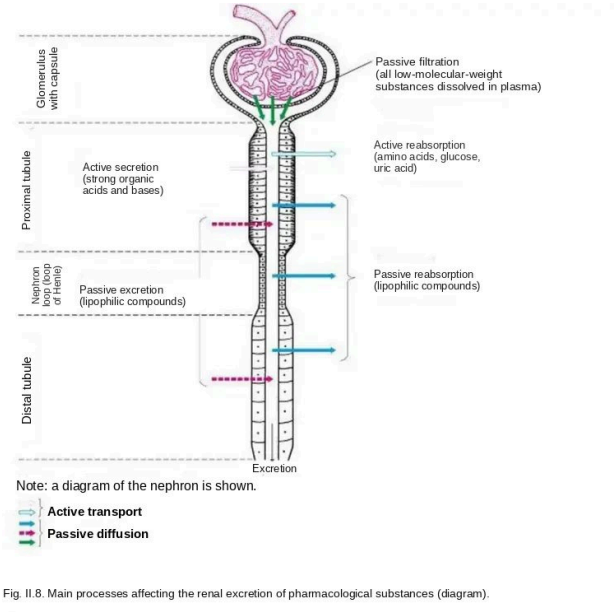


Figure 3.3. *Processes of drug excretion in the nephron: filtration, secretion, reabsorption (teaching material of the TMA Department of Pharmacology)*

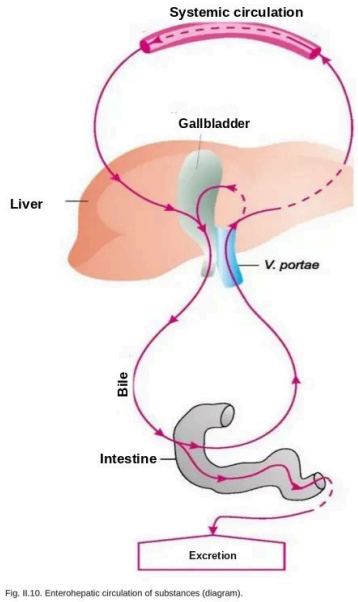


Figure 3.4. *Enterohepatic circulation (teaching material of the TMA Department of Pharmacology)*

Enterohepatic circulation — a conjugate excreted into the intestine with the bile is broken down by intestinal bacteria, and the drug is reabsorbed. This prolongs the action of drugs (digitoxin, estrogens, morphine). Antibiotics can alter the intestinal flora and reduce the enterohepatic circulation of estrogen-containing contraceptives.

CLINICAL SIGNIFICANCE

In renal failure, the doses of drugs excreted unchanged mainly by the kidneys are reduced according to creatinine clearance: aminoglycosides, vancomycin, digoxin, metformin, lithium, enoxaparin. **In hepatic failure** — drugs that undergo extensive hepatic metabolism (morphine, benzodiazepines, propranolol).

2.7. Main pharmacokinetic parameters

Table 3.5. *Pharmacokinetic parameters*

Parameter	Symbol	Meaning
Maximum concentration	Cmax	The highest concentration in the blood
Time to maximum concentration	Tmax	Reflects the rate of absorption
Area under the curve	AUC	The total amount of drug that has entered the body
Bioavailability	F	AUC (oral) / AUC (intravenous) × 100%
Volume of distribution	Vd	Vd = Dose / C_0
Clearance	Cl	The volume of plasma completely cleared of the drug per unit of time (ml/min): Cl = Vd × kel
Elimination rate constant	kel	The fraction of drug eliminated per unit of time
Half-life	t½	The time it takes for the plasma concentration to fall by half: t½ = 0.693 × Vd / Cl
Steady-state concentration	Css	The state in which intake and elimination are balanced

REMEMBER

The "4–5 half-lives" rule: with repeated use at a constant dose, steady-state concentration (Css) is reached after about **4–5 × t½**; when the drug is stopped, it also takes **4–5 × t½** for it to be almost completely (94–97%) eliminated from the body. 1 t½ → 50%, 2 → 75%, 3 → 87.5%, 4 → 93.75%, 5 → 96.9%.

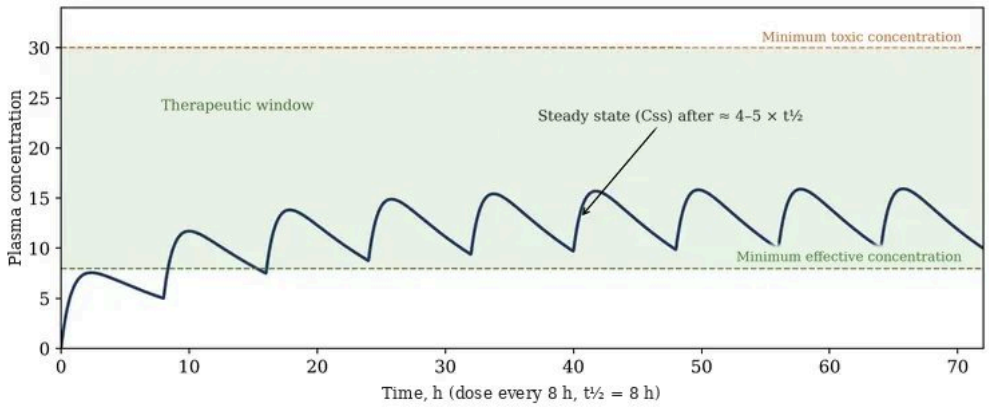


Figure 3.5. *Change in plasma concentration with repeated use: steady state and the therapeutic window (compiler's diagram)*

Loading dose — a large initial dose for drugs with a long t½ (amiodarone, digoxin) to reach Css quickly: **Dload = Css × Vd / F**. **Maintenance dose** — to maintain Css: **Dmaint = Css × Cl × τ / F** (τ — dosing interval).

Elimination kinetics:

- **First-order** (most drugs) — a **constant fraction** of the drug is eliminated per unit of time; t½ is constant.
- **Zero-order (saturation)** — a **constant amount** is eliminated per unit of time; the enzymes are saturated. Examples: **ethanol, phenytoin, high-dose salicylates**. A small increase in dose leads to a sharp rise in concentration.

MEMORY AID

Zero-order kinetics: **“PEA”** — **P**henytoin, **E**thanol, **A**spirin (at high doses). Zero order is a “saturated pump” that removes the same amount every hour.

3. Pharmacodynamics

3.1. Types of action

Table 3.6. *Types of drug action*

Classification	Types	Example
Site of action	Local — resorptive (systemic)	Lidocaine in infiltration anesthesia — local; diazepam — resorptive
Mode of realization	Direct — reflex	Digoxin acts on the heart directly; inhaling ammonia stimulates breathing reflexly
Clinical purpose	Main — adverse	Atropine: antispasmodic (main), dry mouth (adverse)
Reversibility	Reversible — irreversible	Neostigmine — reversible; organophosphorus compounds inhibit cholinesterase irreversibly
Selectivity	Selective — nonselective	Metoprolol (beta-1) — selective; propranolol (beta-1, beta-2) — nonselective
Type of therapy	Etiotropic, pathogenetic, symptomatic, replacement, preventive	Antibiotic — etiotropic; insulin — replacement; analgesic — symptomatic

3.2. Molecular targets of drug action

1. **Receptors** — specialized proteins that bind endogenous ligands (mediators, hormones).
2. **Ion channels** — local anesthetics (Na^+ channels), calcium antagonists (L-type Ca^{2+} channels), sulfonylureas (ATP-dependent K^+ channels).
3. **Enzymes** — NSAIDs (cyclooxygenase), ACE inhibitors, neostigmine (acetylcholinesterase), statins (HMG-CoA reductase).
4. **Transport systems** — omeprazole (H^+/K^+-ATPase), digoxin (Na^+/K^+-ATPase), furosemide (Na^+-K^+-2Cl– cotransporter), antidepressants (serotonin and norepinephrine reuptake).
5. **Nucleic acids and other targets** — cytostatics (DNA), antacids (chemical neutralization), osmotic diuretics (physicochemical action).

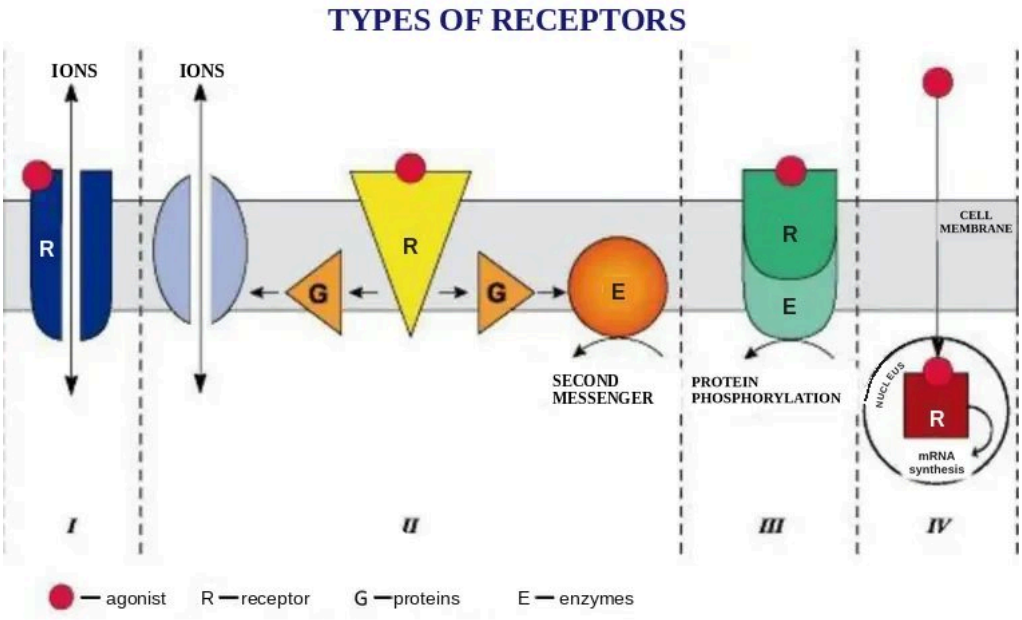


Figure 3.6. *The four main types of receptors: I — ion channel; II — G protein-coupled; III — enzymatic (tyrosine kinase); IV — intracellular (nuclear) receptors (teaching material of the TMA Department of Pharmacology)*

Table 3.7. *Types of receptors*

Type	Speed of action	Mechanism	Examples
I. Ionotropic (ion channel)	Milliseconds	The channel opens when the ligand binds	N-cholinergic receptor, GABA-A, NMDA, 5-HT_3
II. G protein-coupled (metabotropic)	Seconds	G_s (adenylate cyclase ↑, cAMP ↑), G_i (cAMP ↓), G_q (phospholipase C ↑, IP_3, Ca^{2+} ↑)	M-cholinergic, adrenergic, opioid, histamine and dopamine receptors
III. Enzymatic (kinase-linked)	Minutes	Activation of tyrosine kinase	Insulin receptor, growth factor receptors
IV. Intracellular (nuclear) receptors	Hours	Alter gene transcription	Glucocorticoid, mineralocorticoid, sex hormone, thyroid and vitamin D receptors

MEMORY AID

G proteins and receptors: **"QISS"** — **Q**q (α_1, M_1, M_3, H_1), **I**i (α_2, M_2, D_2), **S**s (β_1, β_2, D_1, H_2, V_2). Odd-numbered M receptors (M_1, M_3) — G_q; even (M_2) — G_i.

3.3. Agonists and antagonists

Affinity — the ability of a substance to bind to a receptor. **Intrinsic activity** — the ability, after binding, to activate the receptor and produce a response.

Table 3.8. *Receptor ligands*

Ligand	Affinity	Intrinsic activity	Example
Full agonist	Present	Maximal (1)	Morphine (mu-opioid), salbutamol (beta-2)
Partial agonist	Present	Greater than 0, less than 1	Buprenorphine, pindolol (intrinsic sympathomimetic activity)

Ligand	Affinity	Intrinsic activity	Example
Antagonist (blocker)	Present	0	Atropine, naloxone, propranolol
Inverse agonist	Present	Negative (reduces basal activity)	Some H_1 blockers and beta-blockers
Agonist-antagonist	Present	Agonist at one receptor, antagonist at another	Butorphanol, pentazocine (kappa agonist, mu antagonist)

Types of antagonism: competitive (can be overcome by increasing the agonist dose — atropine and acetylcholine), noncompetitive (cannot be overcome), **physiological** (opposite effects through different receptors — epinephrine against histamine in anaphylaxis), **chemical** (direct binding — protamine sulfate neutralizes heparin), **physical** (adsorption — activated charcoal).

3.4. The dose–effect relationship

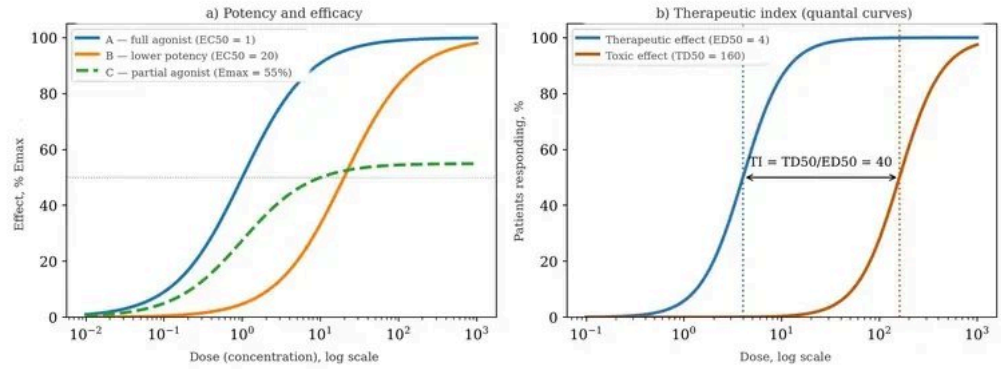


Figure 3.7. *The dose–effect relationship: a — potency (EC50) and efficacy (Emax), b — therapeutic index (compiler's diagram)*

- **Potency** — the smaller the dose that produces the desired effect, the more potent the drug. Its measure is **EC50** (the concentration that produces 50% of the maximal effect).
- **Efficacy** — the **maximal** effect a drug can produce (**Emax**). Clinically, efficacy is more important than potency.
- **Therapeutic index (TI)** = TD50 / ED50 (in animals LD50 / ED50). Drugs with a small TI (**"with a narrow therapeutic range"**): digoxin, lithium, theophylline, warfarin, aminoglycosides, phenytoin, carbamazepine — their blood concentrations are monitored.

> **CAUTION**
> Do not confuse potency with efficacy: furosemide 40 mg is **more effective** than hydrochlorothiazide (25 mg) (it produces a stronger diuresis), even though it is less potent per mg. Bumetanide (1 mg) is **more potent** than furosemide (40 mg), but their efficacy is the same.

3.5. Phenomena on repeated use

Table 3.9. *Phenomena observed with repeated drug use*

Phenomenon	Definition	Example
Cumulation (material)	Accumulation of the drug in the body	Digoxin, digitoxin, amiodarone
Cumulation (functional)	Accumulation of the effect (even after the substance has been eliminated)	Ethanol (in alcoholism), organophosphorus compounds
Tolerance (habituation)	A decrease in effect with repeated use	Nitrates, opioids, benzodiazepines

Phenomenon	Definition	Example
Tachyphylaxis	Tolerance that develops very quickly (after a few doses)	Ephedrine and other indirect adrenomimetics, naphazoline (Naphthyzin)
Sensitization	An allergic reaction on repeated use	Penicillins, sulfonamides
Drug dependence (psychological and physical)	An irresistible need to take the drug	Opioids, benzodiazepines, alcohol, nicotine
Abstinence (withdrawal) syndrome	Disturbance of functions when the drug is stopped abruptly	Opioids, alcohol, benzodiazepines
“Rebound” phenomenon	Symptoms return more strongly than before when the drug is stopped	Clonidine (hypertensive crisis), beta-blockers (angina), proton pump inhibitors

3.6. Drug interactions

- **Pharmaceutical** — outside the body, in one syringe or solution (precipitate, inactivation).
- **Pharmacokinetic** — absorption (antacids reduce the absorption of tetracyclines and fluoroquinolones), distribution (displacement from proteins), metabolism (induction, inhibition), excretion.
- **Pharmacodynamic** — **synergism** (additive — the sum; potentiation — more than the sum) and **antagonism**.

> **CLINICAL SIGNIFICANCE**
> Examples of dangerous interactions: MAO inhibitors + tyramine-containing foods (cheese) → hypertensive crisis; SSRIs + tramadol or MAO inhibitors → **serotonin syndrome**; sildenafil + nitrates → severe hypotension; warfarin + NSAIDs → bleeding; ACE inhibitors + potassium-sparing diuretics → hyperkalemia.

3.7. Factors affecting drug action

- **Age.** In newborns, liver enzymes (glucuronosyltransferase) are immature — chloramphenicol (Levomycetin) causes the “gray baby syndrome”; the BBB is permeable; renal function is low. In the elderly, renal and hepatic clearance decrease, adipose tissue increases (the Vd of lipophilic drugs rises), and CNS sensitivity increases.
- **Pregnancy and lactation.** Teratogenic drugs: thalidomide, isotretinoin, warfarin, valproate, ACE inhibitors, tetracyclines, methotrexate.
- **Pharmacogenetics.** Slow and fast **acetylators** (isoniazid, hydralazine, procainamide); **G6PD deficiency** (primaquine, sulfonamides, nitrofurantoin → hemolysis); the atypical form of **pseudocholinesterase** (suxamethonium → prolonged apnea); CYP2D6 ultrarapid metabolizers (codeine → much more morphine is formed — respiratory arrest in children).
- **Diseases** (renal, hepatic, heart failure), diet, circadian rhythms, the placebo effect.

3.8. Adverse effects

- **Type A** (dose-related, predictable) — an exaggeration of the pharmacological effect: insulin — hypoglycemia, anticoagulant — bleeding.
- **Type B** (not dose-related, unexpected) — **allergic reactions** (anaphylaxis, Stevens–Johnson syndrome) and **idiosyncrasy** (an unusual response associated with a hereditary enzyme defect).

- **Toxic effects** — nephrotoxicity (aminoglycosides), ototoxicity (aminoglycosides, furosemide), hepatotoxicity (paracetamol, isoniazid), cardiotoxicity (doxorubicin), myelotoxicity (cytostatics, chloramphenicol).
- **Teratogenic, embryotoxic, fetotoxic, mutagenic and carcinogenic** effects.
- **Dysbiosis, superinfection** — with broad-spectrum antibiotics.

3.9. General principles of treating acute drug poisoning

1. **Stopping absorption:** gastric lavage (within 1 hour, in a conscious patient, in the absence of corrosive substances), **activated charcoal** (1 g/kg), washing the skin and eyes.
2. **Speeding up the elimination of absorbed poison:** forced diuresis, alkalinization of the urine (salicylates, phenobarbital), hemodialysis (methanol, ethylene glycol, lithium, salicylates), hemosorption.
3. **Antidote therapy** (specific neutralization).
4. **Symptomatic support of vital functions:** airway, breathing, circulation, stopping seizures.

Table 3.10. *Main antidotes*

Poison	Antidote
Opioids	Naloxone
Benzodiazepines	Flumazenil
Paracetamol	N-acetylcysteine
Organophosphorus compounds (OPCs)	Atropine + cholinesterase reactivators (pralidoxime, dipyroxime)
M-cholinergic blockers (atropine)	Physostigmine
Warfarin	Vitamin K (phytomenadione), prothrombin complex concentrate
Heparin	Protamine sulfate
Methanol, ethylene glycol	Ethanol, fomepizole
Cyanides	Hydroxocobalamin, sodium thiosulfate, nitrites
Iron preparations	Deferoxamine
Heavy metals (lead, mercury, arsenic)	Dimercaprol, unithiol, EDTA (calcium disodium edetate), penicillamine
Beta-blockers	Glucagon
Digoxin	Digoxin-specific antibodies (Fab fragments), unithiol
Local anesthetics (systemic toxicity)	20% lipid emulsion
Methemoglobin-forming agents (nitrites)	Methylene blue

4. Drugs acting on the afferent nervous system

Afferent (sensory) nerves — fibers that carry impulses from receptors to the CNS. The drugs of this group are divided into two large groups.

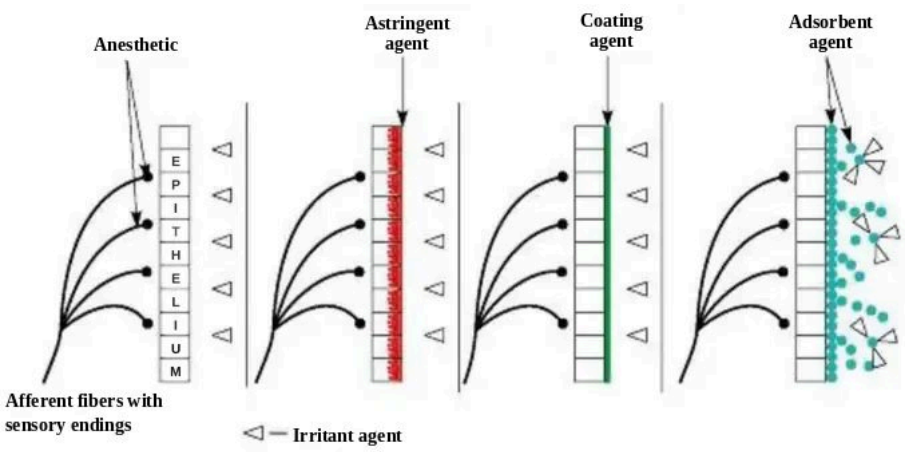


Fig. 1.2. Main direction of action of agents that prevent excitation of sensory nerve endings.

Figure 3.8. *Direction of action of drugs acting on afferent nerve endings: anesthetics, astringents, coating agents and adsorbents (teaching material of the TMA Department of Pharmacology)*

Table 3.11. *Classification of drugs acting on afferent innervation*

Group	Subgroup	Main preparations
A. Drugs that reduce the excitability of sensory nerve endings or prevent their excitation	Local anesthetics	Procaine (Novocain), lidocaine, articaine, bupivacaine, ropivacaine, benzocaine (Anesthesin), tetracaine (Dicaine)
	Astringents	Tannin, oak bark decoction, basic bismuth nitrate, bismuth subcitrate, zinc oxide, silver nitrate (at low concentration)
	Coating agents	Starch mucilage, flaxseed mucilage, aluminum and magnesium hydroxide gels
	Adsorbents	Activated charcoal, smectite, Polyphepan, talc, white clay
B. Drugs that stimulate sensory nerve endings	Irritants	Menthol, ammonia solution (spirit of ammonia), mustard plaster, purified turpentine, capsaicin (pepper), snake and bee venom ointments
	Bitters (appetite stimulants)	Bitter wormwood tincture, dandelion root
	Reflex expectorants	Thermopsis, polygala root (topic 9)
	Laxatives (intestinal irritants)	Bisacodyl, senna (topic 12)

4.1. Local anesthetics

Local anesthetics — drugs that **reversibly** abolish pain sensation (and later other sensations) at the site of application. They do not cause loss of consciousness.

History. In 1879, the Russian scientist **V.K. Anrep** discovered the local anesthetic effect of cocaine; in 1884, K. Koller introduced it into ophthalmology. In 1905, A. Einhorn synthesized **procaine (Novocain)**, which is less toxic than cocaine; in 1943, N. Löfgren created the first amide anesthetic — **lidocaine**.

Chemical structure. The molecule consists of three parts: **a lipophilic aromatic ring — an intermediate chain (ester or amide bond) — a hydrophilic amine group**. Two groups are distinguished according to the intermediate bond.

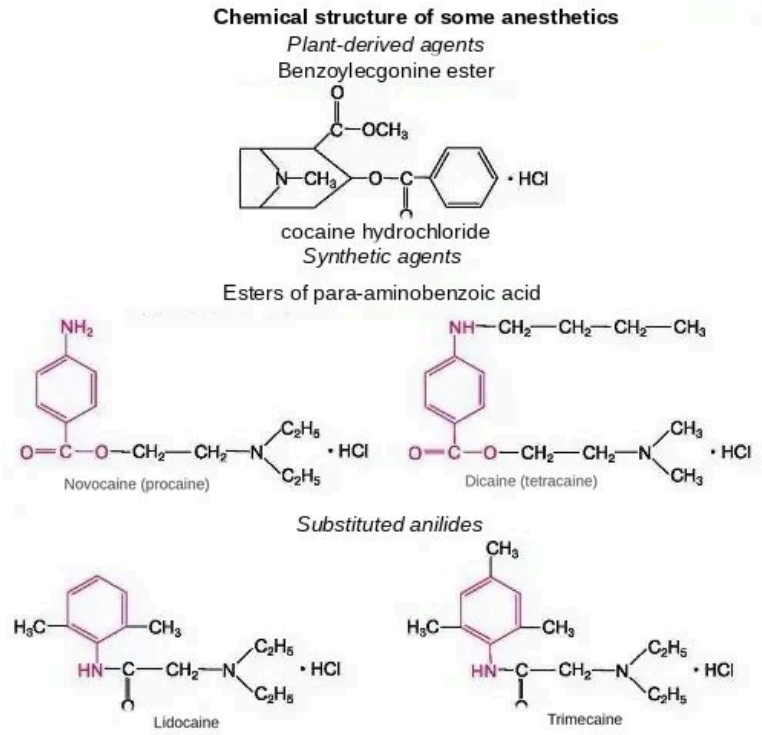


Figure 3.9. *Chemical structure of local anesthetics: cocaine, novocaine, dicaine (esters), lidocaine and trimecaine (amides) (teaching material of the TMA Department of Pharmacology)*

Table 3.12. *Differences between the ester and amide groups*

Feature	Esters (procaine, tetracaine, benzocaine, cocaine)	Amides (lidocaine, articaine, bupivacaine, ropivacaine, mepivacaine)
Breakdown	In the blood by pseudocholinesterase (butyrylcholinesterase), rapid	In the liver by microsomal enzymes, slower
Duration of action	Usually short (except cocaine, tetracaine)	Intermediate and long
Allergic reactions	More frequent (metabolite — para-aminobenzoic acid, PABA)	Very rare
Stability	Less stable in solution	Stable, can be autoclaved

MEMORY AID

Amide anesthetics have two "i"s in their names: l**i**docai**i**ne, art**i**cai**i**ne, bup**i**vacai**i**ne, rop**i**vacai**i**ne, mep**i**vacai**i**ne. Esters have one "i": procaine, tetracaine, benzocaine, cocaine.

Mechanism of action. Local anesthetics block the **voltage-gated sodium channels** of the nerve fiber membrane from the inside. Sodium ions stop entering the cell, no action potential is generated, and impulse conduction is blocked.

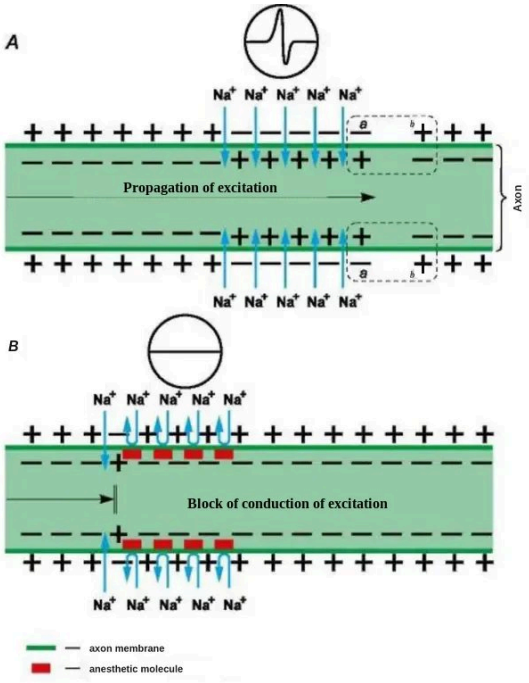


Fig. 1.1. Effect of anesthetics on the conduction of excitation along a nerve fiber.

Figure 3.10. *Effect of anesthetics on conduction of excitation along a nerve fiber: A — normal Na^+ entry; B — anesthetic molecules block the sodium channels (teaching material of the TMA Department of Pharmacology)*

Important details of the mechanism:

- The anesthetic crosses the membrane **in the non-ionized (base) form**, and inside the cell it blocks the channel **in the ionized (cationic) form**.
- In inflamed tissue (pus), the medium is **acidic**, the anesthetic is more ionized and crosses the membrane poorly — anesthesia is **weak**.
- Channels are blocked better in the **open and inactivated state** (**"use-dependent block"**) — pain fibers that conduct impulses frequently are blocked first.
- **Sensitivity of fibers:** thin unmyelinated C fibers and A-delta fibers (pain, temperature) are blocked first, then touch and pressure, and **motor** (thick A-alpha) fibers last. Order of loss of sensations: **pain → temperature → touch → deep pressure → motor function**.

CLINICAL SIGNIFICANCE

Adding a vasoconstrictor. If **epinephrine** (adrenaline, usually 1:100,000 – 1:200,000) is added to a local anesthetic solution: 1) the blood vessels constrict, the anesthetic is absorbed slowly — the effect is **prolonged**; 2) less enters the blood — **systemic toxicity decreases**, and the maximum dose is increased; 3) bleeding decreases. However, epinephrine-containing solutions are **not used** for anesthesia of the **fingers, auricle, tip of the nose and penis** (end arteries — risk of necrosis), in severe heart disease, thyrotoxicosis and in patients taking MAO inhibitors.

4.2. Types of local anesthesia

Table 3.13. *Types of local anesthesia and the anesthetics used*

Type of anesthesia	Essence	Anesthetics used
Surface (terminal, topical)	Applying, instilling or spraying onto the mucous membrane or wound surface	Lidocaine (2–10%), tetracaine (Dicaine, 0.25–2%), benzocaine (Anesthesin), cocaine, oxybuprocaine (in the eye)

Type of anesthesia	Essence	Anesthetics used
Infiltration	“Soaking” the tissues layer by layer with the anesthetic solution	Procaine (0.25–0.5%), lidocaine (0.25–0.5%), articaine
Conduction (regional, nerve and plexus block)	Injection around a nerve trunk or plexus	Lidocaine (1–2%), articaine (in dentistry), bupivacaine, ropivacaine
Spinal (subarachnoid)	Into the cerebrospinal fluid (below L_2–L_3)	Bupivacaine (0.5%, hyperbaric), lidocaine
Epidural (peridural)	Into the space above the dura mater	Bupivacaine, ropivacaine, lidocaine
Intravenous regional (Bier block)	Into a vein of a limb with a tourniquet applied	Lidocaine (0.5%) — bupivacaine is strictly contraindicated

CLINICAL SIGNIFICANCE

Lidocaine is also given intravenously as an **antiarrhythmic** drug (class IB) for ventricular arrhythmias (topic 10).

4.3. Main local anesthetics

Table 3.14. *Comparison of local anesthetics*

Preparation	Group	Relative potency / toxicity (procaine = 1)	Duration	Main use
Cocaine	Ester	2–3 / 3–4	30–60 min (surface)	Surface anesthesia only (ENT, ophthalmology); stimulates the CNS, causes dependence; inhibits norepinephrine reuptake — constricts the vessels
Procaine (Novocain)	Ester	1 / 1	30–60 min	Infiltration, conduction; weak surface effect; for dissolving other drugs (penicillin)
Tetracaine (Dicaine)	Ester	10–15 / 10	2–3 hours	Surface anesthesia only (eye, nasopharynx, bronchoscopy); very toxic
Benzocaine (Anesthesin)	Ester	Insoluble in water, not absorbed	—	Surface: ointment, powder, suppository, tablet (for stomach pain)
Lidocaine	Amide	2–4 / 1,5–2	60–120 min	All types of anesthesia; antiarrhythmic
Articaine (Ultracain)	Amide (with a thiophene ring and an ester group)	5 / 1,5	1–3 hours	Dentistry; rapidly broken down in the blood, diffuses well into tissue
Bupivacaine	Amide	8–16 / 6–8	3–8 hours	Spinal, epidural, conduction; the most cardiotoxic
Ropivacaine	Amide	≈ bupivacaine	3–6 hours	Epidural (in labor), regional; less cardiotoxic than bupivacaine; less motor block

Table 3.15. *Maximum doses of local anesthetics (for adults, approximate)*

Preparation	Without epinephrine	With epinephrine
Procaine	7 mg/kg (≈ 500 mg)	10 mg/kg (≈ 750–1000 mg)
Lidocaine	4.5 mg/kg (≈ 300 mg)	7 mg/kg (≈ 500 mg)
Articaine	—	7 mg/kg (≈ 500 mg)
Bupivacaine	2 mg/kg (≈ 150 mg)	2.5 mg/kg (≈ 175–225 mg)
Ropivacaine	3 mg/kg (≈ 225 mg)	— (epinephrine is usually not added)

CAUTION

Maximum doses differ slightly between sources (especially for procaine). Always calculate the dose in **mg/kg** and reduce it according to the patient's condition (old age, liver disease, pregnancy, heart failure). The doses of anesthetics injected into different sites at the same time **add up**.

4.4. Systemic toxicity of local anesthetics

When the anesthetic accidentally enters a **vessel** or the dose is exceeded, a systemic toxic effect develops (LAST — local anesthetic systemic toxicity).

- **CNS signs** (first): numbness of the lips and tongue, a metallic taste in the mouth, tinnitus, dizziness, restlessness, speech disturbance, muscle twitching → **seizures** → CNS depression, coma, respiratory arrest.
- **Cardiovascular signs:** bradycardia, AV block, hypotension, ventricular arrhythmias, **cardiac arrest** (typical of bupivacaine — it dissociates from sodium channels very slowly).

REMEMBER

Management of systemic toxicity: 1) stop injecting the anesthetic, call for help; 2) secure the airway, 100% oxygen; 3) for seizures — **benzodiazepines** (midazolam, diazepam); 4) **20% lipid emulsion** intravenously: 1.5 ml/kg bolus, then an infusion of 0.25 ml/kg/min; 5) in cardiac arrest — resuscitation (epinephrine in small doses; lidocaine is **not used** as an antiarrhythmic).

Allergic reactions are typical mainly of esters (PABA metabolite). Allergy to amides is very rare; it is usually due to a preservative (methylparaben) or sulfites (in epinephrine-containing solutions).

CAUTION

The procaine metabolite **PABA reduces** the antibacterial effect of sulfonamides (competitive antagonism). Because procaine is broken down by pseudocholinesterase, its effect and toxicity increase in patients with an atypical form of this enzyme and in those taking anticholinesterase agents.

4.5. Astringents

Astringents partly and reversibly denature proteins on the surface of a mucous membrane or wound and form a thin protective film. As a result, sensory nerve endings are protected, pain decreases, the vessels constrict, and exudation and inflammation decrease.

- **Organic:** tannin (5–10% solution — for rinsing the mouth, in burns), oak bark decoction, infusions of chamomile, sage and St John's wort, blackcurrant berries.
- **Inorganic (metal salts):** basic bismuth nitrate, bismuth tripotassium dicitrate (De-Nol — for gastric ulcer, topic 12), zinc sulfate (eye drops), zinc oxide, aluminum potassium sulfate (alum), lead acetate, silver nitrate (astringent at low concentration, caustic at high concentration).

CLINICAL SIGNIFICANCE

The effect of metal salts depends on concentration: at **low** concentration — astringent, at **moderate** — irritant, at **high** — caustic (necrosis). For example, silver nitrate 0.05–0.1% is astringent, 2–5% is caustic (excess granulation tissue, warts).

4.6. Coating agents

Coating agents form colloidal solutions in water, cover the mucous membrane and protect it from irritation. They have no resorptive effect.

- starch mucilage, flaxseed mucilage, marshmallow root mucilage, salep;
- aluminum hydroxide, magnesium trisilicate, Almagel (topic 12). Uses: gastritis, gastric ulcer, enteritis, colitis; together with irritant drugs (chloral hydrate) — to soften their local effect; in poisoning — to reduce absorption.

4.7. Adsorbents (enterosorbents)

Adsorbents — water-insoluble, chemically inert powders with a large surface area. They bind gases, toxins, bacteria and drug substances on their surface.

Table 3.16. *Main adsorbents*

Preparation	Property	Use
Activated charcoal	The surface of 1 g is up to 1000 m^2	Acute poisoning (1 g/kg), dyspepsia, flatulence; binds lithium, iron, alcohols, acids and alkalis poorly
Diosmectite (Smecta)	A natural aluminosilicate; also coats the mucous membrane	Acute diarrhea (especially in children)
Polyphepan, lignin	Made from plant lignin	Poisoning, dyspepsia
Talc, white clay (bolus alba)	For dusting	Sweating of the skin, dermatitis (in powders and pastes)
Polymethylsiloxane (Enterosgel)	A gel-like sorbent	Poisoning, diarrhea

CAUTION

Adsorbents also bind other drugs and reduce their absorption. Therefore, other drugs are taken **at least 1–2 hours before or after** the adsorbent.

4.8. Irritants

Irritants stimulate sensory nerve endings in the skin and mucous membranes. Their effect is mainly **reflex**:

- **“Distracting” effect** — excitation from the skin competes in the spinal cord with pain impulses coming from internal organs (**gate control theory**), thereby reducing pain; endorphins and enkephalins are released.
- **Trophic effect** — blood circulation and trophism improve in the internal organs belonging to the same segment as the irritated skin area (according to the **Zakharyin–Head zones**).
- **Reflex stimulating effect** — ammonia vapor stimulates the trigeminal nerve endings in the nasal mucosa and reflexly stimulates the respiratory and vasomotor centers (in fainting).

Table 3.17. *Main irritants*

Preparation	Action and use
Menthol (peppermint oil)	Stimulates cold receptors (TRPM8); reflexly dilates the vessels. Part of Validol — in mild cases of angina; for muscle pain, rhinitis; a “migraine pencil” for headache
Ammonia solution 10% (spirit of ammonia)	Inhaled in fainting (a few drops on cotton wool); washing the surgeon's hands (0.5%, an old method)
Mustard plaster	Releases allyl isothiocyanate; as a distracting agent in bronchitis, myositis, neuralgia
Purified turpentine	In liniments for neuralgia, myositis, arthritis
Capsaicin (hot pepper)	Stimulates TRPV1 receptors and, with repeated use, depletes substance P — neuropathic pain (patch, cream)
Bee and snake venom ointments (Apizartron, Viprosal)	A local irritant that improves blood circulation; for muscle and joint pain

5. Clinical part

5.1. Choosing a local anesthetic

Table 3.18. *Choosing a local anesthetic according to the clinical situation*

Situation	Appropriate choice
Suturing a small wound (other than on a finger)	Lidocaine 1% with epinephrine (infiltration)
Finger wound (Oberst conduction anesthesia)	Lidocaine 1–2% without epinephrine
Tooth extraction, dental procedures	Articaine 4% with epinephrine
Eye examination, foreign body removal	Oxybuprocaine 0.4% or tetracaine 0.5% drops
Pain relief in labor	Epidural ropivacaine or bupivacaine (at low concentration)
History of allergy to esters	Amide anesthetics (lidocaine)
Pharyngeal anesthesia before gastroscopy	Lidocaine 10% spray
Catheterization, cystoscopy	Lidocaine 2% gel

5.2. Practical application of clinical pharmacokinetics

CLINICAL SIGNIFICANCE

Knowing the time to reach steady state has practical importance. The t½ of digoxin is ≈ 36–48 hours — at an ordinary dose, steady state is reached after 7–10 days; if a rapid effect is needed, a **loading dose** is given. The t½ of amiodarone is several weeks, so treatment is started with a loading dose, and its effect persists for months after it is stopped. In contrast, drugs with a short t½ (adenosine — less than 10 seconds) are used only as a bolus, for an immediate effect.

Practical skills

Skill 1. Algorithm for calculating the maximum volume of a local anesthetic

1. Determine the patient's weight (kg).

2. Choose the maximum dose (mg/kg) of the anesthetic for the solution without or with epinephrine.
3. Maximum dose (mg) = weight × mg/kg (but not exceeding the absolute maximum dose).
4. Convert the solution concentration to mg/ml (percentage × 10).
5. Maximum volume (ml) = maximum dose (mg) / concentration (mg/ml). **Example:** a 60 kg patient, 1% lidocaine without epinephrine. 60 × 4.5 = 270 mg; 1% = 10 mg/ml; 270 / 10 = **27 ml**.

Skill 2. Calculating the time to reach steady-state concentration

1. Determine the t½ of the drug.
2. Time to reach Css ≈ 4–5 × t½.
3. Time for complete elimination after the drug is stopped ≈ 4–5 × t½. **Example:** t½ = 6 hours → Css after 24–30 hours.

Skill 3. Key prescriptions

PRESCRIPTION 3.1. LIDOCAINE (FOR INFILTRATION ANESTHESIA)
Rp.: Solutionis Lidocaini hydrochloridi 1% — 10 ml
D.t.d. N. 10 in ampullis
S. For infiltration anesthesia (maximum dose without epinephrine 4.5 mg/kg).

PRESCRIPTION 3.2. PROCAINE (NOVOCAIN)
Rp.: Solutionis Procaini 0.5% — 200 ml
D.S. For infiltration anesthesia.

PRESCRIPTION 3.3. TETRACAINE (DICAINE) EYE DROPS
Rp.: Solutionis Tetracaini hydrochloridi 0.5% — 5 ml
D.S. Instill 1–2 drops into the eye (before an ophthalmic procedure).

PRESCRIPTION 3.4. BENZOCAINE (ANESTHESIN) TABLETS
Rp.: Benzocaini 0.3
D.t.d. N. 10 in tabulettis
S. Take 1 tablet 3 times a day for stomach pain.

PRESCRIPTION 3.5. LIDOCAINE SPRAY
Rp.: Aerosoli Lidocaini 10% — 38.0
D.S. Spray onto the pharyngeal mucosa 1–3 times (before endoscopy).

PRESCRIPTION 3.6. TANNIN (ASTRINGENT)
Rp.: Solutionis Tannini 2% — 100 ml
D.S. For rinsing the mouth 4–5 times a day.

PRESCRIPTION 3.7. ACTIVATED CHARCOAL
Rp.: Carbonis activati 0.25
D.t.d. N. 30 in tabulettis
S. Take 4–8 tablets 3 times a day, 2 hours apart from other drugs.

PRESCRIPTION 3.8. DIOSMECTITE (SMECTA)

Rp.: Pulveris Diosmectiti 3.0
D.t.d. N. 10
S. Mix 1 sachet in 1/2 glass of water and take 3 times a day.

PRESCRIPTION 3.9. AMMONIA SOLUTION

Rp.: Solutionis Ammonii caustici 10% — 10 ml
D.S. In fainting, put a few drops on cotton wool and let the patient inhale.

PRESCRIPTION 3.10. OILY SOLUTION OF MENTHOL

Rp.: Solutionis Mentholi oleosae 1% — 10 ml
D.S. Instill 3–5 drops into the nose 3 times a day.

Review questions

1. What does pharmacology study? What is the difference between pharmacokinetics and pharmacodynamics?
2. Describe the enteral and parenteral routes of drug administration and their advantages and disadvantages.
3. List the mechanisms of absorption. What factors affect passive diffusion?
4. How does the pH of the medium affect the absorption and excretion of weak acids and bases?
5. What are bioavailability and the first-pass effect? Give examples.
6. What does the volume of distribution mean? Which drugs cross the blood–brain barrier?
7. Phases I and II of biotransformation. What is a prodrug?
8. List the inducers and inhibitors of microsomal enzymes. What is their clinical significance?
9. Processes of renal excretion. What is enterohepatic circulation?
10. Explain the concepts of t½, Cl, Vd and Css. What does the “4–5 half-lives” rule mean?
11. What is the difference between first-order and zero-order elimination kinetics?
12. What types of receptors are there? Give an example of each.
13. The concepts of agonist, partial agonist, antagonist and inverse agonist.
14. The concepts of potency, efficacy and therapeutic index.
15. What are cumulation, tolerance, tachyphylaxis, dependence and the rebound phenomenon?
16. General principles of treating acute poisoning and the main antidotes.
17. Classification of drugs acting on the afferent nervous system.
18. Mechanism of action of local anesthetics. Why is anesthesia weak in inflamed tissue?
19. The difference between ester and amide anesthetics.
20. Types of anesthesia and the anesthetics for each.
21. Signs of systemic toxicity of local anesthetics and emergency measures.
22. Mechanism and use of astringents, coating agents, adsorbents and irritants.

Test questions

1. What is the fraction of an orally taken drug that reaches the systemic circulation unchanged called?
   A) Bioavailability
   B) Volume of distribution
   C) Cumulation
   D) Clearance
   E) Half-life

2. What is done to speed up the excretion of a weak acid (salicylates) in poisoning?
   A) Giving calcium chloride
   B) Acidifying the urine
   C) Alkalinizing the urine
   D) Restricting fluids
   E) Giving atropine

3. Which drug is a strong inducer of microsomal enzymes?
   A) Rifampicin
   B) Grapefruit juice
   C) Clarithromycin
   D) Ketoconazole
   E) Cimetidine

4. With repeated use at a constant dose, approximately when is steady-state concentration reached?
   A) Immediately after the first dose
   B) 2 × t½
   C) 1 × t½
   D) 10 × t½
   E) 4–5 × t½

5. Zero-order elimination kinetics is typical of which substance?
   A) Ethanol
   B) Digoxin
   C) Lidocaine
   D) Gentamicin
   E) Penicillin

6. To which type does the insulin receptor belong?
   A) Transport protein
   B) Enzymatic, with tyrosine kinase activity
   C) G protein-coupled
   D) Nuclear receptor
   E) Ion channel

7. A substance that binds to the receptor but has zero intrinsic activity:
   A) Partial agonist
   B) Prodrug
   C) Inverse agonist
   D) Full agonist
   E) Antagonist

8. A drug with a narrow therapeutic index:
   A) Penicillin
   B) Amoxicillin
   C) Paracetamol
   D) Lithium carbonate
   E) Loratadine

9. Antidote for opioid poisoning:
   A) Flumazenil
   B) Atropine
   C) N-acetylcysteine
   D) Naloxone
   E) Protamine sulfate

10. Mechanism of action of local anesthetics:
    A) They block potassium channels
    B) They inhibit acetylcholinesterase
    C) They block voltage-gated sodium channels
    D) They activate GABA receptors
    E) They open calcium channels

11. The reason local anesthesia is weak in inflamed tissue:
    A) An alkaline medium forms
    B) Blood flow decreases
    C) In an acidic medium the anesthetic is more ionized and crosses the membrane poorly
    D) Sodium channels disappear
    E) The anesthetic is broken down quickly

12. The most cardiotoxic local anesthetic:
    A) Bupivacaine
    B) Benzocaine
    C) Articaine
    D) Lidocaine
    E) Procaine

13. Which solution is not used for finger anesthesia?
    A) Ropivacaine 0.5%
    B) Procaine 1%
    C) Lidocaine 1% without epinephrine
    D) Lidocaine 1% with epinephrine
    E) Lidocaine 2% without epinephrine

14. Specific treatment for systemic toxicity of local anesthetics:
    A) Atropine
    B) 20% lipid emulsion
    C) Glucagon
    D) Calcium gluconate
    E) Naloxone

15. To which group does activated charcoal belong?
    A) Astringents
    B) Adsorbents
    C) Irritants
    D) Coating agents
    E) Local anesthetics

Clinical cases

Case 1

A 55-year-old patient (70 kg) came to outpatient surgery to have a lipoma removed from the forearm. The surgeon wants to use 1% lidocaine with epinephrine for infiltration anesthesia. After 25 ml had been injected during the procedure, the patient reported numbness of the tongue, a metallic taste in the mouth and ringing in the ears, and then his hands began to tremble.

Questions: 1) What complication has developed? 2) What is the maximum dose for this patient? Has the dose been exceeded? 3) What emergency measures are needed?

Answer: 1) Systemic toxicity of the local anesthetic (early CNS signs) — the solution probably partly entered a vessel. 2) The maximum dose of lidocaine with epinephrine is 7 mg/kg → 70 × 7 = 490 mg ≈ 49 ml of 1% solution. The 250 mg injected did not exceed the maximum dose, so the main cause is intravascular injection (injection without an aspiration test). 3) Stop the injection, call for help, give 100% oxygen, establish intravenous access; if seizures begin — midazolam or diazepam; if the condition worsens — 20% lipid emulsion 1.5 ml/kg bolus and infusion; ECG and BP monitoring. Prevention: an aspiration test before each injection, injecting the solution slowly in small portions.

Case 2

A 32-year-old woman is taking rifampicin for tuberculosis. She also takes a combined oral contraceptive. Two months after the start of treatment, pregnancy was diagnosed.

Questions: 1) What is the pharmacological cause of the pregnancy? 2) What measures should have been taken?

Answer: 1) Rifampicin is a strong inducer of hepatic microsomal enzymes (CYP3A4); it accelerated the metabolism of estrogens and progestins and lowered their blood concentrations — the contraceptive became ineffective. 2) Non-hormonal contraception (intrauterine device, condom) should have been recommended during treatment with rifampicin and for at least 4 weeks after it. This example shows the need to check for pharmacokinetic interactions every time a prescription is written.

Case 3

A 3-year-old child (15 kg) swallowed an unknown number of tablets from his grandfather's medicine box at home; 40 minutes have passed since the incident. The child is conscious and his condition is satisfactory for now. The box contained digoxin and amlodipine tablets.

Questions: 1) According to the general principles of treating poisoning, what are the first measures? 2) What is the specific antidote in digoxin poisoning? 3) Why must such a child be observed for a long time?

Answer: 1) Call emergency services; activated charcoal to reduce absorption (1 g/kg = 15 g, most effective within 1 hour of the incident); monitoring of vital signs (pulse, BP, ECG); determining

blood potassium and digoxin levels. 2) Fab fragments of digoxin-specific antibodies; in addition, correction of arrhythmias and hyperkalemia. 3) Digoxin has a large volume of distribution and saturates the tissues slowly, and amlodipine has a long t½ (30–50 hours) — the toxic effect may intensify after several hours; therefore the child must be monitored in hospital for at least 24 hours.

Quick review table

Table 3.19. *General pharmacology and drugs acting on the afferent nervous system — summary*

Concept / group	Key information
ADME	Absorption — distribution — metabolism — excretion
Bioavailability	Intravenous = 100%; strong first-pass effect: nitroglycerin, propranolol, morphine, lidocaine
pH rule	An acid is absorbed in an acidic medium and excreted in an alkaline medium; bases — the opposite
Vd	Large Vd — accumulates in tissues, dialysis ineffective (digoxin, amiodarone)
CYP inducers	Rifampicin, phenobarbital, carbamazepine, phenytoin, alcohol, smoking, St John's wort
CYP inhibitors	Macrolides, azoles, cimetidine, ciprofloxacin, amiodarone, grapefruit
t½ rule	Css and complete elimination ≈ 4–5 × t½
Zero order	Phenytoin, ethanol, high-dose aspirin
Receptors	Ionotropic (N, $GABA_A$), G protein-coupled (M, alpha, beta), kinase-linked (insulin), nuclear (steroids)
Narrow-TI drugs	Digoxin, lithium, theophylline, warfarin, aminoglycosides, phenytoin
Mechanism of local anesthetics	Na_+ channel blockade (from the inside, cationic form)
Esters	Procaine, tetracaine, benzocaine, cocaine — pseudocholinesterase, PABA, allergy common
Amides	Lidocaine, articaine, bupivacaine, ropivacaine — in the liver, allergy rare
Maximum dose	Lidocaine 4.5 mg/kg (7 with epinephrine); bupivacaine 2 mg/kg
Treatment of LAST	Oxygen, benzodiazepines, 20% lipid emulsion 1.5 ml/kg
Astringents	Tannin, oak bark, bismuth, zinc salts — protein film
Coating agents	Mucilages, aluminum hydroxide
Adsorbents	Activated charcoal (1 g/kg), smectite, Polyphepan
Irritants	Menthol, ammonia, mustard, turpentine, capsaicin — distracting, reflex

TOPIC 4

Drugs acting on M- and M-N-cholinergic receptors

Learning objectives

Drugs acting on the cholinergic system appear in almost every area of clinical medicine: treatment of glaucoma and myasthenia gravis, postoperative intestinal atony, bradycardia, bronchial obstruction, intestinal and renal colic, urinary incontinence, Alzheimer's disease. Poisoning with organophosphorus compounds (insecticides) and atropine poisoning are classic emergencies. This topic is the foundation of autonomic nervous system pharmacology.

The student should know:

- the structure of efferent innervation, the physiology of the cholinergic synapse and the types of cholinergic receptors;
- the classification, mechanism, pharmacological effects, uses and adverse effects of M-cholinomimetics, anticholinesterase agents and M-cholinergic blockers;
- the clinical picture and treatment of poisoning with organophosphorus compounds, muscarine and atropine.

The student should be able to:

- write prescriptions for preparations such as pilocarpine, neostigmine, pyridostigmine, galantamine, atropine, platyphylline, ipratropium and tropicamide;
- distinguish drugs that cause miosis from those that cause mydriasis;
- recognize OPC and atropine poisoning by their clinical signs and draw up a treatment plan.

1. Efferent innervation and the cholinergic synapse

1.1. Structure of the autonomic nervous system

The efferent nervous system consists of the **somatic** (skeletal muscles) and **autonomic** (internal organs, vessels, glands) parts. The autonomic nervous system is divided into the **parasympathetic** and **sympathetic** divisions.

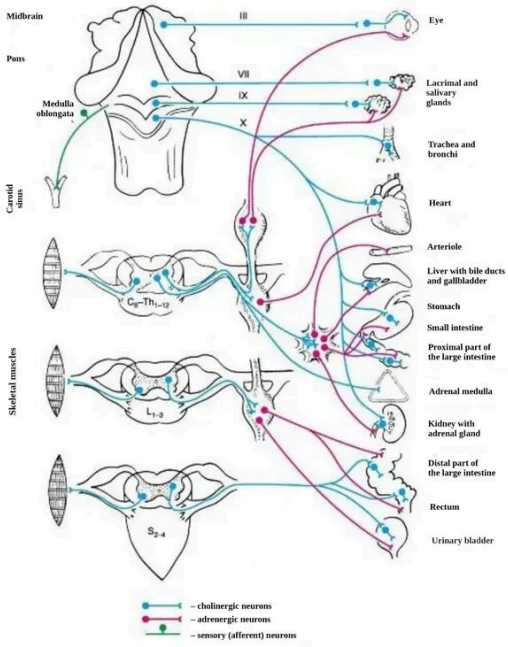


Fig. 3.1. General scheme of cholinergic and adrenergic innervation. C, Th, L, S — segments of the spinal cord.

Figure 4.1. *General scheme of cholinergic and adrenergic innervation (C, Th, L, S — spinal cord segments) (teaching material of the TMA Department of Pharmacology)*

Table 4.1. *Mediators and receptors in efferent nerves*

Synapse	Mediator	Postsynaptic receptor
All autonomic ganglia (sympathetic and parasympathetic)	Acetylcholine	N_N (neuronal)
Adrenal medulla	Acetylcholine	NN
Parasympathetic postganglionic fiber — effector	Acetylcholine	M (muscarinic)
Sympathetic postganglionic fiber — sweat glands	Acetylcholine	M
Sympathetic postganglionic fiber — other organs	Norepinephrine	Alpha, beta
Somatic nerve — skeletal muscle (neuromuscular junction)	Acetylcholine	N_M (muscular)

REMEMBER

Acetylcholine is the mediator of all **preganglionic** fibers, all **parasympathetic postganglionic** fibers, **somatic** nerves and the sympathetic fibers innervating the sweat glands. Norepinephrine is the mediator of most **sympathetic postganglionic** fibers only.

1.2. How the cholinergic synapse works

1. **Synthesis:** in the cytoplasm of the nerve ending, acetylcholine (ACh) is formed from choline and acetyl-CoA with the participation of **choline acetyltransferase**. Choline is taken back up into the cell by a special carrier (hemicholinium blocks this process).
2. **Storage:** ACh is collected into vesicles (vesamicol blocks this).
3. **Release:** when a nerve impulse arrives, Ca^{2+} enters, the vesicles fuse with the membrane and release ACh into the synaptic cleft (exocytosis). **Botulinum toxin** cleaves SNARE proteins and blocks ACh release.
4. **Action on receptors:** ACh stimulates the postsynaptic M or N cholinergic receptors.

5. **Inactivation:** in the synaptic cleft, **acetylcholinesterase** (true cholinesterase) breaks ACh down into choline and acetic acid within a few milliseconds. Blood plasma and the liver contain **butyrylcholinesterase** (pseudocholinesterase).

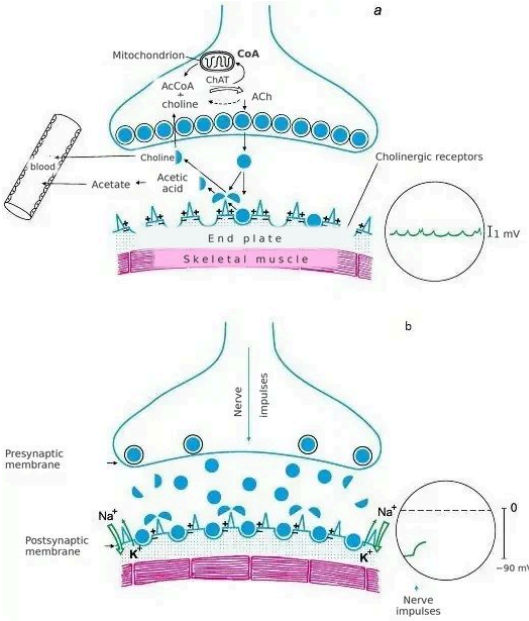


Figure 4.2. *The neuromuscular junction: synthesis and release of acetylcholine and its action on cholinergic receptors (teaching material of the TMA Department of Pharmacology)*

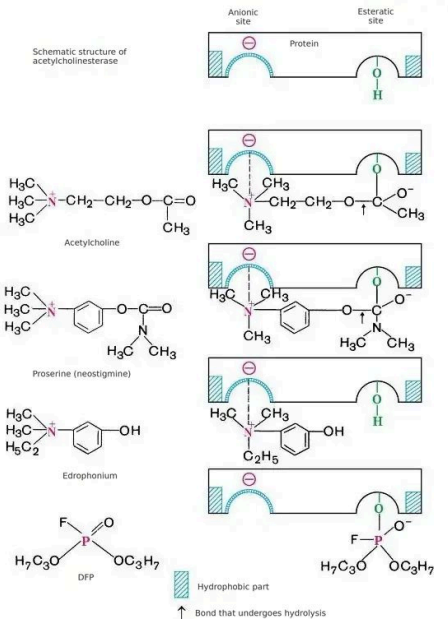


Figure 4.3. *The active site of acetylcholinesterase and the binding of anticholinesterase agents to it (teaching material of the TMA Department of Pharmacology)*

1.3. Cholinergic receptors

Table 4.2. *Types of cholinergic receptors*

Receptor	Location	Coupled mechanism	Main effect when stimulated
M_1	CNS, autonomic ganglia, enterochromaffin cells of the stomach	$G_q \rightarrow IP_3$, DAG, Ca^{2+} ↑	Cognitive functions, gastric secretion ↑
M_2	Heart (sinus and AV nodes, atria), presynaptic	$G_i \rightarrow$ cAMP ↓, K^+ channels open	Bradycardia, AV conduction slows
M_3	Smooth muscles, exocrine glands, vascular endothelium	G_q	Smooth muscle contraction, secretion ↑, vasodilation via NO
M_4, M_5	CNS	G_i / G_q	Modulation of CNS functions
N_N (neuronal)	Autonomic ganglia, adrenal medulla, CNS	Ion channel (Na^+, K^+)	Impulse transmission in ganglia, epinephrine release
N_M (muscular)	Neuromuscular junction of skeletal muscles	Ion channel	Skeletal muscle contraction

Table 4.3. *Changes in organs when M-cholinergic receptors are stimulated*

Organ	Effect
Eye: pupillary constrictor muscle	Contracts — miosis
Eye: ciliary muscle	Contracts — spasm of accommodation (adjustment to near vision), intraocular pressure decreases (outflow through Schlemm's canal improves)
Heart	Bradycardia, AV conduction slows, atrial contraction weakens

Organ	Effect
Blood vessels (endothelium)	NO release — dilation, fall in BP (no innervation; only under the action of drugs)
Bronchi	Constrict (bronchospasm), bronchial gland secretion increases
Gastrointestinal tract	Peristalsis and tone increase, sphincters relax, secretion increases
Bladder	The detrusor contracts, the sphincter relaxes — urination
Glands (salivary, sweat, bronchial, gastric, lacrimal)	Secretion increases

MEMORY AID
M-cholinomimetic effect — **"DUMBELS"**: **D**iarrhea, **U**rination, **M**iosis, **B**radycardia and **B**ronchorrhea/**B**ronchospasm, **E**mesis (vomiting), **L**acrimation, **S**alivation and **S**weating (secretion).

1.4. Classification of drugs acting on the cholinergic synapse

Table 4.4. *Classification of drugs acting on cholinergic synapses*

Group	Subgroup	Preparations
M-cholinomimetics	—	Pilocarpine, aceclidine, bethanechol, muscarine (toxin)
M-N-cholinomimetics (direct)	—	Acetylcholine, carbachol
Indirect cholinomimetics (anticholinesterase agents)	Reversible	Neostigmine (Proserin), pyridostigmine (Kalymin), physostigmine, galantamine, rivastigmine, donepezil, ipidacrine (Amiridin), edrophonium
	Irreversible (OPCs)	Armin, ecothiopate (drugs); trichlorfon (chlorophos), dichlorvos, malathion (karbofos) (insecticides); sarin, soman, VX (chemical warfare agents)
M-cholinergic blockers	Natural alkaloids	Atropine, scopolamine (hyoscine), platyphylline
	Semisynthetic and synthetic	Homatropine, tropicamide, cyclopentolate, metocinium (metocinium iodide), hyoscine butylbromide (Buscopan), ipratropium, tiotropium, pirenzepine, oxybutynin, tolterodine, solifenacin, trihexyphenidyl
N-cholinomimetics, ganglionic blockers, muscle relaxants	—	Covered in topic 5

2. M-cholinomimetics

M-cholinomimetics — drugs that stimulate M-cholinergic receptors **directly**. They reproduce the muscarinic effects of ACh.

2.1. Pilocarpine

Pilocarpine — an alkaloid of the South American plant Pilocarpus jaborandi, a tertiary amine. It is used mainly **topically in ophthalmology**.

- **Miosis** — the pupillary constrictor muscle contracts.
- **Lowering of intraocular pressure** — the iris becomes thinner and stretched, the anterior chamber angle opens, and outflow of fluid into Schlemm's canal improves; the ciliary muscle

contracts and pulls the trabecular meshwork.

- **Spasm of accommodation** — the eye adjusts to near vision, and distance vision temporarily worsens. **Uses:** glaucoma (especially in an attack of angle-closure glaucoma), retinal artery thrombosis, constriction of a pupil dilated by atropine; orally — for **xerostomia** (dry mouth) in Sjögren's syndrome and after radiation.

CLINICAL SIGNIFICANCE

Acute attack of angle-closure glaucoma — an emergency: severe eye pain, redness, sharp loss of vision, “colored halos”, headache, vomiting. Treatment: pilocarpine 1–2% drops (every 15 minutes), acetazolamide orally or intravenously, osmotic diuretics, beta-blocker drops (timolol); then laser iridotomy. **M-cholinergic blockers (atropine) are absolutely contraindicated** in such patients.

2.2. Aceclidine and bethanechol

Aceclidine — a synthetic M-cholinomimetic. It is less irritating than pilocarpine. It is used as eye drops in glaucoma and given subcutaneously for postoperative **atony of the intestine and bladder**.

Bethanechol — is not broken down by cholinesterase and acts mainly on the intestine and bladder; for postoperative urinary retention.

CAUTION

General **contraindications** to M-cholinomimetics and anticholinesterase agents: bronchial asthma, bradycardia and AV block, ischemic heart disease, gastric ulcer, epilepsy, pregnancy (uterine tone increases), **mechanical** obstruction of the urinary and intestinal tracts (peritonitis, intestinal obstruction).

2.3. Muscarine poisoning

Some mushrooms (Inocybe and Clitocybe species; little in fly agaric) contain **muscarine**. Poisoning begins after 15–30 minutes: profuse salivation, sweating, lacrimation, miosis, bronchorrhea, bronchospasm, nausea, vomiting, diarrhea, colicky abdominal pain, bradycardia, hypotension. **Antidote — atropine** (intravenously or intramuscularly, until the signs disappear).

3. Anticholinesterase agents (indirect cholinomimetics)

Anticholinesterase agents inhibit acetylcholinesterase. As a result, ACh is not broken down in the synapses, it accumulates, and the action on **all** cholinergic receptors (M and N) increases. Their effect therefore consists of **muscarinic** (DUMBELS) and **nicotinic** (fibrillary twitching of skeletal muscles, improved neuromuscular transmission, at high doses — depolarizing block) components.

3.1. Reversible anticholinesterase agents

Table 4.5. *Reversible anticholinesterase agents*

Preparation	Structure	Crossing the BBB	Main use
Neostigmine (Proserin)	Quaternary ammonium	Does not cross	Myasthenia gravis; postoperative atony of the intestine and bladder; reversal of nondepolarizing muscle relaxants; glaucoma
Pyridostigmine (Kalymin)	Quaternary ammonium	Does not cross	Myasthenia gravis (the main drug — longer acting than neostigmine, 3–6 hours)
Distigmine (Ubretid)	Quaternary ammonium	Does not cross	Bladder atony, myasthenia gravis (acts for up to 24 hours)

Preparation	Structure	Crossing the BBB	Main use
Edrophonium	Quaternary ammonium	Does not cross	Diagnosis of myasthenia gravis (Tensilon test; acts for 5–10 minutes)
Physostigmine (Eserine)	Tertiary amine	Crosses	Antidote in anticholinergic (atropine) poisoning; glaucoma
Galantamine	Tertiary amine	Crosses	Alzheimer's disease; sequelae of poliomyelitis and neuritis
Rivastigmine	Tertiary amine (carbamate)	Crosses	Dementia in Alzheimer's and Parkinson's disease (capsule, transdermal patch)
Donepezil	Piperidine derivative	Crosses	Alzheimer's disease (once a day)
Ipidacrine (Amiridin, Neuromidin)	Aminoquinoline	Crosses	Peripheral nerve diseases, myasthenia gravis, intestinal atony; also blocks K^+ channels

REMEMBER

Quaternary ammonium compounds (neostigmine, pyridostigmine, metocinium, ipratropium, hyoscine butylbromide) carry a permanent positive charge: they are **poorly absorbed** from the intestine, **do not cross the BBB** and do not act on the CNS. **Tertiary amines** (physostigmine, galantamine, atropine, scopolamine) are well absorbed and act **on the CNS**.

CLINICAL SIGNIFICANCE

Myasthenia gravis — an autoimmune disease in which autoantibodies against N_M cholinergic receptors are formed: pathological fatigability of the extraocular muscles (ptosis, diplopia), the muscles of swallowing and speech, and the limb muscles. Treatment: pyridostigmine (symptomatic), glucocorticoids and immunosuppressants, thymectomy. In a severe **myasthenic crisis** — respiratory failure, mechanical ventilation, plasmapheresis or immunoglobulin.

CAUTION

Myasthenic and cholinergic crisis — both present with muscle weakness and respiratory failure. In a myasthenic crisis, the anticholinesterase drug is **insufficient**; in a cholinergic crisis — **excessive** (DUMBELS signs: miosis, salivation, bradycardia, diarrhea, fasciculations). They used to be distinguished by the edrophonium test; now the airway is secured first and anticholinesterase agents are temporarily stopped.

3.2. Irreversible anticholinesterase agents — organophosphorus compounds (OPCs)

OPCs **phosphorylate** the esteratic site of acetylcholinesterase; the bond is very strong. Over time, the enzyme–OPC complex **"ages"** (an alkyl group is cleaved off) and can then no longer be restored even by reactivators. Enzyme activity recovers only when new enzyme is synthesized (weeks).

Of the OPCs, only **Armin** and **ecothiopate** (in glaucoma) have been used as drugs; their main importance lies in **poisoning**: agricultural insecticides (trichlorfon, malathion, dichlorvos, parathion), household insecticides and chemical warfare agents (sarin, soman, tabun, VX).

3.3. Acute OPC poisoning

OPCs are rapidly absorbed through the skin, mucous membranes, and the respiratory and gastrointestinal tracts.

Table 4.6. *Clinical picture of OPC poisoning*

Effect	Signs
Muscarinic (M)	Miosis (“pinpoint pupils”), blurred vision, profuse salivation, sweating, lacrimation, bronchorrhea and bronchospasm (resembling pulmonary edema), bradycardia, vomiting, diarrhea, abdominal colic, involuntary urination
Nicotinic (N)	Muscle fasciculations (fibrillary twitching), seizures, then muscle weakness and paralysis, paralysis of the respiratory muscles; tachycardia and hypertension (through the ganglia) may occur
Central (CNS)	Restlessness, fear, headache, ataxia, seizures, coma, paralysis of the respiratory center

REMEMBER

Causes of death in OPC poisoning — respiratory failure: bronchorrhea and bronchospasm + paralysis of the respiratory muscles + depression of the respiratory center.

Treatment:

1. **Removing the poison:** remove the victim from the contaminated environment, take off the clothes, wash the skin with soap and water (or with 2–5% sodium bicarbonate solution); if ingested — gastric lavage, activated charcoal. The medical worker must protect himself or herself (gloves).
2. **Securing breathing:** suctioning the airway, oxygen, if necessary intubation and mechanical ventilation. Suxamethonium is **not used** (prolonged apnea).
3. **Atropine** — the main treatment (eliminates the M effect): for adults **2–4 mg intravenously**, then every 5–10 minutes **doubling the dose** until signs of **“atropinization”** appear (cessation of bronchial secretion, disappearance of crackles in the lungs, heart rate above 80, dry skin). In severe poisoning, the daily dose may reach tens or even hundreds of milligrams. Miosis is not a criterion of atropinization.
4. **Cholinesterase reactivators** — pralidoxime, dipyroxime (trimedoxime), alloxime: they detach the phosphoryl group from the enzyme and restore it. They also act on the **N effect** (muscle weakness). They are effective before the enzyme “ages”, i.e. **as early as possible** (in the first hours).
5. **Against seizures** — diazepam or midazolam.

MEMORY AID

Treatment of OPC poisoning — **“A-R-D”**: **A**tropine (blocks M), **R**eactivator (restores the enzyme), **D**iazepam (stops seizures). Atropine does not eliminate the N effect (muscle paralysis) — only a reactivator can do this.

4. M-cholinergic blockers (M-cholinolytics, antimuscarinic agents)

M-cholinergic blockers block M-cholinergic receptors **competitively** and eliminate the effects of ACh and M-cholinomimetics. Their effect resembles “switching off” the parasympathetic innervation.

4.1. Atropine

Atropine — an alkaloid of plants of the nightshade family: belladonna (Atropa belladonna), henbane (Hyoscyamus niger), jimsonweed (Datura stramonium). A tertiary amine; it nonselectively blocks all

types of M-cholinergic receptors.

Table 4.7. *Pharmacological effects of atropine*

Organ	Effect	Mechanism
Eye	Mydriasis (pupil dilates), paralysis of accommodation (cycloplegia — inability to see near objects), intraocular pressure may rise	The pupillary constrictor and ciliary muscles relax; the iris thickens and closes the anterior chamber angle
Heart	Tachycardia, AV conduction improves (transient bradycardia may occur at low doses)	The inhibitory effect of the vagus nerve on the heart is eliminated (M_2)
Bronchi	Dilate, secretion decreases	M_3 blockade
Gastrointestinal tract	Tone and peristalsis decrease (antispasmodic), salivary and gastric secretion decrease	M_3 (M_1) blockade
Urinary tract	The bladder detrusor and urinary tract relax — urinary retention may occur	M_3 blockade
Glands	Secretion of the salivary, sweat, bronchial and lacrimal glands decreases sharply; loss of sweating raises body temperature	M_3 blockade
CNS	At therapeutic doses — little; at toxic doses — excitation, hallucinations, delirium, then depression	Blockade of central M receptors

REMEMBER

Dose dependence of atropine effects: the most sensitive are the salivary, bronchial and sweat glands (0.5 mg) → eye and heart (1–2 mg) → bladder and intestine (2–5 mg) → gastric secretion (the highest dose; that is why atropine is ineffective in gastric ulcer and has many adverse effects). At doses above 10 mg — CNS disturbances.

Uses of atropine:

- **Bradycardia and AV block** (with increased vagal tone, in the first hours of myocardial infarction): 0.5–1 mg intravenously every 3–5 minutes, up to 3 mg in total;
- **premedication** — before anesthesia, to reduce salivary and bronchial secretion and prevent reflex bradycardia (glycopyrronium is now used more often);
- **OPC and muscarine poisoning**, overdose of anticholinesterase agents;
- **smooth muscle spasms** — renal and biliary colic, intestinal colic (platyphylline and hyoscine butylbromide are often preferred);
- **ophthalmology** — examining the fundus and determining refraction (in children), **in iritis and iridocyclitis** (to dilate the pupil, prevent synechiae and rest the ciliary muscle);
- when reversing nondepolarizing muscle relaxants with neostigmine — for protection against the M effects of neostigmine.

CAUTION

A pupil dilated by atropine stays dilated for **7–10 days** (and even longer); for diagnostic purposes, short-acting tropicamide (4–6 hours) or cyclopentolate (up to 24 hours) is now preferred. **Contraindications** to atropine and other M-cholinergic blockers: **angle-closure glaucoma**, prostatic hypertrophy (urinary retention), tachyarrhythmias, intestinal atony and paralytic ileus, severe heart failure, hot climate and febrile children (hyperthermia).

4.2. Atropine poisoning

Poisoning is often seen in children who eat the berries and seeds of belladonna, jimsonweed and henbane, and also with drug overdose.

Clinical picture: dryness of the skin and mucous membranes, difficulty swallowing, hoarseness, **flushing** of the skin of the face and neck, **hyperthermia** (especially in children), marked **mydriasis** and photophobia, inability to see near objects, **tachycardia**, urinary retention, intestinal atony; CNS excitation — restlessness, speech and motor excitation, **hallucinations**, delirium, seizures; in severe cases — coma, paralysis of the respiratory center.

> **MEMORY AID**
> Atropine poisoning, in the classic English phrase: **"Hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter, full as a flask"** (hyperthermia, dry skin, flushing, mydriasis and inability to see near objects, delirium, urinary retention).

Treatment:

1. Gastric lavage (useful even late, because atropine slows intestinal motility), activated charcoal.
2. **Antidote — physostigmine** (a tertiary amine that enters the CNS) 0.5–2 mg intravenously, slowly; it eliminates central and peripheral signs. Neostigmine acts only on peripheral signs.
3. For agitation and seizures — **diazepam** (phenothiazines are contraindicated — they themselves have M-blocking activity).
4. For hyperthermia — physical cooling (ice, wet sponging).
5. For urinary retention — catheterization; for respiratory disturbance — mechanical ventilation.

4.3. Other M-cholinergic blockers

Table 4.8. *Comparison of M-cholinergic blockers*

Preparation	Property	Main use
Scopolamine (hyoscine hydrobromide)	Tertiary amine; calms the CNS (sedation, amnesia), strong mydriasis; weaker effect on the heart	Prevention of sea and air sickness (transdermal patch behind the ear), premedication
Platyphylline	An alkaloid of Senecio (Senecio platyphylloides); M blocker + direct myotropic antispasmodic; mildly depresses the CNS	Spasms of the gastrointestinal, biliary and urinary tracts, colic; vascular spasm
Hyoscine butylbromide (Buscopan)	Quaternary ammonium; does not enter the CNS, poorly absorbed from the intestine	Intestinal, biliary and urinary colic, dysmenorrhea
Metocinium (metocinium iodide)	Quaternary ammonium; stronger peripheral effect than atropine, no effect on the CNS	Spasms, premedication
Ipratropium bromide	Quaternary ammonium; when inhaled, hardly absorbed into the blood	COPD and bronchial asthma (bronchodilator); rhinorrhea (nasal spray)
Tiotropium bromide	Long-acting (24 hours), selective for M_3 and M_1	Maintenance therapy of COPD (once a day)
Pirenzepine	Acts selectively on M_1	Gastric and duodenal ulcer (rarely used now)
Tropicamide	Short-acting (mydriasis for 4–6 hours)	Examination of the fundus

Preparation	Property	Main use
Cyclopentolate, homatropine	Intermediate duration (1–2 days)	Diagnostic mydriasis, cycloplegia (refraction in children)
Oxybutynin, tolterodine, solifenacin, darifenacin	Selective for M_3 receptors of the bladder	Overactive bladder, urge urination, enuresis
Trihexyphenidyl (Cyclodol), biperiden	Central M blockers	Parkinsonism, extrapyramidal disorders caused by neuroleptics

CLINICAL SIGNIFICANCE

"Anticholinergic burden" in the elderly. Many drugs have M-blocking properties: tricyclic antidepressants (amitriptyline), first-generation antihistamines (diphenhydramine, Dimedrol), phenothiazines, oxybutynin, trihexyphenidyl. In the elderly, their combined effect causes **cognitive impairment, delirium, falls, urinary retention, constipation** and attacks of glaucoma. Such drugs should be prescribed to the elderly as little as possible (Beers criteria).

4.4. Drugs causing mydriasis: the difference between cholinergic blockers and adrenomimetics

Table 4.9. *Drugs that change the pupil*

Group	Pupil	Accommodation	Intraocular pressure
M-cholinomimetics, anticholinesterase agents	Miosis	Spasm (adjustment to near vision)	Decreases
M-cholinergic blockers	Mydriasis	Paralysis (cycloplegia)	May increase
Alpha-1 adrenomimetics (phenylephrine)	Mydriasis (the radial muscle contracts)	Unchanged	Almost unchanged
Alpha-2 adrenomimetics, beta-blockers (timolol)	Almost unchanged	Unchanged	Decreases (production of fluid decreases)

MEMORY AID

Eye muscles: **"Circular — parasympathetic, radial — sympathetic"**. The circular (sphincter) muscle constricts the pupil through M receptors; the radial (dilator) muscle dilates it through alpha-1. M blockers cause **both mydriasis and cycloplegia**; phenylephrine — mydriasis only.

5. Laboratory class: studying the antispasmodic effect of M-cholinergic blockers

This section corresponds to the 5th-semester class "Laboratory study of the antispasmodic effect of M-cholinergic blockers".

Essence of the experiment. An isolated segment of intestine (usually guinea pig ileum) is placed in warm (37 °C) oxygenated Tyrode's solution, and its contractions are recorded.

1. Baseline contractions are recorded.
2. **Acetylcholine** is added to the bath — the intestine contracts strongly (tone and amplitude increase).
3. After washing, **atropine** is added to the bath and then acetylcholine is given again — the contracting effect of ACh **weakens or disappears**.
4. It is shown that the block can be overcome by increasing the ACh dose — a sign of **competitive antagonism** (the dose–effect curve shifts to the right, Emax is unchanged).

5. For comparison, a spasm caused by **barium chloride** (a direct myotropic contractor) is not abolished by atropine but is abolished by **papaverine** (a myotropic antispasmodic).

REMEMBER

Conclusion of the experiment: atropine abolishes only spasm induced **through M-cholinergic receptors** (**neurotropic antispasmodic**). Papaverine and drotaverine act directly on the smooth muscle cell (**myotropic antispasmodics**) and relax spasm of any cause. Platyphylline has both mechanisms.

6. Clinical part

Table 4.10. *Choosing cholinergic drugs according to the clinical situation*

Clinical situation	Choice	Comment
Open-angle glaucoma	Prostaglandin analogues, timolol (first line); pilocarpine — add-on	Pilocarpine is poorly tolerated by young people because of spasm of accommodation
Attack of angle-closure glaucoma	Pilocarpine + acetazolamide + timolol	M blockers are strictly contraindicated
Myasthenia gravis	Pyridostigmine	Neostigmine — short-acting, more for diagnosis/emergencies
Postoperative intestinal atony (no mechanical obstruction)	Neostigmine subcutaneously	Mechanical causes of intestinal obstruction must be ruled out
Alzheimer's disease (mild to moderate)	Donepezil, rivastigmine, galantamine	Monitor for bradycardia and nausea
Sinus bradycardia (symptomatic)	Atropine intravenously	Ineffective after heart transplantation (denervation)
Renal colic	Hyoscine butylbromide, drotaverine + analgesic (NSAID)	Atropine has many adverse effects
COPD	Tiotropium (long-acting), ipratropium (short-acting)	Teach the inhalation technique
Sea sickness	Scopolamine patch	Drowsiness, dry mouth
Overactive bladder	Solifenacin, tolterodine	Monitor cognitive effects in the elderly

CLINICAL SIGNIFICANCE

The combination of neostigmine + atropine is used to reverse the residual effect of nondepolarizing muscle relaxants (rocuronium, vecuronium): neostigmine accumulates ACh at the neuromuscular junction and restores muscle strength, while atropine (or glycopyrronium) prevents its M effects such as bradycardia, bronchorrhea and salivation. The modern specific antidote for rocuronium is **sugammadex**.

Practical skills

Skill 1. Algorithm of emergency care for a patient poisoned with OPCs

1. Protect yourself (gloves, mask) and remove the patient from the contaminated environment.
2. Take off the clothes, wash the skin with soap and plenty of water.
3. Clear the airway and give oxygen; if necessary, intubate (without suxamethonium).
4. Establish intravenous access; **atropine 2–4 mg intravenously**, doubling the dose every 5–10 minutes until atropinization (dry bronchi, HR > 80).
5. Give a **cholinesterase reactivator** (pralidoxime or dipyroxime) as early as possible.

6. For seizures — diazepam 10 mg intravenously.
7. If ingested (after the airway has been secured) — gastric lavage, activated charcoal.
8. Determine serum cholinesterase activity and observe for at least 24–48 hours (delayed conditions, “intermediate syndrome”).

Skill 2. Key prescriptions

PRESCRIPTION 4.1. PILOCARPINE EYE DROPS
Rp.: Solutionis Pilocarpini hydrochloridi 1% — 10 ml
D.S. Instill 1–2 drops into each eye 3–4 times a day.

PRESCRIPTION 4.2. NEOSTIGMINE (IN AMPOULES)
Rp.: Solutionis Neostigmini methylsulfatis 0.05% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml subcutaneously 1–2 times a day (in postoperative intestinal atony).

PRESCRIPTION 4.3. NEOSTIGMINE (TABLETS)
Rp.: Neostigmini bromidi 0.015
D.t.d. N. 20 in tabulettis
S. Take 1 tablet 2–3 times a day 30 minutes before meals.

PRESCRIPTION 4.4. PYRIDOSTIGMINE
Rp.: Pyridostigmini bromidi 0.06
D.t.d. N. 50 in tabulettis
S. Take 1 tablet 3 times a day (in myasthenia gravis; the dose is selected individually by the neurologist).

PRESCRIPTION 4.5. GALANTAMINE
Rp.: Capsulas Galantamini 0.008 N. 28
D.S. Take 1 capsule once a day in the morning with food.

PRESCRIPTION 4.6. ATROPINE (IN AMPOULES)
Rp.: Solutionis Atropini sulfatis 0.1% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 0.5–1 ml intravenously (in bradycardia).

PRESCRIPTION 4.7. ATROPINE EYE DROPS
Rp.: Solutionis Atropini sulfatis 1% — 5 ml
D.S. Instill 1 drop into the affected eye 2 times a day (in iridocyclitis).

PRESCRIPTION 4.8. TROPICAMIDE EYE DROPS
Rp.: Solutionis Tropicamidi 1% — 5 ml
D.S. Instill 1 drop into each eye 15–20 minutes before examination of the fundus.

PRESCRIPTION 4.9. PLATYPHYLLINE
Rp.: Solutionis Platyphyllini hydrotartratis 0.2% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml subcutaneously (in colic).

PRESCRIPTION 4.10. HYOSCINE BUTYLBROMIDE

Rp.: Hyoscini butylbromidi 0.01
D.t.d. N. 20 in tabulettis obductis
S. Take 1–2 tablets 3 times a day (in intestinal colic).

PRESCRIPTION 4.11. IPRATROPIUM BROMIDE AEROSOL

Rp.: Aerosoli Ipratropii bromidi 0.00002/dosi — 200 dosi
D.S. Take 2 inhalations 4 times a day.

PRESCRIPTION 4.12. TIOTROPIUM BROMIDE (CAPSULES FOR A DRY POWDER INHALER)

Rp.: Tiotropii bromidi 0.000018
D.t.d. N. 30 in capsulis
S. Inhale the contents of 1 capsule with the inhaler once a day at the same time.

PRESCRIPTION 4.13. SOLIFENACIN

Rp.: Tabulettas Solifenacini 0.005 obductas N. 30
D.S. Take 1 tablet once a day.

PRESCRIPTION 4.14. RECTAL SUPPOSITORIES WITH BELLADONNA EXTRACT

Rp.: Suppositorium cum extracto Belladonnae 0.015
D.t.d. N. 10
S. Insert 1 suppository into the rectum 2 times a day (for pain of hemorrhoids or anal fissure).

Review questions

1. How is the autonomic nervous system organized? At which synapses is acetylcholine the mediator?
2. How do the synthesis, release and inactivation of acetylcholine take place at the cholinergic synapse?
3. Types, location and signaling mechanisms of M and N cholinergic receptors.
4. What changes occur in the organs when M-cholinergic receptors are stimulated?
5. Explain the effect of pilocarpine on the eye and its mechanism in glaucoma.
6. Classification and mechanism of anticholinesterase agents. Why do they have M and N effects?
7. Differences and uses of neostigmine, pyridostigmine, physostigmine and galantamine.
8. How can a myasthenic crisis be distinguished from a cholinergic crisis?
9. Clinical picture and principles of treatment of OPC poisoning. Why must reactivators be given early?
10. Pharmacological effects of atropine and their dose dependence.
11. Uses and contraindications of atropine.
12. Clinical picture and treatment of atropine poisoning. Why is physostigmine the antidote?
13. Properties of scopolamine, platyphylline, hyoscine butylbromide, ipratropium and tiotropium.
14. Which drugs cause miosis and which cause mydriasis? How does mydriasis due to phenylephrine differ from that due to atropine?
15. Pharmacokinetic differences between quaternary ammonium compounds and tertiary amines.
16. How does the antispasmodic effect of atropine differ from that of papaverine in the laboratory experiment?

Test questions

1. Which type of cholinergic receptor is located mainly in the heart?
   A) M_2
   B) NN
   C) M_1
   D) NM
   E) M_3

2. By what mechanism does pilocarpine lower intraocular pressure in glaucoma?
   A) It dilates the pupil
   B) It reduces the production of fluid
   C) It contracts the pupillary and ciliary muscles and improves the outflow of fluid
   D) It blocks beta-adrenergic receptors
   E) It inhibits carbonic anhydrase

3. An anticholinesterase agent that does not cross the BBB:
   A) Rivastigmine
   B) Physostigmine
   C) Neostigmine
   D) Galantamine
   E) Donepezil

4. The main anticholinesterase agent for long-term treatment of myasthenia gravis:
   A) Physostigmine
   B) Edrophonium
   C) Pyridostigmine
   D) Atropine
   E) Armin

5. A nicotinic sign in organophosphorus compound poisoning:
   A) Salivation
   B) Diarrhea
   C) Bronchorrhea
   D) Muscle fasciculations
   E) Miosis

6. By what criterion is the atropine dose titrated in OPC poisoning?
   A) Until muscle strength recovers
   B) Until bronchial secretion stops and the heart rate exceeds 80 (atropinization)
   C) Until the patient regains consciousness
   D) Until pupil size normalizes
   E) Only once, 1 mg

7. A drug that restores cholinesterase (a reactivator):
   A) Naloxone
   B) Pralidoxime
   C) Physostigmine
   D) Neostigmine
   E) Atropine

8. A sign typical of atropine poisoning:
   A) Profuse sweating
   B) Dry and flushed skin
   C) Miosis
   D) Bronchorrhea
   E) Bradycardia

9. The antidote that eliminates both central and peripheral signs of atropine poisoning:
   A) Naloxone
   B) Pralidoxime
   C) Chlorpromazine (aminazine)
   D) Neostigmine
   E) Physostigmine

10. An absolute contraindication to M-cholinergic blockers:
   A) Bradycardia
   B) Bronchial asthma
   C) Angle-closure glaucoma
   D) Diarrhea
   E) Renal colic

11. An M-cholinergic blocker used as a transdermal patch to prevent sea sickness:
   A) Platyphylline
   B) Ipratropium
   C) Tropicamide
   D) Pirenzepine
   E) Scopolamine

12. An M-cholinergic blocker inhaled once a day as maintenance therapy for COPD:
   A) Metocinium
   B) Atropine
   C) Platyphylline
   D) Ipratropium
   E) Tiotropium

13. The shortest-acting mydriatic for examining the fundus:
   A) Homatropine
   B) Atropine
   C) Pilocarpine
   D) Scopolamine
   E) Tropicamide

14. An M-cholinergic blocker used for overactive bladder:
   A) Ipratropium
   B) Neostigmine
   C) Solifenacin
   D) Pirenzepine
   E) Galantamine

15. Which drug abolishes the spasm caused by barium chloride in an isolated intestine?
    A) Papaverine
    B) Metocinium
    C) Atropine
    D) Pirenzepine
    E) Ipratropium

Clinical cases

Case 1

A 45-year-old farmer was brought to the emergency department 2 hours after spraying insecticide in a cotton field. Complaints: blurred vision, shortness of breath, nausea. On examination: pinpoint pupils, profuse salivation and sweating, numerous moist crackles in the lungs, HR 48/min, BP 90/60 mm Hg, fibrillary twitching of the face and arm muscles.

Questions: 1) Your diagnosis? 2) Divide the signs into M, N and central effects. 3) Draw up a treatment plan.

Answer: 1) Acute poisoning with an organophosphorus compound (insecticide). 2) M signs: miosis, salivation and sweating, bronchorrhea (moist crackles), bradycardia, hypotension, nausea; N signs: muscle fasciculations; central: mild so far (seizures and coma may follow). 3) Remove the clothes, wash the skin with soap; oxygen, airway toilet; atropine 2–4 mg intravenously, doubling the dose every 5–10 minutes until bronchial secretion stops and HR > 80; pralidoxime (or dipyroxime) as early as possible; diazepam for seizures; monitoring in the intensive care unit.

Case 2

A 4-year-old child ate jimsonweed seeds growing in the village. An hour later the child is restless and hallucinating; the skin is dry and red, temperature 39.2 °C, the pupils are dilated and unreactive to light, HR 160/min, and he has not urinated for 6 hours.

Questions: 1) What kind of poisoning is this? 2) The antidote and the reason for choosing it. 3) Which drugs are contraindicated?

Answer: 1) Poisoning with atropine-like (anticholinergic) alkaloids. 2) Physostigmine — being a tertiary amine, it crosses the BBB and eliminates central (delirium, hallucinations) and peripheral signs; in children 0.02 mg/kg intravenously, slowly (under ECG monitoring). In addition: gastric lavage, activated charcoal, physical cooling, bladder catheterization, diazepam for agitation. 3) Phenothiazines (chlorpromazine), diphenhydramine (Dimedrol), tricyclic antidepressants — they themselves have M-cholinergic blocking properties and worsen the poisoning.

Case 3

A 70-year-old man was prescribed diphenhydramine (Dimedrol) for insomnia and oxybutynin for overactive bladder; in addition, he takes amitriptyline for depression. In recent days, his relatives have noticed confusion, memory impairment, constipation and difficulty urinating. He has a history of prostatic adenoma.

Questions: 1) What is the cause of this condition? 2) What would you recommend?

Answer: 1) Anticholinergic burden — all three drugs (diphenhydramine, oxybutynin, amitriptyline) have M-cholinergic blocking activity; in the elderly, their combined effect has caused delirium, cognitive impairment, constipation and, against the background of prostatic adenoma, urinary

retention. 2) Stop diphenhydramine (sleep hygiene, if necessary drugs with little M-blocking activity); replace amitriptyline with an SSRI (sertraline); consider the beta-3 adrenomimetic mirabegron, which has no M-blocking activity, instead of oxybutynin; urologist consultation for the prostatic adenoma; regular review of all prescribed drugs.

Quick review table

Table 4.11. *M- and M-N-cholinergic drugs — summary*

Group	Preparations	Mechanism	Main use	Important adverse effect / note
M-cholinomimetics	Pilocarpine, aceclidine, bethanechol	Stimulates M receptors directly	Glaucoma, xerostomia, intestinal and bladder atony	DUMBELS; contraindicated in asthma and bradycardia
Reversible anticholinesterases	Neostigmine, pyridostigmine, physostigmine, galantamine, donepezil, rivastigmine	Reversibly inhibit AChE, ACh accumulates	Myasthenia gravis, atony, reversal of muscle relaxants, Alzheimer's disease, atropine poisoning	Cholinergic crisis, bradycardia
OPCs	Trichlorfon (chlorophos), malathion (karbofos), sarin	Irreversibly phosphorylate AChE	Insecticides, poisoning	Treatment: atropine + reactivator + diazepam
Reactivators	Pralidoxime, dipyroxime	Restore AChE	OPC poisoning (early)	Ineffective after “aging”
M-cholinergic blockers (natural)	Atropine, scopolamine, platyphylline	Competitively block M receptors	Bradycardia, premedication, OPC poisoning, spasms, ophthalmology, sea sickness	Dryness, tachycardia, mydriasis, urinary retention, attack of glaucoma
Quaternary M blockers	Metocinium, hyoscine butylbromide, ipratropium, tiotropium	Do not enter the CNS	Colic, COPD, asthma	Dry mouth
Selective M blockers	Pirenzepine (M_1), solifenacin, tolterodine (bladder)	—	Gastric ulcer (rarely), overactive bladder	Cognitive impairment in the elderly
Mydriatics	Atropine (7–10 days), cyclopentolate (1 day), tropicamide (4–6 hours)	—	Diagnosis, iridocyclitis	Contraindicated in angle-closure glaucoma

TOPIC 5

Drugs acting on N-cholinergic receptors

Learning objectives

N-cholinergic receptors play a decisive role in two places: **in the autonomic ganglia** (NN) and **at the neuromuscular junction of skeletal muscles** (NM). The drugs acting on these receptors are the main drugs of modern anesthesiology (muscle relaxants), smoking cessation drugs (cytisine, varenicline) and ganglionic blockers of historical importance. Working with muscle relaxants requires full control of breathing and circulation, so knowing their mechanism and complications is mandatory for every physician.

The student should know:

- the structure and types of N-cholinergic receptors (neuronal and muscular);
- the effects and uses of N-cholinomimetics (nicotine, cytisine, lobeline, varenicline) and the toxicology of nicotine;
- the mechanism, effects, uses and complications of ganglionic blockers;
- the classification of peripheral muscle relaxants, the difference between the mechanisms of nondepolarizing and depolarizing agents, and their antagonists;
- the complications of muscle relaxants (prolonged apnea, malignant hyperthermia, hyperkalemia) and their management.

The student should be able to:

- write prescriptions for preparations such as cytisine, azamethonium, rocuronium, suxamethonium, neostigmine and sugammadex;
- match the choice of muscle relaxant to the clinical situation;
- draw up a pharmacotherapy plan for smoking cessation.

1. N-cholinergic receptors

The N-cholinergic receptor is a **ligand-gated ion channel** (ionotropic receptor). It consists of five subunits (alpha, beta, gamma or epsilon, delta), with a channel in the center that conducts Na^+ and K^+ ions. When two molecules of acetylcholine bind to **the two alpha subunits**, the channel opens and the membrane depolarizes.

Table 5.1. *Types of N-cholinergic receptors*

Type	Location	When stimulated	Blocking agents
N_N (neuronal)	Sympathetic and parasympathetic ganglia, adrenal medulla, carotid bodies, CNS	Impulse transmission in ganglia, epinephrine release, reflex stimulation of breathing	Ganglionic blockers
N_M (muscular)	End plate of skeletal muscles	Skeletal muscle contraction	Peripheral muscle relaxants

Table 5.2. *Classification of drugs acting on N-cholinergic receptors*

Group	Preparations
N-cholinomimetics (ganglionic stimulants)	Nicotine, cytisine (Tabex), lobeline, varenicline (partial agonist)

Group	Preparations
N-cholinergic blockers: ganglionic blockers	Azamethonium (Pentamin), hexamethonium (Benzohexonium), Hygronium, trimetaphan, pempidine (Pirilen)
N-cholinergic blockers: peripheral muscle relaxants — nondepolarizing (competitive)	Tubocurarine, pancuronium, pipecuronium, vecuronium, rocuronium, atracurium, cisatracurium, mivacurium
N-cholinergic blockers: peripheral muscle relaxants — depolarizing	Suxamethonium (Ditilin, Listenon)
Antagonists of muscle relaxants	Neostigmine (anticholinesterase), sugammadex (binds rocuronium and vecuronium)

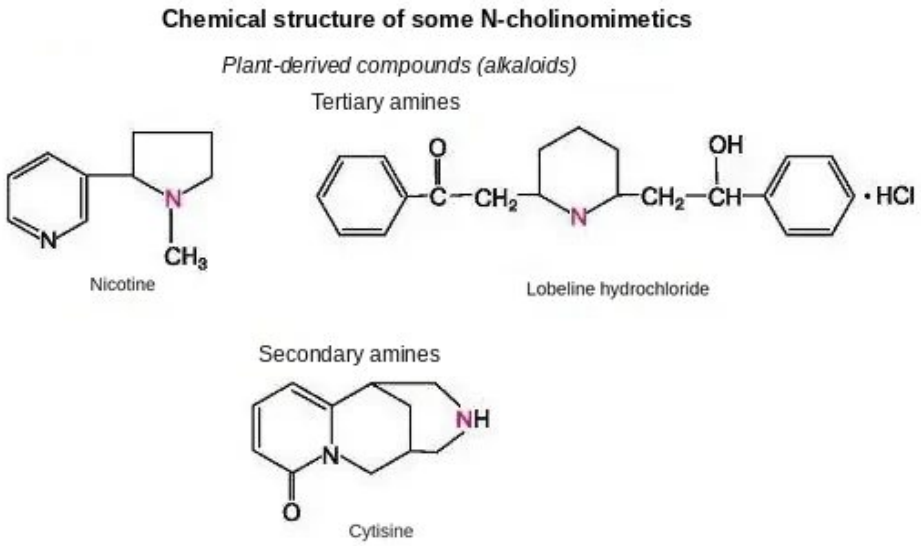


Figure 5.1. *Chemical structure of some N-cholinomimetics: nicotine, lobeline, cytisine (teaching material of the TMA Department of Pharmacology)*

2. N-cholinomimetics (ganglionic stimulants)

2.1. Nicotine

Nicotine — an alkaloid of tobacco (Nicotiana tabacum), a liquid tertiary amine; it is absorbed very quickly from the skin, mucous membranes and lungs. As a drug, it is used only in **nicotine replacement therapy** (smoking cessation).

Biphasic action: at small doses it **stimulates** N-cholinergic receptors, at large doses it **blocks** them through prolonged depolarization.

Table 5.3. *Effects of nicotine (at doses seen in smokers)*

System	Effect
CNS	Excitation, improved attention and mood (dopamine release in the mesolimbic system — the basis of dependence), reduced appetite; at large doses — tremor, seizures
Cardiovascular	Stimulation of the sympathetic ganglia and adrenal medulla — tachycardia, rise in BP, constriction of peripheral vessels, increased myocardial oxygen demand
Gastrointestinal tract	Stimulation of the parasympathetic ganglia — increased salivation and peristalsis, nausea (with the first cigarette)
Respiration	Reflex stimulation of breathing through the carotid bodies

System	Effect
Blood	Platelet aggregation increases

The harm of smoking is due mainly not to nicotine but to **more than 4000** substances in tobacco smoke (tars, benzpyrene and other carcinogens, carbon monoxide, cyanides): lung cancer, COPD, atherosclerosis, myocardial infarction, stroke, impaired fetal development.

Acute nicotine poisoning (when children swallow cigarettes or nicotine liquid, e-cigarette liquid): nausea, vomiting, salivation, abdominal pain, diarrhea, dizziness, tachycardia, then bradycardia and hypotension, seizures, **paralysis of the respiratory muscles**. Treatment: activated charcoal, respiratory support, diazepam for seizures, atropine for M signs.

2.2. Cytisine and lobeline

Cytisine — an alkaloid of thermopsis and laburnum (Cytisus laburnum, the “golden rain” shrub); it stimulates N-cholinergic receptors, but more weakly and more selectively than nicotine. **Tabex** tablets — for smoking cessation: by partly occupying nicotinic receptors, it softens withdrawal symptoms and reduces the “pleasure” obtained from smoking.

Lobeline — an alkaloid of lobelia. It was formerly used as a reflex respiratory analeptic; it is hardly used now.

CAUTION

Older textbooks recommended cytisine (Cytiton) and lobeline as **reflex respiratory analeptics** (carbon monoxide poisoning, asphyxia). In modern practice, respiratory failure is treated with oxygen and mechanical ventilation; analeptics are considered ineffective and dangerous. The main current importance of cytisine is **smoking cessation**.

2.3. Pharmacotherapy of smoking cessation

Table 5.4. *Drugs for smoking cessation*

Drug	Mechanism	Features of use
Nicotine replacement therapy (patch, gum, lozenge, inhaler, spray)	Delivers nicotine slowly, without smoke	8–12 weeks, with gradual dose reduction
Cytisine (Tabex)	Partial agonist of N-cholinergic receptors	25-day course; stop smoking completely by day 5
Varenicline	Partial agonist of alpha-4-beta-2 N receptors	The most effective monotherapy; 12 weeks; nausea, unusual dreams
Bupropion	Dopamine and norepinephrine reuptake inhibitor (antidepressant)	Contraindicated in patients prone to seizures

CLINICAL SIGNIFICANCE

Smoking cessation is the most effective preventive measure for reducing cardiovascular disease and lung cancer. Pharmacotherapy **combined with brief counseling** increases the chance of quitting 2–3-fold. Every GP should offer smoking cessation to a smoking patient at every visit.

3. Ganglionic blockers

Ganglionic blockers block the N_N-cholinergic receptors in the autonomic ganglia and simultaneously stop impulse transmission in **sympathetic and parasympathetic** ganglia; they also

reduce epinephrine release from the adrenal medulla.

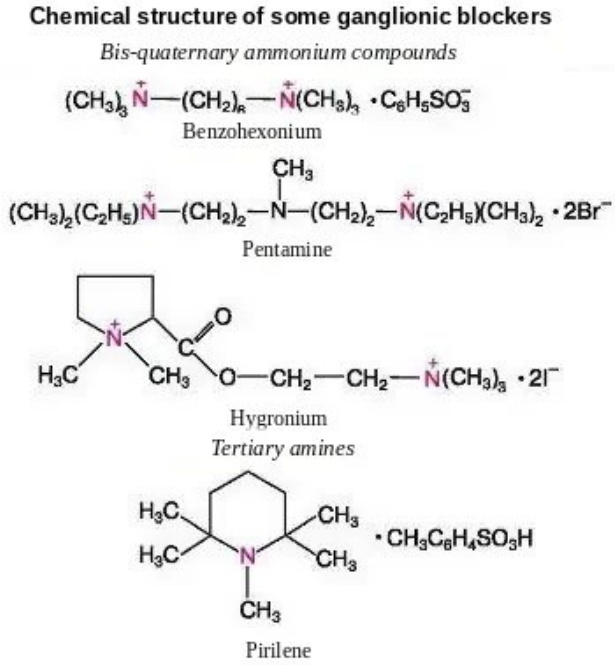


Figure 5.2. *Ganglionic blockers with a bis-quaternary ammonium structure: Benzohexonium, Pentamin, Hygronium; tertiary amine — Pirilen (teaching material of the TMA Department of Pharmacology)*

The resulting effect in each organ depends on which tone predominates:

Table 5.5. *Effects of ganglionic blockers on organs*

Organ	Predominant tone	Effect of the ganglionic blocker
Arterioles	Sympathetic	Dilation, fall in BP (especially in the upright position — orthostatic collapse)
Veins	Sympathetic	Dilation, reduced venous return
Heart	Parasympathetic	Tachycardia
Eye (pupil, accommodation)	Parasympathetic	Mydriasis, paralysis of accommodation
Gastrointestinal tract	Parasympathetic	Decreased tone and peristalsis, constipation, paralytic ileus
Bladder	Parasympathetic	Urinary retention
Salivary glands	Parasympathetic	Dry mouth
Sweat glands	Sympathetic (cholinergic)	Reduced sweating

REMEMBER

Ganglionic blockers **“switch off the sympathetic and parasympathetic systems at once”**: in the vessels — loss of sympathetic tone (hypotension); in most other organs — loss of parasympathetic tone (atropine-like effects). That is why they have many adverse effects.

Table 5.6. *Main ganglionic blockers*

Preparation	Duration	Use
Trimetaphan	Very short (10–15 minutes), by intravenous drip	Controlled hypotension (during surgery), hypertensive crisis (historically)
Hygronium	Short (10–20 minutes)	Controlled hypotension
Azamethonium (Pentamin)	Intermediate (2–4 hours)	Hypertensive crisis, pulmonary edema (with high BP), peripheral vasospasm
Hexamethonium (Benzohexonium)	Intermediate	Hypertensive crisis, gastric ulcer and colic (historically)
Pempidine (Pirilen)	Long, oral	Arterial hypertension (historically)

Uses (mainly historical):

- hypertensive crisis (intravenously or intramuscularly, in the horizontal position);
- **controlled (deliberate) hypotension** — during surgery to reduce blood loss;
- pulmonary edema against the background of arterial hypertension (they “unload” the pulmonary circulation);
- peripheral vasospasm (Raynaud's disease). **Adverse effects: orthostatic collapse** (the most dangerous), tachycardia, dry mouth, disturbed accommodation, constipation and paralytic ileus, urinary retention, development of tolerance.

> **CAUTION**
> After a ganglionic blocker has been given, the patient must lie **horizontally for 2 hours**; if he gets up suddenly, orthostatic collapse develops. Today ganglionic blockers are hardly used to treat hypertension — they have been replaced by modern antihypertensive drugs with fewer adverse effects (topic 11).

4. Peripheral muscle relaxants (curare-like agents)

Peripheral muscle relaxants block N_M-cholinergic receptors at the neuromuscular junction of skeletal muscles and cause **muscle relaxation (paralysis)**. They do not act on the CNS — the patient does not lose consciousness and feels pain. Therefore muscle relaxants are used **only during general anesthesia** and with mechanical ventilation.

History. Curare is the arrow poison of South American Indians. In 1942, Griffith and Johnson first used tubocurarine during surgery; this was a revolution in anesthesiology.

Order of muscle relaxation: first the **facial and eye** muscles, then the **fingers**, limbs, neck and trunk muscles, the **intercostal muscles** and, last of all, the **diaphragm**. Recovery is in the reverse order.

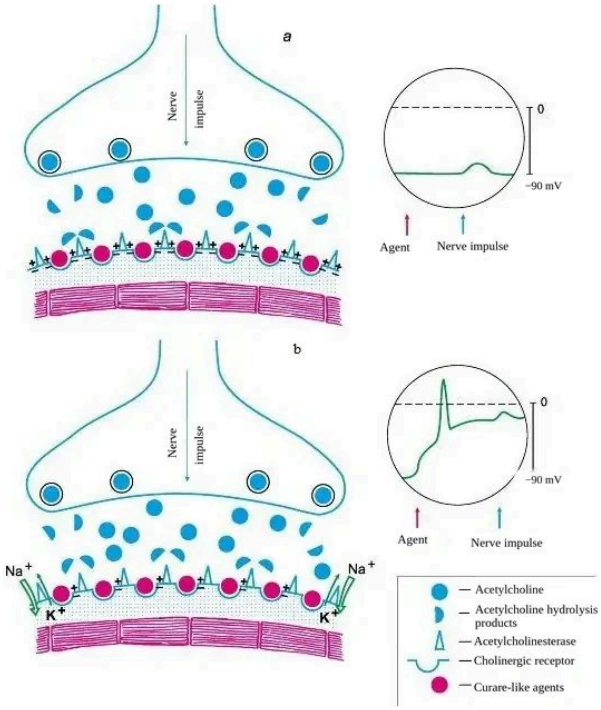


Figure 5.3. *Mechanism of action of curare-like agents: a — nondepolarizing (competitive) block; b — depolarizing block (teaching material of the TMA Department of Pharmacology)*

4.1. Nondepolarizing (competitive) muscle relaxants

Mechanism: they block N_M receptors **competitively** with ACh and do not cause depolarization themselves. No end-plate potential is generated. Increasing the ACh concentration in the synapse (anticholinesterase agents — **neostigmine**) reverses the block.

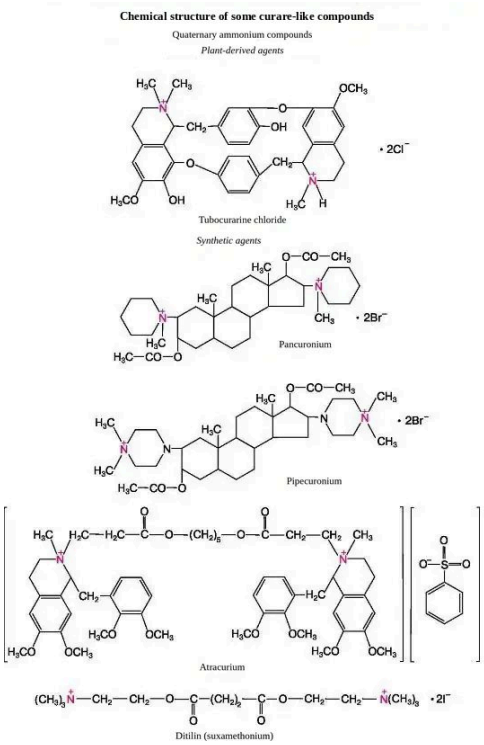


Figure 5.4. *Chemical structure of some curare-like agents: tubocurarine, pancuronium, pipecuronium, atracurium, Ditilin (teaching material of the TMA Department of Pharmacology)*

Table 5.7. *Nondepolarizing muscle relaxants*

Preparation	Onset / duration of action	Elimination	Features
Tubocurarine	3–5 / 40–60 minutes	Kidney	Releases histamine (hypotension, bronchospasm), ganglionic blocker; hardly used now
Pancuronium	3–5 / 60–90 minutes	Kidney	Vagolytic — tachycardia, rise in BP
Pipecuronium	3–5 / 60–120 minutes	Kidney	Little cardiovascular effect
Vecuronium	2–3 / 30–40 minutes	Liver (bile)	No cardiovascular effect
Rocuronium	1–1.5 / 30–40 minutes	Liver	The fastest-acting nondepolarizing agent; can replace suxamethonium in rapid sequence intubation; reversed by sugammadex
Atracurium	2–3 / 30–40 minutes	Hofmann elimination (spontaneous breakdown in plasma) and esterases	Convenient in renal and hepatic failure; releases some histamine
Cisatracurium	3–5 / 40–50 minutes	Hofmann elimination	Does not release histamine; for prolonged relaxation in intensive care
Mivacurium	2–3 / 15–20 minutes	Plasma pseudocholinesterase	The shortest-acting nondepolarizing agent

MEMORY AID

Hofmann elimination — “ATRA and CIS”: **atra**curium and **cis**atracurium break down spontaneously in the blood independently of organ function — the agents of choice in renal and hepatic failure.

Factors that enhance a competitive block: inhalational anesthetics (sevoflurane, isoflurane), **aminoglycosides** (gentamicin — reduces ACh release), polymyxins, lincosamides, magnesium sulfate, calcium antagonists, hypothermia, acidosis, hypokalemia, myasthenia gravis (these patients are very sensitive to competitive relaxants).

Antagonists:

- **Neostigmine** (0.04–0.07 mg/kg intravenously) — together with **atropine** or glycopyrronium (to prevent M effects). Effective only after partial recovery has begun; ineffective in a deep block.
- **Sugammadex** — a modified gamma-cyclodextrin; it **encapsulates** (“wraps up”) **rocuronium** and vecuronium molecules in the blood, forming an inactive complex. It rapidly (within 2–3 minutes) reverses a block of any depth, without M effects.

4.2. Depolarizing muscle relaxant — suxamethonium (Ditilin)

Suxamethonium — a compound of two ACh molecules. It **stimulates** N_M receptors and causes **prolonged depolarization** of the end plate. First, brief **muscle fasciculations** (disorganized contractions) are seen, then the membrane remains depolarized and does not respond to new impulses — **paralysis** (phase I block). With prolonged or repeated administration, the block may become competitive in character (**phase II** block).

- The action begins **very quickly** (**30–60 seconds**) and is **short** (5–10 minutes).

- It is rapidly broken down in the blood by **pseudocholinesterase** (butyrylcholinesterase).
- Uses: **rapid sequence intubation** (full stomach, emergency surgery), short procedures (laryngospasm, endoscopy, electrical cardioversion).

CAUTION

Neostigmine does not reverse a suxamethonium block but, on the contrary, enhances it — because it accumulates ACh and also inhibits pseudocholinesterase. There is no specific antagonist of a depolarizing block; its effect wears off quickly on its own.

Table 5.8. *Adverse effects of suxamethonium*

Adverse effect	Cause and risk group
Prolonged apnea (hours)	Hereditary atypical form or deficiency of pseudocholinesterase (liver disease, pregnancy, OPC poisoning, anticholinesterase agents)
Malignant hyperthermia	Hereditary (RYR1 gene mutation) — suxamethonium and inhalational anesthetics trigger uncontrolled release of Ca^{2+} from the sarcoplasmic reticulum: muscle rigidity, hypercapnia, tachycardia, a sharp rise in temperature
Hyperkalemia (cardiac arrest)	Burns, extensive injuries, prolonged immobilization, denervation, muscular dystrophies — the number of receptors increases
Muscle pain	After surgery, as a result of fasciculations
Increased intraocular, intragastric and intracranial pressure	Fasciculations and contraction of the extraocular muscles; dangerous in penetrating eye injuries
Bradyarrhythmias	Stimulates cardiac M_2 receptors (especially in children and with repeated doses)

REMEMBER

Treatment of malignant hyperthermia: immediately stop all triggering agents (inhalational anesthetics, suxamethonium), hyperventilation with 100% oxygen, **dantrolene** intravenously (2.5 mg/kg, repeated up to 10 mg/kg; blocks calcium release from the sarcoplasmic reticulum), physical cooling, correction of acidosis and hyperkalemia.

4.3. Differences between nondepolarizing and depolarizing muscle relaxants

Table 5.9. *Comparison of the two types of muscle relaxants*

Feature	Nondepolarizing (rocuronium)	Depolarizing (suxamethonium)
Mechanism	Competitive block, no depolarization	Prolonged depolarization
Fasciculations	Absent	Present (before relaxation)
Onset of action	1–5 minutes	30–60 seconds
Duration	15–90 minutes	5–10 minutes
Effect of neostigmine	Reverses the block	Enhances the block
Specific antagonist	Sugammadex (rocuronium, vecuronium)	Absent
Breakdown	Liver, kidney, Hofmann elimination	Pseudocholinesterase
Main risk	Residual curarization (respiratory failure after surgery)	Prolonged apnea, malignant hyperthermia, hyperkalemia

CLINICAL SIGNIFICANCE

Residual curarization — incomplete resolution of the muscle relaxant effect after surgery: weak swallowing and cough reflexes, risk of aspiration and hypoxia. Therefore neuromuscular transmission is monitored (acceleromyography, TOF test), and antagonists are given if necessary. Mechanical ventilation continues until the patient recovers.

4.4. Other drugs affecting neuromuscular transmission

- **Botulinum toxin A** — blocks ACh release from the presynaptic nerve ending (cleaves SNARE proteins). It is injected locally: blepharospasm, spasticity, dystonias, hyperhidrosis, overactive bladder, chronic migraine, cosmetology. Its effect lasts 3–6 months.
- **Dantrolene** — blocks calcium release from the sarcoplasmic reticulum inside skeletal muscle; for malignant hyperthermia and neuroleptic malignant syndrome.
- **Central muscle relaxants** (inhibit polysynaptic reflexes in the CNS): baclofen (GABA-B agonist), tizanidine (alpha-2 agonist), tolperisone — for spasticity, muscle spasm, low back pain; benzodiazepines (diazepam).

5. Clinical part

Table 5.10. *Choosing a muscle relaxant*

Clinical situation	Choice	Reason
Rapid sequence intubation (full stomach)	Suxamethonium or rocuronium (1.2 mg/kg)	Fastest action; rocuronium can be reversed with sugammadex
Burns, trauma, denervation, hyperkalemia	Rocuronium	Suxamethonium — risk of hyperkalemia and cardiac arrest
Renal and hepatic failure	Cisatracurium, atracurium	Hofmann elimination
Short procedure	Mivacurium, suxamethonium	Short action
Long operations	Pipecuronium, rocuronium (repeated doses)	Intermediate and long action
History of malignant hyperthermia	Nondepolarizing agents + intravenous anesthesia (propofol)	Suxamethonium and inhalational anesthetics are absolutely contraindicated
Myasthenia gravis	Small doses of nondepolarizing agents, with monitoring; sugammadex	Extremely sensitive to competitive relaxants, resistant to suxamethonium

CLINICAL SIGNIFICANCE

Before surgery, it is mandatory to ask about cases in **relatives of prolonged failure to wake up after anesthesia** or of **high fever of unknown cause**: this may indicate pseudocholinesterase deficiency or a predisposition to malignant hyperthermia.

Practical skills

Skill 1. Smoking cessation counseling (the 5 A's rule)

1. **Ask** — ask about and record smoking status at every visit.
2. **Advise** — explain the need to stop smoking clearly and in a personalized way.
3. **Assess** — assess the patient's readiness to quit.

4. **Assist** — set a quit date and prescribe pharmacotherapy (nicotine replacement therapy, cytisine or varenicline).
5. **Arrange follow-up** — schedule follow-up visits after 1–2 weeks and after 1 month.

Skill 2. Key prescriptions

PRESCRIPTION 5.1. CYTISINE (TABEX)
Rp.: Cytisini 0.0015
D.t.d. N. 100 in tabulettis
S. According to the schedule: on days 1–3, 1 tablet every 2 hours (6 a day), then gradually reduce the dose over 25 days; stop smoking completely from day 5.

PRESCRIPTION 5.2. NICOTINE TRANSDERMAL PATCH
Rp.: Systematis therapeutici transdermalis Nicotini 15 mg/16 hours
D.t.d. N. 7
S. Apply one patch to clean skin every morning and remove it in the evening.

PRESCRIPTION 5.3. VARENICLINE
Rp.: Tabulettas Vareniclini 0.0005 obductas N. 11
D.S. On days 1–3, 1 tablet once a day; on days 4–7, 1 tablet 2 times a day; then switch to 1 mg tablets.

PRESCRIPTION 5.4. AZAMETHONIUM (PENTAMIN)
Rp.: Solutionis Azamethonii bromidi 5% — 1 ml
D.t.d. N. 5 in ampullis
S. Inject 0.5–1 ml intramuscularly (in hypertensive crisis; the patient must remain lying down for 2 hours).

PRESCRIPTION 5.5. ROCURONIUM BROMIDE
Rp.: Solutionis Rocuronii bromidi 1% — 5 ml
D.t.d. N. 10 in ampullis
S. 0.6 mg/kg intravenously (for intubation), under the supervision of an anesthesiologist.

PRESCRIPTION 5.6. SUXAMETHONIUM (DITILIN)
Rp.: Solutionis Suxamethonii chloridi 2% — 5 ml
D.t.d. N. 10 in ampullis
S. 1–1.5 mg/kg intravenously (for intubation), with mechanical ventilation available.

PRESCRIPTION 5.7. SUGAMMADEX
Rp.: Solutionis Sugammadexi 10% — 2 ml
D.t.d. N. 10
S. 2–4 mg/kg intravenously (to reverse a rocuronium block; 16 mg/kg in a deep block).

PRESCRIPTION 5.8. DANTROLENE
Rp.: Dantroleni natrii 0.02
D.t.d. N. 10
S. Dissolve each vial in 60 ml of water for injection and give 2.5 mg/kg intravenously (in malignant hyperthermia), repeat if necessary.

Review questions

1. Structure and types of the N-cholinergic receptor. Where are N_N and N_M receptors located?
2. Classification of drugs acting on N-cholinergic receptors.
3. Effects of nicotine and their biphasic character. What does the harm of smoking consist of?
4. Clinical picture and treatment of acute nicotine poisoning.
5. Which drugs are used for smoking cessation? Their mechanism.
6. Mechanism of action of ganglionic blockers. Why do they dilate the vessels but reduce the tone of the intestine and bladder?
7. Uses and adverse effects of ganglionic blockers. Prevention of orthostatic collapse.
8. Classification of peripheral muscle relaxants. The order of muscle relaxation.
9. Mechanism of nondepolarizing muscle relaxants and their antagonists.
10. How does the mechanism of action of sugammadex differ from that of neostigmine?
11. Mechanism, features and uses of suxamethonium.
12. Why does neostigmine not reverse a suxamethonium block?
13. Adverse effects of suxamethonium. Causes of prolonged apnea and hyperkalemia.
14. Pathogenesis and treatment of malignant hyperthermia.
15. What is Hofmann elimination and which muscle relaxants are characterized by it?
16. Which antibiotics enhance the effect of muscle relaxants?

Test questions

1. Which receptor is located at the neuromuscular junction of skeletal muscles?
   A) NN
   B) Alpha-1
   C) NM
   D) M_2
   E) M_3

2. A partial agonist of N-cholinergic receptors used for smoking cessation:
   A) Varenicline
   B) Atropine
   C) Neostigmine
   D) Suxamethonium
   E) Azamethonium

3. The most dangerous adverse effect of ganglionic blockers:
   A) Bradycardia
   B) Orthostatic collapse
   C) Miosis
   D) Diarrhea
   E) Bronchospasm

4. What change is seen in the heart under the action of ganglionic blockers?
    A) Bradycardia
    B) Asystole
    C) Tachycardia
    D) No change
    E) AV block

5. An anticholinesterase agent that reverses the effect of nondepolarizing muscle relaxants:
    A) Pralidoxime
    B) Physostigmine
    C) Atropine
    D) Neostigmine
    E) Dantrolene

6. A drug that encapsulates rocuronium and forms an inactive complex:
    A) Sugammadex
    B) Protamine
    C) Flumazenil
    D) Dantrolene
    E) Neostigmine

7. A muscle relaxant whose action is preceded by muscle fasciculations:
    A) Rocuronium
    B) Atracurium
    C) Suxamethonium
    D) Pipecuronium
    E) Vecuronium

8. The cause of prolonged apnea after suxamethonium:
    A) Renal failure
    B) Increased acetylcholinesterase activity
    C) Hypokalemia
    D) Induction of liver enzymes
    E) Hereditary pseudocholinesterase deficiency

9. The specific agent in the treatment of malignant hyperthermia:
    A) Paracetamol
    B) Dantrolene
    C) Atropine
    D) Diazepam
    E) Neostigmine

10. A muscle relaxant broken down by Hofmann elimination:
    A) Tubocurarine
    B) Rocuronium
    C) Suxamethonium
    D) Pancuronium
    E) Cisatracurium

11. Which antibiotic enhances the effect of muscle relaxants?
    A) Gentamicin
    B) Ceftriaxone
    C) Azithromycin
    D) Penicillin
    E) Metronidazole

12. Which muscle relaxant is not recommended for intubation in a patient with extensive burns?
    A) Suxamethonium
    B) Cisatracurium
    C) Rocuronium
    D) Atracurium
    E) Vecuronium

13. A muscle relaxant that causes hypotension and bronchospasm by releasing histamine:
    A) Pipecuronium
    B) Vecuronium
    C) Tubocurarine
    D) Rocuronium
    E) Cisatracurium

14. A substance that blocks acetylcholine release from the presynaptic nerve ending:
    A) Nicotine
    B) Botulinum toxin
    C) Sugammadex
    D) Suxamethonium
    E) Neostigmine

15. A mandatory condition when working with peripheral muscle relaxants:
    A) General anesthesia and the availability of mechanical ventilation
    B) No antidote being given
    C) Oral prescription
    D) Use only in children
    E) The patient being conscious

Clinical cases

Case 1

A 28-year-old patient underwent general anesthesia for appendectomy; suxamethonium was used for intubation. The operation ended in 30 minutes, but 2 hours later spontaneous breathing had still not returned. The patient's uncle had also previously “not woken up for a long time” after anesthesia.

Questions: 1) What is the probable cause? 2) Can neostigmine be given in this situation? 3) Management.

Answer: 1) A hereditary atypical form (deficiency) of pseudocholinesterase — suxamethonium is not broken down and acts for a long time. 2) No — neostigmine also inhibits pseudocholinesterase and may enhance the depolarizing block. 3) Keep the patient on mechanical ventilation under sedation and wait until the block wears off on its own (several hours); if necessary, fresh frozen plasma (a source of pseudocholinesterase); test the patient and relatives (dibucaine number), avoid suxamethonium and mivacurium in the future, and make a note in the medical records.

Case 2

A 19-year-old man was given sevoflurane anesthesia and suxamethonium for intubation during knee arthroscopy. After 20 minutes, rigidity of the masticatory muscles, tachycardia of 150/min, a sharp rise in end-tidal CO_2 and a rise in temperature to 39.5 °C were observed.

Questions: 1) Your diagnosis? 2) Pathogenesis? 3) Emergency treatment.

Answer: 1) Malignant hyperthermia. 2) Because of a hereditary (RYR1) defect, suxamethonium and the inhalational anesthetic triggered uncontrolled release of calcium from the sarcoplasmic reticulum in skeletal muscle — hypermetabolism, muscle rigidity, excessive production of CO_2 and heat, rhabdomyolysis, hyperkalemia. 3) Stop sevoflurane immediately, hyperventilate with 100% oxygen, dantrolene 2.5 mg/kg intravenously (repeated), physical cooling, correction of acidosis and hyperkalemia, support of diuresis; observation in intensive care; warn the patient and his family about future anesthesia.

Case 3

A 52-year-old man has smoked 1 pack of cigarettes a day for 30 years. He had a myocardial infarction two years ago. He wants to stop smoking, but previous attempts have failed (irritability, insomnia, strong craving to smoke).

Questions: 1) What pharmacotherapy would you offer? 2) What causes the withdrawal symptoms?

Answer: 1) Together with counseling — varenicline (12 weeks) or cytisine (25 days), or combined nicotine replacement therapy (patch + gum). In stable ischemic heart disease, nicotine replacement therapy is much safer than smoking (no carbon monoxide or other harmful substances from smoke). 2) Chronic exposure to nicotine changes the number and sensitivity of N-cholinergic receptors in the CNS, and dopamine release in the mesolimbic system becomes nicotine-dependent; when nicotine is stopped, the dopamine deficit causes irritability, depression, insomnia and strong craving. Partial agonists stimulate the receptors moderately and soften these symptoms.

Quick review table

Table 5.11. *Drugs acting on N-cholinergic receptors — summary*

Group	Preparations	Mechanism	Use	Main risk / note
N-cholinomimetics	Nicotine, cytisine, lobeline	Stimulate N receptors (block them at large doses)	Smoking cessation	Nicotine poisoning: vomiting, seizures, respiratory paralysis
Partial agonist	Varenicline	alpha-4-beta-2 partial agonist	Smoking cessation (the most effective)	Nausea, unusual dreams
Ganglionic blockers	Azamethonium, hexamethonium, Hygronium, trimetaphan	N_N blockade — sympathetic and parasympathetic ganglia	Hypertensive crisis, controlled hypotension (historically)	Orthostatic collapse, ileus, urinary retention
Competitive muscle relaxants	Rocuronium, vecuronium, pipecuronium, atracurium, cisatracurium, mivacurium	Competitive N_M blockade	Intubation, relaxation during surgery	Residual curarization; potentiated by aminoglycosides

Group	Preparations	Mechanism	Use	Main risk / note
Depolarizing	Suxamethonium (Ditilin)	Prolonged depolarization	Rapid intubation, short procedures	Fasciculations, prolonged apnea, malignant hyperthermia, hyperkalemia
Antagonists	Neostigmine (+ atropine), sugammadex	Accumulation of ACh / encapsulation of rocuronium	Reversal of a competitive block	Neostigmine enhances a suxamethonium block
Others	Botulinum toxin, dantrolene	Blockade of ACh release / blockade of Ca_{2+} release	Spasticity, dystonia / malignant hyperthermia	—

TOPIC 6

Drugs acting on adrenergic receptors

Learning objectives

Drugs acting on the adrenergic system are the main drugs of emergency medicine (anaphylactic shock, cardiac arrest, cardiogenic and septic shock) and cardiology (arterial hypertension, ischemic heart disease, heart failure, arrhythmias). Stopping anaphylaxis with epinephrine and choosing beta-blockers correctly are skills every GP must master.

The student should know:

- the structure of the adrenergic synapse and the synthesis, release and inactivation of catecholamines;
- the types of adrenergic receptors, their location and the effects of their stimulation;
- the classification, mechanism, uses and adverse effects of adrenomimetics, sympathomimetics, adrenergic blockers and sympatholytics;
- the algorithm of emergency care in anaphylactic shock.

The student should be able to:

- write prescriptions for preparations such as epinephrine, norepinephrine, phenylephrine, salbutamol, xylometazoline, prazosin, tamsulosin, metoprolol, bisoprolol, carvedilol and timolol;
- calculate and give the dose of epinephrine in anaphylaxis;
- choose a beta-blocker according to the clinical situation.

1. The adrenergic synapse

1.1. Synthesis, release and inactivation of catecholamines

1. **Synthesis** (in the nerve ending): **tyrosine** → (tyrosine hydroxylase — the slowest, rate-limiting step) → **DOPA** → (DOPA decarboxylase) → **dopamine** → (dopamine beta-hydroxylase in the vesicles) → **norepinephrine**. In the adrenal medulla, norepinephrine is converted into **epinephrine** by phenylethanolamine N-methyltransferase.
2. **Storage:** norepinephrine is stored in vesicles together with ATP and chromogranins (VMAT carrier; **reserpine** blocks it).
3. **Release:** Ca^{2+}-dependent exocytosis when a nerve impulse arrives.
4. **Action on receptors:** postsynaptic alpha and beta receptors, presynaptic alpha-2 (negative feedback — reduces release) and beta-2 (enhances release) receptors.
5. **Inactivation:** mainly (70–90%) by **neuronal reuptake** (NET, “uptake 1”; blocked by cocaine and tricyclic antidepressants); the rest — breakdown by the enzymes mitochondrial **MAO** (in the nerve ending) and **COMT** (in the effector cell and the liver). Metabolites — vanillylmandelic acid and metanephrines (measured in the diagnosis of pheochromocytoma).

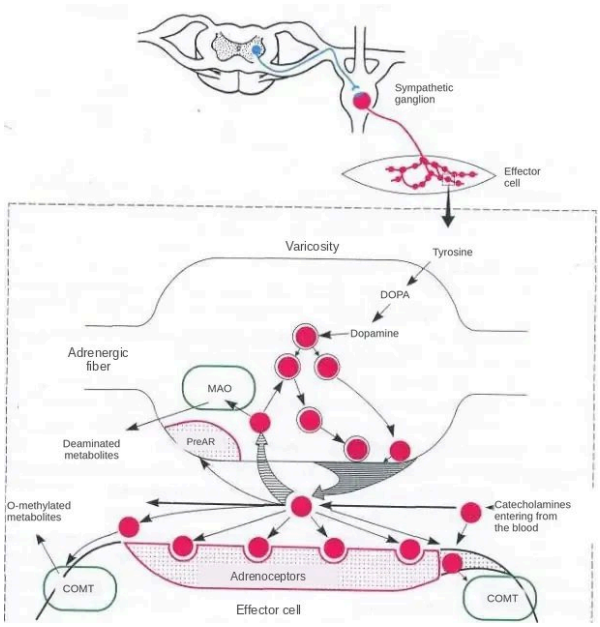


Figure 6.1. *Diagram of the adrenergic synapse: synthesis, release and reuptake of norepinephrine and its breakdown by MAO and COMT (teaching material of the TMA Department of Pharmacology)*

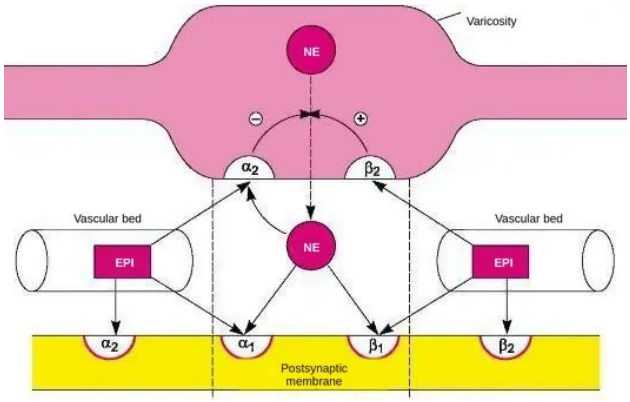


Figure 6.2. *Action of norepinephrine (NE) and epinephrine (EPI) on presynaptic and postsynaptic adrenergic receptors (teaching material of the TMA Department of Pharmacology)*

1.2. Adrenergic receptors

Table 6.1. *Types of adrenergic receptors and the effects of their stimulation*

Receptor	G protein	Location	Effect when stimulated
Alpha-1	G_q	Vessels of the skin, mucous membranes and internal organs; radial muscle of the eye; bladder sphincter and prostate; liver	Vasoconstriction (rise in BP), mydriasis, contraction of sphincters, glycogenolysis
Alpha-2	G_i	Presynaptic nerve endings; CNS (medulla oblongata); pancreatic beta cells; platelets	Reduced NE release, central decrease in sympathetic tone, reduced insulin secretion, platelet aggregation
Beta-1	G_s	Heart; juxtaglomerular cells of the kidney	Increased force and rate of contraction, conduction and automaticity of the heart; renin release
Beta-2	G_s	Bronchi, skeletal muscle and coronary vessels, uterus, liver	Bronchodilation, vasodilation, uterine relaxation (tocolysis), glycogenolysis, entry of K^+ into cells, muscle tremor
Beta-3	G_s	Adipose tissue, bladder detrusor	Lipolysis, detrusor relaxation
D_1 (dopamine)	G_s	Renal, mesenteric and coronary vessels	Vasodilation

MEMORY AID

Remember the beta receptor numbers by the number of organs: **beta-1 — 1 heart**, **beta-2 — 2 lungs**. Alpha-1 — "**constriction**" (vessels, sphincters), alpha-2 — "**brake**" (self-limitation, central sympathetic tone).

1.3. Classification of adrenergic drugs

Table 6.2. *Classification of drugs acting on adrenergic synapses*

Group	Subgroup	Preparations
A. Adrenomimetics (direct)	Alpha- and beta-adrenomimetics	Epinephrine (adrenaline), norepinephrine (noradrenaline)
	Alpha-1 adrenomimetics	Phenylephrine (Mesaton), midodrine
	Alpha-2 adrenomimetics	Xylometazoline, naphazoline (Naphthyzin), oxymetazoline, tetrahydrozoline; brimonidine (for the eye); clonidine, methyldopa (central — topic 11)
	Beta-1 and beta-2 adrenomimetics	Isoprenaline (Isadrin)
	Beta-1 adrenomimetics	Dobutamine
	Beta-2 adrenomimetics	Short-acting: salbutamol, fenoterol, terbutaline; long-acting: salmeterol, formoterol, indacaterol, olodaterol; tocolytic: hexoprenaline
	Agonist of dopamine receptors and adrenergic receptors	Dopamine
B. Sympathomimetics (indirect)	—	Ephedrine (mixed action), amphetamine, tyramine; cocaine (blocks reuptake)
C. Adrenergic blockers	Alpha-1 and alpha-2 blockers	Phentolamine, Tropaphen, phenoxybenzamine, dihydroergotoxine, nicergoline
	Alpha-1 blockers	Prazosin, doxazosin, terazosin; uroselective: tamsulosin, alfuzosin, silodosin
	Alpha-2 blockers	Yohimbine
	Beta-1 and beta-2 blockers (nonselective)	Propranolol, nadolol, timolol, sotalol; with intrinsic sympathomimetic activity: pindolol
	Beta-1 blockers (cardioselective)	Metoprolol, atenolol, bisoprolol, nebivolol, betaxolol, esmolol
	Alpha- and beta-blockers	Labetalol, carvedilol
D. Sympatholytics	—	Reserpine, guanethidine (Octadine)

2. Adrenomimetics

2.1. Epinephrine (adrenaline)

Epinephrine — a hormone of the adrenal medulla; it directly stimulates all **alpha and beta** receptors. It is destroyed in the gastrointestinal tract — it is used only parenterally, by inhalation and topically. Its action is short (intravenously — 5 minutes, subcutaneously — 30 minutes) because of reuptake and breakdown by COMT/MAO.

Table 6.3. *Pharmacological effects of epinephrine*

Organ / system	Effect	Receptor
Heart	The force and rate of contraction increase, conduction and automaticity increase; myocardial oxygen demand increases; arrhythmias are possible	Beta-1

Organ / system	Effect	Receptor
Vessels	Vessels of the skin, mucous membranes and kidneys constrict; skeletal muscle and coronary vessels dilate	Alpha-1 / beta-2
BP	Systolic pressure rises; at small doses diastolic pressure may fall (beta-2), at large doses it rises (alpha-1)	—
Bronchi	Dilate; mucosal edema decreases (alpha-1)	Beta-2, alpha-1
Eye	Mydriasis; intraocular pressure decreases	Alpha-1
Metabolism	Hyperglycemia (glycogenolysis, gluconeogenesis, reduced insulin secretion), lipolysis, hypokalemia	Beta-2, alpha-2, beta-3
Mast cells	Reduces the release of histamine and other mediators	Beta-2

REMEMBER

Dale's phenomenon (vasomotor reversal). If epinephrine is given against the background of an alpha-blocker, BP **does not rise but falls**: the alpha receptors are blocked and only beta-2-mediated vasodilation remains. Therefore epinephrine is not given for hypotension caused by alpha-blockers or neuroleptics (chlorpromazine has alpha-blocking properties) — **norepinephrine** or phenylephrine is chosen.

Uses of epinephrine:

- **anaphylactic shock and severe allergic reactions** — the drug of first choice;
- **cardiac arrest** (asystole, pulseless electrical activity, ventricular fibrillation refractory to defibrillation) — 1 mg intravenously every 3–5 minutes;
- adding to local anesthetic solutions (vasoconstriction);
- stopping bleeding from the nose and small vessels (topically);
- in an attack of bronchial asthma — now only when other agents are unavailable (beta-2 agonists are preferred);
- open-angle glaucoma (historically).

CLINICAL SIGNIFICANCE

Epinephrine dose in anaphylaxis: for adults **0.5 mg** (0.5 ml of 0.1% solution), for children **0.01 mg/kg** (maximum 0.5 mg; 6–12 years — 0.3 mg, under 6 years — 0.15 mg), **intramuscularly into the anterolateral thigh**; if there is no effect, it is repeated every **5 minutes**. Intravenous administration only in a resuscitation setting, diluted and under monitoring (risk of arrhythmia). In anaphylaxis, epinephrine is given **before** all other drugs; antihistamines and glucocorticoids are only adjuncts.

Adverse effects: tachycardia, arrhythmias (especially with halothane), angina, hypertension, headache, fear, tremor, hyperglycemia; with repeated subcutaneous injection — necrosis.

2.2. Norepinephrine (noradrenaline)

Norepinephrine stimulates alpha-1, alpha-2 and beta-1 receptors; **its effect on beta-2 is very weak**. As a result, all vessels constrict and **systolic and diastolic BP rise**. Although it directly stimulates beta-1 receptors in the heart, **reflex bradycardia** develops through the baroreceptors.

- **Uses: septic shock** (the vasopressor of first choice), other types of shock and severe hypotension (after adequate fluid replacement); only **by intravenous drip**, under BP monitoring.
- **Complication:** if the solution leaks out of the vessel — **tissue necrosis**. Treatment: infiltration of the area with **phentolamine** (an alpha-blocker) solution.

2.3. Dopamine and dobutamine

Dopamine — a precursor of norepinephrine; its effects depend on the dose.

Table 6.4. *Dose-dependent effects of dopamine*

Dose (mcg/kg/min)	Main receptors	Effect
1–3 (low)	D_1	Dilation of renal and mesenteric vessels, increased diuresis
3–10 (medium)	Beta-1	Increased force of cardiac contraction (inotropic)
Above 10 (high)	Alpha-1	Vasoconstriction, rise in BP

> **CAUTION**
> **"Renal-dose dopamine"** (1–3 mcg/kg/min) was formerly widely prescribed to prevent acute renal failure. Clinical studies have shown that it does not protect the kidneys or reduce mortality; it is now **not recommended** for this purpose. In septic shock, dopamine causes more arrhythmias than norepinephrine, so norepinephrine is preferred.

Dobutamine — mainly a **beta-1** adrenomimetic; it increases the force of cardiac contraction, speeds up the heart rate less and hardly changes BP. Uses: **cardiogenic shock** and acute decompensated heart failure (low cardiac output), stress echocardiography.

2.4. Alpha-adrenomimetics

Phenylephrine (Mesaton) — an alpha-1 agonist; it is not broken down by COMT and acts longer than epinephrine. It constricts the vessels, raises BP and causes reflex bradycardia. Uses: hypotension (during anesthesia), as a **mydriatic** (does not affect accommodation), rhinitis (in nasal drops).

Midodrine — orally for orthostatic hypotension.

Imidazoline derivatives — decongestants (xylometazoline, oxymetazoline, naphazoline, tetrahydrozoline) — alpha-2 (and alpha-1) agonists; they constrict the vessels of the nasal mucosa and reduce swelling. They are instilled into the nose in rhinitis, sinusitis and otitis.

> **CAUTION**
> Using decongestants for **more than 5–7 days** leads to **tachyphylaxis** and **"rebound"** swelling — **drug-induced rhinitis** develops (naphazoline dependence). In young children, imidazolines are absorbed into the blood and depress the CNS: drowsiness, **bradycardia, hypothermia, coma** (naphazoline poisoning). Only pediatric concentrations (xylometazoline 0.05%) are used in children; naphazoline 0.1% is contraindicated in children.

2.5. Beta-adrenomimetics

Isoprenaline (Isadrin) — a beta-1 and beta-2 agonist. Tachycardia and bronchodilation. Rarely used now (temporarily in AV block, in bradycardia).

Beta-2 adrenomimetics — the main drugs for bronchial asthma and COPD (in detail in topic 9):

- **short-acting** (4–6 hours; for relieving attacks): salbutamol, fenoterol, terbutaline;
- **long-acting** (12 hours): salmeterol, formoterol (formoterol has a rapid onset); **ultra-long-acting** (24 hours): indacaterol, olodaterol, vilanterol;
- **tocolytics** (relaxing the uterus when there is a risk of preterm labor): hexoprenaline (Ginipral), salbutamol. **Adverse effects:** hand tremor (beta-2 of skeletal muscles), tachycardia and

arrhythmias (at high doses they also act on beta-1), **hypokalemia**, hyperglycemia, headache; tachyphylaxis.

2.6. Indirect sympathomimetics

Ephedrine — an alkaloid of the ephedra plant; it has a **mixed** action: it enhances norepinephrine release from the nerve ending (the main action) and also stimulates alpha and beta receptors directly. It resembles epinephrine but acts more weakly and **longer**; it is active orally; it stimulates the CNS. With repeated doses, **tachyphylaxis** develops (norepinephrine stores are depleted). Uses: hypotension during spinal anesthesia, narcolepsy (historically). As a precursor of narcotic substances (methamphetamine), it is under strict control.

Amphetamine, **methylphenidate** — CNS stimulants (topic 8).

Tyramine — found in cheese, wine and smoked products; in patients taking MAO inhibitors it is not broken down, releases a large amount of norepinephrine and causes a **hypertensive crisis** ("cheese reaction").

3. Adrenergic blockers

3.1. Alpha-blockers

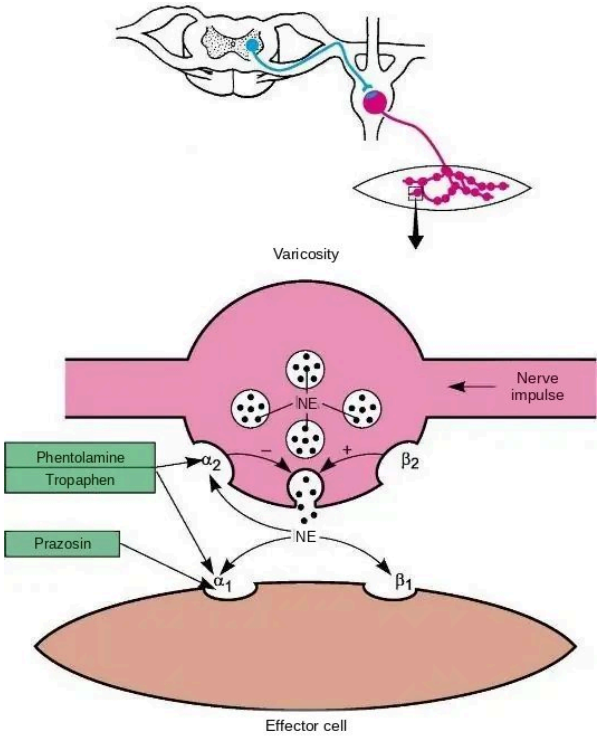


Figure 6.3. *Sites of action of adrenergic blockers: phentolamine, Tropaphen — alpha-1 and alpha-2; prazosin — alpha-1 (teaching material of the TMA Department of Pharmacology)*

Nonselective alpha-1 and alpha-2 blockers (phentolamine, Tropaphen, phenoxybenzamine): they dilate the vessels and lower BP. Because presynaptic alpha-2 receptors are also blocked, norepinephrine release increases and **tachycardia** is marked.

- **Phentolamine** — short-acting; hypertensive crises during the diagnosis and surgery of **pheochromocytoma**, norepinephrine extravasation.
- **Phenoxybenzamine** — irreversible (covalent) block; preparation of pheochromocytoma for surgery.

- **Dihydroergotoxine, nicergoline** (derivatives of ergot alkaloids) — for cerebral and peripheral circulatory disorders.

Selective alpha-1 blockers (prazosin, doxazosin, terazosin): little reflex tachycardia (alpha-2 remains open), slightly improve the lipid profile.

- **Uses:** arterial hypertension (as an add-on drug, especially together with prostatic hyperplasia), **benign prostatic hyperplasia** (urination improves), Raynaud's syndrome.
- **“First-dose phenomenon”** — marked **orthostatic hypotension** and fainting after the first dose. Prevention: start with a small dose, **at bedtime**. **Uroselective alpha-1A blockers** (tamsulosin, silodosin, alfuzosin) act selectively on the smooth muscle of the prostate and bladder neck; they have little effect on BP. Used for prostatic hyperplasia and to ease the passage of urinary stones. Adverse effects: retrograde ejaculation, “floppy iris” syndrome during cataract surgery.

CLINICAL SIGNIFICANCE

In a patient with **pheochromocytoma** (a catecholamine-producing tumor), a beta-blocker must not be given **without an alpha-blocker**: beta-2 vasodilation is lost and alpha vasoconstriction is left unopposed — **a severe hypertensive crisis**. An alpha-blocker (phenoxybenzamine, doxazosin) is given first, and only then is a beta-blocker added for tachycardia.

3.2. Beta-blockers

Beta-blockers competitively block the action of catecholamines on beta receptors. This is one of the most important groups in cardiology.

Table 6.5. *Main effects of beta-blockers*

Effect	Mechanism	Clinical result
Negative chronotropic and inotropic	Blockade of cardiac beta-1	Heart rate and contractility decrease, myocardial oxygen demand decreases (antianginal)
Negative dromotropic	Conduction in the AV node slows	Antiarrhythmic (class II)
Hypotensive	Cardiac output decreases, renin secretion decreases (beta-1), central sympathetic tone decreases	BP falls
Cardioprotective	The toxic effect of catecholamines on the myocardium and remodeling decrease	Reduce mortality in heart failure and after infarction
Lowering of intraocular pressure	Fluid production by the ciliary body epithelium decreases	Glaucoma (timolol drops)
Effect on the bronchi (nonselective agents)	Beta-2 blockade	Bronchospasm
Effect on metabolism	Glycogenolysis and lipolysis are inhibited	Mask the signs of hypoglycemia (tachycardia, tremor) and slow recovery from hypoglycemia

Table 6.6. *Classification and properties of beta-blockers*

Preparation	Selectivity	Additional property	Feature of use
Propranolol	Beta-1 and beta-2	Lipophilic (enters the CNS), membrane-stabilizing	Tremor, migraine prophylaxis, thyrotoxicosis, portal hypertension, somatic signs of anxiety, hemangiomas (in children)

Preparation	Selectivity	Additional property	Feature of use
Timolol	Beta-1 and beta-2	—	Glaucoma (eye drops)
Sotalol	Beta-1 and beta-2	Class III antiarrhythmic (K^+ channel blockade)	Arrhythmias (topic 10)
Pindolol	Beta-1 and beta-2	Intrinsic sympathomimetic activity	Patients prone to bradycardia (rarely used)
Metoprolol	Beta-1	Lipophilic	Hypertension, angina, infarction; metoprolol succinate (extended-release) — heart failure
Atenolol	Beta-1	Hydrophilic, excreted by the kidneys	Hypertension (now less recommended)
Bisoprolol	Beta-1 (highly selective)	Once a day	Hypertension, angina, heart failure
Nebivolol	Beta-1 (the most selective)	Dilates the vessels by releasing NO	Hypertension, heart failure (the elderly)
Esmolol	Beta-1	Ultra-short (t½ ≈ 9 minutes, blood esterases)	Intravenously — for tachycardia and hypertension during surgery
Carvedilol	Beta-1, beta-2 + alpha-1	Vasodilation, antioxidant	Hypertension, heart failure
Labetalol	Beta + alpha-1	Vasodilation	Hypertension in pregnancy, hypertensive crisis

MEMORY AID

The three beta-blockers that reduce mortality in heart failure — "BMC": **B**isoprolol, **M**etoprolol succinate, **C**arvedilol (+ nebivolol in the elderly). Cardioselective ones run alphabetically **"A–M"**: **A**tenolol, **B**isoprolol, **B**etaxolol, **E**smolol, **M**etoprolol (all begin with letters from A to M) + nebivolol.

Uses:

- arterial hypertension;
- ischemic heart disease: stable angina, the period after myocardial infarction (reduce mortality);
- chronic heart failure (in a stable state, increasing gradually from a small dose);
- tachyarrhythmias (rate control in atrial fibrillation, extrasystoles);
- hypertrophic cardiomyopathy, dissecting aortic aneurysm;
- signs of thyrotoxicosis (tachycardia, tremor), pheochromocytoma (after an alpha-blocker);
- migraine prophylaxis, essential tremor, "stage fright";
- glaucoma (timolol, betaxolol drops);
- portal hypertension in liver cirrhosis (prevention of bleeding from esophageal varices — propranolol, carvedilol).

Table 6.7. *Contraindications to beta-blockers*

Absolute	Relative (with caution)
Bronchial asthma (nonselective agents, including eye drops)	COPD (cardioselective agents in small doses are possible)
Second- or third-degree AV block, sick sinus syndrome (without a pacemaker)	Diabetes mellitus (hypoglycemia is masked)
Symptomatic bradycardia (HR below 50)	Peripheral arterial disease, Raynaud's syndrome

Absolute	Relative (with caution)
Acute decompensated heart failure, cardiogenic shock	Depression, psoriasis
Symptomatic hypotension	Pregnancy (labetalol and metoprolol are possible; atenolol — slows fetal growth)
Untreated pheochromocytoma	Cocaine poisoning

CAUTION

A beta-blocker must not be stopped abruptly — "rebound" syndrome: the number of receptors has increased (up-regulation), and when the drug is stopped, sensitivity to catecholamines rises sharply — tachycardia, angina attacks, **myocardial infarction**, hypertensive crisis. The dose is reduced gradually over 1–2 weeks.

Adverse effects: bradycardia, AV block, worsening of heart failure (in decompensation), hypotension, bronchospasm, cold extremities, fatigue and reduced exercise tolerance, sleep disturbances and nightmares (with lipophilic agents), erectile dysfunction, masking of hypoglycemia, worsening of the lipid profile (with nonselective agents).

REMEMBER

Beta-blocker poisoning (bradycardia, hypotension, cardiogenic shock, hypoglycemia, seizures): atropine (often of little effect), **glucagon** (activates adenylate cyclase independently of beta receptors), high-dose insulin + glucose, catecholamines (epinephrine, isoprenaline), cardiac pacing; in propranolol poisoning (wide QRS) — sodium bicarbonate.

4. Sympatholytics

Sympatholytics reduce norepinephrine stores in adrenergic nerve endings or stop its release; they do not act on receptors.

- **Reserpine** — an alkaloid of Rauwolfia; it blocks the uptake of norepinephrine, dopamine and serotonin into vesicles (VMAT), the mediators are broken down by MAO, and the stores are depleted. The hypotensive effect develops slowly (over 1–2 weeks). Adverse effects: **depression** (risk of suicide), parkinsonism, nasal congestion, bradycardia, exacerbation of gastric ulcer (parasympathetic predominance). It is hardly used now; it remains in some combination tablets (Adelphane).
- **Guanethidine (Octadine)** — blocks norepinephrine release; does not enter the CNS. It causes severe orthostatic hypotension. It is of historical importance.

CAUTION

Older textbooks gave reserpine and Octadine (guanethidine) as the main drugs for hypertension. In modern recommendations (topic 11), they are **not used**: they have many adverse effects, and a reduction in cardiovascular complications has not been proven.

5. Clinical part

5.1. Emergency care in anaphylactic shock

Table 6.8. *Algorithm of care in anaphylaxis (adults)*

Step	Action
1	Stop contact with the allergen (stop the drug infusion); call for help

Step	Action
2	Epinephrine 0.5 mg (0.1% — 0.5 ml) intramuscularly into the anterolateral thigh; may be repeated every 5 minutes
3	Lay the patient down with the legs raised (half-sitting if breathing is difficult); do not let him stand up suddenly
4	Secure the airway, high-flow oxygen
5	Establish intravenous access and rapidly infuse 500–1000 ml of crystalloids (0.9% NaCl) (for children 10–20 ml/kg)
6	If bronchospasm persists — salbutamol inhalation
7	Additionally: an H_1 antihistamine (for allergic skin signs), a glucocorticoid (to prevent a delayed reaction — its effectiveness has not been proven)
8	If there is no response to two doses of epinephrine — an epinephrine infusion in the intensive care unit; in patients taking beta-blockers who do not respond to epinephrine — glucagon
9	Observation for at least 6–12 hours (biphasic reaction); an epinephrine auto-injector and allergist consultation for the patient

CLINICAL SIGNIFICANCE

The most common error in anaphylaxis is giving epinephrine **late** or not at all and giving only prednisolone and diphenhydramine instead. The effect of glucocorticoids begins after hours, and they do not stop shock. Only epinephrine given in time saves life.

5.2. Choosing a vasopressor and an inotrope

Table 6.9. *Adrenergic drugs in types of shock*

Type of shock	First choice	Additional
Anaphylactic	Epinephrine (intramuscularly, then an infusion)	Fluids, glucagon (with a beta-blocker)
Septic	Norepinephrine (after fluids)	Vasopressin, epinephrine; dobutamine (in myocardial dysfunction)
Cardiogenic	Dobutamine (when BP is relatively preserved); norepinephrine (in hypotension)	Mechanical support
Hypovolemic	Fluid and blood replacement	Vasopressors — only temporarily
Hypotension during spinal anesthesia	Phenylephrine, ephedrine	Fluids

Practical skills

Skill 1. Giving epinephrine in anaphylaxis

1. Open an ampoule of 0.1% (1 mg/ml) epinephrine.
2. Calculate the dose: 0.01 mg/kg = 0.01 ml/kg (maximum 0.5 ml for adults, 0.3 ml for children).
3. Draw it into a syringe; the needle must be long enough to reach the muscle.
4. Inject **intramuscularly** into the middle third of the anterolateral thigh; this can be done through clothing.
5. Record the time; if there is no effect after 5 minutes, repeat.

Skill 2. Algorithm for choosing a beta-blocker

1. Determine the indication (hypertension, angina, heart failure, arrhythmia, migraine, glaucoma).
2. Check for contraindications (asthma, AV block, bradycardia, decompensation).
3. In heart failure — only bisoprolol, metoprolol succinate, carvedilol or nebivolol; start with a very small dose and double it once every 2 weeks.
4. In COPD and diabetes — a cardioselective preparation (bisoprolol, nebivolol).
5. In pregnancy — labetalol or metoprolol.
6. Warn the patient not to stop the drug abruptly.

Skill 3. Key prescriptions

PRESCRIPTION 6.1. EPINEPHRINE
Rp.: Solutionis Epinephrini hydrochloridi 0.1% — 1 ml
D.t.d. N. 10 in ampullis
S. In anaphylaxis, inject 0.5 ml into the thigh muscle; if there is no effect, repeat after 5 minutes.

PRESCRIPTION 6.2. NOREPINEPHRINE
Rp.: Solutionis Norepinephrini hydrotartratis 0.2% — 4 ml
D.t.d. N. 10 in ampullis
S. Dilute 4 ml in 250 ml of 5% glucose solution and give by intravenous drip under BP monitoring.

PRESCRIPTION 6.3. PHENYLEPHRINE (MESATON)
Rp.: Solutionis Phenylephrini hydrochloridi 1% — 1 ml
D.t.d. N. 10 in ampullis
S. Dilute 0.1–0.5 ml in 20 ml of isotonic solution and inject slowly intravenously (in hypotension).

PRESCRIPTION 6.4. XYLOMETAZOLINE
Rp.: Solutionis Xylometazolini hydrochloridi 0.1% — 10 ml
D.S. Instill 1–2 drops into each nostril 2–3 times a day, for not more than 5 days.

PRESCRIPTION 6.5. SALBUTAMOL AEROSOL
Rp.: Aerosoli Salbutamoli 0.0001/dosi — 200 dosi
D.S. Take 1–2 inhalations during an attack of breathlessness.

PRESCRIPTION 6.6. DOBUTAMINE
Rp.: Dobutamini 0.25
D.t.d. N. 5
S. Dissolve the powder in the vial in 250 ml of 5% glucose solution and give intravenously at a rate of 2.5–10 mcg/kg/min.

PRESCRIPTION 6.7. DOXAZOSIN
Rp.: Doxazosini 0.002
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day in the evening at bedtime.

PRESCRIPTION 6.8. TAMSULOSIN
Rp.: Capsulas Tamsulosini 0.0004 N. 30
D.S. Take 1 capsule once a day after breakfast (without chewing).

PRESCRIPTION 6.9. BISOPROLOL
Rp.: Bisoprololi 0.005
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 6.10. METOPROLOL SUCCINATE
Rp.: Tabulettas Metoprololi succinatis 0.025 N. 30
D.S. Take 1 tablet once a day (starting dose in heart failure).

PRESCRIPTION 6.11. CARVEDILOL
Rp.: Carvediloli 0.00625
D.t.d. N. 30 in tabulettis
S. Take 1 tablet 2 times a day with food.

PRESCRIPTION 6.12. PROPRANOLOL
Rp.: Propranololi 0.04
D.t.d. N. 50 in tabulettis
S. Take 1 tablet 2 times a day (for migraine prophylaxis).

PRESCRIPTION 6.13. TIMOLOL EYE DROPS
Rp.: Solutionis Timololi 0.5% — 5 ml
D.S. Instill 1 drop into each eye 2 times a day.

Review questions

1. Stages of catecholamine synthesis. How is norepinephrine inactivated at the synapse?
2. Types and location of adrenergic receptors and the effects of their stimulation.
3. Classification of adrenergic drugs.
4. Effects of epinephrine on the heart, vessels, bronchi and metabolism.
5. What is Dale's phenomenon? Its practical importance.
6. The dose and route of epinephrine and other emergency measures in anaphylactic shock.
7. How does norepinephrine differ from epinephrine? Why does norepinephrine cause reflex bradycardia?
8. Dose-dependent effects of dopamine. When is dobutamine used?
9. Alpha-adrenomimetics and decongestants. Drug-induced rhinitis and naphazoline poisoning in children.
10. Uses and adverse effects of beta-2 adrenomimetics.
11. Mechanism of action of ephedrine. Why does tachyphylaxis develop?
12. Classification and uses of alpha-blockers. The first-dose phenomenon.
13. Why must a beta-blocker not be given alone in pheochromocytoma?
14. Classification, effects and uses of beta-blockers.
15. Which beta-blockers are used in heart failure?

16. Contraindications and adverse effects of beta-blockers. Rebound syndrome.
17. Management of beta-blocker poisoning.
18. Mechanism of action and adverse effects of reserpine and guanethidine.

Test questions

1. The enzyme that catalyzes the rate-limiting (slowest) step of catecholamine synthesis:
   A) DOPA decarboxylase
   B) COMT
   C) Tyrosine hydroxylase
   D) Dopamine beta-hydroxylase
   E) MAO

2. Stimulation of which receptor causes bronchodilation?
   A) Alpha-2
   B) M_3
   C) Beta-2
   D) Beta-1
   E) Alpha-1

3. The first dose and route of epinephrine in adults in anaphylactic shock:
   A) 0.1 mg subcutaneously
   B) 0.5 mg orally
   C) 5 mg intramuscularly
   D) 1 mg as an intravenous push
   E) 0.5 mg intramuscularly

4. The vasopressor of first choice in septic shock:
   A) Norepinephrine
   B) Dopamine
   C) Isoprenaline
   D) Phenylephrine
   E) Ephedrine

5. To prevent necrosis when norepinephrine solution leaks into the tissue, the following is injected:
   A) Atropine
   B) Propranolol
   C) Calcium chloride
   D) Phentolamine
   E) Epinephrine

6. What is the fall in BP when epinephrine is given against the background of an alpha-blocker called?
   A) Tachyphylaxis
   B) Rebound phenomenon
   C) Dale's phenomenon (vasomotor reversal)
   D) First-dose phenomenon
   E) Cumulation

7. A mainly beta-1 agonist used to increase the force of cardiac contraction in cardiogenic shock:
   A) Salbutamol
   B) Clonidine
   C) Dobutamine
   D) Phenylephrine
   E) Xylometazoline

8. An indirect sympathomimetic that causes tachyphylaxis with repeated doses:
   A) Dobutamine
   B) Ephedrine
   C) Norepinephrine
   D) Phenylephrine
   E) Epinephrine

9. A uroselective alpha-1A blocker used for benign prostatic hyperplasia:
   A) Phentolamine
   B) Tamsulosin
   C) Yohimbine
   D) Phenoxybenzamine
   E) Propranolol

10. A complication seen after the first dose of prazosin and doxazosin:
    A) Orthostatic hypotension
    B) Hypoglycemia
    C) Bronchospasm
    D) Hypertensive crisis
    E) Bradycardia

11. A beta-blocker proven to reduce mortality in heart failure:
    A) Propranolol
    B) Pindolol
    C) Bisoprolol
    D) Sotalol
    E) Atenolol

12. Which preparation is absolutely contraindicated in a patient with bronchial asthma?
    A) Budesonide
    B) Ipratropium
    C) Salbutamol
    D) Montelukast
    E) Propranolol

13. An antidote in beta-blocker poisoning that acts independently of beta receptors:
    A) Atropine
    B) Protamine
    C) Naloxone
    D) Glucagon
    E) Flumazenil

14. An alpha- and beta-blocker used for arterial hypertension in pregnancy:
    A) Carvedilol
    B) Atenolol
    C) Propranolol
    D) Labetalol
    E) Phentolamine

15. The main dangerous adverse effect of reserpine:
    A) Hypoglycemia
    B) Hyperthermia
    C) Bronchospasm
    D) Depression
    E) Tachycardia

Clinical cases

Case 1

Ten minutes after taking amoxicillin at the dentist's, a 25-year-old woman developed itching, urticaria, swelling of the lips and eyelids, hoarseness, shortness of breath and dizziness. BP 70/40 mm Hg, HR 128/min.

Questions: 1) Your diagnosis? 2) The first-line drug, its dose and route? 3) Further measures.

Answer: 1) Anaphylactic shock (to amoxicillin). 2) Epinephrine 0.5 mg (0.5 ml of 0.1% solution) intramuscularly into the anterolateral thigh; if there is no effect, repeat every 5 minutes. 3) Call for help, lay the patient down with the legs raised (half-sitting if breathing is difficult), oxygen, establish intravenous access and rapidly infuse 0.9% NaCl; if bronchospasm persists — salbutamol; additionally — an antihistamine and a glucocorticoid; observation for at least 12 hours; record the penicillin allergy in the medical records and provide an epinephrine auto-injector.

Case 2

A 62-year-old patient takes propranolol for arterial hypertension and angina. It turned out that the patient also has bronchial asthma; in recent weeks the attacks of breathlessness have become more frequent. In addition, the patient stopped taking propranolol 3 days ago because he ran out of the drug, and yesterday he had a severe angina attack.

Questions: 1) What caused the attacks of breathlessness to become more frequent? 2) What caused the angina attack? 3) How should the treatment be changed?

Answer: 1) Propranolol is a nonselective beta-blocker; it blocks beta-2 receptors in the bronchi and causes bronchospasm — it is contraindicated in asthma. 2) Abrupt withdrawal of the beta-blocker — rebound syndrome (increased sensitivity to catecholamines against the background of receptor up-regulation). 3) If a beta-blocker is needed in a patient with asthma — a highly cardioselective one (bisoprolol, nebivolol) in a small dose and under the supervision of a pulmonologist; or an alternative to beta-blockers for angina (verapamil, diltiazem or a dihydropyridine calcium antagonist + nitrates); explain to the patient never to stop the drug abruptly.

Case 3

A 2-year-old child's mother instilled 0.1% naphazoline drops intended for adults into his nose many times for nasal congestion. A few hours later the child is drowsy, the skin is pale and cold, HR 70/min, body temperature 35.4 °C, and breathing has become infrequent.

Questions: 1) What is this condition? 2) Its mechanism? 3) Treatment and prevention.

Answer: 1) Poisoning with an imidazoline derivative (naphazoline). 2) In a young child, a large amount of naphazoline absorbed from the nasal mucosa crosses the BBB and stimulates central alpha-2 adrenergic receptors — sympathetic tone falls: CNS depression, bradycardia, hypothermia, respiratory depression (similar to clonidine poisoning). 3) Urgent hospitalization, monitoring of breathing and circulation, warming, atropine for bradycardia, oxygen and respiratory support for respiratory failure; recovery usually occurs within 12–24 hours. Prevention: only pediatric concentrations for children (xylometazoline 0.05%), not more than 2–3 times a day and for not more than 3–5 days; keep drugs out of the reach of children.

Quick review table

Table 6.10. *Adrenergic drugs — summary*

Group	Preparations	Main effect	Use	Important note
Alpha + beta agonist	Epinephrine	Heart ↑, bronchodilation, vasoconstriction	Anaphylaxis (0.5 mg i/m), cardiac arrest (1 mg i/v)	Dale's phenomenon; arrhythmia
Alpha-1, alpha-2, beta-1 agonist	Norepinephrine	BP ↑, reflex bradycardia	Septic shock	Necrosis on extravasation — phentolamine
Dopamine	Dopamine	Dose-dependent: D_1 → beta-1 → alpha-1	Shock (second line)	The “renal dose” is ineffective
Beta-1 agonist	Dobutamine	Inotropic	Cardiogenic shock	—
Alpha-1 agonist	Phenylephrine, midodrine	Vasoconstriction, mydriasis	Hypotension, mydriasis, rhinitis	Reflex bradycardia
Decongestants	Xylometazoline, naphazoline, oxymetazoline	Constriction of nasal vessels	Rhinitis	Not more than 5–7 days; poisoning in children
Beta-2 agonists	Salbutamol, fenoterol, salmeterol, formoterol	Bronchodilator, tocolytic	Asthma, COPD, tocolysis	Tremor, tachycardia, hypokalemia
Sympathomimetic	Ephedrine	NE release + direct action	Hypotension during spinal anesthesia	Tachyphylaxis
Alpha-1 + alpha-2 blockers	Phentolamine, phenoxybenzamine	Vasodilation, tachycardia	Pheochromocytoma	—
Alpha-1 blockers	Prazosin, doxazosin, tamsulosin	Vasodilation, prostate relaxation	Prostatic hyperplasia, hypertension	First-dose phenomenon
Cardioselective beta-blockers	Bisoprolol, metoprolol, nebivolol, atenolol, esmolol	HR ↓, BP ↓, renin ↓	Hypertension, IHD, heart failure, arrhythmias	Must not be stopped abruptly
Nonselective beta-blockers	Propranolol, timolol, sotalol	+ beta-2 blockade	Migraine, tremor, glaucoma, portal hypertension	Contraindicated in asthma
Alpha + beta blockers	Carvedilol, labetalol	+ vasodilation	Heart failure; hypertension in pregnancy	—

Group	Preparations	Main effect	Use	Important note
Sympatholytics	Reserpine, guanethidine	Depletes NE stores	Hypertension (historically)	Depression, orthostatic hypotension

TOPIC 7

General anesthetics. Neuroleptics. Anxiolytics

Learning objectives

This topic brings together three important groups that depress the central nervous system (CNS). **General anesthetics** made surgery painless and safe. **Neuroleptics (antipsychotics)** revolutionized the treatment of schizophrenia and other psychoses. **Anxiolytics (tranquilizers)** are among the most frequently prescribed drugs, but they carry a high risk of dependence. The GP must recognize the adverse effects of these drugs (neuroleptic malignant syndrome, extrapyramidal disorders, benzodiazepine dependence) and provide help in poisoning.

The student should know:

- the classification of drugs acting on the CNS, the theories of anesthesia and the stages of anesthesia;
- the properties of inhalational and non-inhalational anesthetics and the principles of combined anesthesia and premedication;
- the classification of neuroleptics (typical and atypical), their mechanism of action, uses and adverse effects;
- the mechanism and uses of anxiolytics (benzodiazepines and others), dependence and poisoning.

The student should be able to:

- write prescriptions for preparations such as propofol, ketamine, haloperidol, risperidone, olanzapine, diazepam, alprazolam and buspirone;
- determine the stages of anesthesia in a laboratory experiment;
- provide help in benzodiazepine poisoning and in acute adverse effects of neuroleptics.

1. General classification of drugs acting on the CNS

Table 7.1. *Drugs acting on the CNS*

Direction of action	Groups
Depressant	General anesthetics, hypnotics, ethyl alcohol, antiepileptics, antiparkinsonian drugs, analgesics, neuroleptics, anxiolytics, sedatives, mood stabilizers (lithium salts)
Stimulant	Psychostimulants, antidepressants, nootropics, analeptics, adaptogens

2. General anesthetics

General anesthesia (narcosis) (Gr. narkosis — numbness) — a **reversible** state caused by drugs: loss of consciousness, loss of pain sensation (analgesia), reduction of reflexes and skeletal muscle tone, but **the activity of the vital centers (respiration and circulation) is preserved**.

> **REMEMBER**
> Components of modern general anesthesia: **hypnosis** (loss of consciousness), **analgesia**, **amnesia**, **muscle relaxation** and **suppression of unwanted autonomic reflexes**. No single drug can safely provide all of these, so **combined (balanced) anesthesia** is used.

History. On October 16, 1846, in Boston, W. Morton demonstrated the first public ether anesthesia. In 1847, J. Simpson used chloroform, and N.I. Pirogov used ether anesthesia in military field conditions.



Figure 7.1. *The first public demonstration of ether anesthesia, Boston, 1846 (D.A. Kharkevich, Pharmacology, 2010)*

2.1. Stages of anesthesia

The stages of anesthesia are well observed in classic ether anesthesia (according to Guedel). With modern agents they pass quickly and are not clearly seen, but they remain important in assessing the depth of anesthesia.

Table 7.2. *Stages of anesthesia (using ether anesthesia as an example)*

Step	Signs	Mechanism
I. Analgesia (stupor)	Consciousness preserved but confused; pain sensation reduced; reflexes and muscle tone preserved	Depression of the cerebral cortex and the reticular formation begins
II. Excitement (delirium)	Consciousness lost; speech and motor excitement, irregular breathing, tachycardia, rise in BP, vomiting, increased muscle tone, dilated pupils	The inhibitory influence of the cortex is lost and subcortical centers are “released” (disinhibition)
III. Surgical anesthesia (4 planes)	No consciousness or pain, reflexes gradually disappear, muscles relax, breathing is rhythmic; the pupils first constrict and then dilate as anesthesia deepens	The spinal cord is depressed; III1–III3 — surgery can be performed, III4 — excessively deep
IV. Awakening or agonal stage	In overdose — paralysis of the respiratory and vasomotor centers; the pupils are dilated and do not react to light	Depression of the medulla oblongata

CAUTION

The therapeutic range of anesthesia (the difference between the surgical stage and stage IV) is relatively narrow for inhalational anesthetics. The excitement stage is dangerous (aspiration of vomit, arrhythmias), so in modern anesthesia, anesthesia is started with fast-acting intravenous agents (**induction**), and stage II is hardly seen.

2.2. Classification of general anesthetics

Table 7.3. *Classification of general anesthetics*

Group	Preparations
Inhalational: volatile liquids	Sevoflurane, isoflurane, desflurane; halothane (Fluothane), enflurane; ether (historically)
Inhalational: gases	Nitrogen (I) oxide (nitrous oxide), xenon
Non-inhalational (intravenous)	Propofol, thiopental sodium, ketamine, etomidate, sodium oxybate, midazolam; opioids (fentanyl) — in combination

2.3. Inhalational anesthetics

The potency of an inhalational anesthetic is assessed by the **minimum alveolar concentration (MAC)**: the alveolar concentration at which 50% of patients lose the motor response to a skin incision. **The smaller the MAC, the more potent the anesthetic.** The speed of induction and awakening depends on the **solubility of the anesthetic in blood**: those poorly soluble in blood (desflurane, sevoflurane, nitrous oxide, xenon) act quickly and are quickly eliminated.

Table 7.4. *Inhalational anesthetics*

Preparation	MAC, %	Advantages	Disadvantages
Sevoflurane	≈ 2	Pleasant smell, does not irritate the airway — mask induction (in children); fast induction and recovery; bronchodilator	Emergence delirium in children; a trigger of malignant hyperthermia
Isoflurane	≈ 1,2	Cheap, stable, sensitizes the heart little	Pungent smell (unsuitable for induction)
Desflurane	≈ 6	The fastest awakening (obese patients, long operations)	Irritates the airway, tachycardia
Halothane (Fluothane)	≈ 0,75	Potent, pleasant smell	Sensitizes the myocardium to catecholamines — arrhythmias (contraindicated with epinephrine); halothane hepatitis; hardly used now
Nitrous oxide (N_2O)	≈ 105	Potent analgesic; rapid action; in labor and dentistry (with oxygen 50:50)	Weak anesthetic (does not produce deep anesthesia alone); expands gas in closed cavities (pneumothorax, intestinal obstruction); inactivates B_{12} with prolonged use; “diffusion hypoxia” — 100% oxygen must be given at the end
Xenon	≈ 70	Inert, the safest, no effect on hemodynamics	Very expensive

MEMORY AID

“Sevo is a child's friend”: sevoflurane is pleasant and non-irritating, so it is the main agent for mask induction in children. **“Halothane is the enemy of the heart and liver”**: arrhythmias and hepatitis.

2.4. Intravenous (non-inhalational) anesthetics

Table 7.5. *Comparison of intravenous anesthetics*

Preparation	Onset / duration of action	Main properties	Adverse effects
Propofol	30 seconds / 5–10 minutes	The most commonly used induction agent; rapid and "clear-headed" awakening; antiemetic effect; maintenance of anesthesia by infusion (TIVA), sedation	Hypotension, respiratory depression, pain on injection; propofol infusion syndrome (acidosis, rhabdomyolysis) with prolonged high-dose infusion
Thiopental sodium	30 seconds / 5–10 minutes	Barbiturate; lowers intracranial pressure; anticonvulsant	Hypotension, respiratory depression, laryngospasm; necrosis if injected into an artery; contraindicated in porphyria; no analgesic effect
Ketamine	30–60 seconds (i/v), 3–5 minutes (i/m) / 10–15 minutes	NMDA receptor antagonist; dissociative anesthesia; strong analgesia; raises BP and heart rate; bronchodilator; breathing and protective reflexes are preserved	Hallucinations and nightmares on awakening (reduced by a benzodiazepine); salivation; raises intracranial and intraocular pressure
Etomidate	30 seconds / 5 minutes	Hardly affects hemodynamics — induction in heart disease and shock	Suppresses adrenal cortex function (cortisol synthesis), myoclonus, vomiting
Sodium oxybate (GHB)	30–60 minutes / 1.5–3 hours	A GABA metabolite; antihypoxic effect	Hypokalemia, motor excitement; it is abused
Midazolam	1–3 minutes / 30–60 minutes	Benzodiazepine; premedication, sedation, induction; amnesia	Respiratory depression (especially with opioids)

CLINICAL SIGNIFICANCE

Ketamine is valuable in emergency medicine and in the field: in hemodynamically unstable patients (shock, blood loss) it does not lower BP, preserves breathing, can also be given intramuscularly, and is convenient in bronchial asthma. In small (subanesthetic) doses it is a strong analgesic and a fast-acting antidepressant (esketamine — nasal spray, treatment-resistant depression).

2.5. Combined anesthesia and premedication

Premedication — preparation with drugs before surgery:

- **anxiolytic and sedative** — benzodiazepines (midazolam);
- **analgesic** — opioids (fentanyl, Promedol);
- **vagolytic** (reducing salivary and bronchial secretion and reflex bradycardia) — atropine, glycopyrronium;
- **antiemetic and antacid** (risk of aspiration) — ondansetron, dexamethasone, H_2 blockers or proton pump inhibitors;
- **antihistamine** (in an allergic history).

Balanced anesthesia scheme: induction (propofol) → analgesia (fentanyl) → muscle relaxation (rocuronium) and intubation → maintenance of anesthesia (sevoflurane or propofol infusion + fentanyl) → awakening (muscle relaxant antagonist). **Neuroleptanalgesia** — a combination of a

neuroleptic (droperidol) + a potent opioid (fentanyl); the patient is conscious but indifferent and pain-free.

3. Laboratory class: studying the stages of anesthesia

This section corresponds to the 5th-semester class “Laboratory study of the stages of general anesthetics”.

Course of the experiment (in a mouse or rat):

1. The animal's baseline state is recorded: motor activity, respiratory rate, response to a painful stimulus (tail pinch), corneal and righting reflexes, muscle tone.
2. The animal is placed under a glass bell jar and a cotton ball soaked in the anesthetic (ether or sevoflurane) is placed inside.
3. Signs are recorded every 30–60 seconds: **stage I** — reduced activity, impaired coordination; **stage II** — motor excitement, irregular breathing; **stage III** — loss of the righting reflex, no response to pain, relaxed muscles, rhythmic and slow breathing.
4. If anesthesia deepens (breathing slows sharply), the animal is immediately taken out into fresh air and the order of awakening is observed.
5. The results are entered into a table and compared with an intravenous agent (thiopental): with it, stage II is hardly seen.

> **REMEMBER**
> Main criteria of the depth of anesthesia in the laboratory: the **righting reflex** (consciousness), **response to pain**, the **corneal reflex**, **muscle tone**, and **respiratory rate and pattern**. A sharp slowing of breathing is a sign of the danger of passing into stage IV.

4. Neuroleptics (antipsychotic agents)

Neuroleptics — drugs that eliminate **psychoses** (hallucinations, delusions, psychomotor agitation, thought disorders). They do not cause loss of consciousness and do not produce anesthesia even at large doses.

History. In 1952, in France, Delay and Deniker introduced chlorpromazine (aminazine) into psychiatry — the “psychopharmacological revolution” began.

4.1. Mechanism of action

According to the **dopamine theory** of psychoses, the positive symptoms of schizophrenia are associated with increased dopamine activity in the **mesolimbic** pathway. Neuroleptics **block D_2 dopamine receptors**. Depending on which dopaminergic system is blocked, both therapeutic and adverse effects arise.

Table 7.6. *Dopaminergic pathways and the effects of neuroleptics*

Route	Function	Result of D_2 blockade
Mesolimbic	Emotions, motivation	Antipsychotic effect (positive symptoms decrease)
Mesocortical	Cognitive functions, volition	Worsening of negative symptoms and cognitive impairment
Nigrostriatal	Motor control (extrapyramidal system)	Extrapyramidal disorders (parkinsonism, dystonia, akathisia, tardive dyskinesia)

Route	Function	Result of D_2 blockade
Tuberoinfundibular	Inhibition of prolactin secretion	Hyperprolactinemia (galactorrhea, amenorrhea, gynecomastia, impotence)
Chemoreceptor trigger zone of the vomiting center	Vomiting	Antiemetic effect

MEMORY AID

The 4 dopamine pathways — **"Limbic — Cortex — Substantia nigra — Pituitary"**: meso**limbic** (psychosis decreases), meso**cortical** (negative symptoms), nigrostriatal — **substantia nigra** (movement disorders), tuberoinfundibular — **pituitary** (prolactin increases). The therapeutic effect is only in the first; the others are a source of adverse effects.

Most neuroleptics also block other receptors, which determines additional effects: **alpha-1** (hypotension, orthostatic collapse), **M-cholinergic receptors** (dry mouth, constipation, tachycardia, urinary retention), **H_1** (sedation, weight gain), **5-HT_{2A}** (in atypical neuroleptics — reduces extrapyramidal disorders).

4.2. Classification of neuroleptics

Table 7.7. *Classification of neuroleptics*

Generation	Chemical group	Preparations	Property
Typical (first generation)	Phenothiazines	Chlorpromazine (aminazine), levomepromazine (Tisercin), trifluoperazine (Triftazin), fluphenazine, thioridazine	Chlorpromazine — strongly sedative, alpha-blocker (hypotension); trifluoperazine — strongly antipsychotic, many extrapyramidal disorders
	Butyrophenones	Haloperidol, droperidol	Strongly antipsychotic, little sedation; the most extrapyramidal disorders; haloperidol decanoate — a depot preparation (once a month)
	Thioxanthenes	Chlorprothixene, zuclopenthixol, flupenthixol	Antidepressant component
	Benzamides	Sulpiride, tiapride	At small doses — antidepressant, at large doses — antipsychotic
Atypical (second generation)	—	Risperidone, olanzapine, quetiapine, clozapine, aripiprazole, ziprasidone, paliperidone, lurasidone	D_2 + 5-HT_{2A} blockade (aripiprazole — a partial D_2 agonist); few extrapyramidal disorders; also act on negative symptoms

4.3. Pharmacological effects and uses

- **Antipsychotic** — elimination of hallucinations and delusions;
- **sedative**, stopping psychomotor agitation;
- **antiemetic** (D_2 blockade in the trigger zone; droperidol and metoclopramide also have this mechanism);
- **potentiation** of the effects of anesthetics, analgesics and hypnotics;
- **hypothermic** (depression of the thermoregulatory center);

- **anti-hiccup** (chlorpromazine). **Uses:** schizophrenia, mania in bipolar disorder, acute psychoses (alcoholic delirium, organic psychoses), psychomotor agitation, delirium (haloperidol), vomiting, neuroleptanalgesia (droperidol), Tourette's syndrome, as an adjunct in treatment-resistant depression (atypical agents).

4.4. Adverse effects

Table 7.8. *Extrapyramidal adverse effects of neuroleptics*

Disorder	When it develops	Signs	Treatment
Acute dystonia	First hours to days (often in young people)	Spasm of the neck, tongue and extraocular muscles (torticollis, oculogyric crisis)	Biperiden or trihexyphenidyl, diphenhydramine intravenously or intramuscularly
Akathisia	First weeks	Inner restlessness, inability to sit still	Dose reduction, propranolol, a benzodiazepine
Drug-induced parkinsonism	Weeks to months	Hypokinesia, rigidity, tremor	Dose reduction, central M-cholinergic blockers (not levodopa)
Tardive dyskinesia	Months to years	Involuntary movements of the face, lips, tongue and limbs; often irreversible	Change of neuroleptic (clozapine), VMAT2 inhibitors; anticholinergics make it worse

CAUTION

Neuroleptic malignant syndrome (NMS) — a rare but life-threatening complication (mortality up to 10%): **high fever**, **"lead-pipe" muscle rigidity**, impaired consciousness, autonomic instability (tachycardia, fluctuating BP, sweating), a **sharp rise in blood creatine kinase**, leukocytosis, rhabdomyolysis and renal failure. **Treatment:** stop the neuroleptic immediately, intensive care, cooling, fluids, **dantrolene** and/or **bromocriptine** (a dopamine agonist), benzodiazepines.

Other adverse effects:

- **metabolic syndrome** (weight gain, hyperglycemia, dyslipidemia) — especially with olanzapine and clozapine;
- **hyperprolactinemia** — with risperidone and typical agents;
- **QT prolongation** and torsades de pointes arrhythmia — thioridazine, haloperidol (intravenous), ziprasidone;
- **orthostatic hypotension** (alpha-1 blockade) — chlorpromazine, clozapine, quetiapine;
- anticholinergic effects, sedation, lowering of the seizure threshold;
- **agranulocytosis** — **clozapine** (up to 1%; regular monitoring of blood counts is mandatory), myocarditis;
- skin photosensitization, cholestatic jaundice (chlorpromazine);
- increased risk of stroke and death in elderly patients with dementia.

CLINICAL SIGNIFICANCE

Clozapine is the most effective neuroleptic in treatment-resistant schizophrenia and does not cause tardive dyskinesia, but because of the risk of agranulocytosis it is prescribed only with weekly (later monthly) monitoring of the white cell count.

5. Anxiolytics (tranquilizers)

Anxiolytics — drugs that reduce anxiety, fear and emotional tension. They do not eliminate psychoses.

5.1. Benzodiazepines

Mechanism of action. Benzodiazepines bind to a special **benzodiazepine (allosteric) binding site** of the **GABA-A receptor–chloride channel complex** and increase the affinity of GABA for the receptor. As a result, the **frequency of chloride channel opening** under the action of GABA increases, and the neuron is hyperpolarized — inhibition is enhanced. Benzodiazepines do not open the channel by themselves in the absence of GABA — that is why they are relatively safe when used alone.

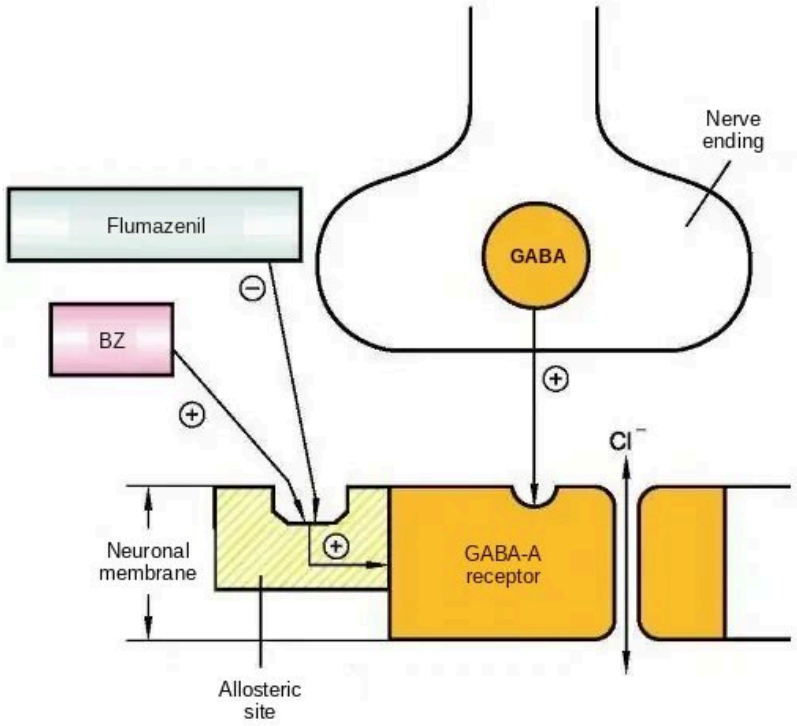


Figure 7.2. *Action of benzodiazepines (BZ) and flumazenil on the GABA-A receptor: BZs bind to the allosteric site and enhance the action of GABA, flumazenil competitively blocks them (D.A. Kharkevich, Pharmacology, 2010)*

> **REMEMBER**
> **Benzodiazepines** increase the **frequency** of chloride channel opening, while **barbiturates** increase its **duration**. At large doses, barbiturates open the channel even without GABA — that is why their therapeutic range is narrow and they can paralyze the respiratory center.

The five main effects of benzodiazepines:

1. **anxiolytic** (reducing anxiety);
2. **sedative and hypnotic**;
3. **muscle relaxant** (central, at the spinal cord level);
4. **anticonvulsant**;
5. **amnestic** (anterograde amnesia).

Table 7.9. *Main benzodiazepines*

Preparation	t½ (with active metabolites)	Main use
Diazepam (Sibazon, Relanium)	Long (20–100 hours)	Anxiety, status epilepticus and seizures, alcohol withdrawal, muscle spasm, premedication
Chlordiazepoxide (Elenium)	Long	Alcohol withdrawal
Phenazepam	Long	Anxiety, insomnia (in CIS countries)
Clonazepam	Long	Seizures, panic disorder
Lorazepam	Intermediate (10–20 hours), no active metabolites	Status epilepticus, anxiety; convenient in liver disease (conjugation)
Alprazolam	Intermediate (12 hours)	Panic disorder, anxiety; high risk of dependence
Oxazepam	Short to intermediate, no active metabolites	The elderly, liver disease
Midazolam	Short (2–3 hours)	Premedication, sedation, induction, seizures (intranasal, intramuscular)
Medazepam (Rudotel), tofisopam (Grandaxin)	—	“Daytime tranquilizers” — little sedation and muscle relaxation

Adverse effects: drowsiness, lethargy, reduced reaction speed (dangerous for drivers), ataxia and **falls and hip fractures in the elderly**, anterograde amnesia, paradoxical excitement (in children and the elderly); **tolerance, psychological and physical dependence**; **withdrawal** on abrupt cessation (anxiety, insomnia, tremor, seizures); with alcohol, opioids and other CNS depressants — respiratory depression.

> **CAUTION**
> Benzodiazepines are prescribed in **short courses** (not more than 2–4 weeks). In chronic anxiety disorders, the first choice is **antidepressants** (SSRIs) and psychotherapy. When long-term benzodiazepines are stopped, the dose must be reduced gradually over several weeks to months. **Combination with opioids** is one of the main causes of overdose deaths.

Benzodiazepine poisoning: drowsiness, ataxia, dysarthria, coma in severe cases; alone, respiratory depression is rare, but it is severe with alcohol and opioids. **Antidote — flumazenil** (a competitive antagonist of benzodiazepine receptors), 0.2 mg intravenously, up to 1 mg. The effect of flumazenil is short (≈ 1 hour), and resedation is possible; in chronic users and in mixed poisoning with tricyclic antidepressants, it may provoke **seizures**.

5.2. Non-benzodiazepine anxiolytics

Table 7.10. *Non-benzodiazepine anxiolytics*

Preparation	Mechanism	Features
Buspirone	Partial 5-HT_{1A} agonist	No sedation, muscle relaxation or dependence; the effect develops in 2–4 weeks; for generalized anxiety disorder
Hydroxyzine (Atarax)	H_1 blocker, M-cholinergic blocker	Sedative, antipruritic; no dependence
Etifoxine (Stresam)	Acts on GABA-A, increases the synthesis of neurosteroids	In adjustment disorders

Preparation	Mechanism	Features
Mebicar (Adaptol), Afobazol (fabomotizole)	Various	Preparations of CIS countries; limited evidence base
SSRIs and SNRIs (sertraline, escitalopram, venlafaxine)	Serotonin (and norepinephrine) reuptake inhibitors	First-line treatment of chronic anxiety disorders, panic disorder and OCD (topic 8)
Pregabalin	Binds to the alpha-2-delta subunit of Ca^{2+} channels	Generalized anxiety disorder (in some countries)

6. Clinical part

Table 7.11. *Choosing a CNS depressant in clinical situations*

Situation	Choice
Induction of anesthesia in a hemodynamically unstable patient	Ketamine or etomidate
Mask induction of anesthesia in children	Sevoflurane
Short outpatient procedure, sedation for endoscopy	Propofol, midazolam
Acute psychomotor agitation (psychosis)	Haloperidol intramuscularly ± lorazepam/diazepam; olanzapine
Delirium in the elderly	Search for and eliminate the cause; if necessary, haloperidol in a small dose
Status epilepticus	Lorazepam or diazepam intravenously (midazolam intranasally/intramuscularly)
Alcohol withdrawal	Diazepam, chlordiazepoxide (in hepatic failure — lorazepam, oxazepam)
Generalized anxiety disorder	SSRIs/SNRIs, buspirone; a benzodiazepine only for a short time
Neuroleptic-induced acute dystonia	Biperiden or diphenhydramine

Practical skills

Skill 1. Management of acute dystonia

1. Establish a history of neuroleptic use (haloperidol, and metoclopramide too!).
2. Biperiden 2.5–5 mg intramuscularly or slowly intravenously (or diphenhydramine 50 mg).
3. If there is no effect within 20–30 minutes, repeat.
4. Prescribe biperiden tablets for the next 2–3 days; review the neuroleptic dose.

Skill 2. Algorithm for withdrawing benzodiazepines

1. Replace a short-acting benzodiazepine with an equivalent dose of diazepam.
2. Reduce the dose by 10–25% every 1–2 weeks.
3. If withdrawal symptoms intensify, slow down the rate of reduction.
4. Treat the anxiety with psychotherapy and an SSRI.

Skill 3. Key prescriptions

PRESCRIPTION 7.1. PROPOFOL
Rp.: Emulsi Propofoli 1% — 20 ml
D.t.d. N. 5 in ampullis
S. 1.5–2.5 mg/kg intravenously for induction of anesthesia (by an anesthesiologist).

PRESCRIPTION 7.2. KETAMINE
Rp.: Solutionis Ketamini 5% — 2 ml
D.t.d. N. 10 in ampullis
S. 1–2 mg/kg intravenously or 4–6 mg/kg intramuscularly (for general anesthesia).

PRESCRIPTION 7.3. HALOPERIDOL (IN AMPOULES)
Rp.: Solutionis Haloperidoli 0.5% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml intramuscularly (in psychomotor agitation).

PRESCRIPTION 7.4. RISPERIDONE
Rp.: Risperidoni 0.002
D.t.d. N. 30 in tabulettis
S. Take 1 tablet 1–2 times a day.

PRESCRIPTION 7.5. OLANZAPINE
Rp.: Tabulettas Olanzapini 0.01 obductas N. 28
D.S. Take 1 tablet once a day in the evening.

PRESCRIPTION 7.6. CHLORPROMAZINE (AMINAZINE)
Rp.: Solutionis Chlorpromazini hydrochloridi 2.5% — 2 ml
D.t.d. N. 10 in ampullis
S. Inject 1–2 ml deep intramuscularly (the patient should then lie down for 1–2 hours).

PRESCRIPTION 7.7. DIAZEPAM (TABLETS)
Rp.: Diazepami 0.005
D.t.d. N. 20 in tabulettis
S. Take 1 tablet 2 times a day (for not more than 2 weeks).

PRESCRIPTION 7.8. DIAZEPAM (IN AMPOULES)
Rp.: Solutionis Diazepami 0.5% — 2 ml
D.t.d. N. 10 in ampullis
S. Inject 2 ml slowly intravenously (in seizures).

PRESCRIPTION 7.9. ALPRAZOLAM
Rp.: Alprazolami 0.00025
D.t.d. N. 30 in tabulettis
S. Take 1 tablet 3 times a day.

PRESCRIPTION 7.10. BUSPIRONE
Rp.: Buspironi hydrochloridi 0.01
D.t.d. N. 60 in tabulettis
S. Take 1 tablet 2 times a day.

PRESCRIPTION 7.11. FLUMAZENIL
Rp.: Solutionis Flumazenili 0.01% — 5 ml
D.t.d. N. 5 in ampullis
S. 0.2 mg (2 ml) intravenously, if necessary 0.1 mg every minute up to 1 mg.

PRESCRIPTION 7.12. BIPERIDEN
Rp.: Solutionis Biperideni lactatis 0.5% — 1 ml
D.t.d. N. 5 in ampullis
S. Inject 1 ml intramuscularly or slowly intravenously (in acute dystonia).

Review questions

1. Classification of drugs acting on the CNS.
2. What is general anesthesia? The components of modern general anesthesia.
3. Describe the stages of anesthesia and their signs.
4. Inhalational anesthetics. What is MAC? What does the speed of induction depend on?
5. Advantages and disadvantages of sevoflurane, isoflurane, halothane and nitrous oxide.
6. Comparison of propofol, thiopental, ketamine and etomidate.
7. Dissociative anesthesia with ketamine and features of its use.
8. Principles of premedication and balanced anesthesia.
9. How is the depth of anesthesia assessed in a laboratory experiment?
10. Mechanism of action of neuroleptics. Dopaminergic pathways and the results of their blockade.
11. Differences between typical and atypical neuroleptics, with examples.
12. Extrapyramidal adverse effects of neuroleptics and their treatment.
13. Signs and treatment of neuroleptic malignant syndrome.
14. Properties and risks of clozapine.
15. Mechanism of action and main effects of benzodiazepines. Difference from barbiturates.
16. Adverse effects of benzodiazepines, dependence and withdrawal.
17. Benzodiazepine poisoning and flumazenil.
18. Properties of buspirone.

Test questions

1. At which stage of anesthesia are speech and motor excitement seen?
   A) Awakening
   B) Excitement
   C) Analgesia
   D) Agonal
   E) Surgical anesthesia

2. The most convenient inhalational anesthetic for mask induction in children:
   A) Halothane
   B) Isoflurane
   C) Ester
   D) Sevoflurane
   E) Desflurane

3. An anesthetic that sensitizes the myocardium to epinephrine and causes arrhythmias:
   A) Propofol
   B) Nitrous oxide
   C) Xenon
   D) Halothane
   E) Sevoflurane

4. An NMDA receptor antagonist that raises BP and causes dissociative anesthesia:
   A) Etomidate
   B) Ketamine
   C) Thiopental sodium
   D) Midazolam
   E) Propofol

5. An induction anesthetic that suppresses adrenal cortex function:
   A) Etomidate
   B) Nitrous oxide
   C) Ketamine
   D) Propofol
   E) Sevoflurane

6. With D_2 blockade in which pathway is the antipsychotic effect of neuroleptics associated?
   A) Tuberoinfundibular
   B) Trigger zone
   C) Hypothalamic-pituitary
   D) Nigrostriatal
   E) Mesolimbic

7. Hyperprolactinemia caused by neuroleptics results from blockade of which pathway?
   A) Vomiting center
   B) Mesocortical
   C) Nigrostriatal
   D) Tuberoinfundibular
   E) Mesolimbic

8. The neuroleptic that causes extrapyramidal disorders most often:
   A) Haloperidol
   B) Clozapine
   C) Aripiprazole
   D) Olanzapine
   E) Quetiapine

9. A neuroleptic that requires regular blood monitoring because of the risk of agranulocytosis:
   A) Risperidone
   B) Haloperidol
   C) Clozapine
   D) Chlorpromazine (aminazine)
   E) Sulpiride

10. A sign typical of neuroleptic malignant syndrome:
   A) Bradycardia and miosis
   B) Muscle hypotonia
   C) High fever and muscle rigidity
   D) Hypoglycemia
   E) Hypothermia

11. Given to stop acute dystonia:
   A) Haloperidol
   B) Neostigmine
   C) Epinephrine
   D) Biperiden
   E) Levodopa

12. How do benzodiazepines act on the GABA-A receptor?
   A) They block the channel
   B) They open the channel without GABA
   C) They increase the duration of chloride channel opening
   D) They increase the frequency of chloride channel opening
   E) They enhance GABA synthesis

13. Antidote in benzodiazepine poisoning:
   A) Physostigmine
   B) Bemegride
   C) Atropine
   D) Flumazenil
   E) Naloxone

14. An anxiolytic that is a partial 5-HT_{1A} agonist and causes neither sedation nor dependence:
   A) Buspirone
   B) Hydroxyzine
   C) Alprazolam
   D) Phenazepam
   E) Diazepam

15. The first choice for stopping status epilepticus:
   A) Lorazepam or diazepam intravenously
   B) Carbamazepine orally
   C) Buspirone
   D) Caffeine
   E) Haloperidol

Clinical cases

Case 1

A 30-year-old man injured in a road accident was brought to surgery with intra-abdominal bleeding. BP 75/40 mm Hg, HR 130/min. Induction of anesthesia is needed for emergency laparotomy.

Questions: 1) Which induction anesthetic is appropriate and why? 2) Which agent is dangerous?

Answer: 1) Ketamine (or etomidate): ketamine activates the sympathetic system and maintains or raises BP and heart rate; etomidate hardly affects hemodynamics. In parallel, blood and fluid replacement. 2) Propofol and thiopental, because of vasodilation and myocardial depression, can cause severe hypotension and cardiac arrest in a patient in hemorrhagic shock.

Case 2

A 22-year-old man was given haloperidol intramuscularly for an acute episode of schizophrenia. After 6 hours, the patient's head was pulled back and to the side, his eyes became fixed in an upward gaze, the tongue protruded, and speaking was difficult.

Questions: 1) What complication is this? 2) Its mechanism? 3) Treatment.

Answer: 1) Acute dystonia (oculogyric crisis, torticollis) — an extrapyramidal disorder. 2) Haloperidol blocks D_2 receptors in the nigrostriatal pathway; the dopamine–acetylcholine balance is disturbed and cholinergic influence predominates. 3) Biperiden 2.5–5 mg or diphenhydramine 50 mg intramuscularly/intravenously (central M-cholinergic blockers restore the balance); then oral biperiden for several days; consider switching to an atypical neuroleptic with few extrapyramidal effects.

Case 3

A 75-year-old woman has taken phenazepam every day for 3 years for insomnia. In the last month she has fallen twice at night, and her forgetfulness has increased. Her daughter wants to stop the drug immediately.

Questions: 1) The link between the symptoms and the drug? 2) Is stopping the drug immediately correct? 3) Recommendations.

Answer: 1) A benzodiazepine with a long t½ accumulates in the elderly and causes ataxia, falls and cognitive impairment (a wrong choice for the elderly). 2) No — after 3 years of use there is physical dependence; abrupt cessation may cause a withdrawal syndrome (anxiety, insomnia, seizures, delirium). 3) Reduce the dose gradually over several weeks to months (by 10–25% every 2 weeks); cognitive behavioral therapy for insomnia, sleep hygiene; if there is depression or anxiety — an antidepressant; measures to reduce the risk of falls.

Quick review table

Table 7.12. *General anesthetics, neuroleptics, anxiolytics — summary*

Group	Preparations	Mechanism / feature	Main risk
Inhalational anesthetics	Sevoflurane, isoflurane, desflurane, halothane	MAC; solubility in blood determines speed	Halothane — arrhythmia, hepatitis; malignant hyperthermia
Gases	Nitrous oxide, xenon	N_2O — strong analgesic, weak anesthetic	Diffusion hypoxia, closed cavities

Group	Preparations	Mechanism / feature	Main risk
Intravenous	Propofol, thiopental, ketamine, etomidate	Propofol — rapid, clear-headed awakening; ketamine — NMDA, BP ↑; etomidate — hemodynamically stable	Propofol — hypotension; ketamine — hallucinations; etomidate — adrenal gland
Typical neuroleptics	Chlorpromazine, haloperidol, trifluoperazine	D_2 blockade	Extrapyramidal disorders, NMS, hyperprolactinemia
Atypical neuroleptics	Risperidone, olanzapine, quetiapine, clozapine, aripiprazole	D_2 + 5-HT_{2A} blockade	Metabolic syndrome; clozapine — agranulocytosis
Benzodiazepines	Diazepam, lorazepam, alprazolam, midazolam, phenazepam	$GABA_A$ — frequency of chloride channel opening ↑	Dependence, falls, respiratory depression with alcohol/opioids; antidote — flumazenil
Non-benzodiazepine anxiolytics	Buspirone, hydroxyzine	Partial 5-HT_{1A} agonist / H_1 blocker	No dependence
Treatment of extrapyramidal disorders	Biperiden, trihexyphenidyl	Central M-cholinergic blockers	Worsen tardive dyskinesia
Treatment of NMS	Dantrolene, bromocriptine	—	Stop the neuroleptic

TOPIC 8

Psychostimulants. Antidepressants

Learning objectives

Depression is one of the leading causes of disability in the world; it often hides behind somatic complaints (fatigue, insomnia, pain) at the GP's appointment. Antidepressants are among the most frequently prescribed drugs, but their delayed onset of action, serotonin syndrome and the risk of tricyclic antidepressant poisoning require sound knowledge. Psychostimulants are used to treat attention deficit hyperactivity disorder and narcolepsy, but they carry a high risk of abuse.

The student should know:

- the classification of CNS stimulants;
- the mechanism, effects, uses and adverse effects of psychostimulants (caffeine, amphetamine, methylphenidate, modafinil, mesocarb);
- the monoamine theory of depression and the classification, mechanism, uses and adverse effects of antidepressants;
- the clinical picture and treatment of serotonin syndrome and tricyclic antidepressant poisoning.

The student should be able to:

- write prescriptions for preparations such as caffeine sodium benzoate, methylphenidate, amitriptyline, sertraline, escitalopram, fluoxetine, venlafaxine and mirtazapine;
- choose an antidepressant for a patient and explain the rules for taking it.

1. Classification of CNS stimulants

Table 8.1. *CNS stimulants*

Group	Preparations
Psychostimulants	Methylxanthines (caffeine), phenylalkylamines (amphetamine, methamphetamine, dexamfetamine), methylphenidate, mesocarb (Sydnocarb), modafinil, atomoxetine (not a stimulant)
Antidepressants	TCAs, SSRIs, SNRIs, MAO inhibitors, atypical antidepressants (section 3)
Nootropics	Piracetam, Phenotropil, citicoline, gamma-aminobutyric acid and others (a topic for independent study)
Analeptics	Caffeine, nikethamide (Cordiamine), bemegride, camphor (a topic for independent study)
Adaptogens and general tonics	Ginseng, eleutherococcus, schisandra (6th-semester independent study)

2. Psychostimulants

Psychostimulants increase mental and physical activity, temporarily reduce fatigue and drowsiness, lift the mood and improve attention.

2.1. Caffeine and methylxanthines

Caffeine — an alkaloid of tea leaves, coffee beans, cocoa and the kola nut; a methylxanthine. With sodium benzoate, as **caffeine sodium benzoate**, it is readily soluble in water and suitable for

injection.

Mechanism of action:

1. **Blockade of adenosine receptors (A_1 and A_{2A})** — the main mechanism. Adenosine is an inhibitory modulator in the CNS; its blockade leads to excitation.
2. **Inhibition of phosphodiesterase** (at high concentrations) — accumulation of cAMP.
3. Enhanced release of Ca^{2+} from the sarcoplasmic reticulum (at very high doses).

Table 8.2. *Effects of caffeine*

System	Effect
CNS	Stimulates the cerebral cortex: alertness, reduced fatigue, increased mental performance; at large doses — restlessness, insomnia, tremor, seizures; stimulates the respiratory and vasomotor centers
Heart	The force and rate of contraction increase (tachycardia and arrhythmia at large doses)
Vessels	Constricts cerebral vessels (reduces pain in migraine); directly dilates other vessels
Bronchi	Dilate (theophylline is stronger)
Kidney	Diuresis increases slightly
Stomach	Hydrochloric acid secretion increases

Uses: mental and physical fatigue, drowsiness; in combination with analgesics for headache and migraine (Ascofen, Citramon, Cofetamin); hypotension; mild poisoning with CNS depressants; **apnea** in preterm infants (caffeine citrate — the first choice).

> **CAUTION**
> The effect of caffeine **depends on the initial state of the CNS**: it is strong in a tired person and weak in a rested one. Consuming more than 400 mg a day (about 4 cups of coffee) leads to anxiety, insomnia and tachycardia; when regular use is stopped abruptly, headache and lethargy (withdrawal) occur. Contraindications: anxiety disorders, insomnia, severe hypertension, tachyarrhythmias, glaucoma, gastric ulcer.

2.2. Amphetamine, methylphenidate and others

Amphetamine — a phenylalkylamine; an indirect sympathomimetic. It enhances the **release of dopamine and norepinephrine** from nerve endings and inhibits their reuptake. A strong psychostimulant; it reduces appetite and raises BP. It is under strict control as a **narcotic drug**; it causes dependence and psychoses.

Methylphenidate — blocks the reuptake of dopamine and norepinephrine. The main drug for treating **attention deficit hyperactivity disorder** (ADHD); paradoxically, it calms hyperactive children and improves attention. It is also used in narcolepsy.

Modafinil — reduces daytime sleepiness in narcolepsy and shift work disorder; the risk of dependence is lower than with amphetamine.

Mesocarb (Sydnocarb) — used in CIS countries for asthenic states; its effect develops gradually.

Atomoxetine — a selective norepinephrine reuptake inhibitor; not a psychostimulant, the second choice in ADHD, does not cause dependence.

Table 8.3. *Adverse effects of psychostimulants*

Adverse effect	Comment
Insomnia, restlessness, anxiety	Take the drug in the first half of the day
Reduced appetite, weight loss, growth retardation in children	Measure weight and height regularly
Tachycardia, rise in BP, arrhythmias	Screen for heart disease beforehand
Psychological dependence, tolerance	Prescriptions on a special form
Psychoses (at large doses)	—
“Debt” phenomenon — fatigue and depression when the effect wears off	The body's resources are spent and no rest is given

CLINICAL SIGNIFICANCE

Cocaine — a local anesthetic and a strong psychostimulant: it blocks the reuptake of dopamine, norepinephrine and serotonin. Cocaine poisoning causes hyperthermia, tachycardia, hypertension, coronary spasm (**myocardial infarction in a young person**), seizures and psychosis. Treatment: **benzodiazepines** (the main treatment), cooling, nitrates and calcium antagonists; nonselective **beta-blockers are contraindicated** (unopposed alpha vasoconstriction).

3. Depression and antidepressants

3.1. The monoamine theory of depression

Depression — a disease with a depressed mood or loss of interest (anhedonia) lasting at least 2 weeks, accompanied by disturbed sleep and appetite, fatigue, feelings of guilt, reduced concentration and thoughts of death.

According to the **monoamine theory**, depression is associated with a deficiency of **serotonin** and **norepinephrine** (and also dopamine) in the CNS. Antidepressants increase the amount of these mediators in the synapse. However, although mediator concentrations rise within a few hours, the clinical effect appears only **after 2–4 weeks** — this is explained by adaptation of receptor sensitivity (down-regulation), neuroplasticity and increased production of BDNF (brain-derived neurotrophic factor).

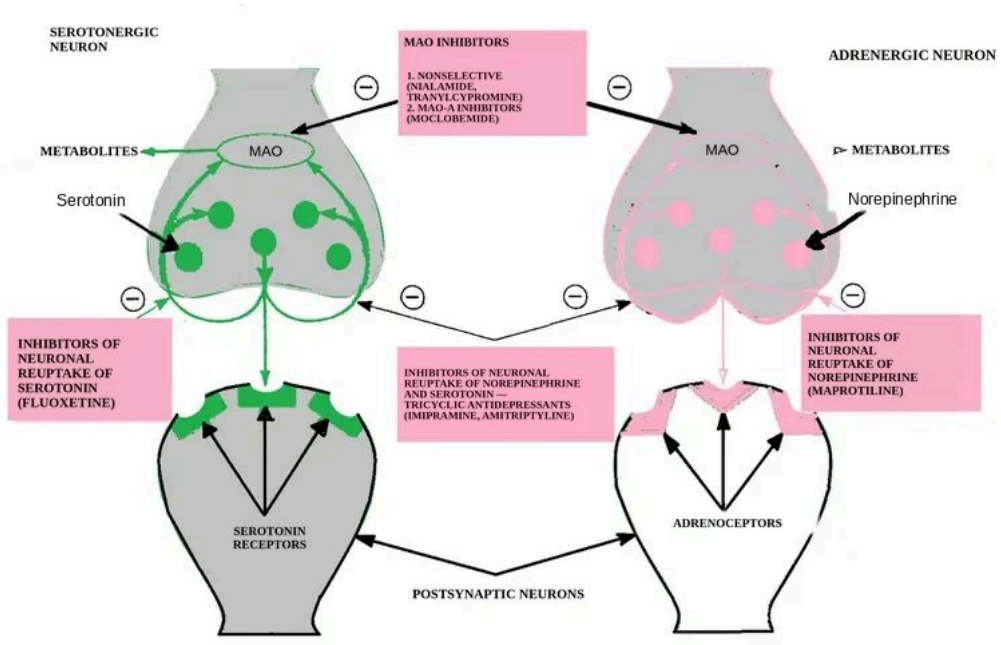


Figure 8.1. *Main directions of action of antidepressants: MAO inhibitors, blockers of neuronal reuptake of serotonin and norepinephrine (teaching material of the TMA Department of Pharmacology)*

3.2. Classification of antidepressants

Table 8.4. *Classification of antidepressants*

Group	Mechanism	Preparations
Tricyclic antidepressants (TCAs)	Nonselectively block the reuptake of norepinephrine and serotonin; + M, H_1 and alpha-1 blockade	Amitriptyline, imipramine (Imizin), clomipramine, nortriptyline
Tetracyclic	Norepinephrine reuptake	Maprotiline, mianserin
Selective serotonin reuptake inhibitors (SSRIs)	Serotonin reuptake (SERT) blockade	Sertraline, escitalopram, citalopram, fluoxetine, paroxetine, fluvoxamine
Serotonin and norepinephrine reuptake inhibitors (SNRIs)	SERT and NET blockade	Venlafaxine, duloxetine, milnacipran
MAO inhibitors — irreversible, nonselective	MAO-A and MAO-B	Phenelzine, tranylcypromine (rarely now)
Reversible inhibitors of MAO-A	MAO-A	Moclobemide, pirlindole (Pyrazidol)
Noradrenergic and specific serotonergic (NaSSA)	Presynaptic alpha-2 blockade; 5-HT_2, 5-HT_3 and H_1 blockade	Mirtazapine
Dopamine and norepinephrine reuptake inhibitor	DAT and NET	Bupropion
Serotonin antagonists and reuptake inhibitors	5-HT_{2A} blockade + SERT	Trazodone
Melatonergic	MT1/MT2 agonist + 5-HT_{2C} antagonist	Agomelatine
Multimodal	SERT blockade + various 5-HT receptors	Vortioxetine

Group	Mechanism	Preparations
NMDA antagonist	Glutamate system	Esketamine (nasal spray) — treatment-resistant depression
Herbal preparation	Various	St John's wort extract — mild depression; a CYP inducer!

3.3. Tricyclic antidepressants (TCAs)

Amitriptyline — the most commonly used TCA; it has a strong **sedative and anxiolytic** component (H_1 blockade). **Imipramine** — has a stimulating component. **Clomipramine** — serotonergic, effective in obsessive-compulsive disorder.

Uses: depression (now more often as second line), **chronic pain syndromes** — neuropathic pain, fibromyalgia, tension-type headache and migraine prophylaxis (in small doses), nocturnal enuresis in children (imipramine), insomnia.

Table 8.5. *Adverse effects of TCAs and their mechanisms*

Receptor / mechanism	Adverse effect
M-cholinergic blockade	Dry mouth, constipation, urinary retention, tachycardia, disturbed accommodation, attack of glaucoma, delirium in the elderly
H_1 blockade	Sedation, weight gain
Alpha-1 blockade	Orthostatic hypotension, dizziness, falls
Sodium channel blockade (quinidine-like effect)	QRS widening, AV block, ventricular arrhythmias
Lowering of the seizure threshold	Seizures

CAUTION

TCA poisoning — one of the most dangerous poisonings in suicide attempts (narrow therapeutic range: a 10–15-day supply can be fatal). Signs — **"3 C's"**: **C**oma, **C**onvulsions (seizures), **C**ardiotoxicity (wide QRS, arrhythmias, hypotension) + anticholinergic syndrome. **Treatment:** secure the airway, activated charcoal, intravenous **sodium bicarbonate** (if QRS > 100 ms — the sodium load and alkalinization reduce cardiotoxicity), benzodiazepines for seizures. Physostigmine and flumazenil are **contraindicated** (risk of seizures, asystole). A depressed patient should not be prescribed a large quantity of TCAs at one time.

3.4. Selective serotonin reuptake inhibitors (SSRIs)

SSRIs are **first-line** drugs for depression, anxiety disorders, panic disorder, obsessive-compulsive disorder (OCD), post-traumatic stress disorder, bulimia and premenstrual dysphoria. They are safer: they hardly act on M, H_1 and alpha-1 receptors and are of low toxicity in overdose.

Table 8.6. *Properties of SSRI preparations*

Preparation	Property
Sertraline	Safer in heart disease; few interactions; relatively suitable in pregnancy and lactation
Escitalopram	The most selective; well tolerated; slightly prolongs QT
Fluoxetine	The longest t½ (active metabolite — 1–2 weeks); activating; little withdrawal syndrome; a strong CYP2D6 inhibitor

Preparation	Property
Paroxetine	Sedative, some anticholinergic effect; strong withdrawal syndrome; not recommended in pregnancy
Fluvoxamine	A strong CYP1A2 inhibitor (interactions with theophylline and clozapine)

Adverse effects of SSRIs: nausea and dyspepsia (in the first weeks), headache, insomnia or drowsiness, **sexual dysfunction** (libido and orgasm), increased anxiety in the first weeks, hyponatremia (in the elderly), increased risk of bleeding (platelet serotonin decreases — especially with NSAIDs and anticoagulants), QT prolongation (citalopram, escitalopram), withdrawal syndrome.

> **CAUTION**
> **In children, adolescents and people under 25**, the risk of suicidal thoughts and behavior may increase in the first weeks of antidepressant treatment. The patient and relatives are warned, and frequent follow-up is arranged in the first month of treatment.

3.5. Serotonin syndrome

Serotonin syndrome — an excessive increase of serotonin in the CNS. Causes: combinations of serotonergic drugs — SSRIs/SNRIs + **MAO inhibitors** (the most dangerous), tramadol, triptans, linezolid, methylene blue, lithium, St John's wort, MDMA (ecstasy).

Table 8.7. *Signs of serotonin syndrome (triad)*

Group	Signs
Mental status	Restlessness, agitation, confusion
Autonomic	Hyperthermia, tachycardia, sweating, mydriasis, diarrhea, fluctuating BP
Neuromuscular	Clonus (especially in the legs, ocular clonus), hyperreflexia, tremor, muscle rigidity

Treatment: stop the serotonergic drugs, benzodiazepines (for agitation and muscle hyperactivity), cooling, fluids; in severe cases **cyproheptadine** (a 5-HT_{2A} antagonist), intensive care.

> **MEMORY AID**
> **Distinguishing serotonin syndrome from NMS**: serotonin syndrome — **rapid** (hours), **clonus and hyperreflexia**, diarrhea, mydriasis (a serotonergic drug); NMS — **slow** (days), **"lead-pipe" rigidity**, hyporeflexia (a neuroleptic).

3.6. SNRIs, MAO inhibitors and atypical antidepressants

SNRIs (venlafaxine, duloxetine) — severe depression, anxiety; **duloxetine** — also for the pain of diabetic neuropathy and fibromyalgia. Adverse effects: rise in BP (venlafaxine, at high doses), nausea, sweating, strong withdrawal syndrome.

MAO inhibitors. Irreversible nonselective agents (phenelzine) are now used only in treatment-resistant cases. While taking them, eating **tyramine-rich foods** (aged cheese, smoked fish and meat, red wine, beer, soy sauce) causes a **hypertensive crisis** ("cheese reaction"). A **2-week** break is needed before switching to an SSRI (5 weeks when switching from fluoxetine to an MAOI). **Moclobemide** (reversible, MAO-A) — low risk of the "cheese reaction".

Mirtazapine — sedative, increases appetite, little effect on sexual function; convenient in depression with insomnia and weight loss and in the elderly. **Bupropion** — does not affect sexual function, causes no weight gain, also used for smoking cessation; lowers the seizure threshold. **Trazodone** —

in small doses for insomnia; rarely priapism. **Agomelatine** — restores the circadian rhythm; liver enzymes must be monitored.

3.7. Principles of antidepressant treatment

REMEMBER

Basic rules of antidepressant treatment: 1. The effect begins in **2–4 weeks**, the full effect in 6–8 weeks; the patient must be warned about this. 2. Start with a small dose and increase to the effective dose over 1–2 weeks. 3. If there is no effect after 4–6 weeks, increase the dose or switch to another group. 4. After remission, treatment is continued for at least **6–12 months** (in recurrent episodes — for years). 5. Withdrawal — **gradually** (over several weeks): abrupt cessation causes dizziness, "electric shock" sensations, nausea, anxiety and sleep disturbance. 6. Assess suicide risk; if there is a risk — refer to a psychiatrist.

In **bipolar disorder**, antidepressants are given only together with mood stabilizers (lithium, valproate, lamotrigine) — otherwise there is a risk of switching to **mania**.

4. Clinical part

Table 8.8. *Choosing an antidepressant according to the clinical situation*

Situation	Preferred choice	To be avoided
First episode, no comorbidity	Sertraline, escitalopram	—
Ischemic heart disease, after infarction	Sertraline	TCAs (arrhythmia), venlafaxine at high doses
Insomnia and weight loss	Mirtazapine	Fluoxetine, bupropion (activating)
Obesity, concern about sexual dysfunction	Bupropion	Mirtazapine, paroxetine
Chronic pain, diabetic neuropathy	Duloxetine, amitriptyline (in a small dose)	—
The elderly	Sertraline, escitalopram, mirtazapine	TCAs, paroxetine (anticholinergic effect)
Pregnancy	Sertraline (individual decision)	Paroxetine
Epilepsy	SSRIs with caution	Bupropion, maprotiline, clomipramine
High risk of suicide	SSRIs (safer in overdose), prescribed in small quantities	TCAs

CLINICAL SIGNIFICANCE

Screening for the GP: the **PHQ-2** questions — "Over the last 2 weeks, have you been bothered by feeling down, depressed or hopeless?" and "Have you lost interest or pleasure in doing things?". A "yes" to both requires a more thorough assessment (PHQ-9).

Practical skills

Skill 1. Instructions given to the patient when prescribing an antidepressant

1. The drug does not improve the mood immediately — the effect begins in 2–4 weeks.
2. Nausea and headache in the first days usually pass within 1–2 weeks.

3. Do not stop the drug yourself, even if you feel better — it is continued for at least 6–12 months.
4. Avoid alcoholic drinks.
5. Take other drugs (tramadol, migraine drugs, St John's wort) only after agreeing with the physician.
6. If your condition worsens or thoughts of harming yourself appear — seek help immediately.

Skill 2. Key prescriptions

PRESCRIPTION 8.1. CAFFEINE SODIUM BENZOATE (IN AMPOULES)
Rp.: Solutionis Coffeini-natrii benzoatis 20% — 1 ml
D.t.d. N. 10 in ampullis
S. Inject 1 ml subcutaneously 1–2 times a day (in hypotension).

PRESCRIPTION 8.2. CAFFEINE SODIUM BENZOATE (TABLETS)
Rp.: Coffeini-natrii benzoatis 0.1
D.t.d. N. 10 in tabulettis
S. Take 1 tablet 2 times a day (in the first half of the day).

PRESCRIPTION 8.3. METHYLPHENIDATE
Rp.: Tabulettas Methylphenidati 0.018 prolongatas N. 30
D.S. Take 1 tablet once a day in the morning (ADHD; on a special form).

PRESCRIPTION 8.4. AMITRIPTYLINE
Rp.: Amitriptylini 0.025
D.t.d. N. 50 in tabulettis
S. Take 1 tablet in the evening; increase the dose gradually.

PRESCRIPTION 8.5. SERTRALINE
Rp.: Tabulettas Sertralini 0.05 obductas N. 28
D.S. Take 1 tablet once a day in the morning after a meal.

PRESCRIPTION 8.6. ESCITALOPRAM
Rp.: Escitaloprami 0.01
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet once a day (first week — half a tablet).

PRESCRIPTION 8.7. FLUOXETINE
Rp.: Capsulas Fluoxetini 0.02 N. 30
D.S. Take 1 capsule once a day in the morning.

PRESCRIPTION 8.8. VENLAFAXINE
Rp.: Capsulas Venlafaxini 0.075 prolongatas N. 28
D.S. Take 1 capsule once a day with food.

PRESCRIPTION 8.9. MIRTAZAPINE
Rp.: Mirtazapini 0.015
D.t.d. N. 30 in tabulettis obductis
S. Take 1 tablet in the evening at bedtime.

PRESCRIPTION 8.10. DULOXETINE
Rp.: Capsulas Duloxetini 0.06 N. 28
D.S. Take 1 capsule once a day (in diabetic neuropathy).

Review questions

1. Classification of CNS stimulants.
2. Mechanism of action and effects of caffeine. Dependence of the effect of caffeine on the state of the CNS.
3. Uses, adverse effects and contraindications of caffeine.
4. Mechanism, uses and risks of amphetamine and methylphenidate.
5. Why are beta-blockers contraindicated in cocaine poisoning?
6. The monoamine theory of depression. Why is the effect of antidepressants delayed?
7. Classification of antidepressants.
8. Mechanism, uses and adverse effects of TCAs.
9. Clinical picture and treatment of TCA poisoning. The role of sodium bicarbonate.
10. Advantages and adverse effects of SSRIs.
11. Causes, signs and treatment of serotonin syndrome. Difference from NMS.
12. MAO inhibitors and the “cheese reaction”.
13. Properties of mirtazapine, bupropion, trazodone and agomelatine.
14. Principles of antidepressant treatment.
15. What risk associated with antidepressants exists in young patients?

Test questions

1. The main mechanism of action of caffeine:
   A) Blocking sodium channels
   B) Inhibiting MAO
   C) Activating GABA-A receptors
   D) Blocking dopamine receptors
   E) Blocking adenosine receptors
2. Used to treat apnea in preterm infants:
   A) Caffeine citrate
   B) Bemegride
   C) Amphetamine
   D) Diazepam
   E) Methylphenidate
3. The main psychostimulant for treating attention deficit hyperactivity disorder:
   A) Caffeine
   B) Amitriptyline
   C) Mesocarb
   D) Methylphenidate
   E) Haloperidol

4. The main therapeutic agents in cocaine poisoning:
   A) Nonselective beta-blockers
   B) Benzodiazepines
   C) Flumazenil
   D) Neuroleptics
   E) Physostigmine

5. A tricyclic antidepressant with a strong sedative effect:
   A) Moclobemide
   B) Fluoxetine
   C) Amitriptyline
   D) Bupropion
   E) Sertraline

6. Given in TCA poisoning when the QRS widens:
   A) Calcium chloride
   B) Flumazenil
   C) Sodium bicarbonate
   D) Potassium chloride
   E) Physostigmine

7. The group of first-line antidepressants in depression:
   A) SSRIs
   B) Irreversible MAO inhibitors
   C) TCAs
   D) Psychostimulants
   E) Neuroleptics

8. The SSRI with the longest half-life:
   A) Fluoxetine
   B) Escitalopram
   C) Fluvoxamine
   D) Sertraline
   E) Paroxetine

9. A sign typical of serotonin syndrome:
   A) Hyporeflexia
   B) Hypothermia
   C) Miosis
   D) Clonus and hyperreflexia
   E) Constipation

10. The agent that most increases the risk of serotonin syndrome when given with an SSRI:
   A) Omeprazole
   B) Metformin
   C) Amoxicillin
   D) An MAO inhibitor
   E) Paracetamol

11. Develops when a patient taking MAO inhibitors eats aged cheese:
    A) Hypothermia
    B) Bradycardia
    C) Hypoglycemia
    D) Hypertensive crisis
    E) Hemolysis

12. An antidepressant convenient in depression with insomnia and weight loss:
    A) Fluoxetine
    B) Moclobemide
    C) Venlafaxine
    D) Mirtazapine
    E) Bupropion

13. An antidepressant that does not impair sexual function and is also used for smoking cessation:
    A) Bupropion
    B) Sertraline
    C) Amitriptyline
    D) Clomipramine
    E) Paroxetine

14. When does the clinical effect of antidepressants usually begin?
    A) After the first dose
    B) After 3 months
    C) In 2–4 weeks
    D) In 1–2 days
    E) After 6 months

15. An SNRI also used for the pain of diabetic neuropathy:
    A) Agomelatine
    B) Fluoxetine
    C) Trazodone
    D) Duloxetine
    E) Mirtazapine

Clinical cases

Case 1

A 19-year-old female student was brought in by her relatives: 2 hours ago she took 30 of her mother's amitriptyline tablets. The patient is drowsy, her speech is slurred, the pupils are dilated, the skin is dry and hot, HR 135/min, BP 90/55 mm Hg. The ECG shows QRS 130 ms. She had one seizure during transport.

Questions: 1) Diagnosis? 2) Mechanism of the signs? 3) Treatment.

Answer: 1) Acute poisoning with a tricyclic antidepressant (amitriptyline). 2) Mydriasis, dry hot skin, tachycardia — anticholinergic effect; hypotension — alpha-1 blockade; QRS widening — sodium channel blockade (cardiotoxicity); seizures and drowsiness — CNS effect. 3) Secure the airway (intubation if necessary), activated charcoal; sodium bicarbonate 8.4% 1–2 mmol/kg intravenously (until the QRS narrows, blood pH 7.45–7.55); for seizures — benzodiazepines; for

hypotension — fluids and norepinephrine; ECG monitoring; physostigmine and flumazenil are contraindicated. After recovery — psychiatric consultation (suicide attempt).

Case 2

A 45-year-old woman has taken sertraline for depression for 3 months. An orthopedic trauma surgeon prescribed tramadol for low back pain. Two days later the patient developed restlessness, profuse sweating, tremor, diarrhea, a temperature of 38.8 °C, HR 118/min, and clonus and hyperreflexia in the legs.

Questions: 1) Your diagnosis? 2) The cause? 3) Treatment.

Answer: 1) Serotonin syndrome. 2) Sertraline (an SSRI) and tramadol (an opioid + inhibits serotonin and norepinephrine reuptake) — a combination of two serotonergic agents. 3) Stop both drugs, benzodiazepines (for agitation and muscle hyperactivity), cooling, fluids; cyproheptadine for marked signs; in severe hyperthermia — intensive care. Later, choose a non-serotonergic analgesic for the pain (paracetamol, NSAIDs).

Case 3

A 70-year-old man (history of prostatic adenoma, angle-closure glaucoma, ischemic heart disease) presented with signs of depression: insomnia, loss of appetite, a weight loss of 6 kg in 3 months.

Questions: Which antidepressant would you choose, and which should be avoided?

Answer: Mirtazapine (improves sleep, increases appetite, has no anticholinergic effect, relatively safe for the heart) or sertraline. TCAs (amitriptyline) are absolutely not recommended: the anticholinergic effect causes urinary retention in prostatic adenoma, an attack of glaucoma and delirium in the elderly; sodium channel blockade — arrhythmia in IHD; alpha-1 blockade — falls. Paroxetine is also inappropriate because of its anticholinergic effect. In an elderly patient, the sodium level should be monitored (SSRIs — hyponatremia).

Quick review table

Table 8.9. *Psychostimulants and antidepressants — summary*

Group	Preparations	Mechanism	Use	Main risk
Methylxanthines	Caffeine	Adenosine receptor blockade	Fatigue, hypotension, migraine (in combination), apnea in newborns	Insomnia, tachycardia
Phenylalkylamines	Amphetamine, methylphenidate	Release / reuptake of DA and NE	ADHD, narcolepsy	Dependence, psychosis
TCAs	Amitriptyline, imipramine, clomipramine	Reuptake of NE and 5-HT + M, H_1, alpha-1, Na_+	Depression, chronic pain, enuresis	Poisoning: coma, seizures, wide QRS — sodium bicarbonate
SSRIs	Sertraline, escitalopram, fluoxetine, paroxetine	5-HT reuptake	Depression, anxiety, OCD	Serotonin syndrome, sexual dysfunction, bleeding
SNRIs	Venlafaxine, duloxetine	5-HT and NE	Depression, neuropathic pain	BP ↑, withdrawal syndrome

Group	Preparations	Mechanism	Use	Main risk
MAO inhibitors	Phenelzine, moclobemide	Inhibition of MAO	Resistant depression	“Cheese reaction”, serotonin syndrome
Atypical agents	Mirtazapine, bupropion, trazodone, agomelatine	Various	Individual choice	Mirtazapine — weight gain; bupropion — seizures

TOPIC 9

Drugs acting on the respiratory system

Learning objectives

Cough, sputum and bronchial obstruction are among the most common complaints at the GP's appointment. Bronchial asthma and chronic obstructive pulmonary disease (COPD) affect millions of people, and correctly chosen inhalation therapy radically changes their quality of life. At the same time, errors such as using antitussives together with expectorants, or treating asthma only with a “reliever”, are widespread. This topic covers respiratory pharmacology on the basis of modern clinical guidelines.

The student should know:

- the classification of drugs acting on the respiratory system;
- respiratory stimulants (analeptics) and their current place;
- the mechanism and uses of antitussive (central and peripheral), expectorant and mucolytic agents;
- bronchodilators (beta-2 adrenomimetics, M-cholinergic blockers, methylxanthines) and anti-inflammatory agents used in bronchial asthma;
- the principles of stepwise treatment of bronchial asthma and COPD, and the treatment of pulmonary edema.

The student should be able to:

- write prescriptions for preparations such as salbutamol, budesonide/formoterol, tiotropium, aminophylline (Euphylline), acetylcysteine, ambroxol, codeine, butamirate and montelukast;
- provide emergency care in an attack of bronchial asthma;
- choose a drug according to the type of cough.

1. Classification

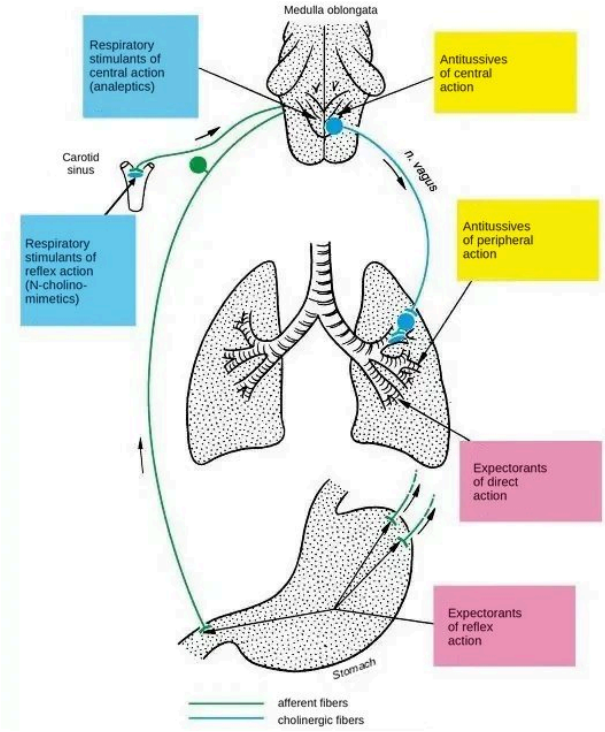


Figure 9.1. *Sites of action of drugs acting on the respiratory system: central and reflex respiratory stimulants, central and peripheral antitussives, direct and reflex expectorants (teaching material of the TMA Department of Pharmacology)*

Table 9.1. *Classification of drugs acting on the respiratory system*

Group	Subgroup	Preparations
Respiratory stimulants	Centrally acting analeptics	Doxapram, nikethamide (Cordiamine), bemegride, caffeine, camphor; carbogen (CO_2 + O_2)
	Reflex-acting	Lobeline, cytisine (historically)
Antitussives	Central — opioid	Codeine, ethylmorphine, dextromethorphan
	Central — non-opioid	Butamirate, glaucine, oxeladin
	Peripheral	Prenoxdiazine (Libexin), levodropropizine
Expectorants	Reflex-acting	Thermopsis, polygala, marshmallow (Althaea), licorice root, ivy leaf extract
	Directly (resorptively) acting	Potassium iodide, sodium bicarbonate, ammonium chloride, essential oils
Mucolytics	—	Acetylcysteine, carbocisteine, bromhexine, ambroxol, dornase alfa (in cystic fibrosis)
Bronchodilators	Beta-2 adrenomimetics	Salbutamol, fenoterol, terbutaline; salmeterol, formoterol, indacaterol, vilanterol, olodaterol
	M-cholinergic blockers	Ipratropium; tiotropium, glycopyrronium, umeclidinium, aclidinium
	Methylxanthines	Theophylline, aminophylline (Euphylline)
Anti-inflammatory and antiallergic agents used in bronchial asthma	Inhaled glucocorticoids (ICS)	Budesonide, fluticasone, beclometasone, mometasone, ciclesonide

Group	Subgroup	Preparations
	Leukotriene receptor antagonists	Montelukast, zafirlukast
	Mast cell membrane stabilizers	Sodium cromoglicate, nedocromil
	Biologics (monoclonal antibodies)	Omalizumab (anti-IgE), mepolizumab, benralizumab (anti-IL-5), dupilumab (anti-IL-4R), tezepelumab
Drugs used in pulmonary edema	—	Oxygen, furosemide, nitroglycerin, morphine; antifoaming agents (ethyl alcohol vapor, Antifomsilan)
Surfactants	—	Poractant alfa, beractant (respiratory distress syndrome of newborns)

2. Respiratory stimulants (analeptics)

Analeptics stimulate the respiratory center in the medulla oblongata directly (bemegride, nikethamide, caffeine, doxapram) or reflexly through the chemoreceptors of the carotid bodies (lobeline, cytisine; doxapram as well).

Carbogen (a mixture of 5–7% CO_2 and 93–95% O_2) — a physiological stimulant of the respiratory center.

> **CAUTION**
> Older textbooks recommended analeptics for restoring breathing in poisoning with anesthetics, hypnotics and alcohol. In modern practice, respiratory failure is treated by **securing the airway, oxygen and mechanical ventilation**. The effect of analeptics is short and their therapeutic range is narrow; in a severely depressed CNS they are ineffective, and at large doses they cause **seizures** and increase oxygen demand. Doxapram is used by specialists in some cases (respiratory depression after anesthesia, hypercapnia in COPD); the specific antidote in opioid poisoning is **naloxone**.

3. Antitussives

Cough is a protective reflex that clears the airways. Therefore suppressing cough is appropriate only for a **dry, unproductive, troublesome** cough (when the cough disturbs sleep and quality of life).

Table 9.2. *Antitussives*

Preparation	Mechanism	Features
Codeine	Inhibits the cough center (a mu-opioid agonist; converted to morphine by CYP2D6)	Strong; constipation, drowsiness, respiratory depression, dependence; contraindicated in children under 12 (respiratory arrest in CYP2D6 ultrarapid metabolizers)
Dextromethorphan	Central (NMDA antagonist, sigma receptors); no analgesic effect	Little dependence, but hallucinogenic at high doses; serotonin syndrome with SSRIs and MAOIs
Butamirate (Sinecod)	Central non-opioid; bronchodilator and anti-inflammatory effect	Does not depress breathing, no dependence; used in children
Glaucine	Central non-opioid; weak alpha-blocker	May cause hypotension

Preparation	Mechanism	Features
Oxeladin	Central non-opioid	—
Prenoxdiazine (Libexin)	Peripheral — anesthetizes the sensory receptors of the airways, weak bronchodilator	The tablet must not be chewed (numbness of the mouth)
Levodropropizine	Peripheral	Convenient in children

CAUTION

Antitussives and expectorants (mucolytics) must not be given together: the mucolytic thins the sputum and increases its volume, while the antitussive suppresses the reflex that expels it — sputum accumulates in the bronchi (**"bogging"**), and the risk of pneumonia and obstruction increases.

4. Expectorants and mucolytics

4.1. Expectorants

Reflex-acting agents irritate receptors in the gastric mucosa and, through the vagus nerve, reflexly enhance bronchial gland secretion and the activity of the ciliated epithelium (gastropulmonary reflex). At large doses they cause **vomiting** (proximity to the vomiting center). Preparations: thermopsis herb, polygala root, marshmallow root (also provides mucilage), licorice root, ivy leaf extract, terpin hydrate.

Directly (resorptively) acting agents, after absorption, are excreted through the bronchial glands, increase secretion and thin the sputum: potassium and sodium iodide (contraindicated in thyrotoxicosis and iodine allergy), sodium bicarbonate (inhalation), ammonium chloride.

4.2. Mucolytics

Table 9.3. *Mucolytics*

Preparation	Mechanism	Features
Acetylcysteine	Free SH groups break the disulfide bonds of mucoproteins; a precursor of glutathione	Oral, inhalation, injection; antidote for paracetamol poisoning; may cause bronchospasm (by inhalation)
Carbocisteine	Normalizes the composition of sputum (mucoregulator)	Caution in gastric ulcer
Bromhexine	Increases serous gland secretion, depolymerizes mucopolysaccharides; converted to ambroxol in the body	—
Ambroxol	The active metabolite of bromhexine; increases surfactant synthesis, improves mucociliary clearance	Oral, inhalation, intravenous
Dornase alfa	Recombinant DNase; breaks down neutrophil DNA in purulent sputum	Cystic fibrosis (inhalation)
Hypertonic sodium chloride (3–7%)	Osmotic effect	Cystic fibrosis, bronchiolitis (nebulizer)

CLINICAL SIGNIFICANCE

In cough due to acute bronchitis and viral infections, the effectiveness of mucolytics and expectorants is limited; **adequate fluid intake** and air humidity are important. In children under 2 years, mucolytics may increase sputum accumulation in the bronchi — they are used with caution.

5. Bronchodilators

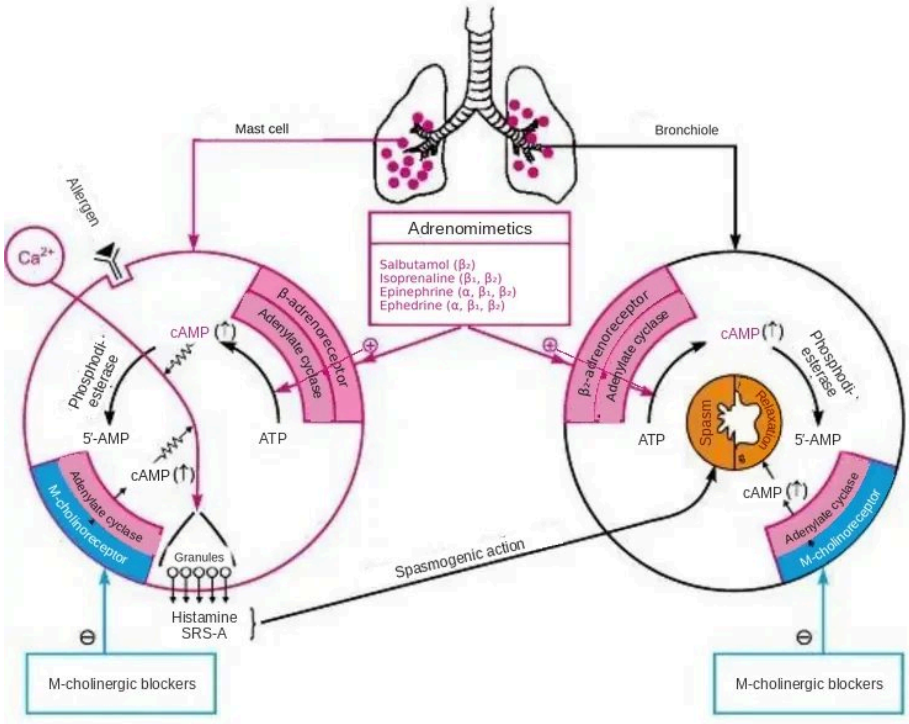


Figure 9.2. *Mechanism of action of bronchodilators: beta-2 adrenomimetics increase cAMP, M-cholinergic blockers reduce cGMP formation and the spasmogenic effect (teaching material of the TMA Department of Pharmacology)*

5.1. Beta-2 adrenomimetics

They stimulate beta-2 receptors in bronchial smooth muscle: G_s → adenylate cyclase → **cAMP** ↑ → relaxation of the muscle through protein kinase A; in addition, they reduce mediator release from mast cells and improve mucociliary clearance.

Table 9.4. *Beta-2 adrenomimetics*

Group	Preparations	Onset / duration of action	Role
Short-acting (SABA)	Salbutamol, fenoterol, terbutaline	3–5 minutes / 4–6 hours	Relief of attacks, prevention before exercise
Long-acting (LABA)	Salmeterol	20–30 minutes / 12 hours	In maintenance therapy — only together with an ICS
	Formoterol	3–5 minutes / 12 hours	Maintenance therapy + relief of attacks (together with budesonide)
Ultra-long-acting	Indacaterol, olodaterol, vilanterol	5 minutes / 24 hours	COPD, asthma (with an ICS)

CAUTION

In bronchial asthma, using a LABA as monotherapy without an ICS is forbidden — the risk of death from asthma increases (the inflammation is not treated and the symptoms are masked). In addition, current GINA recommendations (since 2019) **do not recommend treating asthma with a SABA (salbutamol) alone** in adults and adolescents; every patient should receive ICS-containing treatment.

5.2. M-cholinergic blockers

Ipratropium (short-acting, 6–8 hours) and tiotropium, glycopyrronium and umeclidinium (long-acting, 24 hours) block M_3 receptors in the bronchi and reduce the bronchoconstrictor tone of the vagus nerve. In COPD, parasympathetic tone is the main reversible component of bronchial obstruction, so **long-acting M-cholinergic blockers (LAMAs) are the main bronchodilators in COPD**. In asthma — an add-on drug in severe cases. Adverse effects: dry mouth, rarely urinary retention, glaucoma (if it gets into the eye through a nebulizer mask).

5.3. Methylxanthines

Theophylline and **aminophylline** (**Euphylline**; theophylline + ethylenediamine, readily soluble in water): inhibit phosphodiesterase (cAMP ↑), block adenosine receptors and improve diaphragmatic contractility.

CAUTION

Theophylline has a **narrow therapeutic range** (effective blood concentration 5–15 mg/l, above 20 mg/l — toxic): nausea, vomiting, tachycardia, **arrhythmias**, **seizures**. It is metabolized by CYP1A2: **ciprofloxacin, clarithromycin and fluvoxamine** increase its concentration; smoking and rifampicin decrease it. Aminophylline (Euphylline) must be given intravenously **very slowly** (over at least 10–20 minutes). In modern recommendations, methylxanthines are only reserve drugs in asthma and COPD.

6. Anti-inflammatory agents used in bronchial asthma

6.1. Inhaled glucocorticoids (ICS)

ICS are the most effective maintenance therapy for bronchial asthma. They reduce allergic inflammation in the bronchial wall (eosinophils, mast cells, T lymphocytes, cytokines), reduce bronchial hyperreactivity and restore the sensitivity of beta-2 receptors. Their effect develops gradually (days to weeks) — **they do not stop an attack**.

Preparations: budesonide, fluticasone propionate and furoate, beclometasone dipropionate, mometasone, ciclesonide. Combinations: **budesonide/formoterol**, beclometasone/formoterol, fluticasone/salmeterol, fluticasone furoate/vilanterol.

Adverse effects: oral and pharyngeal candidiasis, hoarseness (dysphonia) — prevention: rinsing the mouth after inhalation, a spacer; with prolonged use at high doses — systemic effects (slight growth retardation in children, osteoporosis, cataract), risk of pneumonia in COPD.

6.2. Other anti-inflammatory agents

- **Montelukast** — a cysteinyl leukotriene (CysLT1) receptor antagonist; one tablet a day. Aspirin-induced asthma, exercise-induced asthma, asthma with allergic rhinitis. Adverse effects: neuropsychiatric disorders (sleep, mood, suicidal thoughts) — warning.
- **Cromones** (sodium cromoglicate, nedocromil) — stabilize the mast cell membrane and prevent degranulation; prophylactic only; rarely used now.

- **Systemic glucocorticoids** (oral prednisolone, intravenous methylprednisolone) — a short course (5–7 days) in severe attacks.
- **Biologics** — in severe uncontrolled asthma: omalizumab (allergic), mepolizumab, benralizumab (eosinophilic), dupilumab, tezepelumab.

7. Principles of treating bronchial asthma and COPD

7.1. Bronchial asthma (based on GINA)

Table 9.5. *Stepwise treatment of bronchial asthma (adults and adolescents over 12, GINA, preferred track)*

Step	Maintenance therapy	For attacks
1–2	Low-dose ICS-formoterol as needed (budesonide/formoterol)	The same inhaler
3	Regular low-dose ICS-formoterol (MART — one inhaler for maintenance and relief)	The same inhaler
4	Medium-dose ICS-formoterol (MART)	The same inhaler
5	+ a long-acting M-cholinergic blocker, a biologic according to phenotype, high-dose ICS-formoterol	The same inhaler

REMEMBER

The main rule of asthma control: “Every asthmatic should receive an ICS”. Budesonide/formoterol in a single inhaler serves both as maintenance anti-inflammatory therapy and as a fast-acting bronchodilator — this reduces severe attacks and hospitalizations. Alternative approach: ICS (or ICS-LABA) maintenance therapy + salbutamol for attacks.

CAUTION

Treatment approaches are constantly being updated: older textbooks recommended starting asthma treatment with salbutamol and using aminophylline and cromones as maintenance drugs. Rely on the current GINA reports (updated every year) and the national clinical protocols of Uzbekistan.

7.2. Emergency care in an attack of bronchial asthma

1. **Salbutamol** 2.5–5 mg via a nebulizer or 4–10 puffs (100 mcg) through a **spacer**, every 20 minutes in the first hour.
2. **Oxygen** (maintain SpO_2 at 93–95%).
3. In a severe attack — **ipratropium** (together with salbutamol in a nebulizer).
4. **Systemic glucocorticoid** — prednisolone 40–50 mg orally (or methylprednisolone/hydrocortisone intravenously), for 5–7 days.
5. In a life-threatening attack — **magnesium sulfate** intravenously (2 g over 20 minutes), intensive care, intubation and mechanical ventilation.
6. **Sedatives are contraindicated** (they depress the respiratory center); beta-blockers are contraindicated.

7.3. COPD

In COPD, the main bronchodilators are **LAMAs (tiotropium)** and **LABAs**, often a **LAMA + LABA** combination. An ICS is added only in frequently exacerbating COPD with high blood eosinophils.

Stopping smoking is the only effective measure that slows the progression of the disease. In exacerbations — a SABA + a short-acting M-cholinergic blocker, a systemic glucocorticoid (5 days), and an antibiotic if there are signs of bacterial infection.

8. Drugs used in pulmonary edema

Cardiogenic pulmonary edema (left ventricular failure) is an emergency.

Table 9.6. *Management of pulmonary edema*

Measure	Mechanism
Sit the patient up (with the legs lowered)	Reduces venous return
Oxygen; if necessary, noninvasive ventilation (CPAP)	Correction of hypoxemia, reduction of fluid leakage into the alveoli
Nitroglycerin (sublingual, intravenous) — if BP > 110 mm Hg	Dilates the veins and reduces the load on the heart
Furosemide 40–80 mg intravenously	Venodilation (immediately) and diuresis
Morphine (in a small dose, with caution)	Reduces anxiety and dyspnea, venodilation; but depresses breathing — routine use is now not recommended
In hypotension — inotropes (dobutamine), vasopressors	Increase cardiac output
Antifoaming agents (oxygen through 30–40% ethyl alcohol vapor, Antifomsilan)	Reduce the surface tension of the foam (mainly a historical method)

In pulmonary edema against a background of arterial hypertension, rapid lowering of BP is also used (ganglionic blockers — historically; now nitrates).

9. Clinical part

Table 9.7. *Choosing a drug according to the type of cough*

Type of cough	Choice	Comment
Dry, troublesome (viral infection)	Butamirate, dextromethorphan; for a short time	Stop if sputum appears
Thick sputum that is hard to expectorate	Acetylcysteine, ambroxol, carbocisteine; fluids	Not together with antitussives
Cough with bronchial obstruction	Bronchodilators (salbutamol), ICS in asthma	—
Prolonged cough (more than 8 weeks)	Determine the cause: asthma, GERD, postnasal drip, ACE inhibitors, tuberculosis, tumor	ACE inhibitor cough — change the drug
Opioid antitussives	Adults only, for a short time	Codeine is contraindicated in children

CLINICAL SIGNIFICANCE

ACE inhibitors (enalapril, lisinopril) cause a **dry cough** in 5–20% of patients (accumulation of bradykinin). Such a cough is not treated with antitussives — the ACE inhibitor is replaced with an angiotensin receptor blocker (losartan).

Practical skills

Skill 1. Teaching the use of a metered-dose inhaler with a spacer

1. Shake the inhaler and remove the cap.
2. Attach the inhaler to the spacer.
3. Breathe out fully.
4. Seal the lips tightly around the spacer mouthpiece (a mask in young children).
5. Press the canister once and take a slow, deep breath (in children — 5–6 normal breaths).
6. Hold your breath for 5–10 seconds.
7. If another dose is needed, repeat after 30–60 seconds.
8. After an ICS, rinse your mouth with water.

Skill 2. Key prescriptions

PRESCRIPTION 9.1. SALBUTAMOL (FOR A NEBULIZER)
Rp.: Solutionis Salbutamoli 0.1% — 2.5 ml
D.t.d. N. 20
S. Inhale 2.5 ml through a nebulizer (in an attack, up to 3 times every 20 minutes in the first hour).

PRESCRIPTION 9.2. BUDESONIDE/FORMOTEROL (DRY POWDER INHALER)
Rp.: Pulveris pro inhalationibus “Budesonidum + Formoterolum” 160/4.5 mcg/dosi — 120 dosi
D.S. 1 inhalation 2 times a day and 1 additional inhalation when symptoms appear (up to 12 inhalations a day); then rinse the mouth.

PRESCRIPTION 9.3. FLUTICASONE (AEROSOL)
Rp.: Aerosoli Fluticasoni propionatis 0.000125/dosi — 120 dosi
D.S. 2 inhalations 2 times a day through a spacer; then rinse the mouth.

PRESCRIPTION 9.4. BUDESONIDE (SUSPENSION FOR A NEBULIZER)
Rp.: Suspensionis Budesonidi pro inhalationibus 0.5 mg/ml — 2 ml
D.t.d. N. 20
S. 2 ml 2 times a day through a nebulizer (maintenance therapy in children).

PRESCRIPTION 9.5. MONTELUKAST
Rp.: Montelukasti 0.01
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet once a day in the evening.

PRESCRIPTION 9.6. AMINOPHYLLINE (EUPHYLLINE)
Rp.: Solutionis Aminophyllini 2.4% — 10 ml
D.t.d. N. 10 in ampullis
S. Dilute 10 ml in 10 ml of isotonic solution and inject intravenously very slowly (over 10–20 minutes).

PRESCRIPTION 9.7. PREDNISOLONE (IN AN ASTHMA ATTACK)
Rp.: Prednisoloni 0.005
D.t.d. N. 50 in tabulettis
S. Take 8–10 tablets (40–50 mg) once a day in the morning for 5 days.

PRESCRIPTION 9.8. ACETYLCYSTEINE
Rp.: Tabulettas effervescentes Acetylcysteini 0.6 N. 10
D.S. Dissolve 1 tablet in 1 glass of water and take once a day.

PRESCRIPTION 9.9. AMBROXOL
Rp.: Ambroxoli hydrochloridi 0.03
D.t.d. N. 20 in tabulettis
S. Take 1 tablet 3 times a day after meals.

PRESCRIPTION 9.10. BUTAMIRATE
Rp.: Sirupi Butamirati 0.15% — 200 ml
D.S. Take 15 ml 3 times a day before meals (for a dry cough).

PRESCRIPTION 9.11. PRENOXDIAZINE
Rp.: Prenoxdiazini 0.1
D.t.d. N. 20 in tabulettis
S. Swallow 1 tablet without chewing 3 times a day.

PRESCRIPTION 9.12. INFUSION OF THERMOPSIS HERB
Rp.: Infusi herbae Thermopsidis 0.6 — 180 ml
D.S. Take 1 tablespoon 3 times a day.

Review questions

1. Classification of drugs acting on the respiratory system.
2. Mechanism of analeptics and their place in modern practice.
3. Classification of antitussives. The difference between codeine and butamirate.
4. Why must antitussives and mucolytics not be given together?
5. Mechanism of reflex-acting and directly acting expectorants.
6. Mucolytics: acetylcysteine, ambroxol, bromhexine, dornase alfa.
7. Classification of beta-2 adrenomimetics. Why must a LABA not be given without an ICS in asthma?
8. The place of M-cholinergic blockers in COPD.
9. Mechanism, risks and interactions of theophylline and aminophylline (Euphylline).
10. Effects and adverse effects of inhaled glucocorticoids. Their prevention.
11. The place of montelukast, cromones and biologics.
12. Stepwise treatment of asthma according to GINA. The concept of MART.
13. Algorithm of emergency care in an asthma attack.
14. Principles of COPD treatment.
15. Measures of care in cardiogenic pulmonary edema.

Test questions

1. A peripherally acting antitussive:
   A) Prenoxdiazine
   B) Glaucine
   C) Dextromethorphan
   D) Butamirate
   E) Codeine

2. An antitussive contraindicated in children under 12:
   A) Codeine
   B) Levodropropizine
   C) Ambroxol
   D) Butamirate
   E) Prenoxdiazine

3. A drug that breaks the disulfide bonds of mucoproteins and is used as an antidote in paracetamol poisoning:
   A) Acetylcysteine
   B) Ambroxol
   C) Bromhexine
   D) Thermopsis
   E) Potassium iodide

4. A mucolytic that increases surfactant synthesis:
   A) Codeine
   B) Acetylcysteine
   C) Ambroxol
   D) Dornase alfa
   E) Carbocisteine

5. The fastest-acting drug for stopping an attack of bronchial asthma:
   A) Salbutamol
   B) Budesonide
   C) Salmeterol
   D) Montelukast
   E) Sodium cromoglicate

6. The most effective maintenance (anti-inflammatory) treatment of bronchial asthma:
   A) Antihistamines
   B) Short-acting beta-2 agonists
   C) Antibiotics
   D) Inhaled glucocorticoids
   E) Mucolytics

7. A common local adverse effect of inhaled glucocorticoids:
   A) Hypokalemia
   B) Tachycardia
   C) Oral candidiasis
   D) Hypoglycemia
   E) Bronchospasm

8. The main bronchodilator in COPD:
    A) Prednisolone
    B) Montelukast
    C) Cromoglicate
    D) Tiotropium
    E) Theophylline

9. An antibiotic that increases theophylline concentration and the risk of toxicity:
    A) Metronidazole
    B) Ceftriaxone
    C) Doxycycline
    D) Amoxicillin
    E) Ciprofloxacin

10. A leukotriene receptor antagonist:
    A) Nedocromil
    B) Ipratropium
    C) Omalizumab
    D) Montelukast
    E) Fluticasone

11. Which treatment approach is considered dangerous in asthma?
    A) ICS + LAMA
    B) ICS + montelukast
    C) Use of a LABA as monotherapy without an ICS
    D) ICS monotherapy
    E) Budesonide/formoterol in the MART regimen

12. An additional drug given intravenously in a life-threatening asthma attack:
    A) Furosemide
    B) Propranolol
    C) Codeine
    D) Diazepam
    E) Magnesium sulfate

13. For venodilation in cardiogenic pulmonary edema when BP is normal or high:
    A) Aminophylline (Euphylline)
    B) Nitroglycerin
    C) Epinephrine
    D) Salbutamol
    E) Prednisolone

14. The correct approach to a dry cough caused by ACE inhibitors:
    A) Prescribe an antibiotic
    B) Replace the ACE inhibitor with an angiotensin receptor blocker
    C) Increase the dose
    D) Prescribe a mucolytic
    E) Prescribe codeine

15. An anti-IL-5 monoclonal antibody used in severe eosinophilic asthma:
    A) Infliximab
    B) Rituximab
    C) Mepolizumab
    D) Omalizumab
    E) Adalimumab

Clinical cases

Case 1

A 24-year-old woman has had bronchial asthma for 2 years. She uses only a salbutamol inhaler — in the last month 4–5 times a week, and she woke up choking at night twice. She used up 2 inhalers in one month.

Questions: 1) Is her asthma controlled? 2) What is the error in treatment? 3) Recommendations.

Answer: 1) Uncontrolled asthma (symptoms more than twice a week, night awakenings, frequent need for salbutamol). 2) Treatment with a SABA alone — the inflammation is not treated; heavy SABA use is associated with a risk of severe attacks and death. 3) Low-dose budesonide/formoterol in the MART regimen (one inhaler for maintenance and attacks) or ICS maintenance therapy; check inhaler technique, eliminate triggers, a written action plan; follow-up after 2–3 months.

Case 2

A 58-year-old man with COPD uses tiotropium once a day. Because of an exacerbation of bronchitis, the doctor prescribed ciprofloxacin; at home he has also been taking aminophylline (Euphylline) tablets for a long time. After 3 days he developed nausea, vomiting, palpitations and hand tremor.

Questions: 1) The cause of the symptoms? 2) Management?

Answer: 1) Theophylline toxicity: ciprofloxacin inhibited CYP1A2, slowed theophylline metabolism, and its blood concentration reached a toxic level (narrow therapeutic range). 2) Stop aminophylline, measure blood theophylline and potassium, ECG monitoring; in severe cases activated charcoal (repeated doses), treatment of arrhythmias and seizures. Replace the antibiotic with a drug that does not affect CYP1A2 (amoxicillin/clavulanate); optimize COPD maintenance therapy with a LAMA + LABA.

Case 3

A 3-year-old child has a dry cough during an ARVI. The mother bought a codeine cough syrup and an ambroxol syrup at the pharmacy at the same time.

Questions: Explain the mother's mistakes and give the correct advice.

Answer: 1) Codeine is not allowed in children under 12 — in CYP2D6 ultrarapid metabolizers a lot of morphine is formed and there is a risk of respiratory arrest. 2) Giving an antitussive and a mucolytic together — sputum accumulation (“swamping” of the bronchi). 3) Advice: plenty of warm fluids, humidifying the room air, rinsing the nasal passages with saline; for a dry troublesome cough — an age-appropriate non-opioid agent (butamirate drops) on a pediatrician's advice; mucolytics with caution in children under 2; if there is shortness of breath or prolonged fever — a pediatrician's examination.

Quick review table

Table 9.8. *Drugs acting on the respiratory system — summary*

Group	Preparations	Mechanism	Key note
Analeptics	Doxapram, nikethamide, bemegride, caffeine	Stimulate the respiratory center	Oxygen and artificial ventilation are preferred in modern practice
Central antitussives	Codeine, dextromethorphan, butamirate, glaucine	Inhibit the cough center	Codeine — not for children
Peripheral antitussives	Prenoxdiazine, levodropropizine	Reduce receptor sensitivity	Not together with a mucolytic
Expectorants	Thermopsis, marshmallow (Althaea), potassium iodide	Reflex / direct secretion ↑	Vomiting at large doses
Mucolytics	Acetylcysteine, ambroxol, bromhexine, dornase alfa	Thin the sputum	Acetylcysteine — paracetamol antidote
SABA	Salbutamol, fenoterol	Beta-2 → cAMP ↑	For attacks; monotherapy not recommended
LABA	Salmeterol, formoterol, indacaterol	Beta-2 (long-acting)	In asthma only with an ICS
M-cholinergic blockers	Ipratropium, tiotropium	M_3 blockade	The main bronchodilator in COPD
Methylxanthines	Theophylline, aminophylline (Euphylline)	PDE ↓, adenosine blockade	Narrow TI, CYP1A2 interactions
ICS	Budesonide, fluticasone, beclometasone	Anti-inflammatory	The main maintenance therapy of asthma; rinse the mouth
Antileukotrienes	Montelukast	$CysLT_1$ blockade	Aspirin-induced asthma, rhinitis
Biologics	Omalizumab, mepolizumab, dupilumab	IgE, IL-5, IL-4R	Severe asthma

PART II. PRACTICAL CLASSES OF THE 6TH SEMESTER

Drugs acting on the cardiovascular, digestive, urinary and blood systems. Hormones. Anti-inflammatory and chemotherapeutic agents

TOPIC 10

Antiarrhythmic drugs. Antianginal drugs

Learning objectives

Cardiac arrhythmias and ischemic heart disease (IHD) are the main forms of cardiovascular disease, which ranks first among causes of death. Atrial fibrillation, supraventricular tachycardia, an attack of angina and acute coronary syndrome are conditions that GPs and emergency physicians face every day. Paradoxically, antiarrhythmic drugs can themselves cause arrhythmias (proarrhythmic effect), so choosing them requires a deep knowledge of their mechanisms. In line with the 6th-semester laboratory class, this topic also covers **cardiotonic drugs (cardiac glycosides)**.

The student should know:

- the phases of the action potential of cardiac cells and the mechanisms of arrhythmias;
- the classification of antiarrhythmic drugs (Vaughan Williams), the mechanism, uses and adverse effects of each class;
- the pathogenesis of angina and the mechanism of antianginal drugs (nitrates, beta-blockers, calcium antagonists, others);
- the mechanism and uses of cardiac glycosides and non-glycoside cardiotonics, and glycoside toxicity.

The student should be able to:

- write prescriptions for drugs such as amiodarone, verapamil, adenosine, lidocaine, nitroglycerin, isosorbide mononitrate, amlodipine, ivabradine and digoxin;
- provide emergency care in an attack of angina and in supraventricular tachycardia;
- recognize glycoside toxicity and draw up a plan of care.

1. Electrophysiology of the heart

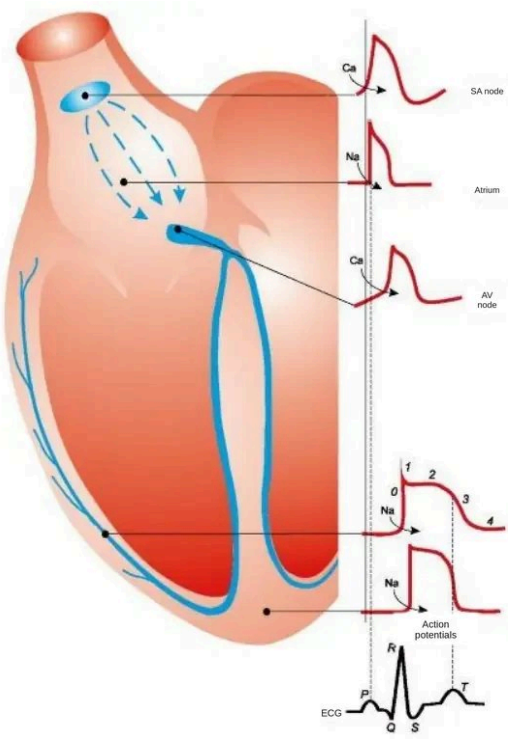


Figure 10.1. *Action potentials in different parts of the heart and their relation to the ECG (teaching material of the TMA Department of Pharmacology)*

Table 10.1. *Phases of the action potential of a ventricular cardiomyocyte*

Phase	Process	Ion current	Antiarrhythmics acting here
0 — rapid depolarization	The membrane depolarizes rapidly	Na^+ influx (fast sodium channels)	Class I
1 — early repolarization	Brief repolarization	K^+ efflux	—
2 — plateau	Depolarization is maintained	Balance of Ca^{2+} influx (L-type) and K^+ efflux	Class IV (partly)
3 — rapid repolarization	The resting potential is restored	K^+ efflux	Class III
4 — rest (diastole)	Stable in working myocardium; spontaneous slow diastolic depolarization in the sinus and AV nodes (automaticity)	If (“funny”) current, Ca^{2+}	Classes II and IV, ivabradine

In the sinus and AV nodes, phase 0 depends on the **calcium** current — that is why beta-blockers and calcium antagonists slow conduction in them.

Mechanisms of arrhythmias:

1. **Disturbed impulse formation** — enhanced automaticity (ischemia, catecholamines, hypokalemia) and **triggered activity** (early and delayed afterdepolarizations; glycoside toxicity, long QT).
2. **Disturbed impulse conduction** — **re-entry**: with a unidirectional block and slow conduction, the impulse circulates and re-excites the tissue. The main mechanism of most tachyarrhythmias.

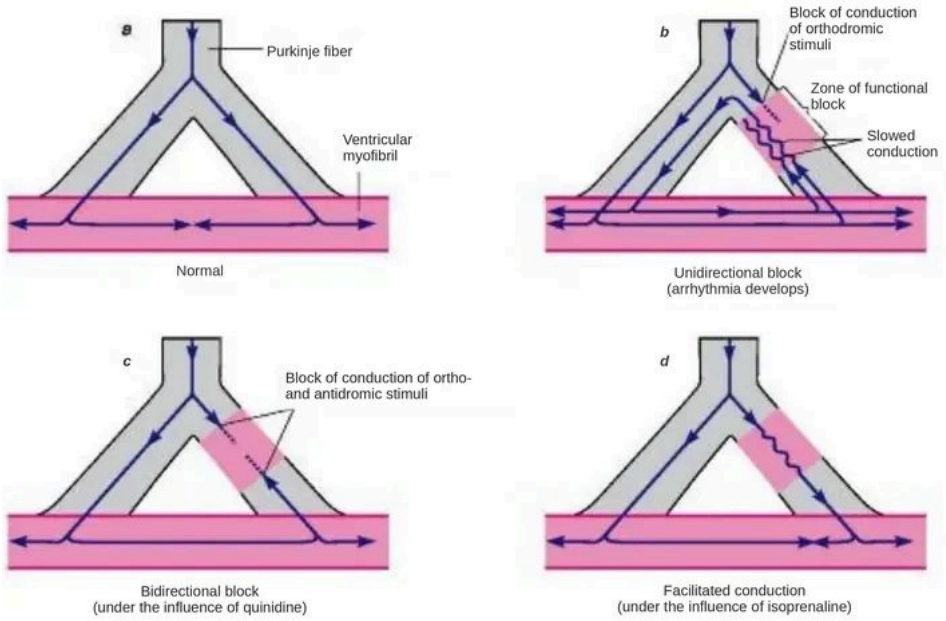


Figure 10.2. *The re-entry mechanism: a — normal; b — unidirectional block and arrhythmia; c — bidirectional block (antiarrhythmic effect); d — facilitated conduction (teaching material of the TMA Department of Pharmacology)*

REMEMBER

Antiarrhythmics stop re-entry in two ways: **converting a unidirectional block into a bidirectional one** (slowing conduction even further — class I) or **prolonging the effective refractory period** (when the impulse returns, the tissue is still unexcitable — class III).

2. Antiarrhythmic drugs

2.1. Classification (Vaughan Williams)

Table 10.2. *Classification of antiarrhythmic drugs*

Class	Mechanism	Preparations	ECG changes
IA	Na^+ channel blockade (moderate) + K^+ channel blockade; the refractory period is prolonged	Quinidine, procainamide (Novocainamide), disopyramide, ajmaline	QRS and QT are prolonged
IB	Na^+ channel blockade (weak, rapid dissociation); the refractory period is shortened; mainly on ischemic ventricular myocardium	Lidocaine, mexiletine, phenytoin	Unchanged or QT shortened
IC	Na^+ channel blockade (strong, slow dissociation)	Flecainide, propafenone, ethacizine, allapinine	QRS is markedly widened
II	Beta-blockers: reduce sympathetic influence, slow phase 4	Propranolol, metoprolol, bisoprolol, esmolol	PR prolonged, bradycardia
III	K^+ channel blockade: repolarization and the refractory period are prolonged	Amiodarone, sotalol, dronedarone, ibutilide, dofetilide	QT prolonged
IV	L-type Ca^{2+} channel blockade (in the sinus and AV nodes)	Verapamil, diltiazem	PR prolonged

Class	Mechanism	Preparations	ECG changes
Others	Various	Adenosine (A_1 receptors, opens K^+ channels), digoxin (vagal activation), magnesium sulfate, atropine (in bradyarrhythmias), ivabradine (If)	—

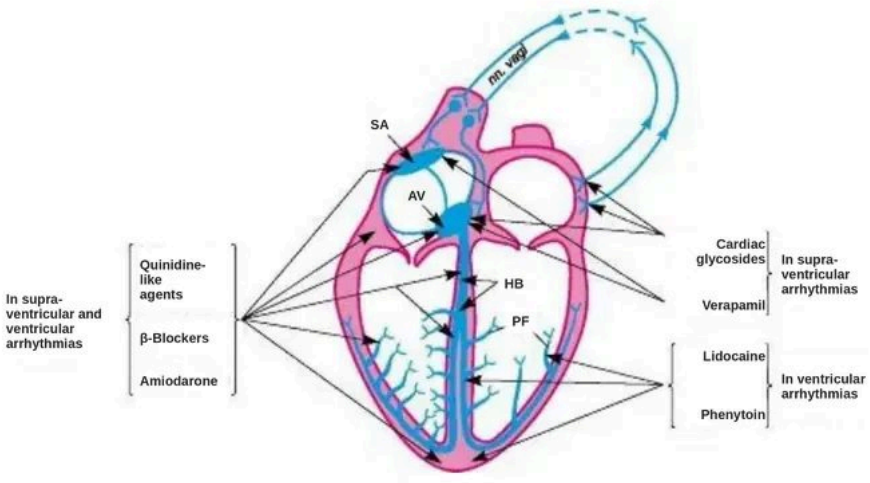


Figure 10.3. *Main sites of action of antiarrhythmic drugs in the heart: in supraventricular and ventricular arrhythmias (teaching material of the TMA Department of Pharmacology)*

MEMORY AID

Vaughan Williams classes — **"Sodium, Beta, Potassium, Calcium"** (1, 2, 3, 4). To remember IA, IB, IC: **"IA — Affects both QRS and QT (prolongs both); IB — Best in ischemia; IC — Causes marked QRS widening"**.

2.2. Class I — sodium channel blockers

Quinidine — an alkaloid of the cinchona tree (an isomer of quinine). The classic IA antiarrhythmic; now hardly used: M-cholinergic blocking action (it may speed up AV conduction and increase the ventricular rate in atrial flutter), **QT prolongation and torsades de pointes** (quinidine syncope), diarrhea, cinchonism (tinnitus, headache, visual disturbances), thrombocytopenia; it increases digoxin concentration.

Procainamide (Novocainamide) — given intravenously to restore sinus rhythm in atrial fibrillation and in ventricular tachycardia. Adverse effects: hypotension, QT prolongation, with prolonged use a **lupus-like syndrome** (especially in slow acetylators), agranulocytosis.

Lidocaine — selectively blocks sodium channels in ischemically damaged (depolarized) ventricular myocardium; it has almost no effect on the atria. Only **intravenously** (strong first-pass effect). Uses: **ventricular arrhythmias** (especially in acute ischemia), defibrillation-resistant ventricular fibrillation when amiodarone is not available. Toxicity — CNS signs (Topic 3).

Class IC (flecainide, propafenone) — for restoring rhythm in atrial fibrillation and preventing it (the **"pill-in-the-pocket"** approach), in patients **without** structural heart disease.

CAUTION

The CAST trial (1989): in patients with ventricular extrasystoles after myocardial infarction, flecainide and encainide reduced the extrasystoles but **increased mortality** (proarrhythmic effect). Since then class IC drugs **are contraindicated in IHD, after infarction and in heart failure**. This trial is the classic proof of the principle "treat the patient, not the ECG".

2.3. Class II — beta-blockers

By reducing sympathetic stimulation, they slow sinus node automaticity and AV conduction. They are proven antiarrhythmics **that reduce the risk of sudden death after myocardial infarction**. Uses: sinus tachycardia, rate control in atrial fibrillation, supraventricular tachycardias, catecholamine-dependent ventricular arrhythmias, long QT syndrome. **Esmolol** — ultrashort-acting, intravenous.

2.4. Class III — potassium channel blockers

Amiodarone — the most effective and broad-spectrum antiarrhythmic; it has properties of all four classes (blockade of K^+, Na^+ and Ca^{2+} channels and beta receptors). It may also be used in structural heart disease and heart failure.

- **Pharmacokinetics:** highly lipophilic, a very large volume of distribution; **t½ — 40–60 days**; the effect develops over weeks — treatment is started with a **loading dose**.
- **Uses:** atrial fibrillation (restoring and maintaining rhythm), ventricular tachycardia, defibrillation-resistant ventricular fibrillation **in cardiac arrest** (300 mg intravenously, then 150 mg).

Table 10.3. *Adverse effects of amiodarone (it contains iodine and accumulates in tissues)*

Organ	Adverse effect	Monitoring
Thyroid gland	Hypothyroidism or thyrotoxicosis	TSH every 6 months
Lungs	Interstitial pneumonitis, pulmonary fibrosis (the most dangerous)	Radiography, spirometry
Liver	Raised transaminases, hepatitis	Liver enzymes
Eye	Corneal microdeposits (in almost all patients), optic neuropathy	Ophthalmologist
Skin	Photosensitivity, gray-bluish pigmentation	Sun protection
Heart	Bradycardia, AV block, QT prolongation (torsades is rare)	ECG
Nervous system	Peripheral neuropathy, tremor	—

MEMORY AID
The adverse effects of amiodarone — **“Thyroid, lungs, liver, eyes, skin”** — all the “fatty” organs where it accumulates. Amiodarone increases the concentrations of warfarin and digoxin — their doses must be reduced.

Sotalol — a beta-blocker + class III; prolongs QT, with a risk of torsades (reduce the dose in renal failure). **Dronedarone** — an iodine-free analog of amiodarone; contraindicated in severe heart failure.

2.5. Class IV — calcium channel blockers

Verapamil and **diltiazem** block the calcium current in the sinus and AV nodes and slow AV conduction. Uses: **supraventricular tachycardias** involving the AV node (if adenosine is ineffective), rate control in atrial fibrillation and flutter.

CAUTION

Verapamil and diltiazem: 1) **contraindicated** in atrial fibrillation with **WPW syndrome** — by blocking the AV node they "direct" impulses to the ventricles through the accessory pathway and may cause ventricular fibrillation; 2) contraindicated in **wide-QRS tachycardia** (it may be ventricular tachycardia); 3) not given **intravenously together with beta-blockers** (risk of asystole); 4) heart failure (reduced ejection fraction) — a contraindication.

2.6. Other antiarrhythmics

- **Adenosine** — via A_1 receptors briefly (for seconds) stops conduction in the AV node; t½ < 10 seconds. **The drug of choice for terminating AV nodal re-entrant tachycardia**: 6 mg intravenously as a **rapid bolus** (into a proximal vein, immediately "flushed" with 20 ml of saline); if there is no effect — 12 mg. Adverse effects (brief): chest discomfort, facial flushing, dyspnea, brief asystole. Contraindicated in **bronchial asthma**; theophylline reduces its effect, dipyridamole enhances it.
- **Magnesium sulfate** — the drug of choice for **torsades de pointes ventricular tachycardia** (long QT) (2 g intravenously); hypomagnesemia, glycoside arrhythmias.
- **Digoxin** — slows AV conduction by increasing vagal tone; rate control in atrial fibrillation combined with heart failure (Section 6).
- **Atropine** — in sinus bradycardia and AV blocks (Topic 4).

2.7. Proarrhythmic effect of antiarrhythmics

Almost all antiarrhythmics can themselves cause new or more severe arrhythmias: classes IA and III — QT prolongation and **torsades de pointes**; IC — ventricular tachycardia (especially in ischemia); classes II and IV, digoxin — bradycardia and AV block. Factors that increase the risk of proarrhythmia: **hypokalemia, hypomagnesemia**, bradycardia, other QT-prolonging drugs (macrolides, fluoroquinolones, antipsychotics, ondansetron), renal failure, female sex.

2.8. Principles of treating the main arrhythmias

Table 10.4. *Choice of drugs in arrhythmias*

Arrhythmia	Emergency care	Long-term treatment
AV nodal re-entrant tachycardia (narrow QRS)	Vagal maneuvers (modified Valsalva) → adenosine 6–12 mg → verapamil; with unstable hemodynamics — electrical cardioversion	A beta-blocker or verapamil; radiofrequency ablation
Atrial fibrillation	Unstable — cardioversion; stable — rate control (beta-blocker, verapamil/diltiazem, digoxin) or rhythm restoration (amiodarone, propafenone)	Rate or rhythm control + an anticoagulant (according to CHA_2DS_2-VASc)
Ventricular tachycardia (with a pulse, stable)	Amiodarone 300 mg intravenously; unstable — cardioversion	Implantable cardioverter-defibrillator, amiodarone, a beta-blocker
Ventricular fibrillation, pulseless ventricular tachycardia	Defibrillation, CPR, epinephrine (adrenaline), amiodarone (after the 3rd shock)	ICD
Torsades	Magnesium sulfate, stopping QT-prolonging drugs, correcting potassium	Eliminate the cause
Symptomatic bradycardia	Atropine 0.5–1 mg intravenously; if ineffective — epinephrine/dopamine infusion, pacing	Permanent pacemaker

CLINICAL SIGNIFICANCE

Stroke prevention in atrial fibrillation is more important than the choice of antiarrhythmic. According to the CHA_2DS_2-VASc score (≥ 2 points in men, ≥ 3 in women), an **oral anticoagulant** (apixaban, rivaroxaban, dabigatran or warfarin) is prescribed (Topic 14). Aspirin is ineffective for this purpose.

3. Antianginal drugs

3.1. Pathogenesis of angina

Angina is retrosternal pain caused by a mismatch between the myocardial oxygen **demand** and its **supply** through coronary blood flow. Causes: atherosclerotic narrowing of the coronary arteries (exertional angina), coronary spasm (vasospastic — Prinzmetal angina), thrombus (acute coronary syndrome).

Antianginal drugs work in two ways:

- **reducing the myocardial oxygen demand** — lowering the heart rate, contractile force, preload (venous return) and afterload (BP);
- **increasing the oxygen supply** to the myocardium — dilating the coronary vessels, relieving spasm, prolonging diastole (coronary perfusion takes place in diastole).

Table 10.5. *Classification of antianginal drugs*

Group	Preparations	Main mechanism
Organic nitrates	Nitroglycerin, isosorbide dinitrate, isosorbide-5-mononitrate; molsidomine (NO donor)	NO → cGMP ↑ → mainly dilation of veins (preload ↓), coronary spasm is relieved
Beta-blockers	Bisoprolol, metoprolol, atenolol, nebivolol	Heart rate and contractile force ↓ — oxygen demand ↓
Calcium channel blockers	Dihydropyridines: amlodipine, nifedipine (sustained-release), felodipine; non-dihydropyridines: verapamil, diltiazem	Dilation of arterioles and coronary vessels; verapamil and diltiazem also reduce the heart rate
If current inhibitor	Ivabradine	Reduces only the sinus rate
Metabolic (cytoprotectors)	Trimetazidine, ranolazine, meldonium	Improving myocardial energy metabolism (trimetazidine reduces fatty acid oxidation; ranolazine is an inhibitor of the late sodium current)
Potassium channel openers	Nicorandil	Opening of KATP channels + NO donor

3.2. Organic nitrates

Mechanism. In the body, nitrates release **nitric oxide (NO)**. In smooth muscle cells NO activates guanylate cyclase → **cGMP** ↑ → myosin light chains are dephosphorylated → the vessel wall relaxes.

Hemodynamic effects:

- at small doses they dilate mainly **veins** — venous return and the preload of the heart (diastolic volume and wall tension) decrease → **the myocardial oxygen demand decreases** (the main mechanism);
- at large doses they also dilate arteries — BP (afterload) falls;

- they dilate the large epicardial coronary arteries and **relieve spasm**, improve collateral blood flow and redistribute blood to the subendocardial layers.

Table 10.6. *Nitrate preparations*

Preparation	Shape	Onset / duration of action	Use
Nitroglycerin	Sublingual tablet (0.5 mg), spray (0.4 mg/dose)	1–3 minutes / 20–30 minutes	Stopping an attack of angina and preventing it before exertion
Nitroglycerin	Intravenous infusion	Immediately	Acute coronary syndrome, pulmonary edema, hypertensive crisis
Nitroglycerin	Transdermal patch	Hours / 12–24 hours	Prevention
Isosorbide dinitrate	Sublingual and oral tablets, spray	3–5 minutes (sublingual) / 1–6 hours	Attacks and prevention
Isosorbide-5-mononitrate	Oral (sustained-release)	30–60 minutes / 8–24 hours	Prevention of attacks (no first-pass effect, F ≈ 100%)

REMEMBER

Tolerance to nitrates — with continuous use the effect decreases within 24–48 hours (depletion of sulfhydryl groups, neurohormonal activation). Prevention — **an 8–12-hour "nitrate-free interval" each day** (for example, sustained-release isosorbide mononitrate once a day in the morning; a patch is removed at night).

Adverse effects: headache (dilation of cerebral vessels; the most common, usually decreasing over time), facial flushing, dizziness, **orthostatic hypotension** and fainting, reflex tachycardia, methemoglobinemia (at large doses), rebound on abrupt withdrawal of nitrates.

CAUTION

Nitrates + phosphodiesterase-5 inhibitors (sildenafil, tadalafil, vardenafil) — absolutely contraindicated: both increase cGMP and **life-threatening hypotension** develops. No nitrate is given for 24 hours after sildenafil or 48 hours after tadalafil. Other contraindications: hypotension (systolic BP < 90), right ventricular infarction, severe aortic stenosis, hypertrophic obstructive cardiomyopathy, raised intracranial pressure.

3.3. Beta-blockers and calcium channel blockers

Beta-blockers (Topic 6) — **first-line** drugs for stable exertional angina: by lowering the heart rate they reduce oxygen demand, and by prolonging diastole they improve coronary perfusion. Target resting heart rate — 55–60 beats/min. In **vasospastic angina**, nonselective beta-blockers may worsen spasm.

Calcium channel blockers block L-type calcium channels and reduce Ca^{2+} entry into smooth muscle and cardiomyocytes.

Table 10.7. *Comparison of calcium channel blockers*

Feature	Dihydropyridines (amlodipine, nifedipine, felodipine)	Verapamil (phenylalkylamine)	Diltiazem (benzothiazepine)
Effect on vessels	Strong vasodilation	Intermediate	Intermediate

Feature	Dihydropyridines (amlodipine, nifedipine, felodipine)	Verapamil (phenylalkylamine)	Diltiazem (benzothiazepine)
Effect on the heart (HR, AV conduction, contractility)	Almost none; reflex tachycardia (with short-acting forms)	Strongly reduces	Moderately reduces
Antiarrhythmic effect	Absent	Present	Present
Main use	Hypertension, angina, vasospastic angina, Raynaud's phenomenon	Supraventricular arrhythmias, angina (if a beta-blocker cannot be used)	Angina, arrhythmias
Main adverse effects	Ankle edema, headache, facial flushing, gingival hyperplasia	Constipation, bradycardia, AV block	Bradycardia

CAUTION

Short-acting nifedipine (ordinary capsules) **increases the risk of death** in acute coronary syndrome because of abrupt vasodilation and reflex sympathetic activation; giving it sublingually in a hypertensive crisis is also not recommended. In angina and hypertension only long-acting forms (amlodipine, nifedipine GITS) are used.

3.4. Other antianginal drugs

- **Ivabradine** — by selectively inhibiting the If current in the sinus node, it **reduces only the heart rate**; it does not affect contractility or BP. Effective only **in sinus rhythm** (not in atrial fibrillation). Used when a beta-blocker cannot be used or is insufficient. Adverse effects: **phosphenes** (flashes of light in vision), bradycardia.
- **Trimetazidine** — reduces beta-oxidation of fatty acids and shifts metabolism to glucose oxidation (saves oxygen). An add-on drug; it may cause parkinsonism and movement disorders.
- **Ranolazine** — inhibits the late sodium current and improves diastolic relaxation; slightly prolongs QT.
- **Nicorandil** — a potassium channel opener and NO donor; may cause oral and intestinal ulcers.

3.5. Prognosis-improving treatment

The treatment of angina has two goals: **reducing symptoms** (the antianginal drugs above) and **preventing myocardial infarction and death**:

- **antiplatelet agents** — acetylsalicylic acid 75–100 mg/day (or clopidogrel);
- **statins** — high-intensity (atorvastatin, rosuvastatin), to the target level of LDL cholesterol;
- **ACE inhibitors** (or ARBs) — with concomitant hypertension, diabetes or heart failure;
- **beta-blockers** — after infarction and in heart failure;
- elimination of risk factors: stopping smoking, controlling BP and glucose, physical activity.

CLINICAL SIGNIFICANCE

Care in an attack of angina: stop physical activity, sit down; **nitroglycerin** 0.5 mg sublingually (or 1 spray dose); if the pain does not pass in 5 minutes — a second dose and **call an ambulance**; up to 3 doses in total. If the pain lasts more than 15–20 minutes — acute coronary syndrome is suspected: chew aspirin 150–300 mg (if there is no allergy), ECG, urgent hospitalization.

4. Cardiotonic drugs (cardiac glycosides and non-glycoside inotropes)

This section corresponds to the 6th-semester class “Laboratory study of the effects of antiarrhythmic and cardiotonic drugs on the heart”.

4.1. Cardiac glycosides

Cardiac glycosides are substances obtained from plants, whose molecule consists of an **aglycone (genin)** — a steroid nucleus and a lactone ring — and a **sugar** part. Sources: foxglove (Digitalis lanata — digoxin, Digitalis purpurea — digitoxin), strophanthus (strophanthin), lily of the valley (Corglycon), spring adonis.

Mechanism of action. Glycosides inhibit **Na^+/K^+-ATPase** in the cardiomyocyte membrane → Na^+ increases inside the cell → Ca^{2+} efflux via the Na^+/Ca^{2+} exchanger decreases → **Ca^{2+} increases** in the sarcoplasmic reticulum and cytoplasm → contraction is strengthened.

Table 10.8. *Effects of cardiac glycosides on the heart*

Effect	Essence	Mechanism
Positive inotropic	Systole becomes strong and short, stroke volume increases	Intracellular Ca^{2+} ↑
Negative chronotropic	The heart rate slows, diastole is prolonged	Increased vagal tone (baroreceptor reflex and central action), decreased sympathetic activity
Negative dromotropic	AV conduction slows	Vagal effect, the refractory period in the AV node is prolonged
Positive bathmotropic (at toxic doses)	The excitability and automaticity of the ventricular myocardium increase — arrhythmias	Ca^{2+} overload, delayed afterdepolarizations

Table 10.9. *Pharmacokinetic characteristics of cardiac glycosides*

Preparation	Intestinal absorption	Protein binding	t½	Elimination	Onset of action (IV)
Digitoxin	≈ 95%	≈ 95%	5–7 days	Liver (enterohepatic circulation) — strong cumulation	2–4 hours
Digoxin	60–80%	≈ 25%	36–48 hours	Kidneys (unchanged)	15–30 minutes
Strophanthin K	Almost not absorbed (intravenous only)	Low	≈ 20 hours	Kidney	5–10 minutes
Corglycon	Not absorbed (intravenous)	Low	Short	Kidney	5–10 minutes

REMEMBER
The higher the **lipophilicity** (nonpolarity) of glycosides, the stronger their intestinal absorption, protein binding, hepatic metabolism and **cumulation**: **digitoxin > digoxin > strophanthin**. Because digoxin is excreted by the kidneys, **its dose is reduced in renal failure**.

Uses (modern):

- **atrial fibrillation** — control of the ventricular rate (especially in heart failure and in patients with low mobility);

- **chronic heart failure** (reduced ejection fraction) — when symptoms persist on the main drugs; it reduces hospitalizations but **does not reduce mortality**.

> **CAUTION**
> Older textbooks presented cardiac glycosides as the main drugs for treating heart failure and recommended "digitalization" (loading) schemes. In modern recommendations, the main treatment of heart failure is ACE inhibitors/ARNIs, beta-blockers, mineralocorticoid antagonists and SGLT2 inhibitors (Topic 11); digoxin is an add-on drug used at low doses (blood concentration 0.5–0.9 ng/ml).

4.2. Glycoside toxicity

The therapeutic range is very narrow (the toxic dose is only 2–3 times the therapeutic dose). Factors that increase toxicity: **hypokalemia** (diuretics!), hypomagnesemia, hypercalcemia, old age, **renal failure**, hypothyroidism, hypoxia, drugs (**amiodarone, verapamil, quinidine and clarithromycin** increase digoxin concentration).

Table 10.10. *Signs of glycoside toxicity*

System	Signs
Heart	Ventricular extrasystoles (bigeminy), ventricular tachycardia and fibrillation, AV blocks, bradycardia, atrial tachycardia with block; "trough-shaped" (scooped) ST depression on the ECG (this also occurs at therapeutic doses)
Gastrointestinal	Loss of appetite, nausea, vomiting (the first signs), abdominal pain, diarrhea
CNS and vision	Headache, weakness, confusion (in the elderly), visual disturbances — objects appear yellow-green (xanthopsia), spots before the eyes

Treatment:

1. Stop the glycoside; measure the blood digoxin concentration and **potassium** level, ECG monitoring.
2. **Correction of hypokalemia** — potassium preparations (potassium chloride, Panangin) — only in the absence of AV block and hyperkalemia.
3. **Digoxin-specific antibody Fab fragments** — in life-threatening arrhythmias, hyperkalemia (> 5 mmol/l), severe toxicity.
4. In ventricular arrhythmias — **lidocaine**, phenytoin; magnesium sulfate; in bradycardia and AV block — atropine, temporary pacing.
5. **Unithiol** (a donor of SH groups) — reactivates Na^+/K^+-ATPase (used in CIS countries).
6. In acute poisoning — activated charcoal.

> **CAUTION**
> In glycoside toxicity **calcium preparations are contraindicated** (they worsen arrhythmias and cardiac arrest), and electrical cardioversion is avoided as far as possible (risk of ventricular fibrillation). Intravenous calcium preparations must not be given to a patient taking glycosides.

4.3. Non-glycoside cardiotonics

Table 10.11. *Non-glycoside inotropic drugs*

Group	Preparations	Mechanism	Use
Beta-1 adrenomimetics	Dobutamine; dopamine	cAMP ↑ (adenylate cyclase)	Cardiogenic shock, acute decompensation (short term)

Group	Preparations	Mechanism	Use
Phosphodiesterase-3 inhibitors	Milrinone	cAMP breakdown ↓; inotropy + vasodilation	Acute heart failure (short term); long-term use increases mortality
Calcium sensitizers	Levosimendan	Increases the Ca^{2+} sensitivity of troponin C + opens KATP channels	Acute decompensated heart failure

4.4. Laboratory class: effects of drugs on the isolated heart

Experiment (frog heart, Straub method, or isolated heart):

1. The frog heart is connected to a cannula and perfused with Ringer's solution; contractions are recorded with a kymograph or a sensor.
2. Normal contractions are recorded; then a **"heart failure model"** is created (for example, with a low-calcium solution — the contractions weaken).
3. **Strophanthin** or **digoxin** is added to the solution: the **contraction amplitude increases** (positive inotropy), systole is strong and short, diastole is prolonged, the rhythm slows.
4. When the glycoside dose is increased, a **toxic effect** is seen: arrhythmias, incomplete contractions and **arrest of the heart in systole**. If K^+ in the solution is reduced, the toxic effect appears sooner.
5. **Studying the antiarrhythmic effect**: against the background of an arrhythmia (induced with calcium chloride or epinephrine), lidocaine or verapamil is added and restoration of the rhythm is observed; verapamil also reduces the force of contraction (negative inotropy).

REMEMBER
Conclusions of the experiment: cardiac glycosides have a **positive inotropic, negative chronotropic and dromotropic** effect; at toxic doses — arrhythmias and systolic arrest; **hypokalemia increases toxicity**. Verapamil is antiarrhythmic but negatively inotropic.

5. Clinical part

Table 10.12. *Choice of drugs in clinical situations*

Situation	Choice	To be avoided
Stable angina, high HR	Beta-blocker + nitroglycerin for attacks + aspirin + statin	—
Angina + bronchial asthma	A calcium antagonist (amlodipine, diltiazem), ivabradine, nitrates	Nonselective beta-blockers
Vasospastic angina	Calcium antagonists, nitrates	Nonselective beta-blockers
Angina + heart failure (reduced EF)	A beta-blocker (bisoprolol, carvedilol), ivabradine, amlodipine	Verapamil, diltiazem
Atrial fibrillation + heart failure	A beta-blocker, digoxin; amiodarone for rhythm	Verapamil, diltiazem, class IC
Atrial fibrillation + WPW	Procainamide, cardioversion; ablation	Verapamil, diltiazem, digoxin, adenosine
A patient with angina taking sildenafil for erectile dysfunction	Nitrate-free antianginal therapy	Nitrates

Practical skills

Skill 1. Care in supraventricular (AV nodal) tachycardia

1. Record an ECG (narrow QRS, regular rhythm, 150–250 beats/min) and assess the hemodynamics.
2. Unstable (hypotension, shock, acute heart failure) — synchronized electrical cardioversion.
3. Stable — **the modified Valsalva maneuver** (straining, then lying the patient down and raising the legs); carotid sinus massage (not in the elderly or with a carotid bruit).
4. If ineffective — **adenosine 6 mg** as a rapid intravenous bolus + 20 ml of saline; if there is no effect in 1–2 minutes — 12 mg.
5. If adenosine cannot be used (asthma) or is ineffective — verapamil 5 mg intravenously slowly.
6. After the rhythm is restored, refer to a cardiologist (ablation).

Skill 2. Key prescriptions

PRESCRIPTION 10.1. AMIODARONE (TABLETS)
Rp.: Amiodaroni 0.2
D.t.d. N. 30 in tabulettis
S. Loading dose: 1 tablet 3 times a day for 1 week, then 1 tablet 2 times a day for 1 week, then 1 tablet a day.

PRESCRIPTION 10.2. AMIODARONE (AMPOULES)
Rp.: Solutionis Amiodaroni 5% — 3 ml
D.t.d. N. 10 in ampullis
S. 300 mg (2 ampoules) in 20 ml of 5% glucose intravenously (in cardiac arrest — as a bolus).

PRESCRIPTION 10.3. ADENOSINE
Rp.: Solutionis Adenosini 0.3% — 2 ml
D.t.d. N. 6 in ampullis
S. 6 mg as a rapid intravenous bolus, immediately flushed with 20 ml of 0.9% NaCl; if there is no effect, 12 mg.

PRESCRIPTION 10.4. VERAPAMIL
Rp.: Verapamili hydrochloridi 0.04
D.t.d. N. 50 in tabulettis obductis
S. Take 1 tablet 3 times a day.

PRESCRIPTION 10.5. LIDOCAINE (AS AN ANTIARRHYTHMIC)
Rp.: Solutionis Lidocaini hydrochloridi 2% — 5 ml
D.t.d. N. 10 in ampullis
S. 1–1.5 mg/kg intravenously slowly, then an infusion (in ventricular tachycardia).

PRESCRIPTION 10.6. PROPAFENONE
Rp.: Propafenoni 0.15
D.t.d. N. 40 in tabulettis obductis
S. Take 1 tablet 3 times a day (in the absence of structural heart disease).

PRESCRIPTION 10.7. NITROGLYCERIN (SUBLINGUAL)
Rp.: Nitroglycerini 0.0005
D.t.d. N. 40 in tabulettis
S. In an attack of angina, 1 tablet under the tongue; if it does not pass in 5 minutes, repeat and call an ambulance.

PRESCRIPTION 10.8. NITROGLYCERIN SPRAY
Rp.: Aerosoli Nitroglycerini 0.0004/dosi — 200 dosi
D.S. In an attack, spray 1 dose under the tongue (in a sitting position).

PRESCRIPTION 10.9. ISOSORBIDE-5-MONONITRATE
Rp.: Tabulettas Isosorbidi mononitratis 0.05 retard N. 30
D.S. Take 1 tablet once a day in the morning.

PRESCRIPTION 10.10. AMLODIPINE
Rp.: Amlodipini 0.005
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day.

PRESCRIPTION 10.11. IVABRADINE
Rp.: Ivabradini 0.005
D.t.d. N. 56 in tabulettis obductis
S. Take 1 tablet 2 times a day with meals.

PRESCRIPTION 10.12. TRIMETAZIDINE
Rp.: Tabulettas Trimetazidini 0.035 retard N. 60
D.S. Take 1 tablet 2 times a day with meals.

PRESCRIPTION 10.13. DIGOXIN (TABLETS)
Rp.: Digoxini 0.00025
D.t.d. N. 30 in tabulettis
S. Take 1/2–1 tablet once a day (under pulse and potassium monitoring).

PRESCRIPTION 10.14. DIGOXIN (AMPOULES)
Rp.: Solutionis Digoxini 0.025% — 1 ml
D.t.d. N. 10 in ampullis
S. Dilute 1 ml in 10 ml of 0.9% NaCl and inject intravenously slowly.

PRESCRIPTION 10.15. MAGNESIUM SULFATE
Rp.: Solutionis Magnesii sulfatis 25% — 10 ml
D.t.d. N. 10 in ampullis
S. 8 ml (2 g) in 20 ml of 5% glucose intravenously over 10–15 minutes (in torsades de pointes).

Review questions

1. Phases of the cardiomyocyte action potential and ion currents.
2. The main mechanisms of arrhythmias. How does re-entry arise?
3. Classification of antiarrhythmic drugs (Vaughan Williams).
4. Differences between class IA, IB and IC drugs. The significance of the CAST trial.

5. Mechanism, uses and adverse effects of amiodarone.
6. The antiarrhythmic effect and contraindications of verapamil and diltiazem.
7. Uses of adenosine and the technique of administration.
8. What is a proarrhythmic effect? Treatment of torsades de pointes.
9. Principles of treating atrial fibrillation.
10. Pathogenesis of angina and the main ways antianginal drugs act.
11. Mechanism, hemodynamic effects and preparations of nitrates.
12. Tolerance to nitrates and its prevention. Contraindications to nitrates.
13. Classification of calcium channel blockers and their differences.
14. Mechanism of ivabradine, trimetazidine, ranolazine and nicorandil.
15. Prognosis-improving treatment in angina.
16. Mechanism of action of cardiac glycosides and their effects on the heart.
17. Pharmacokinetic differences between digoxin and digitoxin.
18. Signs of glycoside toxicity, aggravating factors and treatment.
19. Non-glycoside cardiotonics.

Test questions

1. The class of antiarrhythmics that blocks phase 0 of the action potential:
   A) Class III
   B) Class II
   C) Class I
   D) Class IV
   E) Adenosine

2. An antiarrhythmic that blocks sodium channels mainly in ischemic ventricular myocardium and is given only intravenously:
   A) Lidocaine
   B) Quinidine
   C) Amiodarone
   D) Flecainide
   E) Verapamil

3. The class of antiarrhythmics banned in post-infarction patients because it increased mortality:
   A) III
   B) IV
   C) IB
   D) IC
   E) II

4. An antiarrhythmic that causes pulmonary fibrosis and thyroid dysfunction:
   A) Adenosine
   B) Amiodarone
   C) Lidocaine
   D) Propranolol
   E) Verapamil

5. The drug of choice for terminating AV nodal re-entrant tachycardia:
   A) Amiodarone
   B) Lidocaine
   C) Adenosine
   D) Atropine
   E) Digoxin

6. The drug of choice in torsades de pointes ventricular tachycardia:
   A) Amiodarone
   B) Sotalol
   C) Procainamide
   D) Quinidine
   E) Magnesium sulfate

7. Which drug is contraindicated in atrial fibrillation with WPW syndrome?
   A) Ibutilide
   B) Procainamide
   C) Amiodarone
   D) Electrical cardioversion
   E) Verapamil

8. The main mechanism of the antianginal effect of nitrates:
   A) Inhibition of platelet aggregation
   B) Dilating veins and reducing preload
   C) Reducing the heart rate
   D) Blocking beta receptors
   E) Increasing myocardial contractility

9. A drug whose use together with nitrates is absolutely contraindicated:
   A) Sildenafil
   B) Amlodipine
   C) Atorvastatin
   D) Aspirin
   E) Bisoprolol

10. A way to prevent tolerance to nitrates:
    A) Intravenous administration only
    B) Leaving the patch on at night as well
    C) An 8–12-hour nitrate-free interval each day
    D) Constantly increasing the dose
    E) Giving together with a beta-blocker

11. An antianginal drug typically causing ankle edema:
    A) Amlodipine
    B) Ivabradine
    C) Bisoprolol
    D) Trimetazidine
    E) Isosorbide mononitrate

12. A drug that reduces the heart rate by inhibiting the If current only in the sinus node:
    A) Verapamil
    B) Ivabradine
    C) Nicorandil
    D) Digoxin
    E) Ranolazine

13. Mechanism of action of cardiac glycosides:
    A) Blocking calcium channels
    B) Inhibiting Na^+/K^+-ATPase
    C) Increasing the calcium sensitivity of troponin
    D) Stimulating beta-1 receptors
    E) Inhibiting phosphodiesterase-3

14. An electrolyte disturbance that increases glycoside toxicity:
    A) Hypocalcemia
    B) Hypokalemia
    C) Hypernatremia
    D) Hyperkalemia
    E) Hypermagnesemia

15. A visual disturbance typical of glycoside toxicity:
    A) Seeing objects in yellow-green
    B) Accommodation spasm
    C) Phosphenes
    D) Miosis
    E) Diplopia

Clinical cases

Case 1

A 72-year-old woman takes digoxin 0.25 mg/day and furosemide for chronic heart failure and atrial fibrillation. In recent days she has lost her appetite, feels nauseous and vomits, and “everything looks yellowish”. The ECG shows ventricular bigeminy, HR 48 beats/min. Blood potassium is 3.0 mmol/l, creatinine is raised.

Questions: 1) Diagnosis? 2) Aggravating factors? 3) Treatment.

Answer: 1) Glycoside (digoxin) toxicity. 2) Hypokalemia (furosemide), reduced renal function (digoxin is excreted by the kidneys), old age, a relatively large dose. 3) Stop digoxin and furosemide; measure the blood digoxin level; ECG monitoring; potassium preparations intravenously slowly (if there is no AV block); in severe arrhythmias — digoxin-specific Fab fragments, lidocaine; atropine for bradycardia. Calcium preparations are contraindicated. Later the digoxin dose is reduced according to creatinine clearance (0.0625–0.125 mg).

Case 2

A 55-year-old man has had exertional angina for 2 years and takes isosorbide mononitrate and bisoprolol. On a friend's advice he took sildenafil for erectile dysfunction; an hour later he had an attack of angina and also put a nitroglycerin tablet under his tongue. He lost consciousness, BP 60/30 mm Hg.

Questions: 1) The cause of the condition? 2) Care.

Answer: 1) The combination of a nitrate and a PDE-5 inhibitor (sildenafil) — both increase cGMP, causing strong vasodilation and life-threatening hypotension. 2) Lay the patient down and raise the legs, oxygen, rapid intravenous crystalloids; if ineffective — vasopressors (phenylephrine or norepinephrine); ECG monitoring (myocardial ischemia). Do not give nitrates. Later explain to the patient that nitrates and PDE-5 inhibitors must not be taken together, and resolve the question of erectile dysfunction with a cardiologist.

Case 3

A 28-year-old woman suddenly developed palpitations. ECG: narrow QRS, regular tachycardia of 190 beats/min, P waves not visible. BP 115/75 mm Hg, general condition satisfactory. She has a history of bronchial asthma.

Questions: 1) Diagnosis? 2) Algorithm of care and choice of drug.

Answer: 1) Paroxysmal supraventricular (AV nodal re-entrant) tachycardia, hemodynamically stable. 2) First vagal maneuvers (modified Valsalva). Adenosine is usually the drug of choice, but in bronchial asthma adenosine may cause bronchospasm — so verapamil 5 mg intravenously slowly (over 2–3 minutes) is preferred. If unstable — electrical cardioversion. Then a cardiologist's consultation; radiofrequency ablation for recurrent attacks.

Quick review table

Table 10.13. *Antiarrhythmic, antianginal and cardiotonic drugs — summary*

Group	Preparations	Mechanism	Main use	Important risk
IA	Quinidine, procainamide	Na_+ + K_+ blockade	Atrial fibrillation, WPW	QT prolongation, torsades; lupus (procainamide)
IB	Lidocaine, mexiletine	Na_+ (in ischemic tissue)	Ventricular arrhythmias	CNS toxicity
IC	Flecainide, propafenone	Strong Na_+ blockade	Atrial fibrillation (in a healthy heart)	Increases mortality in IHD (CAST)
II	Beta-blockers	Sympathetic influence ↓	Tachyarrhythmias, after infarction	Bradycardia, AV block
III	Amiodarone, sotalol	K_+ blockade, refractory period ↑	All tachyarrhythmias; cardiac arrest	Amiodarone: thyroid, lungs, liver, eyes, skin
IV	Verapamil, diltiazem	Ca_{2+} blockade (AV node)	SVT, rate control	Contraindicated in WPW, wide QRS, HF
Adenosine	Adenosine	A_1 — AV node blockade	Terminating SVT	Contraindicated in asthma
Nitrates	Nitroglycerin, isosorbide mono/dinitrate	NO → cGMP; venodilation	Angina attacks and prevention	Headache, tolerance; contraindicated with PDE-5 inhibitors
Calcium antagonists	Amlodipine, sustained-release nifedipine	Dilation of arterioles	Angina, hypertension	Edema; short-acting nifedipine is dangerous
I_f inhibitor	Ivabradine	HR ↓ (in sinus rhythm)	Angina, heart failure	Phosphenes

Group	Preparations	Mechanism	Main use	Important risk
Metabolic	Trimetazidine, ranolazine	Improved energy metabolism	Add-on antianginal	—
Cardiac glycosides	Digoxin, strophanthin, Corglycon	Na_+/K_+-ATPase ↓ → Ca_{2+} ↑	Atrial fibrillation, HF (add-on)	Narrow TI; hypokalemia worsens it; antidote — Fab
Non-glycoside inotropes	Dobutamine, milrinone, levosimendan	cAMP ↑ / Ca_{2+} sensitization	Acute HF, cardiogenic shock	Arrhythmias

TOPIC 11

Antihypertensive drugs. Hypertensive drugs

Learning objectives

Arterial hypertension (AH) occurs in about one third of adults and is the main risk factor for stroke, myocardial infarction, and heart and kidney failure. Lowering BP to the target level significantly reduces these complications. GPs choose and combine antihypertensive drugs and monitor their adverse effects every day. Emergency care in hypertensive crisis and in hypotensive states (collapse, shock) is also every doctor's duty. At the end of the topic, the modern pharmacotherapy of chronic heart failure is also briefly covered, because its main drugs are the same as the antihypertensives.

The student should know:

- the systems that regulate BP and the sites of action of antihypertensive drugs;
- the classification of antihypertensive drugs, and the mechanism, uses and adverse effects of the first-line drugs (ACE inhibitors, ARBs, calcium antagonists, diuretics) and the additional groups;
- the principles of treating AH, rational combinations, and the choice of drugs in particular groups of patients;
- the treatment of hypertensive crisis and hypertension in pregnancy;
- hypertensive (BP-raising) drugs and their uses.

The student should be able to:

- write prescriptions for drugs such as enalapril, lisinopril, losartan, amlodipine, indapamide, hydrochlorothiazide, moxonidine, methyldopa and spironolactone;
- draw up an antihypertensive treatment plan for a patient and choose a combination;
- provide emergency care in hypertensive crisis.

1. Regulation of blood pressure and classification of antihypertensive drugs

BP = **cardiac output (minute volume)** × **total peripheral vascular resistance**. The main systems that regulate it:

- **the sympathetic nervous system** (vasomotor center, baroreceptors, alpha and beta receptors);
- **the renin-angiotensin-aldosterone system (RAAS)**: renin → angiotensin I from angiotensinogen → ACE → **angiotensin II** (via AT_1 receptors — strong vasoconstriction, aldosterone release, sympathetic activation, remodeling of the heart and vessels);
- excretion of sodium and water by the **kidneys** (circulating volume);
- local factors: NO, endothelin, prostacyclin, bradykinin, natriuretic peptides.

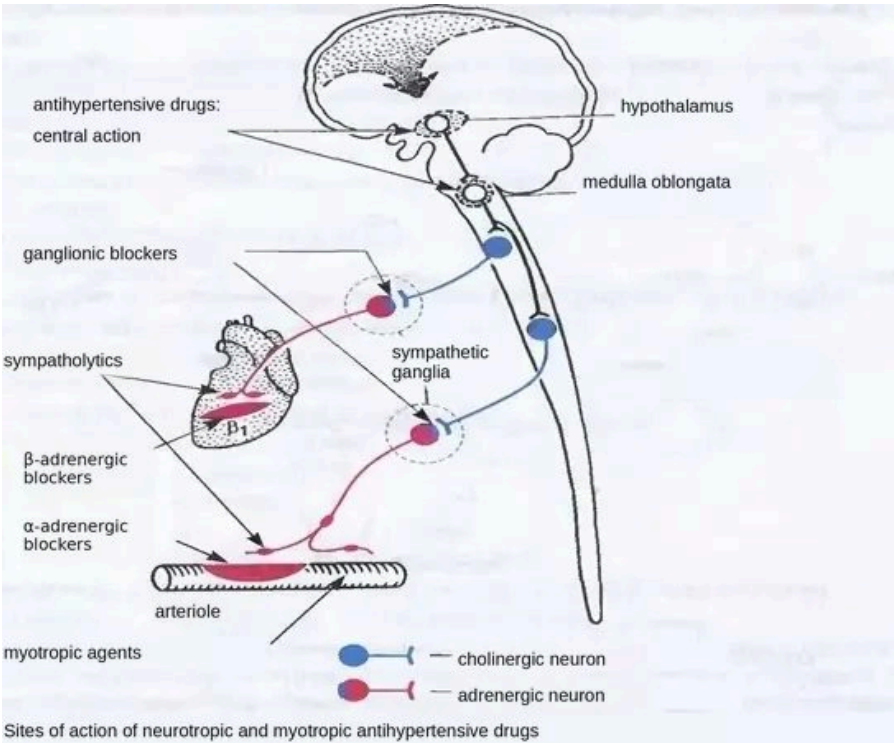


Figure 11.1. *Sites of action of antihypertensive drugs: central, ganglia, sympathetic nerve endings, beta and alpha adrenoceptors, myotropic agents (teaching material of the TMA Department of Pharmacology)*

Table 11.1. *Classification of antihypertensive drugs*

Group	Subgroup	Preparations
A. Drugs acting on the RAAS	ACE inhibitors	Enalapril, lisinopril, ramipril, perindopril, captopril, fosinopril, zofenopril
	Angiotensin II receptor (AT_1) blockers (ARBs, sartans)	Losartan, valsartan, irbesartan, telmisartan, candesartan, olmesartan, azilsartan
	Direct renin inhibitor	Aliskiren
	Mineralocorticoid receptor antagonists	Spironolactone, eplerenone, finerenone
	ARNI (angiotensin receptor–neprilysin inhibitor)	Sacubitril/valsartan (in heart failure)
B. Calcium channel blockers	Dihydropyridines	Amlodipine, lercanidipine, felodipine, nifedipine (sustained-release)
	Non-dihydropyridines	Verapamil, diltiazem
C. Diuretics	Thiazides and thiazide-like diuretics	Hydrochlorothiazide, chlorthalidone, indapamide
	Loop diuretics	Furosemide, torasemide
D. Neurotropic agents	Beta-blockers	Bisoprolol, metoprolol, nebivolol, carvedilol
	Alpha-1 blockers	Doxazosin, prazosin
	Centrally acting	Clonidine (Clofelin), methyldopa, moxonidine, rilmenidine
	Ganglionic blockers, sympatholytics	Azamethonium, reserpine (historically)
E. Myotropic vasodilators	Arteriolar	Hydralazine, minoxidil, diazoxide, bendazol (Dibazol), magnesium sulfate

Group	Subgroup	Preparations
	Mixed (arteries + veins)	Sodium nitroprusside

REMEMBER

The 5 groups of first-line (main) drugs in AH: **ACE inhibitors, ARBs, calcium antagonists, thiazide/thiazide-like diuretics** and (when there are specific indications) **beta-blockers**. They have been proven to reduce cardiovascular complications and mortality.

2. Drugs acting on the renin-angiotensin-aldosterone system

2.1. ACE inhibitors

Mechanism. They inhibit angiotensin-converting enzyme (ACE, kininase II):

- **angiotensin II formation decreases** → vasodilation, reduced aldosterone release (sodium and water are excreted, potassium is retained), reduced sympathetic activity;
- **bradykinin breakdown decreases** → bradykinin accumulates → release of NO and prostacyclin increases (additional vasodilation) — but **dry cough** and **angioedema** are also related to this. **Organ-protective effects:** reduction of left ventricular hypertrophy and vascular remodeling; **kidney protection** — by dilating the efferent arteriole they reduce intraglomerular pressure and **reduce proteinuria** (diabetic nephropathy).

Table 11.2. *Features of ACE inhibitors*

Preparation	Property	Duration of action
Captopril	Active form (not a prodrug), contains an SH group; rapid action	Short (2–3 times a day) — sublingually in hypertensive crisis
Enalapril	Prodrug → enalaprilat (in the liver); enalaprilat intravenously	1–2 times a day
Lisinopril	Active, not metabolized in the liver, excreted by the kidneys	Once a day
Ramipril, perindopril	Prodrugs; penetrate tissues well	Once a day
Fosinopril	Eliminated by two routes (kidneys + liver)	Convenient in renal failure

Uses: arterial hypertension (especially with diabetes, proteinuria, left ventricular hypertrophy), **chronic heart failure** (reduced ejection fraction — reduces mortality), after myocardial infarction, diabetic and nondiabetic nephropathy.

Table 11.3. *Adverse effects of ACE inhibitors*

Adverse effect	Mechanism / note
Dry cough (5–20%)	Bradykinin accumulation; switch to an ARB
Angioedema (rare but dangerous)	Bradykinin; more common in Black patients; never prescribed again
Hyperkalemia	Reduced aldosterone; especially with renal failure, potassium-sparing diuretics, NSAIDs
“First-dose hypotension”	In hypovolemic patients taking diuretics

Adverse effect	Mechanism / note
Worsening of renal function	In bilateral renal artery stenosis — the efferent arteriole dilates and the filtration pressure falls sharply
Teratogenicity (2nd–3rd trimesters)	Impaired fetal kidney development, oligohydramnios — absolutely contraindicated in pregnancy
Taste disturbance, rash, neutropenia	More common with captopril (SH group)

MEMORY AID

Adverse effects of ACE inhibitors — **"CAPTOPRIL"**: **C**ough (dry, irritating), **A**ngioedema, **P**roteinuria (rare, captopril), **T**aste disturbance, **O**rthostatic (first-dose) hypotension, **P**regnancy — contraindicated, **R**enal artery stenosis — contraindicated, **I**ncreased potassium (hyperkalemia), **L**eukopenia (neutropenia).

2.2. Angiotensin II receptor blockers (ARBs, sartans)

Mechanism. Selective blockade of AT_1 receptors — all the harmful effects of angiotensin II are abolished (including angiotensin II formed by pathways other than ACE). They do not affect bradykinin breakdown — so they **do not cause cough**, and angioedema is very rare.

Uses: the same as ACE inhibitors (AH, heart failure, diabetic nephropathy); in patients who cannot tolerate ACE inhibitors. **Losartan** increases uric acid excretion (AH with gout). **Telmisartan** is long-acting.

Adverse effects and contraindications are similar to those of ACE inhibitors: hyperkalemia, worsening of renal function (in renal artery stenosis), **contraindicated in pregnancy**.

CAUTION

The combination **ACE inhibitor + ARB** (or either of them + aliskiren) — "dual RAAS blockade" — **is not recommended**: it gives no additional benefit and increases the risk of hyperkalemia, hypotension and acute kidney injury.

2.3. Mineralocorticoid receptor antagonists and the renin inhibitor

Spironolactone and **eplerenone** block aldosterone receptors (in detail in Topic 13). Uses: **resistant AH** (BP still high on three drugs — the fourth drug), primary hyperaldosteronism, chronic heart failure (reduces mortality), ascites in liver cirrhosis. Adverse effects: **hyperkalemia**; with spironolactone — **gynecomastia**, menstrual disorders (antiandrogenic effect); eplerenone is more selective.

Aliskiren inhibits renin directly; now rarely used.

3. Calcium channel blockers

Their mechanism and classification were covered in Topic 10. In AH, mainly **long-acting dihydropyridines** (amlodipine, lercanidipine, nifedipine GITS) are used: they dilate arterioles and reduce peripheral resistance.

- **Advantages:** metabolically neutral, effective in the elderly and in isolated systolic AH, good effect in Black patients, effective in stroke prevention, allowed in pregnancy (nifedipine).
- **Adverse effects: ankle edema** (not reduced by a diuretic, reduced by an ACE inhibitor/ARB), headache, facial flushing, gingival hyperplasia; with verapamil — constipation and bradycardia.

4. Diuretics

Their mechanism is covered in detail in Topic 13. In AH, mainly **thiazide and thiazide-like** diuretics are used: at first they reduce the circulating volume, and later they lower peripheral resistance through a decrease in sodium in the vessel wall.

- **Indapamide** and **chlorthalidone** (thiazide-like) — long-acting, with few metabolic adverse effects; proven to reduce cardiovascular complications.
- **Hydrochlorothiazide** — often in combination tablets (12.5–25 mg).
- **Loop diuretics** (furosemide, torasemide) are used in AH only with **renal failure** (GFR < 30 ml/min) and in heart failure with edema. **Metabolic adverse effects of thiazides:** hypokalemia, hyponatremia, **hyperuricemia** (gout attack), hyperglycemia, dyslipidemia, hypercalcemia — mainly at high doses.

5. Neurotropic antihypertensive drugs

5.1. Centrally acting drugs

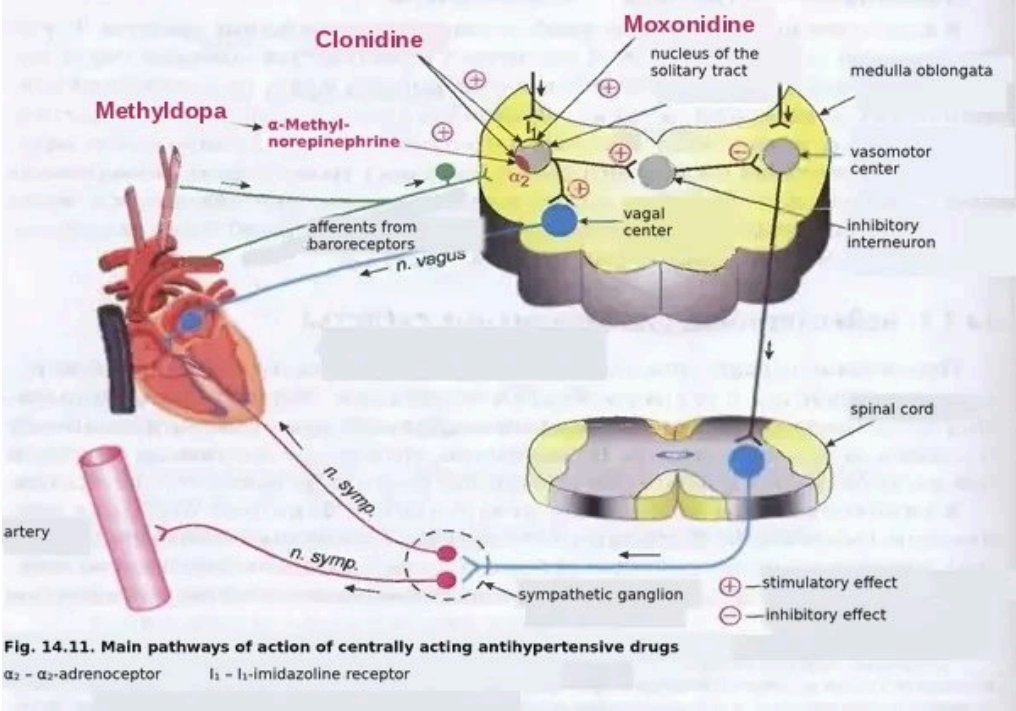


Figure 11.2. *Pathways of action of centrally acting antihypertensive drugs: clonidine (Clofelin) and methyldopa — alpha-2 adrenoceptors, moxonidine — I_1-imidazoline receptors (teaching material of the TMA Department of Pharmacology)*

Table 11.4. *Centrally acting antihypertensive drugs*

Preparation	Mechanism	Features and uses	Adverse effects
Clonidine (Clofelin)	Stimulates alpha-2 and I_1-imidazoline receptors in the medulla oblongata — sympathetic tone falls	Hypertensive crisis (sublingual, intravenous); opioid and alcohol withdrawal	Sedation, dry mouth, bradycardia; rebound hypertensive crisis on abrupt withdrawal; with alcohol — strong CNS depression; in overdose — coma, bradycardia, miosis
Methyldopa	Converted to alpha-methylnorepinephrine and stimulates central alpha-2 receptors	The drug of choice in hypertension of pregnancy	Sedation, depression, Coombs-positive hemolytic anemia, hepatitis, hyperprolactinemia

Preparation	Mechanism	Features and uses	Adverse effects
Moxonidine, rilmenidine	A selective agonist of I_1-imidazoline receptors	AH with metabolic syndrome (improves insulin sensitivity); an add-on drug	Sedation and dry mouth much less than with clonidine; little rebound

CAUTION

Clonidine used to be one of the main drugs for AH in CIS countries. Modern recommendations **do not recommend it for regular treatment** (rebound, sedation, no proven organ-protective effect). Clonidine is also under strict control because it can be used for criminal purposes (combined with alcohol).

5.2. Beta- and alpha-blockers

Beta-blockers (Topic 6) are first-line in AH in the following cases: IHD, after infarction, heart failure, tachyarrhythmias, planning pregnancy, sympathetic hyperactivity in young patients. In the elderly they are slightly weaker than other groups in preventing stroke. **Preferred drugs**: bisoprolol, nebivolol, carvedilol, metoprolol succinate (not atenolol).

Alpha-1 blockers (doxazosin) — an add-on drug in resistant AH, especially together with **prostatic hyperplasia**.

6. Myotropic vasodilators

- **Hydralazine** dilates arterioles directly; strong reflex tachycardia and fluid retention (given with a beta-blocker and a diuretic); with prolonged use — **drug-induced lupus** (in slow acetylators). Given intravenously in severe AH of pregnancy; in Black patients with heart failure — together with isosorbide dinitrate.
- **Minoxidil** opens KATP channels; very potent; used in resistant AH; **hypertrichosis** — that is why its topical solution is used for alopecia.
- **Sodium nitroprusside** — an NO donor; dilates arteries and veins; the effect begins immediately and ends 1–2 minutes after the infusion is stopped. Given **as an intravenous drip** through a light-protected line (it decomposes in light) in **hypertensive crisis** and aortic dissection (with a beta-blocker). With prolonged use — **cyanide toxicity** (antidote — sodium thiosulfate, hydroxocobalamin).
- **Bendazol (Dibazol)** — a weak myotropic antispasmodic; used in CIS countries with papaverine for mild crises; its effectiveness does not meet modern standards.
- **Magnesium sulfate** — **prevention of seizures in preeclampsia and eclampsia** (the main indication); its antihypertensive effect is weak.

7. Principles of treating arterial hypertension

7.1. Target BP and initial treatment

REMEMBER

For most patients the target BP is **below 130/80 mm Hg** (individualized in those over 65 and in frail elderly people, usually systolic 130–139 mm Hg). In most patients, treatment is recommended to start with **a fixed low-dose combination of two drugs (in a single tablet)**; in low-risk grade 1 AH and in frail elderly people — with monotherapy.

Table 11.5. *Stepwise drug treatment of AH (based on European recommendations)*

Step	Treatment
1	ACE inhibitor or ARB + a calcium antagonist or + a thiazide/thiazide-like diuretic (in a single tablet)
2	Three drugs: ACE inhibitor/ARB + calcium antagonist + diuretic (in a single tablet)
3 (resistant AH)	+ spironolactone (or another diuretic, an alpha-blocker, a beta-blocker); specialist consultation
At any step	If indicated (IHD, heart failure, arrhythmia) — a beta-blocker is added

Lifestyle changes at every step: less than 5 g of table salt a day, weight loss, regular physical activity, limiting alcohol, stopping smoking, a diet rich in vegetables and fruit.

Table 11.6. *Rational and irrational combinations*

Rational	Irrational / dangerous
ACE inhibitor/ARB + calcium antagonist	ACE inhibitor + ARB
ACE inhibitor/ARB + thiazide/thiazide-like diuretic	Verapamil or diltiazem + a beta-blocker (bradycardia, AV block)
Calcium antagonist (dihydropyridine) + diuretic	ACE inhibitor/ARB + a potassium-sparing diuretic (except spironolactone in heart failure, with monitoring) — hyperkalemia
Dihydropyridine + beta-blocker	Beta-blocker + a central alpha-2 agonist (clonidine) — bradycardia, rebound

7.2. Choice of drugs in particular groups of patients

Table 11.7. *Choice of antihypertensive drug according to comorbidities*

Comorbidity	Preferred drugs	To be avoided
Diabetes mellitus, proteinuria, chronic kidney disease	ACE inhibitor or ARB (+ SGLT2 inhibitor, finerenone)	High-dose thiazides and beta-blockers (metabolic)
IHD, after infarction	Beta-blocker, ACE inhibitor	Short-acting nifedipine
Chronic heart failure	ACE inhibitor/ARB/ARNI, beta-blocker, MRA, diuretic	Verapamil, diltiazem, alpha-blockers, moxonidine
Gout	Losartan, calcium antagonists	Thiazide and loop diuretics
Bronchial asthma	Calcium antagonists, ACE inhibitor/ARB	Nonselective beta-blockers
Prostatic hyperplasia	Alpha-1 blocker as an add-on	—
Pregnancy	Methyldopa, labetalol, nifedipine (sustained-release)	ACE inhibitors, ARBs, aliskiren (teratogenic); spironolactone; atenolol
Elderly, isolated systolic AH	Calcium antagonists, thiazide-like diuretics	Alpha-blockers (orthostatic hypotension), clonidine
Bilateral renal artery stenosis	Calcium antagonists	ACE inhibitors, ARBs

CLINICAL SIGNIFICANCE

If a woman planning pregnancy or already pregnant is taking an ACE inhibitor or ARB, it is **immediately** replaced with methyldopa, labetalol or nifedipine. When prescribing these drugs to women of reproductive age, advice on contraception must be given.

7.3. Hypertensive crisis

Hypertensive crisis is a sharp rise in BP (usually > 180/120 mm Hg). Two types are distinguished:

- **complicated (emergency) crisis** — with acute target-organ damage: hypertensive encephalopathy, stroke, acute coronary syndrome, pulmonary edema, aortic dissection, eclampsia, acute kidney injury. BP is lowered **stepwise** with **intravenous** drugs (in the first hour mean BP by no more than 25%; in aortic dissection — rapidly, to systolic < 120);
- **uncomplicated crisis** — without organ damage; BP is lowered gradually over 24–48 hours with **oral** drugs.

Table 11.8. *Drugs used in hypertensive crisis*

Situation	Drugs
Uncomplicated crisis	Captopril 25 mg orally or sublingually; sustained-release nifedipine, moxonidine, labetalol, clonidine orally
Hypertensive encephalopathy	Labetalol, nicardipine, nitroprusside intravenously
Acute coronary syndrome	Intravenous nitroglycerin, a beta-blocker (esmolol, metoprolol)
Acute left ventricular failure (pulmonary edema)	Intravenous nitroglycerin or nitroprusside + furosemide
Aortic dissection	Esmolol or labetalol + nitroprusside or nitroglycerin
Pheochromocytoma	Phentolamine; then a beta-blocker
Preeclampsia and eclampsia	Intravenous labetalol or hydralazine, oral nifedipine + magnesium sulfate (seizure prevention)
Crisis caused by cocaine/amphetamine	Benzodiazepines, nitrates, phentolamine; beta-blockers are contraindicated

CAUTION

In a hypertensive crisis, giving **short-acting nifedipine sublingually**, and lowering BP very quickly and sharply, is dangerous: cerebral, cardiac and renal ischemia (stroke, infarction) may develop. In an uncomplicated crisis, urgent hospitalization is usually not required — a calm environment, an oral drug and follow-up within 24–48 hours are enough.

8. Pharmacotherapy of chronic heart failure (brief)

In chronic heart failure (reduced ejection fraction, ≤ 40%), the **“four pillars” that reduce mortality**:

Table 11.9. *The main treatment of heart failure with reduced ejection fraction*

Pillar	Preparations	Comment
1. RAAS inhibitor	ARNI (sacubitril/valsartan) or an ACE inhibitor / ARB	ARNI inhibits neprilysin and increases natriuretic peptides; started 36 hours after an ACE inhibitor (risk of angioedema)
2. Beta-blocker	Bisoprolol, metoprolol succinate, carvedilol, nebivolol	Start with a very small dose, increasing gradually

Pillar	Preparations	Comment
3. Mineralocorticoid antagonist	Spironolactone, eplerenone	Monitoring of potassium and creatinine
4. SGLT2 inhibitor	Dapagliflozin, empagliflozin	Regardless of diabetes; reduces hospitalizations and mortality
Symptomatic	Loop diuretics (furosemide, torasemide)	Reduce edema and congestion but do not affect prognosis
Additional	Ivabradine (in sinus rhythm with HR > 70), digoxin, iron (intravenously in iron deficiency), vericiguat	In selected patients

CLINICAL SIGNIFICANCE

Drugs to avoid in heart failure: NSAIDs (fluid retention, kidneys), verapamil and diltiazem, thiazolidinediones (pioglitazone), most class I antiarrhythmics, dronedarone, moxonidine, alpha-blockers.

9. Hypertensive drugs (BP-raising drugs)

Used in **hypotension** (acute — collapse, shock; chronic — orthostatic and constitutional hypotension).

Table 11.10. *Hypertensive drugs*

Group	Preparations	Mechanism	Use
Adrenomimetics	Norepinephrine, epinephrine, phenylephrine (Mesaton), dopamine, ephedrine, midodrine	Stimulation of alpha-1 (and beta-1) receptors	Shock, collapse, hypotension of spinal anesthesia; midodrine — orthostatic hypotension
Vasopressin and its analogs	Vasopressin, terlipressin	Vasoconstriction via V_1 receptors	Septic shock (in addition to norepinephrine), bleeding from esophageal varices (terlipressin)
Angiotensin II	Angiotensinamide	AT_1 stimulation	Vasodilatory shock (rarely)
Mineralocorticoids	Fludrocortisone	Retains sodium and water and increases volume	Orthostatic hypotension, adrenal insufficiency
Glucocorticoids	Hydrocortisone, prednisolone	Increase sensitivity to catecholamines	Adrenal crisis, septic shock
Analeptics	Caffeine sodium benzoate, nikethamide (Cordiamine), camphor	Stimulate the vasomotor center	Mild hypotension (historically)
Volume-expanding solutions	Crystalloids (0.9% NaCl, Ringer's lactate), colloids (albumin)	Increase the circulating volume	Hypovolemic shock — the first measure
Tonics (adaptogens)	Schisandra, ginseng, eleutherococcus, modern — caffeine	Increase in general tone	Chronic hypotension, asthenic states (6th-semester independent study)

REMEMBER

In acute hypotension, the cause must be identified first: hypovolemia (blood loss, dehydration) — fluids; anaphylaxis — epinephrine; septic shock — fluids + norepinephrine; cardiogenic shock — inotropes; bradycardia — atropine; adrenal insufficiency — hydrocortisone. Vasopressors are ineffective and harmful in hypovolemia without adequate fluid replacement.

Practical skills

Skill 1. Drawing up a treatment plan for a patient newly diagnosed with AH

1. Measure BP correctly (repeatedly, on both arms) and teach home measurement.
2. Assess cardiovascular risk and target-organ damage (ECG, creatinine, GFR, potassium, glucose, lipids, urine albumin).
3. Give advice on lifestyle changes.
4. Choose a drug taking comorbidities into account (Table 11.7).
5. In most patients, start with a fixed combination of an ACE inhibitor/ARB + a calcium antagonist or a diuretic.
6. Check creatinine and potassium 1–2 weeks after starting an ACE inhibitor/ARB.
7. Follow-up at 4 weeks; if the target is not reached — a combination of three drugs.

Skill 2. Key prescriptions

PRESCRIPTION 11.1. ENALAPRIL
Rp.: Enalaprili 0.01
D.t.d. N. 30 in tabulettis
S. Take 1 tablet 1–2 times a day.

PRESCRIPTION 11.2. LISINOPRIL
Rp.: Lisinoprili 0.01
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 11.3. CAPTOPRIL
Rp.: Captoprili 0.025
D.t.d. N. 20 in tabulettis
S. When BP rises sharply, chew 1 tablet or place it under the tongue.

PRESCRIPTION 11.4. LOSARTAN
Rp.: Losartani kalii 0.05
D.t.d. N. 30 in tabulettis obductis
S. Take 1 tablet once a day.

PRESCRIPTION 11.5. AMLODIPINE + PERINDOPRIL (FIXED COMBINATION)
Rp.: Tabulettas “Amlodipinum + Perindoprilum” 5 mg/5 mg N. 30
D.S. Take 1 tablet once a day in the morning.

PRESCRIPTION 11.6. INDAPAMIDE
Rp.: Tabulettas Indapamidi 0.0015 retard N. 30
D.S. Take 1 tablet once a day in the morning.

PRESCRIPTION 11.7. HYDROCHLOROTHIAZIDE
Rp.: Hydrochlorothiazidi 0.025
D.t.d. N. 20 in tabulettis
S. Take 1/2–1 tablet once a day in the morning.

PRESCRIPTION 11.8. SPIRONOLACTONE
Rp.: Spironolactoni 0.025
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day (under potassium monitoring).

PRESCRIPTION 11.9. MOXONIDINE
Rp.: Moxonidini 0.0002
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 11.10. METHYLDOPA (IN PREGNANCY)
Rp.: Methyldopae 0.25
D.t.d. N. 50 in tabulettis
S. Take 1 tablet 2–3 times a day.

PRESCRIPTION 11.11. SACUBITRIL/VALSARTAN
Rp.: Tabulettas “Sacubitrilum + Valsartanum” 49 mg/51 mg N. 28
D.S. Take 1 tablet 2 times a day (in heart failure).

PRESCRIPTION 11.12. SODIUM NITROPRUSSIDE
Rp.: Natrii nitroprussidi 0.05
D.t.d. N. 5
S. Dissolve the contents of the ampoule in 500 ml of 5% glucose and infuse intravenously through a light-protected line at 0.3–3 mcg/kg/min, under BP monitoring.

PRESCRIPTION 11.13. MIDODRINE
Rp.: Midodrini hydrochloridi 0.0025
D.t.d. N. 30 in tabulettis
S. Take 1 tablet 3 times a day (the last at least 4 hours before going to bed).

Review questions

1. Systems that regulate BP. The structure of the RAAS and the effects of angiotensin II.
2. Classification of antihypertensive drugs. First-line drugs.
3. Mechanism, organ-protective effects and uses of ACE inhibitors.
4. Adverse effects and contraindications of ACE inhibitors.
5. How ARBs differ from ACE inhibitors.
6. Why is giving an ACE inhibitor and an ARB together not recommended?
7. Uses and adverse effects of spironolactone and eplerenone.

8. The place of calcium antagonists and diuretics in AH.
9. Mechanism of centrally acting antihypertensive drugs. Clonidine rebound syndrome.
10. Myotropic vasodilators: hydralazine, minoxidil, sodium nitroprusside.
11. Principles of treating AH and the target BP.
12. Rational and irrational combinations.
13. Which drugs are used in hypertension of pregnancy, and which are contraindicated?
14. Types of hypertensive crisis and their treatment.
15. The “four pillars” of chronic heart failure treatment.
16. Classification and uses of hypertensive drugs.

Test questions

1. The reason ACE inhibitors cause a dry cough:
   A) Accumulation of angiotensin II
   B) Reduced breakdown of bradykinin
   C) Histamine release
   D) Increased aldosterone
   E) Blockade of beta-2 receptors

2. Antihypertensive drugs contraindicated in bilateral renal artery stenosis:
   A) Alpha-blockers
   B) Calcium antagonists
   C) ACE inhibitors
   D) Thiazide-like diuretics
   E) Beta-blockers

3. A drug that blocks AT_1 receptors and does not cause cough:
   A) Spironolactone
   B) Losartan
   C) Enalapril
   D) Aliskiren
   E) Captopril

4. The drug of choice for arterial hypertension in pregnancy:
   A) Losartan
   B) Methyldopa
   C) Enalapril
   D) Spironolactone
   E) Aliskiren

5. A drug that causes a rebound hypertensive crisis on abrupt withdrawal:
   A) Clonidine
   B) Amlodipine
   C) Lisinopril
   D) Indapamide
   E) Losartan

6. A selective agonist of I_1-imidazoline receptors:
   A) Clonidine
   B) Methyldopa
   C) Moxonidine
   D) Prazosin
   E) Hydralazine

7. A drug not recommended in AH accompanied by gout:
   A) Amlodipine
   B) Losartan
   C) Lisinopril
   D) Hydrochlorothiazide
   E) Bisoprolol

8. A characteristic adverse effect of dihydropyridine calcium antagonists:
   A) Ankle edema
   B) Bradycardia
   C) Gynecomastia
   D) Dry cough
   E) Hyperkalemia

9. An antihypertensive drug with a risk of cyanide toxicity with prolonged use:
   A) Hydralazine
   B) Minoxidil
   C) Magnesium sulfate
   D) Sodium nitroprusside
   E) Diazoxide

10. A drug that causes gynecomastia because of its antiandrogenic effect:
    A) Spironolactone
    B) Losartan
    C) Eplerenone
    D) Furosemide
    E) Amlodipine

11. The main drug for preventing seizures in eclampsia:
    A) Clonidine
    B) Furosemide
    C) Magnesium sulfate
    D) Diazepam
    E) Phenytoin

12. An irrational combination:
    A) Perindopril + indapamide
    B) Amlodipine + indapamide
    C) Enalapril + losartan
    D) Lisinopril + amlodipine
    E) Losartan + hydrochlorothiazide

13. A combination that inhibits neprilysin and reduces mortality in chronic heart failure:
    A) Perindopril/indapamide
    B) Amlodipine/valsartan
    C) Hydrochlorothiazide/losartan
    D) Sacubitril/valsartan
    E) Bisoprolol/amlodipine

14. The “fourth pillar” drug group used in chronic heart failure regardless of diabetes:
    A) DPP-4 inhibitors
    B) Insulin
    C) Thiazolidinediones
    D) SGLT2 inhibitors
    E) Sulfonylureas

15. The first-choice hypertensive drug in septic shock after fluid replacement:
    A) Norepinephrine
    B) Caffeine
    C) Fludrocortisone
    D) Midodrine
    E) Cordiamine

Clinical cases

Case 1

A 54-year-old man with type 2 diabetes mellitus, BP 158/96 mm Hg, raised urine albumin/creatinine ratio, normal creatinine, potassium 4.3 mmol/l. Previously untreated.

Questions: 1) Which drug(s) would you start with? 2) What monitoring is needed? 3) Target BP?

Answer: 1) An ACE inhibitor or ARB (nephroprotective — reduces albuminuria) + a calcium antagonist (amlodipine) or a thiazide-like diuretic (indapamide) — in a fixed combination; consider adding an SGLT2 inhibitor for diabetes and kidney protection. 2) Creatinine and potassium after 1–2 weeks (if GFR falls by more than 30% or potassium is > 5.5 — reconsider the dose), home BP monitoring. 3) < 130/80 mm Hg.

Case 2

A 32-year-old woman has been taking enalapril for AH for 2 years. She has found out that she is 10 weeks pregnant.

Questions: 1) Management? 2) Which drugs can she be switched to?

Answer: 1) Stop enalapril immediately — ACE inhibitors (especially in the 2nd–3rd trimesters) impair fetal kidney development and may lead to oligohydramnios, skull hypoplasia and fetal death. When taken in the 1st trimester, the risk is relatively low, but the fetus must be monitored by ultrasound. 2) Methyldopa, labetalol or sustained-release nifedipine; follow-up jointly by an obstetrician-gynecologist and an internist; monitoring for signs of preeclampsia (proteinuria, edema, headache).

Case 3

A 60-year-old patient was brought in with BP 200/120 mm Hg, severe headache, nausea, blurred vision and confusion. No focal neurological signs. Papilledema on the fundus.

Questions: 1) Diagnosis? 2) Treatment approach and rate of BP lowering.

Answer: 1) Complicated hypertensive crisis — hypertensive encephalopathy. 2) Admission to the intensive care unit, BP monitoring; titratable intravenous drugs — labetalol, nicardipine or sodium nitroprusside; in the first hour lower mean BP by no more than 20–25%, over the next 2–6 hours to 160/100, then gradually to normal over 24–48 hours. Abrupt lowering causes cerebral ischemia. Short-acting nifedipine sublingually is contraindicated.

Quick review table

Table 11.11. *Antihypertensive and hypertensive drugs — summary*

Group	Preparations	Mechanism	Main indications	Main risk
ACE inhibitors	Enalapril, lisinopril, ramipril, perindopril	Ang II ↓, bradykinin ↑	AH, HF, diabetic nephropathy, after infarction	Cough, angioedema, hyperkalemia; pregnancy, bilateral stenosis
ARBs	Losartan, valsartan, telmisartan	AT_1 blockade	As ACE inhibitors; when there is cough	Hyperkalemia; pregnancy
MRA	Spironolactone, eplerenone	Aldosterone blockade	Resistant AH, HF, hyperaldosteronism	Hyperkalemia, gynecomastia
ARNI	Sacubitril/valsartan	Neprilysin + AT_1	HF	Hypotension, angioedema
Calcium antagonists	Amlodipine, lercanidipine, sustained-release nifedipine	L-type Ca_{2+} blockade	AH (elderly), angina	Edema, headache
Thiazide-like diuretics	Indapamide, chlorthalidone, hydrochlorothiazide	Blockade of Na_+-Cl_- cotransport	AH	Hypokalemia, hyperuricemia
Beta-blockers	Bisoprolol, nebivolol, carvedilol	Beta-1 blockade, renin ↓	AH + IHD, HF, arrhythmia	Bradycardia, bronchospasm
Centrally acting	Clonidine, methyldopa, moxonidine	Alpha-2 / I_1 agonism	Crisis, pregnancy (methyldopa), metabolic syndrome	Rebound, sedation
Myotropic	Hydralazine, minoxidil, nitroprusside	Direct vasodilation	Resistant AH, crisis	Reflex tachycardia, lupus, cyanide
Hypertensive	Norepinephrine, phenylephrine, midodrine, vasopressin, fludrocortisone	Vasoconstriction / volume ↑	Shock, orthostatic hypotension	Ischemia, arrhythmia

TOPIC 12

Drugs acting on the digestive system

Learning objectives

Gastrointestinal diseases are among the most common reasons for visiting a GP: dyspepsia, gastroesophageal reflux disease (GERD), gastric and duodenal ulcer, constipation, diarrhea, nausea and vomiting, biliary tract and liver diseases. Proton pump inhibitors are among the most prescribed drugs in the world, but they are often used without reason and for too long. Eradication of Helicobacter pylori has radically changed the treatment of peptic ulcer disease. This topic covers the main groups of digestive system pharmacology based on modern recommendations.

The student should know:

- the classification of drugs acting on the digestive system;
- the regulation of gastric secretion and the mechanism of antisecretory drugs (PPIs, H_2 blockers), antacids and gastroprotectors;
- H. pylori eradication regimens;
- emetic and antiemetic drugs, prokinetics, laxatives and antidiarrheals;
- choleretic and cholagogue drugs, hepatoprotectors, pancreatic enzymes and probiotics.

The student should be able to:

- write prescriptions for drugs such as omeprazole, famotidine, Almagel, De-Nol, ondansetron, metoclopramide, domperidone, loperamide, lactulose, bisacodyl, ursodeoxycholic acid and pancreatin;
- draw up an H. pylori eradication regimen;
- give instructions on treating dehydration in diarrhea and vomiting.

1. Classification

Table 12.1. *Classification of drugs acting on the digestive system*

Group	Subgroup	Preparations
Drugs affecting appetite	Appetite stimulants	Bitters (wormwood tincture), megestrol, cyproheptadine
	Appetite suppressants (anorexigenics)	Sibutramine (historically), GLP-1 agonists (semaglutide, liraglutide)
Drugs affecting the function of the gastric glands	Replacement therapy	Dilute hydrochloric acid, pepsin, gastric juice (historically)
	Antisecretory	Proton pump inhibitors (PPIs), H_2-histamine blockers, M_1-cholinergic blockers (pirenzepine), potassium-competitive acid blockers (vonoprazan)
Antacids	Absorbable and nonabsorbable	Sodium bicarbonate, calcium carbonate; aluminum and magnesium hydroxides (Almagel, Maalox), alginates
Gastroprotectors	—	Bismuth tripotassium dicitrate (De-Nol), sucralfate, misoprostol, rebamipide
Emetics and antiemetics	Emetics	Apomorphine, ipecacuanha (historically)

Group	Subgroup	Preparations
	Antiemetics	5-HT_3 antagonists (ondansetron), D_2 antagonists (metoclopramide, domperidone), NK_1 antagonists (aprepitant), H_1 blockers, scopolamine, dexamethasone
Drugs affecting gastrointestinal motility	Prokinetics	Metoclopramide, domperidone, itopride, erythromycin (a motilin agonist)
	Antispasmodics	Drotaverine, papaverine, mebeverine, otilonium bromide, hyoscine butylbromide
	Laxatives	Osmotic, contact (stimulant), bulk-forming, softening
	Antidiarrheals	Loperamide, adsorbents (smectite), rehydration solutions
Drugs acting on the liver and biliary tract	Choleretics and cholagogues	Choleretics (Allochol, bile acids), cholekinetics (magnesium sulfate)
	Gallstone dissolution	Ursodeoxycholic acid
	Hepatoprotectors	Ursodeoxycholic acid, ademetionine, silymarin, essential phospholipids
Drugs affecting pancreatic function	Enzyme preparations	Pancreatin (Creon, Mezym)
	Proteolysis inhibitors	Aprotinin (historically)
Drugs normalizing the intestinal microflora	—	Probiotics, prebiotics (6th-semester independent study)

2. Drugs affecting gastric secretion

2.1. Regulation of gastric acid secretion

The **parietal cells** of the stomach produce hydrochloric acid. Their secretion is regulated by three main stimulants:

- **acetylcholine** (vagus nerve) — via M_3 receptors;
- **gastrin** (antral G cells) — via CCK_2 receptors and by releasing histamine from enterochromaffin-like cells;
- **histamine** (enterochromaffin-like cells) — via **H_2 receptors** (cAMP ↑) — the main final stimulant. All pathways converge on **H^+/K^+-ATPase (the proton pump)** — the **final common step** of acid secretion.

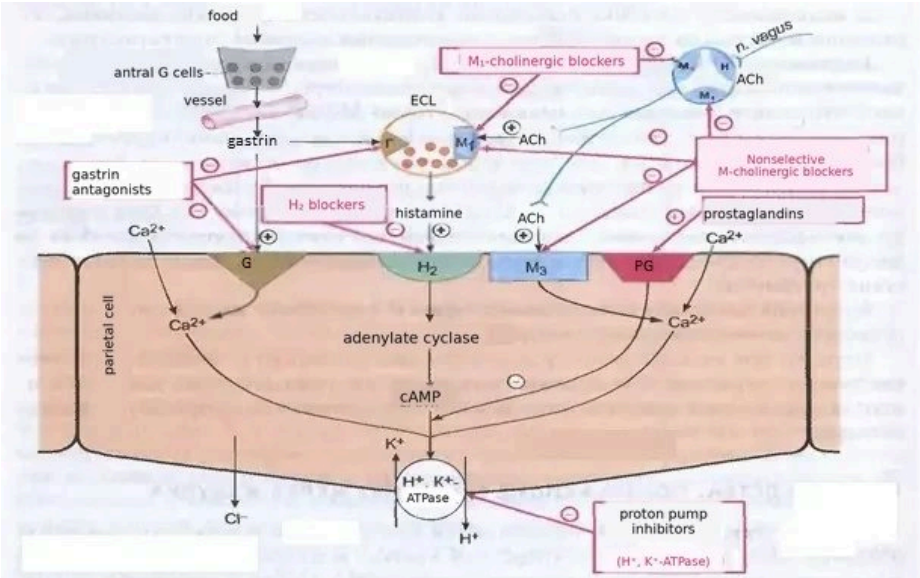


Fig. 15.1. Sites of action of drugs that reduce the production of gastric hydrochloric acid

ECL — enterochromaffin-like cells (paracrine cells); G — receptors that interact with gastrin (CCK-B cholecystokinin receptors); proglumide is a blocker of these receptors; PG — prostaglandin receptors; H_2 — histamine receptors; M — M-cholinergic receptors; N — N-cholinergic receptors; ACh — acetylcholine; G cells contain gastrin; plus — stimulatory effect; minus — blocking effect

Figure 12.1. *Production of hydrochloric acid in the stomach and the sites of action of drugs that reduce it: M_1-cholinergic blockers, H_2 blockers, gastrin antagonists, prostaglandins, proton pump inhibitors (teaching material of the TMA Department of Pharmacology)*

2.2. Proton pump inhibitors (PPIs)

Preparations: omeprazole, esomeprazole, pantoprazole, lansoprazole, rabeprazole.

Mechanism. PPIs are **prodrugs**, unstable in acid, so they are produced in **enteric-coated** capsules or tablets. After absorption from the intestine they reach the parietal cell via the blood; in the acidic environment of the secretory canaliculi they are converted into the active **sulfenamide**, which binds **irreversibly and covalently** to the SH groups of H^+/K^+-ATPase. Acid secretion falls by more than 90%; the effect lasts until new pumps are synthesized (more than 24 hours), although t½ is only 1–2 hours.

> **REMEMBER**
> PPIs block only **actively working** pumps. Therefore they are taken **30–60 minutes before meals** (usually before breakfast) — when food activates the pumps, the drug is at its peak blood concentration. The full effect develops in 3–5 days; a PPI is not a quick “switch-off” (symptomatic, as-needed) drug.

Uses: GERD and reflux esophagitis, gastric and duodenal ulcer, **H. pylori eradication** (with antibiotics), prevention and treatment of NSAID gastropathy, Zollinger–Ellison syndrome, ulcer bleeding (intravenously), functional dyspepsia.

Adverse effects (with long-term use): headache, diarrhea or constipation; increased risk of **Clostridioides difficile infection** and pneumonia; reduced absorption of **vitamin B_{12}, magnesium, iron and calcium** (hypomagnesemia, osteoporosis, bone fractures), interstitial nephritis; hypergastrinemia; **rebound acid hypersecretion** on abrupt withdrawal.

CLINICAL SIGNIFICANCE

Omeprazole and esomeprazole inhibit CYP2C19 and may reduce the conversion of **clopidogrel** into its active metabolite. If a patient taking clopidogrel needs gastroprotection, **pantoprazole** is preferred. In every patient taking a PPI long term, the indication should be reviewed regularly and, where possible, the dose reduced or the drug stopped ("deprescribing").

2.3. H_2-histamine receptor blockers

Preparations: famotidine (the main one), ranitidine (withdrawn from the market in many countries in 2020 because of N-nitrosodimethylamine (NDMA) contamination), cimetidine (outdated).

By competitively blocking the H_2 receptors of parietal cells, they reduce basal secretion, especially at **night**. They are weaker than PPIs, and tolerance develops quickly. Uses: mild GERD, dyspepsia, an add-on to a PPI (nocturnal acid "breakthrough"). **Cimetidine** — a CYP450 inhibitor (many interactions), antiandrogenic effect (gynecomastia) — no longer used.

2.4. Antacids

Antacids chemically neutralize hydrochloric acid in the stomach. The effect begins quickly (minutes) but is short (30–60 minutes on an empty stomach; up to 3 hours if taken 1 hour after a meal).

Table 12.2. *Antacids*

Type	Preparations	Advantage	Disadvantage
Absorbable	Sodium bicarbonate (baking soda), calcium carbonate	Very rapid action	CO_2 formation — belching, gastric distension; acid rebound; systemic alkalosis; sodium load (AH, HF); with calcium — milk-alkali syndrome, hypercalcemia
Nonabsorbable	Aluminum hydroxide, magnesium hydroxide and trisilicate (Almagel, Maalox, Phosphalugel)	Longer action, no rebound; coating and adsorbing effect	Aluminum — constipation, phosphate binding (hypophosphatemia); magnesium — diarrhea (balanced in combinations); aluminum and magnesium accumulate in renal failure
Alginates	Sodium alginate (Gaviscon)	Forms a "raft" in the stomach and prevents reflux	—

CAUTION

Antacids reduce the absorption of many drugs (**tetracyclines, fluoroquinolones**, iron preparations, digoxin, levothyroxine, bisphosphonates). Other drugs are taken **at least 2 hours before or 4 hours after** an antacid.

2.5. Gastroprotectors

- **Bismuth tripotassium dicitrate (De-Nol)** — forms a protective **film** on the ulcer surface in an acidic environment, increases the synthesis of prostaglandins and bicarbonate, and **acts against H. pylori** (in quadruple regimens). Adverse effect: **darkening of the stool and tongue** (harmless); with long-term use at large doses — encephalopathy (contraindicated in renal failure).
- **Sucralfate** — forms a sticky paste in an acidic environment and coats the ulcer base. Constipation; reduces the absorption of other drugs.

- **Misoprostol** — a prostaglandin E_1 analog: reduces acid secretion and increases the production of mucus and bicarbonate. **Prevention of NSAID gastropathy**. Adverse effects: diarrhea; **uterine contractions — absolutely contraindicated in pregnancy** (it causes abortion; it is used for this purpose in obstetrics).

2.6. Eradication of Helicobacter pylori

H. pylori is the main cause of gastric and duodenal ulcer, chronic gastritis, gastric cancer and MALT lymphoma. Eradication sharply reduces ulcer recurrence.

Table 12.3. *H. pylori eradication regimens (based on the Maastricht consensus, 14 days)*

Regimen	Composition	When
Bismuth quadruple therapy	PPI + bismuth + tetracycline + metronidazole	First line in areas with high clarithromycin resistance (> 15%); or second line
Clarithromycin triple therapy	PPI + amoxicillin + clarithromycin	If clarithromycin resistance is low or susceptibility has been confirmed
Non-bismuth quadruple therapy (concomitant)	PPI + amoxicillin + clarithromycin + metronidazole	When bismuth is not available
Levofloxacin-based	PPI + amoxicillin + levofloxacin	Second line

REMEMBER
The effectiveness of eradication is checked **at least 4 weeks** after it (and 2 weeks after stopping the PPI) with a **urea breath test** or a stool antigen test. A serological test (antibodies) is unsuitable for assessing the result of eradication — antibodies persist for months.

3. Emetic and antiemetic drugs

3.1. Mechanism of vomiting

The vomiting center is located in the medulla oblongata and receives impulses from:

- **the chemoreceptor trigger zone** (in the floor of the fourth ventricle, outside the blood-brain barrier; D_2, $5\text{-}HT_3$ and NK_1 receptors) — circulating toxins, drugs (cytostatics, opioids, glycosides), metabolites;
- **the gastrointestinal tract** — vagal afferents ($5\text{-}HT_3$; cytostatics release serotonin from enterochromaffin cells);
- **the vestibular apparatus** (M_1, H_1 receptors) — motion sickness;
- the cerebral cortex (fear, smells, “anticipatory vomiting”).

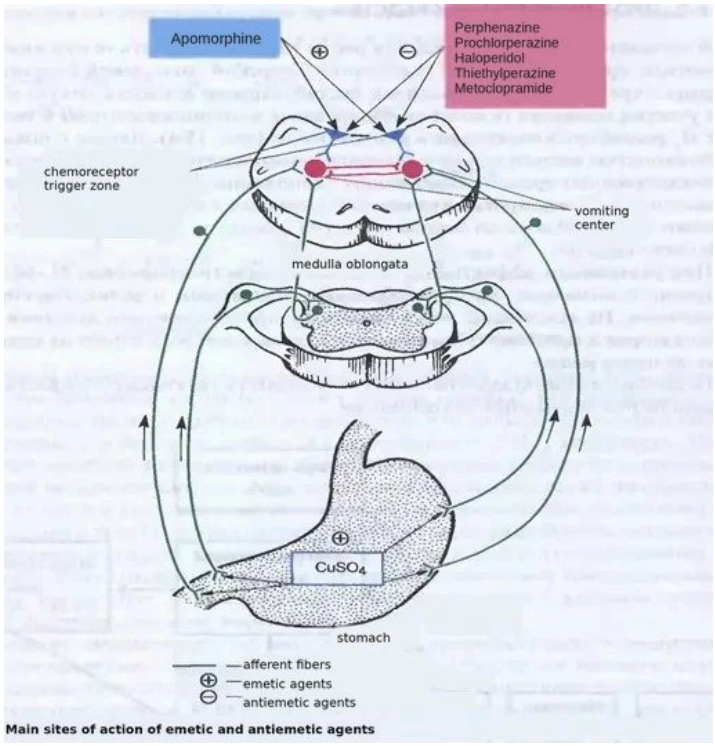


Figure 12.2. *Pathways of vomiting and the sites of action of antiemetics: chemoreceptor trigger zone, vomiting center, afferent fibers (teaching material of the TMA Department of Pharmacology)*

3.2. Emetics

Apomorphine stimulates D_2 receptors in the trigger zone; given subcutaneously. **Ipecacuanha** irritates the stomach reflexly. They were formerly used in poisoning; now **not recommended** (risk of aspiration, unproven effectiveness) — gastric lavage and activated charcoal are preferred in poisoning. Apomorphine is now used in Parkinson's disease.

3.3. Antiemetics

Table 12.4. *Antiemetic drugs*

Group	Preparations	Site of action	Main use	Adverse effects
5-HT_3 antagonists	Ondansetron, granisetron, palonosetron	Trigger zone, vagal afferents	Vomiting from chemotherapy and radiotherapy, postoperative vomiting, gastroenteritis	Constipation, headache, QT prolongation
D_2 antagonists	Metoclopramide	Trigger zone + peripheral prokinetic	Vomiting, gastroparesis, nausea in migraine	Extrapyramidal disorders (acute dystonia, especially in children and young people), hyperprolactinemia, tardive dyskinesia (not recommended for more than 5 days)
	Domperidone	Trigger zone (crosses the BBB poorly) + peripheral	Dyspepsia, nausea; vomiting from levodopa in Parkinson's disease	QT prolongation and risk of sudden death (at large doses); few extrapyramidal effects

Group	Preparations	Site of action	Main use	Adverse effects
	Antipsychotics (droperidol, haloperidol, prochlorperazine)	Trigger zone	Postoperative, palliative vomiting	Sedation, extrapyramidal effects
NK_1 antagonists	Aprepitant, fosaprepitant	Central (substance P)	Highly emetogenic chemotherapy (delayed vomiting)	CYP3A4 interactions
H_1 blockers	Dimenhydrinate, meclizine, promethazine	Vestibular, vomiting center	Motion sickness, vomiting of pregnancy (doxylamine + pyridoxine)	Drowsiness, anticholinergic effects
M-cholinergic blockers	Scopolamine	Vestibular	Motion sickness (patch)	Dry mouth, drowsiness
Glucocorticoids	Dexamethasone	Unknown	In combination for chemotherapy-induced and postoperative vomiting	—
Cannabinoids	Nabilone, dronabinol	CB_1	Refractory chemotherapy-induced vomiting (in some countries)	Psychotropic

MEMORY AID

The drug according to the cause of vomiting: **“Chemo — Ondansetron, Sea — Dimenhydrinate and scopolamine, Slow stomach — Metoclopramide/domperidone, Delayed chemo — Aprepitant”**.

4. Drugs affecting gastrointestinal motility

4.1. Prokinetics

Metoclopramide (D_2 blocker + 5-HT_4 agonist), **domperidone** (peripheral D_2 blocker), **itopride** (D_2 blocker + anticholinesterase) — speed up gastric emptying and increase the tone of the lower esophageal sphincter. Uses: gastroparesis (diabetic), functional dyspepsia, reflux (as an add-on). **Erythromycin** at a small dose stimulates motilin receptors (in gastric bleeding before endoscopy).

4.2. Antispasmodics

- **Myotropic**: drotaverine (No-Spa), papaverine — phosphodiesterase-4 inhibitors (cAMP ↑) that relax smooth muscle directly; **mebeverine** (Na^+ channel blockade) and otilonium bromide — selective for the intestine, in **irritable bowel syndrome**.
- **Neurotropic**: hyoscine butylbromide (Buscopan), platyphylline (Topic 4).
- **Peppermint oil** capsules — in irritable bowel syndrome.

4.3. Laxatives

Table 12.5. *Classification of laxatives*

Group	Preparations	Mechanism	Onset of action	Features
Bulk-forming (ballast)	Bran, flaxseed, psyllium (ispaghula)	Absorb water and increase stool volume	1–3 days	First line in chronic constipation; plenty of water is essential
Osmotic — saline	Magnesium sulfate, sodium sulfate	Retain water in the intestine	1–6 hours	In poisoning, bowel preparation; magnesium is dangerous in renal failure
Osmotic — sugars and alcohols	Lactulose, sorbitol	Broken down by bacteria to form acidic metabolites	1–3 days	Children, pregnant women; in hepatic encephalopathy (reduces ammonia formation and absorption)
Osmotic — macrogols	Polyethylene glycol (macrogol)	Retains water, not absorbed	1–2 days	Chronic constipation, preparation for colonoscopy (in large volumes)
Contact (stimulant)	Bisacodyl, sodium picosulfate, senna (anthraglycosides), castor oil	Stimulate the nerve plexuses of the colon, increase secretion	6–12 hours (oral), 15–60 minutes (suppository)	Short term; with long-term use — tolerance, hypokalemia, intestinal atony (“lazy bowel”), senna — melanosis coli
Softeners	Liquid paraffin, docusate, glycerin suppositories	Soften and lubricate the stool	6–24 hours	Liquid paraffin reduces the absorption of fat-soluble vitamins; lipoid pneumonia with aspiration
Peripheral opioid antagonists	Methylnaltrexone, naloxegol	Block mu receptors in the intestine	Hours	Opioid-induced constipation

CAUTION

Laxatives are **contraindicated** in **intestinal obstruction**, an acute abdomen (suspected appendicitis), and abdominal pain of unknown cause. In treating constipation, diet (fiber), fluids and physical activity come first; of the drugs — bulk-forming and osmotic laxatives. Regular long-term use of contact (stimulant) laxatives is not recommended.

4.4. Antidiarrheal drugs

The main treatment of acute diarrhea is rehydration. Oral rehydration solution (the WHO low-osmolarity ORS: sodium chloride, potassium chloride, sodium citrate, glucose) sharply reduces dehydration and death: glucose enhances the absorption of sodium and water in the intestine (sodium-glucose cotransport, SGLT1). In children, **zinc** preparations (10–14 days) are recommended together with ORS.

- **Loperamide** — a peripheral mu-opioid agonist that hardly crosses the BBB; it slows peristalsis and increases anal sphincter tone. Symptomatic in acute nonspecific diarrhea in adults. **Contraindicated in bloody diarrhea, high fever** (dysentery, invasive infection), C. difficile colitis and **in children under 2** (toxic megacolon, ileus). At large doses — cardiotoxicity (QT prolongation).
- **Adsorbents** — diosmectite (Smecta), activated charcoal (Topic 3).

- **Racecadotril** — an enkephalinase inhibitor; antisecretory effect.
- **Probiotics** (Lactobacillus rhamnosus GG, Saccharomyces boulardii) — slightly shorten the duration of acute diarrhea in children; prevention of antibiotic-associated diarrhea.

CLINICAL SIGNIFICANCE

In acute diarrhea, **antibiotics** are used only when indicated: dysentery (bloody stool + fever), cholera, severe travelers' diarrhea, immunodeficiency. Most acute diarrheas (viral) do not require antibiotics.

5. Drugs affecting liver and biliary function

5.1. Choleretics and cholagogues

- **Choleretics** enhance bile formation in the liver: those containing bile acids (Allochol, Cholenzym), synthetic (hymecromone, osalmid), herbal (corn silk, immortelle, rose hip — Cholosas).
- **Cholekinetics** enhance gallbladder contraction and relaxation of the sphincter of Oddi: magnesium sulfate, sorbitol, xylitol, cholecystokinin.
- **Cholespasmolytics** relax biliary spasm: drotaverine, papaverine, hymecromone, M-cholinergic blockers.

CAUTION

Choleretic drugs (especially cholekinetics) are contraindicated in **gallstone disease** — a stone may become lodged in the bile duct and lead to obstructive jaundice and pancreatitis. They are also not used in acute cholecystitis, acute hepatitis and pancreatitis. The effectiveness of most herbal choleretics has not been proven in modern studies.

5.2. Ursodeoxycholic acid (UDCA)

UDCA is a hydrophilic bile acid. Mechanism: it reduces cholesterol saturation of bile (**dissolves cholesterol stones**), displaces toxic hydrophobic bile acids and protects hepatocytes and cholangiocytes, choleresis and immunomodulation.

Uses: primary biliary cholangitis (the main drug), small (< 15 mm) radiolucent cholesterol stones (with preserved gallbladder function), intrahepatic cholestasis of pregnancy, cholestatic liver diseases, prevention of stone formation after bariatric surgery.

5.3. Hepatoprotectors

Hepatoprotectors are a heterogeneous group intended to protect liver cells from harmful influences: **ademetionine** (Heptral), **silymarin** (milk thistle — Karsil), **essential phospholipids** (Essentiale), ornithine aspartate (Hepa-Merz — reduces ammonia in hepatic encephalopathy).

CAUTION

Hepatoprotectors are very widely prescribed in CIS countries, but the effectiveness of most of them (silymarin, essential phospholipids) **has not been proven** in high-quality clinical trials and they are not included in international recommendations. In liver diseases the main focus is on **treating the cause**: antiviral therapy in viral hepatitis, abstinence from alcohol in alcoholic disease, weight loss in fatty liver disease, **N-acetylcysteine** in paracetamol poisoning.

6. Drugs acting on the pancreas and the intestinal microflora

Pancreatin — a mixture of **lipase, amylase and protease** obtained from the porcine pancreas. Uses: **exocrine pancreatic insufficiency** (chronic pancreatitis, cystic fibrosis, after pancreatectomy). The dose is set in **lipase units** (25,000–50,000 units per main meal) and taken **with food**. Enteric-coated microsphere capsules (Creon) are preferred. In cystic fibrosis, very large doses may cause fibrosing colonopathy.

Probiotics are live microorganisms that confer a benefit when taken in adequate amounts (lactobacilli, bifidobacteria, Saccharomyces boulardii). **Prebiotics** are nondigestible substances that stimulate the growth of beneficial microflora (inulin, lactulose, oligosaccharides). Their effectiveness depends on the strain; proven indications are limited (antibiotic-associated diarrhea, acute gastroenteritis in children).

> **CAUTION**
> The diagnosis of **“dysbacteriosis”** and prescribing probiotics to treat it on the basis of stool analysis is widespread in CIS countries, but international medicine does not recognize it as a separate disease.

7. Clinical part

Table 12.6. *Choice of drugs in gastrointestinal diseases*

Disease	Main treatment	Additional
GERD	PPI (4–8 weeks), then the lowest effective dose or as needed	Alginates, antacids (symptomatic); weight loss, limiting late meals, raising the head of the bed
H. pylori-associated ulcer	Eradication (14 days) + PPI (until the ulcer heals)	Stopping the NSAID
Prevention of NSAID gastropathy (in a risk group)	PPI	Misoprostol; a selective COX-2 inhibitor
Ulcer bleeding	Endoscopic hemostasis + intravenous PPI	Blood replacement
Functional dyspepsia	Eradication if H. pylori is present; PPI; prokinetics	Low-dose TCA
Irritable bowel syndrome	Antispasmodics (mebeverine, peppermint oil); loperamide for diarrhea; psyllium for constipation	Low-dose TCA, diet (FODMAP)
Acute gastroenteritis (children)	ORS + zinc	Ondansetron (for vomiting, once)
Chemotherapy-induced vomiting	Ondansetron + dexamethasone ± aprepitant	—
Opioid-induced constipation	Osmotic and contact laxatives together	Methylnaltrexone
Hepatic encephalopathy	Lactulose, rifaximin	Ornithine aspartate

Practical skills

Skill 1. Teaching oral rehydration at home

1. Dissolve a ready-made ORS sachet in the stated volume (usually 1 liter or 200–500 ml) of clean, boiled, cooled water.
2. Give the child after each loose stool: under 2 years — 50–100 ml, 2–10 years — 100–200 ml, adults — as much as they want.
3. If the child is vomiting — give 1 teaspoon every 1–2 minutes; pause for 10 minutes and continue.
4. Do not stop breastfeeding or normal feeding.
5. Warning signs: repeated vomiting, bloody stool, lethargy, reduced urine output, fever — see a doctor immediately.
6. The prepared solution is discarded after 24 hours.

Skill 2. Key prescriptions

PRESCRIPTION 12.1. OMEPRAZOLE
Rp.: Capsulas Omeprazoli 0.02 N. 28
D.S. Take 1 capsule once a day 30 minutes before breakfast.

PRESCRIPTION 12.2. PANTOPRAZOLE (INTRAVENOUS)
Rp.: Pantoprazoli 0.04
D.t.d. N. 10
S. Dissolve the contents of the vial in 10 ml of 0.9% NaCl and inject intravenously 1–2 times a day.

PRESCRIPTION 12.3. FAMOTIDINE
Rp.: Famotidini 0.02
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet 2 times a day (in the morning and before bed).

PRESCRIPTION 12.4. ANTACID (ALMAGEL)
Rp.: Suspensionis “Almagel” 170 ml
D.S. Take 1–2 measuring spoons 1 hour after meals and before bed; shake before use.

PRESCRIPTION 12.5. BISMUTH TRIPOTASSIUM DICITRATE
Rp.: Bismuthi tripotassii dicitratis 0.12
D.t.d. N. 56 in tabulettis obductis
S. 2 tablets 2 times a day 30 minutes before meals, for 14 days (in an eradication regimen).

PRESCRIPTION 12.6. ONDANSETRON
Rp.: Ondansetroni 0.008
D.t.d. N. 10 in tabulettis obductis
S. 1 tablet 1 hour before chemotherapy, then 1 tablet every 12 hours for 2 days.

PRESCRIPTION 12.7. METOCLOPRAMIDE

Rp.: Solutionis Metoclopramidi 0.5% — 2 ml
D.t.d. N. 5 in ampullis
S. 2 ml intramuscularly or slowly intravenously, up to 3 times a day (for no more than 5 days).

PRESCRIPTION 12.8. DOMPERIDONE

Rp.: Domperidoni 0.01
D.t.d. N. 30 in tabulettis
S. 1 tablet 3 times a day 15–30 minutes before meals, for up to 7 days.

PRESCRIPTION 12.9. DROTAVERINE

Rp.: Drotaverini hydrochloridi 0.04
D.t.d. N. 20 in tabulettis
S. Take 1–2 tablets 2–3 times a day (for pain).

PRESCRIPTION 12.10. MEBEVERINE

Rp.: Capsulas Mebeverini 0.2 retard N. 30
D.S. Take 1 capsule 2 times a day 20 minutes before meals.

PRESCRIPTION 12.11. LACTULOSE

Rp.: Sirupi Lactulosi 66.7% — 200 ml
D.S. Take 15–30 ml once a day in the morning.

PRESCRIPTION 12.12. MACROGOL

Rp.: Pulveris Macrogoli 4000 10.0
D.t.d. N. 20
S. Dissolve 1 sachet in 1 glass of water and take 1–2 times a day.

PRESCRIPTION 12.13. BISACODYL (SUPPOSITORY)

Rp.: Suppositorium cum Bisacodylo 0.01
D.t.d. N. 10
S. 1 suppository rectally once a day (for constipation).

PRESCRIPTION 12.14. LOPERAMIDE

Rp.: Capsulas Loperamidi hydrochloridi 0.002 N. 10
D.S. 2 capsules after the first loose stool, then 1 capsule after each loose stool (no more than 8 a day).

PRESCRIPTION 12.15. ORAL REHYDRATION SOLUTION

Rp.: Pulveris pro solutione “ORS” N. 10
D.S. Dissolve 1 sachet in the stated volume of boiled, cooled water and drink after each loose stool.

PRESCRIPTION 12.16. URSODEOXYCHOLIC ACID

Rp.: Capsulas Acidi ursodeoxycholici 0.25 N. 100
D.S. 10–15 mg/kg/day, taken once in the evening.

PRESCRIPTION 12.17. PANCREATIN

Rp.: Capsulas “Creon 25000” N. 50

D.S. Take 1–2 capsules with a main meal and 1 capsule with a snack.

Review questions

1. Classification of drugs acting on the digestive system.
2. Regulation of gastric acid secretion and the sites of action of antisecretory drugs.
3. Mechanism of PPIs, rules for taking them, uses and adverse effects with long-term use.
4. H_2 blockers and how they differ from PPIs.
5. Classification of antacids, their advantages and disadvantages. Interactions with drugs.
6. Gastroprotectors: bismuth preparations, sucralfate, misoprostol.
7. H. pylori eradication regimens and monitoring their effectiveness.
8. Mechanism of vomiting and classification of antiemetics.
9. Differences and adverse effects of ondansetron, metoclopramide, domperidone and aprepitant.
10. Prokinetics and antispasmodics.
11. Classification, mechanism and contraindications of laxatives.
12. Principles of treating acute diarrhea. When is loperamide contraindicated?
13. Classification of choleretics and cholagogues. Which are contraindicated in gallstone disease?
14. Mechanism and uses of ursodeoxycholic acid.
15. Hepatoprotectors and their evidence base.
16. Rules for dosing and taking pancreatin.

Test questions

1. Drugs that irreversibly block the final step of gastric acid secretion:
   A) H_2 blockers
   B) Prostaglandins
   C) Antacids
   D) M_1-cholinergic blockers
   E) Proton pump inhibitors

2. When should a PPI be taken?
   A) During pain
   B) Together with an antacid
   C) Only at night
   D) 30–60 minutes before a meal
   E) 2 hours after a meal

3. The preferred PPI for gastroprotection in a patient taking clopidogrel:
   A) Esomeprazole
   B) Lansoprazole
   C) Omeprazole
   D) Cimetidine
   E) Pantoprazole

4. An antacid that causes acid rebound and systemic alkalosis:
   A) Sodium alginate
   B) Sucralfate
   C) Sodium bicarbonate
   D) Magnesium hydroxide
   E) Aluminum hydroxide

5. A gastroprotector absolutely contraindicated in pregnancy:
   A) Alginate
   B) Famotidine
   C) Sucralfate
   D) Misoprostol
   E) Almagel

6. A preparation that darkens the stool and tongue and acts against H. pylori:
   A) Bismuth tripotassium dicitrate
   B) Sucralfate
   C) Metronidazole
   D) Famotidine
   E) Omeprazole

7. A test unsuitable for assessing the result of H. pylori eradication:
   A) Urea breath test
   B) Stool antigen test
   C) Detection of antibodies in the blood
   D) Histology of a biopsy
   E) Culture of a biopsy

8. The drug of choice for chemotherapy-induced vomiting:
   A) Dimenhydrinate
   B) Scopolamine
   C) Ondansetron
   D) Apomorphine
   E) Metoclopramide

9. An antiemetic that may cause acute dystonia in children:
   A) Ondansetron
   B) Metoclopramide
   C) Domperidone
   D) Aprepitant
   E) Dexamethasone

10. An osmotic laxative also used in hepatic encephalopathy:
   A) Bisacodyl
   B) Lactulose
   C) Liquid paraffin
   D) Senna
   E) Castor oil

11. A group of laxatives that cause a “lazy bowel” and hypokalemia with long-term use:
    A) Contact (stimulant) laxatives
    B) Lactulose
    C) Bulk-forming agents
    D) Macrogols
    E) Glycerin suppositories

12. A peripheral opioid antagonist used for opioid-induced constipation:
    A) Methylnaltrexone
    B) Tramadol
    C) Oral naloxone
    D) Loperamide
    E) Naltrexone

13. A contraindication to loperamide:
    A) Irritable bowel syndrome (with diarrhea)
    B) Bloody diarrhea and high fever
    C) Watery diarrhea in adults
    D) High output from an ileostomy
    E) Travelers' diarrhea (without fever)

14. Which drug is contraindicated in gallstone disease?
    A) Ursodeoxycholic acid
    B) Magnesium sulfate (as a cholekinetic)
    C) Pantoprazole
    D) Paracetamol
    E) Drotaverine

15. By which enzyme's units is the dose of pancreatin set?
    A) Pepsin
    B) Lactase
    C) Trypsin
    D) Lipase
    E) Amylase

Clinical cases

Case 1

A 42-year-old man complains of epigastric pain that worsens on an empty stomach and at night. Upper GI endoscopy shows an 8 mm ulcer in the duodenal bulb; the urease test is positive. He has taken clarithromycin before. Clarithromycin resistance in the area is high.

Questions: 1) Draw up a treatment regimen. 2) When and how is eradication checked?

Answer: 1) Bismuth quadruple therapy for 14 days: a PPI (for example, esomeprazole 20 mg or omeprazole 20 mg 2 times a day) + bismuth tripotassium dicitrate 240 mg 2 times a day + tetracycline 500 mg 4 times a day + metronidazole 500 mg 3 times a day. Because of his previous clarithromycin use and the local resistance, a clarithromycin regimen is not appropriate. After eradication, the PPI is continued until the ulcer heals (4–8 weeks). 2) At least 4 weeks after the end of treatment, with the PPI stopped for 2 weeks — a urea breath test or a stool antigen test.

Case 2

A 65-year-old woman has been taking omeprazole 20 mg “to protect the stomach” for 5 years; endoscopy shows no pathology, and she does not take NSAIDs. Recently she has had muscle cramps, weakness and numbness in the legs. Blood magnesium is low, B_{12} is low, and densitometry shows osteopenia.

Questions: 1) Are the symptoms related to the PPI? 2) Management?

Answer: 1) Long-term PPI use is associated with hypomagnesemia (muscle cramps), vitamin B_{12} deficiency (neuropathy, numbness) and reduced calcium absorption (osteopenia, fracture risk). 2) There is no indication — stop the PPI gradually (reduce the dose over 2–4 weeks or switch to as-needed use to reduce rebound hypersecretion; temporarily an antacid or famotidine for symptoms); replace magnesium and B_{12}; treat the osteopenia.

Case 3

A 2-year-old child (12 kg) is brought in with watery diarrhea for 2 days (6–8 times a day) and several episodes of vomiting. The child is thirsty and restless, the eyes are slightly sunken, and a skin pinch goes back slowly. No blood in the stool, temperature 37.8 °C. The mother has bought loperamide at the pharmacy.

Questions: 1) Assess the condition. 2) Can loperamide be given? 3) Treatment.

Answer: 1) Acute gastroenteritis (probably viral), moderate dehydration. 2) No — loperamide is contraindicated in children under 2 (risk of intestinal obstruction, toxic megacolon, CNS depression). 3) Oral rehydration: low-osmolarity ORS, about 75 ml/kg over 4 hours (≈ 900 ml), in small, frequent portions; if vomiting is frequent — a single dose of ondansetron (age-appropriate) to make oral rehydration easier; zinc 20 mg/day for 10–14 days (in children over 6 months); continue breastfeeding and feeding; no antibiotics are needed. If the condition worsens or the child refuses to drink — intravenous rehydration.

Quick review table

Table 12.7. *Drugs acting on the digestive system — summary*

Group	Preparations	Mechanism	Main use	Important note
PPI	Omeprazole, pantoprazole, esomeprazole	Irreversible blockade of H_+/K_+-ATPase	GERD, ulcer, H. pylori, NSAID gastropathy	Before meals; with long-term use Mg, B_{12}, fractures, C. difficile
H_2 blockers	Famotidine	H_2 blockade	Mild GERD, nocturnal acid	Tolerance
Antacids	Almagel, Maalox, sodium bicarbonate	Neutralizing acid	Symptomatic	A 2-hour gap from other drugs
Gastroprotectors	De-Nol, sucralfate, misoprostol	Protective film / PG	Ulcer, H. pylori, NSAID prophylaxis	Misoprostol — contraindicated in pregnancy
5-HT_3 antagonists	Ondansetron	Serotonin blockade	Chemotherapy, postoperative vomiting	QT prolongation
D_2 antagonists	Metoclopramide, domperidone	D_2 blockade + prokinetic	Vomiting, gastroparesis	Extrapyramidal (metoclopramide), QT (domperidone)

Group	Preparations	Mechanism	Main use	Important note
Antispasmodics	Drotaverine, mebeverine, Buscopan	Myotropic / neurotropic	Spasm, irritable bowel syndrome	—
Laxatives	Psyllium, lactulose, macrogol, bisacodyl, senna	Bulk / osmotic / contact	Constipation	Contraindicated in intestinal obstruction
Antidiarrheals	ORS, zinc, loperamide, smectite	Rehydration / peristalsis ↓	Acute diarrhea	Loperamide — contraindicated in bloody diarrhea and under 2 years
Bile and liver	UDCA, magnesium sulfate, Allochol	Stone dissolution / choleresis	Primary biliary cholangitis, cholesterol stones	Cholekinetics — contraindicated in gallstone disease
Enzymes	Pancreatin	Lipase, amylase, protease	Exocrine insufficiency	Dosed by lipase, with food

TOPIC 13

Diuretics

Learning objectives

Diuretics are among the most commonly used drugs in cardiology, nephrology and intensive care. They relieve edema and pulmonary congestion in heart failure, reduce complications in arterial hypertension, and are used to treat ascites in liver cirrhosis, cerebral edema and glaucoma. At the same time, diuretics significantly change the electrolyte balance: hypokalemia can lead to glycoside toxicity and arrhythmias, and hyperkalemia to cardiac arrest. Knowing which part of the nephron each diuretic acts on is the key to understanding its effectiveness and adverse effects logically.

The student should know:

- the main mechanisms of sodium and water reabsorption in the nephron;
- the classification of diuretics (by site of action, potency and mechanism);
- the mechanism of each group, its effect on electrolyte metabolism, its uses and adverse effects;
- the principles of rational combination of diuretics and of overcoming diuretic resistance.

The student should be able to:

- write prescriptions for drugs such as furosemide, torasemide, hydrochlorothiazide, indapamide, spironolactone, acetazolamide and mannitol;
- choose a diuretic according to the clinical situation;
- recognize and correct electrolyte disturbances during diuretic therapy.

1. Nephron physiology and the sites of action of diuretics

About **180 liters** of primary urine are filtered in the glomeruli each day, but only 1–1.5 liters are excreted — **more than 99%** of filtered sodium is reabsorbed. Diuretics inhibit sodium reabsorption; water leaves together with sodium.

Table 13.1. *Sodium reabsorption in the segments of the nephron and diuretics*

Segment of the nephron	Reabsorption of filtered Na^+	Main transport mechanism	Diuretics acting here
Proximal tubule	≈ 65%	Na^+/H^+ exchange (with carbonic anhydrase), Na^+-glucose (SGLT2)	Carbonic anhydrase inhibitors, SGLT2 inhibitors, osmotic diuretics
Thick ascending limb of the loop of Henle	≈ 25%	Na^+-K^+-2Cl– cotransporter (NKCC2)	Loop diuretics
Distal convoluted tubule	≈ 5–8%	Na^+-Cl– cotransporter (NCC)	Thiazide and thiazide-like diuretics
Collecting duct (cortical part)	≈ 2–3%	Aldosterone-dependent epithelial Na^+ channels (ENaC), K^+ and H^+ secretion	Potassium-sparing diuretics
Collecting duct (water reabsorption)	—	Vasopressin (V_2) — aquaporin-2	Vaptans (aquaretics)

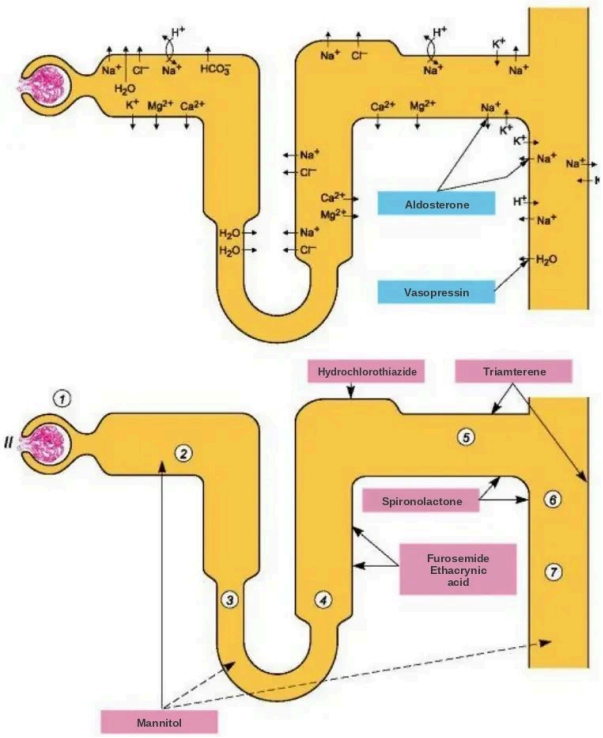


Figure 13.1. *Sites of action of diuretics in the nephron: I — processes involved in urine formation; II — segments on which diuretics act (1 — glomerulus, 2 — proximal tubule, 3–4 — loop of Henle, 5 — distal tubule, 6–7 — collecting duct) (teaching material of the TMA Department of Pharmacology)*

REMEMBER

The **potency** of a diuretic depends on how much sodium is reabsorbed in the segment it acts on: loop diuretics (25%) — **strong**, thiazides (5–8%) — **moderate**, potassium-sparing agents (2–3%) — **weak**. Agents acting on the proximal tubule are also weak, because the later segments "catch" the sodium.

The general mechanism of potassium loss. All diuretics acting before the collecting duct (loop, thiazide, carbonic anhydrase inhibitors, osmotic) deliver more sodium and water to the collecting duct → here more sodium is reabsorbed through aldosterone-dependent channels and, in exchange, **more potassium and hydrogen ions are secreted → hypokalemia and metabolic alkalosis**. In addition, hypovolemia activates the RAAS and increases aldosterone.

2. Classification of diuretics

Table 13.2. *Classification of diuretics*

Group	Preparations	Potency	Onset / duration of action (oral)
Loop diuretics	Furosemide, torasemide, bumetanide, ethacrynic acid	Strong	30–60 minutes / 6 hours (furosemide); 12–18 hours (torasemide); intravenously — 5 minutes
Thiazide diuretics	Hydrochlorothiazide, chlorothiazide	Intermediate	1–2 hours / 12 hours
Thiazide-like diuretics	Indapamide, chlorthalidone, metolazone	Intermediate	1–2 hours / 24 hours or more
Potassium-sparing — aldosterone antagonists	Spironolactone, eplerenone, finerenone	Weak	2–3 days (full effect)

Group	Preparations	Potency	Onset / duration of action (oral)
Potassium-sparing — Na^+ channel blockers	Amiloride, triamterene	Weak	2 hours / 12–24 hours
Carbonic anhydrase inhibitors	Acetazolamide (Diacarb)	Weak	1 hour / 8–12 hours
Osmotic diuretics	Mannitol, urea	Strong (water diuresis)	Intravenously — 15–30 minutes
SGLT2 inhibitors	Dapagliflozin, empagliflozin	Weak (glucosuria, natriuresis)	—
Vasopressin V_2 antagonists (aquaretics)	Tolvaptan, conivaptan	Water excretion (without sodium)	—
Xanthines	Theophylline, aminophylline (Euphylline)	Weak	Historically

3. Loop diuretics

Mechanism. In the thick ascending limb of the loop of Henle they block the **Na^+-K^+-2Cl– cotransporter** from the luminal side. Up to 25% of filtered sodium is reabsorbed here — that is why they are the **most potent** diuretics. In addition:

- the **hyperosmolarity** of the renal medulla is disrupted — the ability to concentrate urine decreases;
- paracellular reabsorption of Ca^{2+} and Mg^{2+} in this segment also decreases — **calcium and magnesium excretion increases** (unlike thiazides!);
- by increasing prostaglandin synthesis in the kidney they increase renal blood flow; **when given intravenously — venodilation before diuresis** (immediate relief in pulmonary edema) — NSAIDs reduce this effect.

Uses:

- **acute left ventricular failure and pulmonary edema** (intravenously);
- edema and congestion in chronic heart failure (torasemide acts longer and has higher bioavailability);
- edema in liver cirrhosis and nephrotic syndrome (with spironolactone);
- edema and AH **in renal failure** (GFR < 30 ml/min — thiazides are ineffective);
- **hypercalcemia** (after fluid replacement);
- forced diuresis (in poisoning; now rarely), hyperkalemia.

Table 13.3. *Adverse effects of loop diuretics*

Adverse effect	Comment
Hypovolemia, hypotension, prerenal azotemia	Strong diuresis; especially in the elderly
Hypokalemia, hypomagnesemia, hypocalcemia, hyponatremia	Arrhythmias, increased glycoside toxicity
Metabolic (hypochloremic) alkalosis	Loss of H^+ and Cl–

Adverse effect	Comment
Ototoxicity (tinnitus, deafness)	With rapid intravenous injection of large doses, together with aminoglycosides; more common with ethacrynic acid
Hyperuricemia, gout	Competition for uric acid secretion
Hyperglycemia, dyslipidemia	Less than with thiazides
Sulfonamide allergy	Furosemide, torasemide and bumetanide have a sulfonamide structure; ethacrynic acid does not (an alternative in sulfonamide allergy)

CAUTION

Giving furosemide intravenously **slowly** (at no more than 4 mg per minute) reduces the risk of ototoxicity. Giving it together with aminoglycosides, vancomycin or cisplatin increases **ototoxicity and nephrotoxicity**. NSAIDs reduce the effect of loop diuretics.

4. Thiazide and thiazide-like diuretics

Mechanism. They block the **Na^+-Cl– cotransporter** in the distal convoluted tubule. In addition, they **increase calcium reabsorption** (urinary calcium decreases) — the opposite of loop diuretics.

Uses:

- **arterial hypertension** — first-line drugs (indapamide and chlorthalidone are preferred; Topic 11);
- mild and moderate edema (heart failure);
- **prevention of kidney stones** — idiopathic hypercalciuria (calcium stones);
- **nephrogenic diabetes insipidus** (paradoxically reduces urine volume — by increasing proximal reabsorption);
- in severe resistant edema, combined with loop diuretics (metolazone — “sequential nephron blockade”).

MEMORY AID

The “opposite relationship” with calcium: **“Loops Lose calcium, Thiazides Take it back”**. That is why loop diuretics are useful in hypercalcemia, and thiazides in calcium kidney stones and a tendency to osteoporosis.

Adverse effects (mainly at high doses): hypokalemia, **hyponatremia** (especially in elderly women — thiazides more than loop diuretics), hypomagnesemia, metabolic alkalosis, **hyperuricemia and gout**, **hyperglycemia** (reduced insulin secretion), dyslipidemia, **hypercalcemia**, photosensitivity, erectile dysfunction, sulfonamide allergy; they increase **lithium** concentration.

CLINICAL SIGNIFICANCE

When **GFR is below 30 ml/min**, thiazides are usually **ineffective** (chlorthalidone is a partial exception) — loop diuretics are used in such patients. There is evidence that long-term, high-dose hydrochlorothiazide slightly increases the risk of non-melanoma skin cancer (especially squamous cell) — sun protection is recommended.

5. Potassium-sparing diuretics

5.1. Aldosterone (mineralocorticoid receptor) antagonists

Spironolactone is a **competitive antagonist** of aldosterone: by blocking mineralocorticoid receptors in collecting duct cells, it reduces the synthesis of Na^+ channels and Na^+/K^+-ATPase. Sodium excretion increases slightly, and **potassium and hydrogen secretion decreases**. Its effect develops slowly (over 2–3 days) and it is **the more effective the higher the aldosterone level**.

Uses:

- **chronic heart failure** (reduced ejection fraction) — reduces mortality;
- **ascites in liver cirrhosis** (secondary hyperaldosteronism) — the diuretic of first choice;
- **primary hyperaldosteronism** (Conn syndrome);
- resistant arterial hypertension;
- with thiazide or loop diuretics — prevention of hypokalemia;
- hyperandrogenism in women (hirsutism, acne) — because of its antiandrogenic effect. **Adverse effects: hyperkalemia** (especially with renal failure, ACE inhibitors/ARBs, potassium preparations, NSAIDs), metabolic acidosis; **antiandrogenic and progestogenic effects** — **gynecomastia** and impotence in men, menstrual disorders and breast pain in women.

Eplerenone is more selective for mineralocorticoid receptors; it does not cause gynecomastia. **Finerenone** is a nonsteroidal antagonist; used in diabetic kidney disease.

5.2. Sodium channel blockers

Amiloride and **triamterene** block the epithelial sodium channels (ENaC) of the collecting duct **directly** — their effect does not depend on aldosterone. Mainly in combination with thiazides to prevent hypokalemia (Triampur — triamterene + hydrochlorothiazide). Amiloride — in **Liddle syndrome** and lithium-induced nephrogenic diabetes insipidus. Adverse effects: hyperkalemia; triamterene — kidney stones, folic acid antagonism.

> **CAUTION**
> **Risk of hyperkalemia**: potassium-sparing diuretics must be used with caution or not at all together with **potassium preparations, ACE inhibitors, ARBs, NSAIDs, trimethoprim and heparin**, and **in renal failure** (GFR < 30). If potassium exceeds 5.5 mmol/l — risk of cardiac arrhythmias and arrest. Regular monitoring of blood potassium and creatinine is mandatory.

6. Other diuretics

6.1. Carbonic anhydrase inhibitors

Acetazolamide (Diacarb) inhibits **carbonic anhydrase** in the proximal tubule → H^+ formation and Na^+/H^+ exchange decrease → excretion of **bicarbonate**, sodium and potassium increases, the urine becomes alkaline, and **metabolic acidosis** develops. The diuretic effect is weak and decreases within 2–3 days as acidosis develops (self-limiting).

Uses (mainly non-diuretic): **glaucoma** (reduces the production of aqueous humor; in an attack of angle-closure glaucoma — orally or intravenously; topical analogs — dorzolamide and brinzolamide drops), **prevention of mountain (altitude) sickness**, idiopathic intracranial hypertension, as an add-on in epilepsy, correction of metabolic alkalosis, alkalinization of urine (in uric acid and cystine stones).

Adverse effects: metabolic acidosis, hypokalemia, paresthesias (numbness of the hands and feet), **calcium phosphate stones**, sulfonamide allergy; in liver cirrhosis — ammonia accumulation (hepatic encephalopathy).

6.2. Osmotic diuretics

Mannitol is filtered but not reabsorbed; by increasing the osmotic pressure in the tubules, it stops **water** reabsorption (mainly in the proximal tubule and the thin limb of the loop of Henle). By also increasing osmotic pressure in the blood, it **draws water from the tissues into the vessels**.

Uses (intravenous only): **cerebral edema and raised intracranial pressure**, **an acute attack of angle-closure glaucoma** (rapid lowering of intraocular pressure), prevention of acute kidney injury with oliguria (rhabdomyolysis — historically).

> **CAUTION**
> Mannitol is **absolutely contraindicated in heart failure and pulmonary edema**: it first increases the circulating volume and worsens pulmonary edema. It is also not used in anuria (no urine output), severe dehydration and the active phase of intracerebral hemorrhage.

6.3. SGLT2 inhibitors and aquaretics

SGLT2 inhibitors (dapagliflozin, empagliflozin) block the sodium-glucose cotransporter in the proximal tubule and cause glucosuria and moderate natriuresis. Although they are not strong diuretics, they reduce hospitalizations and mortality in **heart failure** (regardless of ejection fraction) and **chronic kidney disease** (Topics 11 and 15).

Vaptans (tolvaptan) block vasopressin V_2 receptors in the collecting duct and increase the excretion of **free water** (aquaresis). Uses: **hypervolemic and euvolemic hyponatremia** (syndrome of inappropriate vasopressin secretion — SIADH, heart failure), autosomal dominant polycystic kidney disease. Raising sodium too quickly (> 10 mmol/l/day) carries a risk of **osmotic demyelination**; hepatotoxicity.

7. Effects of diuretics on electrolyte metabolism (summary table)

Table 13.4. *Effects of diuretics on urine composition and blood electrolytes*

Diuretic	Na^+ (in urine)	K^+ (in urine)	Ca^{2+} (in urine)	Mg^{2+}	Acid-base status
Loop	↑↑↑	↑↑	↑↑	↑ (loss)	Metabolic alkalosis
Thiazides	↑↑	↑↑	↓ (retained)	↑ (loss)	Metabolic alkalosis
Potassium-sparing	↑	↓ (retained)	—	↓	Metabolic acidosis
Carbonic anhydrase inhibitors	↑	↑	—	—	Metabolic acidosis (bicarbonate is lost)
Osmotic	↑	↑	↑	↑	— (more water)

> **MEMORY AID**
> **“Diuretics that cause acidosis — A, K, and Stop”**: **A**cetazolamide, **K**-sparing diuretics (spironolactone, amiloride, triamterene) — **S**top, that's all. The other diuretics (loop, thiazide) cause **alkalosis**.

8. Principles of clinical use of diuretics

8.1. Choice of diuretic

Table 13.5. *Choice of diuretic according to the clinical situation*

Situation	Choice
Pulmonary edema, acute decompensated heart failure	Intravenous furosemide (40–80 mg or 1–2.5 times the home dose)
Chronic heart failure (congestion)	A loop diuretic (torasemide, furosemide) at the lowest effective dose + spironolactone + an SGLT2 inhibitor
Arterial hypertension	Indapamide, chlorthalidone (or hydrochlorothiazide in a combination)
Ascites in liver cirrhosis	Spironolactone ± furosemide (in a 100:40 ratio); in large ascites — paracentesis + albumin
Edema in nephrotic syndrome	Loop diuretics (at high doses), with a thiazide if needed
Edema/AH in renal failure	Loop diuretics
Hypercalcemia	Fluid replacement + furosemide
Calcium kidney stones (hypercalciuria)	Thiazides
Nephrogenic diabetes insipidus	Thiazides (+ amiloride when caused by lithium)
Glaucoma, mountain sickness	Acetazolamide
Cerebral edema	Mannitol
Primary hyperaldosteronism	Spironolactone, eplerenone
Hyponatremia (SIADH)	Fluid restriction; tolvaptan

8.2. Diuretic resistance

Causes: high salt intake, taking NSAIDs, low perfusion (hypotension), hypoalbuminemia (furosemide reaches the tubule bound to protein), impaired intestinal absorption (bowel edema), adaptive hypertrophy of the distal nephron (the “braking” phenomenon).

Ways to overcome it: salt restriction; stopping NSAIDs; giving the diuretic intravenously or switching to a continuous infusion; increasing the dose; **sequential nephron blockade** — adding a thiazide (metolazone) or acetazolamide to a loop diuretic; with albumin in hypoalbuminemia.

> **CLINICAL SIGNIFICANCE**
> **Daily weight monitoring** during diuretic therapy is the simplest and most reliable method: a 1 kg decrease in weight corresponds to a loss of about 1 liter of fluid. Patients with heart failure are taught to see a doctor (or to increase the diuretic dose themselves) if their weight increases by more than 2 kg in 2–3 days.

Practical skills

Skill 1. Monitoring electrolytes during diuretic therapy

1. Before starting treatment: sodium, potassium, creatinine (GFR), glucose, uric acid.
2. When starting loop and thiazide diuretics or increasing the dose — potassium and creatinine after 1–2 weeks.

3. When starting spironolactone — potassium and creatinine at 3 days and 1 week, then at 1, 2, 3 and 6 months.
4. Hypokalemia (< 3.5 mmol/l) — add potassium preparations or a potassium-sparing diuretic; especially important in a patient taking glycosides.
5. Hyperkalemia (> 5.5 mmol/l) — reduce the dose of the potassium-sparing diuretic or stop it; > 6.0 — urgent treatment (calcium gluconate, insulin + glucose, salbutamol).

Skill 2. Key prescriptions

PRESCRIPTION 13.1. FUROSEMIDE (TABLETS)
Rp.: Furosemidi 0.04
D.t.d. N. 30 in tabulettis
S. Take 1 tablet in the morning on an empty stomach (under weight and potassium monitoring).

PRESCRIPTION 13.2. FUROSEMIDE (AMPOULES)
Rp.: Solutionis Furosemidi 1% — 2 ml
D.t.d. N. 10 in ampullis
S. Inject 2–4 ml intravenously slowly (in pulmonary edema).

PRESCRIPTION 13.3. TORASEMIDE
Rp.: Torasemidi 0.01
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 13.4. HYDROCHLOROTHIAZIDE
Rp.: Hydrochlorothiazidi 0.025
D.t.d. N. 20 in tabulettis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 13.5. INDAPAMIDE
Rp.: Tabulettas Indapamidi 0.0025 obductas N. 30
D.S. Take 1 tablet once a day in the morning.

PRESCRIPTION 13.6. SPIRONOLACTONE
Rp.: Spironolactoni 0.1
D.t.d. N. 30 in capsulis
S. Take 1 capsule once a day in the morning (for ascites in liver cirrhosis).

PRESCRIPTION 13.7. TRIAMTERENE + HYDROCHLOROTHIAZIDE
Rp.: Tabulettas “Triampur compositum” N. 50
D.S. Take 1–2 tablets once a day in the morning.

PRESCRIPTION 13.8. ACETAZOLAMIDE
Rp.: Acetazolamidi 0.25
D.t.d. N. 24 in tabulettis
S. Take 1 tablet 2 times a day for 3 days, then take a 1-day break; together with a potassium preparation.

PRESCRIPTION 13.9. MANNITOL

Rp.: Solutionis Mannitoli 15% — 200 ml

D.S. 0.5–1 g/kg as an intravenous drip over 15–30 minutes (in cerebral edema, under osmolarity monitoring).

PRESCRIPTION 13.10. POTASSIUM AND MAGNESIUM ASPARAGINATE

Rp.: Tabulettas “Asparcam” N. 50

D.S. Take 1–2 tablets 3 times a day after meals (to prevent hypokalemia).

Review questions

1. Mechanisms of sodium reabsorption in the segments of the nephron.
2. Classification of diuretics. What does the potency of a diuretic depend on?
3. Why do most diuretics cause hypokalemia?
4. Mechanism, uses and adverse effects of loop diuretics.
5. Why furosemide gives immediate relief in pulmonary edema.
6. Mechanism and uses of thiazide and thiazide-like diuretics.
7. The difference between the effects of loop diuretics and thiazides on calcium metabolism, and its clinical significance.
8. Metabolic adverse effects of thiazides.
9. Mechanism, uses and adverse effects of spironolactone.
10. How amiloride and triamterene differ from spironolactone.
11. The risk of hyperkalemia and its prevention.
12. Mechanism and main areas of use of acetazolamide.
13. Mechanism, uses and contraindications of mannitol.
14. The place of SGLT2 inhibitors and vaptans as diuretics.
15. Which diuretics cause metabolic acidosis, and which cause alkalosis?
16. Causes of diuretic resistance and ways to overcome it.

Test questions

1. A diuretic that blocks the Na^+-K^+-2Cl– cotransporter:
   A) Spironolactone
   B) Mannitol
   C) Acetazolamide
   D) Furosemide
   E) Hydrochlorothiazide

2. The group with the strongest diuretic effect:
   A) Xanthines
   B) Thiazides
   C) Potassium-sparing
   D) Loop diuretics
   E) Carbonic anhydrase inhibitors

3. A diuretic that reduces urinary calcium excretion and is used to prevent calcium kidney stones:
   A) Furosemide
   B) Hydrochlorothiazide
   C) Mannitol
   D) Torasemide
   E) Acetazolamide

4. A diuretic used in hypercalcemia after fluid replacement:
   A) Hydrochlorothiazide
   B) Spironolactone
   C) Indapamide
   D) Furosemide
   E) Amiloride

5. A diuretic that increases ototoxicity when given together with aminoglycosides:
   A) Indapamide
   B) Spironolactone
   C) Furosemide
   D) Amiloride
   E) Acetazolamide

6. The diuretic of first choice for treating ascites in liver cirrhosis:
   A) Spironolactone
   B) Acetazolamide
   C) Indapamide
   D) Hydrochlorothiazide
   E) Mannitol

7. A diuretic that causes gynecomastia:
   A) Torasemide
   B) Furosemide
   C) Spironolactone
   D) Amiloride
   E) Eplerenone

8. A diuretic used in glaucoma and to prevent mountain sickness:
   A) Spironolactone
   B) Acetazolamide
   C) Triamterene
   D) Indapamide
   E) Furosemide

9. A diuretic that causes metabolic acidosis:
   A) Hydrochlorothiazide
   B) Torasemide
   C) Acetazolamide
   D) Indapamide
   E) Furosemide

10. An osmotic diuretic used in cerebral edema:
    A) Tolvaptan
    B) Acetazolamide
    C) Furosemide
    D) Mannitol
    E) Spironolactone

11. A diuretic absolutely contraindicated in heart failure and pulmonary edema:
    A) Spironolactone
    B) Mannitol
    C) Hydrochlorothiazide
    D) Furosemide
    E) Torasemide

12. The more effective diuretic in arterial hypertension when GFR is below 30 ml/min:
    A) Acetazolamide
    B) Furosemide
    C) Triamterene
    D) Hydrochlorothiazide
    E) Amiloride

13. A diuretic that paradoxically reduces urine volume in nephrogenic diabetes insipidus:
    A) Spironolactone
    B) Furosemide
    C) Hydrochlorothiazide
    D) Acetazolamide
    E) Mannitol

14. A vasopressin V_2 receptor antagonist (aquaretic):
    A) Eplerenone
    B) Dapagliflozin
    C) Tolvaptan
    D) Desmopressin
    E) Amiloride

15. A diuretic with a high risk of hyperkalemia when given together with an ACE inhibitor:
    A) Spironolactone
    B) Hydrochlorothiazide
    C) Furosemide
    D) Indapamide
    E) Torasemide

Clinical cases

Case 1

A 68-year-old man takes furosemide 80 mg/day, digoxin and enalapril for chronic heart failure and atrial fibrillation. In recent days he has felt weakness, muscle cramps, nausea and an irregular heartbeat. Potassium 2.9 mmol/l, magnesium low.

Questions: 1) The cause of the electrolyte disturbance? 2) Why is it especially dangerous for this patient? 3) Correction.

Answer: 1) The loop diuretic (furosemide) causes large losses of potassium and magnesium (increased Na^+ delivery to the collecting duct, secondary hyperaldosteronism). 2) Hypokalemia and hypomagnesemia increase digoxin toxicity and the risk of ventricular arrhythmias. 3) Potassium and magnesium preparations (slowly intravenously if needed), adding spironolactone (it also reduces mortality in HF and retains potassium), reducing the furosemide dose to the lowest effective dose; monitoring of the digoxin level and ECG; recheck electrolytes in 1 week.

Case 2

A 55-year-old woman takes hydrochlorothiazide 50 mg/day for AH. She has developed acute pain, redness and swelling in the first metatarsophalangeal joint of the big toe. Blood uric acid is raised. In addition, glucose is at a borderline level.

Questions: 1) Are these conditions related to the diuretic? 2) How should the antihypertensive treatment be changed?

Answer: 1) Thiazides (especially at high doses) cause hyperuricemia (competition for tubular secretion of uric acid and increased reabsorption with hypovolemia) and hyperglycemia — an attack of gout has developed. 2) Stop hydrochlorothiazide; for AH choose losartan (it has a uricosuric effect) and/or amlodipine; treat the gout attack (colchicine or an NSAID or a glucocorticoid); later, allopurinol if needed.

Case 3

A 30-year-old mountaineer is planning to climb to an altitude of 4,500 m. On his previous trip he had headache, nausea and insomnia.

Questions: Which diuretic would you recommend for prevention, and by what mechanism?

Answer: Acetazolamide (125 mg 2 times a day, starting 1 day before the ascent and continuing for 2 days after reaching the highest point). Mechanism: by excreting bicarbonate it causes metabolic acidosis — this stimulates the respiratory center, increases ventilation and speeds up acclimatization (the hypoxic ventilatory response). Adverse effects: paresthesias, altered taste of carbonated drinks, frequent urination; contraindicated in sulfonamide allergy. The main prevention is a gradual ascent.

Quick review table

Table 13.6. *Diuretics — summary*

Group	Preparations	Site and mechanism of action	Main use	Main adverse effects
Loop	Furosemide, torasemide, bumetanide, ethacrynic acid	Loop of Henle, NKCC2	Pulmonary edema, HF, renal failure, hypercalcemia	Hypokalemia, hypovolemia, ototoxicity, alkalosis, Ca_{2+} loss
Thiazides	Hydrochlorothiazide, indapamide, chlorthalidone	Distal tubule, NCC	AH, calcium stones, nephrogenic diabetes insipidus	Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hypercalcemia
Aldosterone antagonists	Spironolactone, eplerenone	Collecting duct, mineralocorticoid receptors	HF, ascites, hyperaldosteronism, resistant AH	Hyperkalemia, gynecomastia

Group	Preparations	Site and mechanism of action	Main use	Main adverse effects
Na_+ channel blockers	Amiloride, triamterene	Collecting duct, ENaC	Combination with thiazides	Hyperkalemia
Carbonic anhydrase inhibitors	Acetazolamide	Proximal tubule	Glaucoma, mountain sickness	Metabolic acidosis, paresthesia, stones
Osmotic	Mannitol	Proximal tubule, loop (water)	Cerebral edema, glaucoma attack	Contraindicated in HF and pulmonary edema
SGLT2 inhibitors	Dapagliflozin, empagliflozin	Proximal tubule	HF, chronic kidney disease, diabetes	Genital infections, ketoacidosis
Vaptans	Tolvaptan	Collecting duct, V_2	Hyponatremia (SIADH)	Rapid rise in sodium, hepatotoxicity

TOPIC 14

Drugs acting on the blood system. Antiplatelet agents. Anticoagulants. Fibrinolytic agents

Learning objectives

Thromboses are the main cause of myocardial infarction, ischemic stroke, deep vein thrombosis and pulmonary embolism. Antithrombotic drugs (antiplatelet agents, anticoagulants, fibrinolytics) save millions of lives, but they are also among the drugs that cause the most serious adverse effects (bleeding). Anemia — especially iron deficiency anemia — is also widespread among women and children in Uzbekistan. This topic covers drugs that affect blood coagulation, hemostasis and hematopoiesis.

The student should know:

- the stages of hemostasis: vascular-platelet hemostasis, the coagulation cascade and fibrinolysis;
- the mechanism and uses of antiplatelet agents (acetylsalicylic acid, $P2Y_{12}$ blockers, GP IIb/IIIa blockers);
- direct (heparins, Xa and thrombin inhibitors) and indirect (warfarin) anticoagulants, their monitoring and antidotes;
- fibrinolytics, antifibrinolytics and hemostatics;
- drugs that stimulate hematopoiesis (iron, B_{12}, folic acid, erythropoietin).

The student should be able to:

- write prescriptions for drugs such as acetylsalicylic acid, clopidogrel, heparin, enoxaparin, warfarin, rivaroxaban, apixaban, alteplase, tranexamic acid and iron preparations;
- adjust the warfarin dose according to the INR and provide care in anticoagulant-related bleeding;
- draw up a treatment plan for iron deficiency anemia.

1. The hemostatic system

Hemostasis is the system that stops bleeding when a vessel is damaged and keeps the blood fluid. It consists of three stages:

1. **Vascular-platelet hemostasis.** Collagen and von Willebrand factor under the damaged endothelium cause platelet **adhesion**. Activated platelets release **thromboxane A_2** (TXA_2), **ADP** and serotonin and recruit other platelets. The final common pathway is activation of the **GP IIb/IIIa** receptors on the platelet membrane, which link platelets to each other through fibrinogen (**aggregation**) — a white thrombus forms.
2. **Coagulation hemostasis.** The cascade of plasma clotting factors: tissue factor (with factor VII) → **factor Xa** → prothrombin (II) → **thrombin** (IIa) → fibrinogen → **fibrin** → factor XIIIa stabilizes the fibrin — a red thrombus.
3. **Fibrinolysis.** Under the influence of tissue plasminogen activator (t-PA), plasminogen is converted into **plasmin**, which breaks down fibrin.

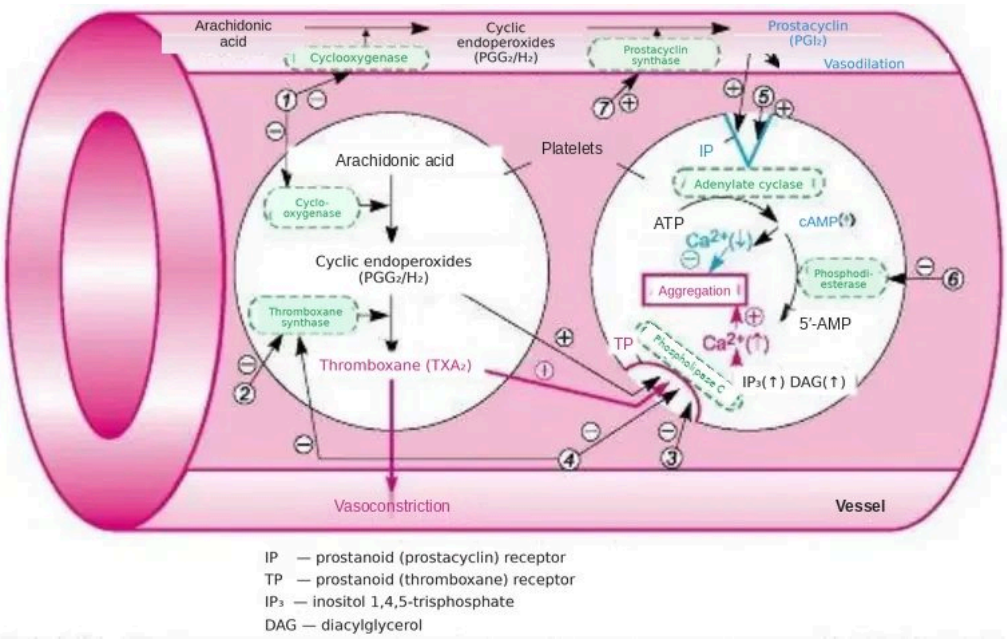


Figure 14.1. *The prostacyclin–thromboxane system and platelet aggregation: thromboxane A_2 enhances aggregation and vasoconstriction, prostacyclin (in the endothelium) reduces them (teaching material of the TMA Department of Pharmacology)*

Natural anticoagulants: antithrombin III (inactivates thrombin and Xa; heparin enhances its action), proteins C and S (vitamin K-dependent), tissue factor pathway inhibitor.

REMEMBER

Vitamin K-dependent clotting factors — II, VII, IX, X (and proteins C and S). To remember them: **"1972"** — 10, 9, 7, 2. Warfarin disrupts the synthesis of exactly these factors.

MEMORY AID

Arterial and venous thrombi: in arteries (high velocity) — a **platelet-rich "white" thrombus → antiplatelet agents** (prevention of infarction and stroke); in veins (slow flow) — a **fibrin- and red cell-rich "red" thrombus → anticoagulants** (deep vein thrombosis, PE, atrial fibrillation).

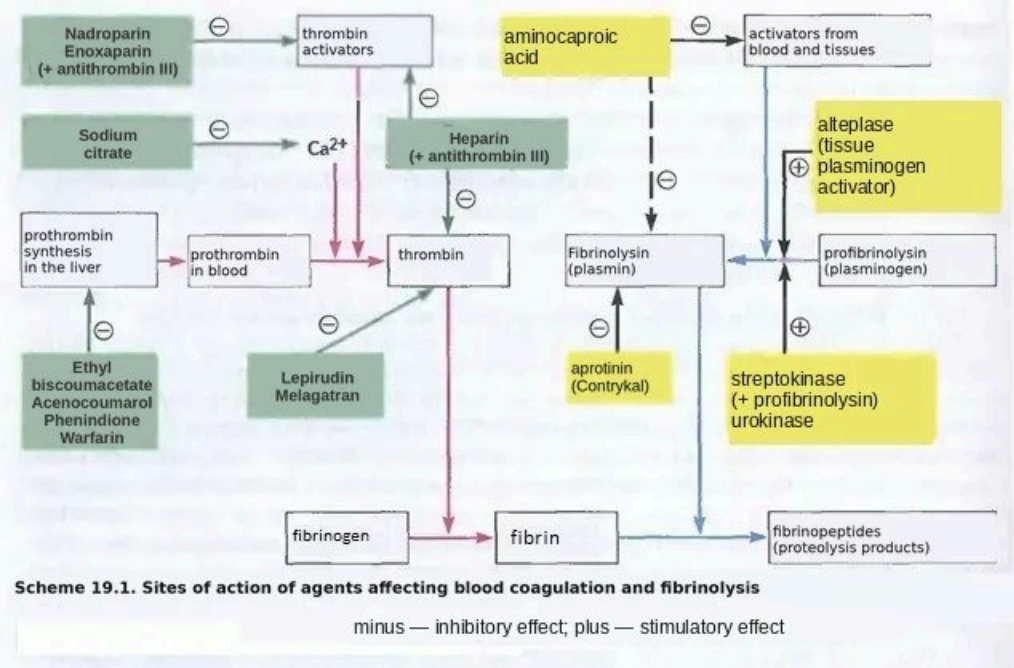


Figure 14.2. *Directions of action of drugs affecting blood coagulation and fibrinolysis (teaching material of the TMA Department of Pharmacology)*

Table 14.1. *Classification of drugs affecting blood coagulation*

Group	Subgroup	Preparations
Antiplatelet agents	Cyclooxygenase inhibitors	Acetylsalicylic acid
	$P2Y_{12}$ (ADP) receptor blockers	Clopidogrel, prasugrel, ticagrelor, cangrelor
	GP IIb/IIIa blockers	Abciximab, eptifibatide, tirofiban
	Phosphodiesterase inhibitors and others	Dipyridamole, cilostazol, pentoxifylline
	PAR-1 antagonist	Vorapaxar
Anticoagulants — direct	Heparins	Unfractionated heparin (UFH), low-molecular-weight heparins (LMWH): enoxaparin, nadroparin, dalteparin
	Selective Xa inhibitors (parenteral)	Fondaparinux
	Oral Xa inhibitors	Rivaroxaban, apixaban, edoxaban
	Direct thrombin inhibitors	Dabigatran (oral), bivalirudin, argatroban (parenteral)
Anticoagulants — indirect	Vitamin K antagonists	Warfarin, acenocoumarol, phenindione (Phenylin)
Fibrinolytics (thrombolytics)	—	Alteplase, tenecteplase, reteplase, streptokinase, urokinase, prourokinase
Agents that stop bleeding	Antifibrinolytics	Tranexamic acid, aminocaproic acid, aprotinin
	Coagulants	Vitamin K (phytomenadione, Vikasol), prothrombin complex concentrate, clotting factors, thrombin, fibrinogen
	Local hemostatics	Hemostatic sponge, thrombin, calcium alginate
	Antidotes	Protamine sulfate, idarucizumab, andexanet alfa

Group	Subgroup	Preparations
	Agents that strengthen the vessel wall	Etamsylate

2. Antiplatelet agents

2.1. Acetylsalicylic acid (aspirin)

Mechanism. It **irreversibly acetylates cyclooxygenase-1 (COX-1)** in platelets → TXA_2 synthesis stops. Platelets have no nucleus and cannot synthesize new enzyme, so the effect lasts for **the whole life of the platelet** (7–10 days). Endothelial cells, in contrast, resynthesize COX and restore prostacyclin production.

> **REMEMBER**
>
> **The antiplatelet dose of aspirin is 75–100 mg/day** (low dose). At high doses (> 300 mg), prostacyclin synthesis in the endothelium is also inhibited and gastrointestinal adverse effects rise sharply, while the antiplatelet effect does not increase. In acute coronary syndrome and stroke, a **loading dose** of 150–300 mg is chewed.

Uses: acute coronary syndrome, stable IHD, ischemic stroke and TIA, peripheral artery disease — **secondary prevention**; after coronary stenting (with a $P2Y_{12}$ blocker — dual antiplatelet therapy); prevention of preeclampsia (in high-risk pregnant women). In **primary prevention** (in people without cardiovascular disease), the risk of bleeding may outweigh the benefit — it is now **not routinely recommended**.

Adverse effects: gastrointestinal erosions and ulcers, **bleeding**, bronchospasm (**"aspirin-induced asthma"** — NSAID-exacerbated respiratory disease), **Reye syndrome** in children (during a viral infection — liver and brain damage; contraindicated for fever under 16 years).

2.2. $P2Y_{12}$ receptor blockers

By blocking **ADP receptors ($P2Y_{12}$)** on platelets, they stop GP IIb/IIIa activation.

Table 14.2. *$P2Y_{12}$ blockers*

Preparation	Features	Recovery
Clopidogrel	A prodrug — activated in the liver via CYP2C19; slow action, large individual variation (genetic "poor metabolizers"); interaction with omeprazole	Irreversible; 5 days after stopping
Prasugrel	A prodrug, but activated quickly and reliably; more potent	Irreversible; 7 days
Ticagrelor	An active drug, binds reversibly; 2 times a day; dyspnea (related to adenosine)	3–5 days
Cangrelor	Intravenous, very rapid and short-acting	Within an hour

Uses: together with aspirin in acute coronary syndrome and after coronary stenting (dual antiplatelet therapy, usually for 12 months), in aspirin allergy or intolerance, in ischemic stroke and peripheral artery disease (clopidogrel).

2.3. Other antiplatelet agents

- **GP IIb/IIIa blockers** (eptifibatide, tirofiban) — intravenous only, during coronary interventions with a large angiographic thrombus burden; the most potent antiplatelet agents; bleeding and

thrombocytopenia.

- **Dipyridamole** — a phosphodiesterase inhibitor that also inhibits adenosine uptake; with aspirin for secondary prevention of stroke; the “steal” phenomenon (may worsen angina in IHD).
- **Cilostazol** — a PDE-3 inhibitor; in peripheral artery disease (claudication); contraindicated in heart failure.
- **Pentoxifylline** — improves red cell deformability; limited effectiveness.

3. Anticoagulants

3.1. Unfractionated heparin (UFH)

Heparin is a sulfated glycosaminoglycan formed in mast cells; a natural anticoagulant. The preparation is obtained from porcine intestinal mucosa.

Mechanism. Heparin binds to **antithrombin III** and increases its activity **1,000-fold**. The complex inactivates **thrombin (IIa)** and **factor Xa** (as well as IXa, XIa and XIIa). To inactivate thrombin, the heparin molecule must be long (18 or more saccharides) — it binds antithrombin and thrombin simultaneously like a “bridge”.

- It acts **in vitro and in vivo**; the effect begins **immediately** (intravenously).
- It is not absorbed from the intestine — only **intravenously** or **subcutaneously** (intramuscular injection is contraindicated because of the risk of hematoma).
- Short action (t½ ≈ 1–1.5 hours); nonlinear pharmacokinetics, variable individual response.
- **Laboratory monitoring: activated partial thromboplastin time (aPTT)** — prolonged to 1.5–2.5 times normal. **Uses:** acute coronary syndrome, PE, deep vein thrombosis, cardiopulmonary bypass, hemodialysis, the hypercoagulable phase of DIC, anticoagulation **in severe renal failure** (instead of LMWH), situations requiring rapid reversal.

Adverse effects:

- **bleeding**;
- **heparin-induced thrombocytopenia (HIT)** — immune (antibodies to the heparin-PF4 complex), on days 5–14; platelets fall by more than 50% and **paradoxical thrombosis** (arterial and venous) develops. Treatment: stop all heparins, switch to a non-heparin anticoagulant (argatroban, bivalirudin, fondaparinux);
- with long-term use — **osteoporosis**, alopecia, hyperkalemia (inhibition of aldosterone synthesis), allergic reactions.

> **REMEMBER**
> **The antidote to heparin is protamine sulfate**: a strongly basic protein that binds heparin directly and inactivates it (chemical antagonism). **1 mg of protamine neutralizes 100 units of heparin** (the amount of heparin given in the last 2–3 hours is taken into account). It neutralizes LMWH only partly (≈ 60%).

3.2. Low-molecular-weight heparins (LMWH)

Enoxaparin, nadroparin, dalteparin — short-chain fractions of heparin. They inactivate mainly **factor Xa** (less effect on thrombin).

Table 14.3. *Differences between unfractionated heparin and LMWH*

Feature	Unfractionated heparin	Low-molecular-weight heparins
Anti-Xa : anti-IIa activity	1 : 1	2–4 : 1

Feature	Unfractionated heparin	Low-molecular-weight heparins
Bioavailability (subcutaneous)	≈ 30%	> 90%
t½	1–1.5 hours	4–6 hours — 1–2 times a day
Dosing	Titration by aPTT	Fixed weight-based dose, without routine monitoring
Laboratory monitoring	aPTT	Not needed (in special cases — anti-Xa activity)
Risk of HIT	High	Much lower
Risk of osteoporosis	High	Low
Elimination	Reticuloendothelial system + kidneys	Kidneys — the dose is reduced in severe renal failure
Neutralization with protamine	Complete	Partial
Outpatient use	Inconvenient	Convenient (patients inject it themselves)

Uses: prevention (surgery, prolonged bed rest, oncology) and treatment of venous thromboembolism, acute coronary syndrome, **anticoagulation in pregnancy** (does not cross the placenta — the drug of choice).

Fondaparinux — a synthetic pentasaccharide; through antithrombin it inactivates only Xa; it does not cause HIT.

3.3. Indirect anticoagulants — warfarin

Mechanism. Warfarin inhibits **vitamin K epoxide reductase (VKORC1)** → the active (reduced) form of vitamin K is not regenerated → gamma-carboxylation of factors **II, VII, IX and X** in the liver is disrupted → inactive factors are formed.

- It acts **only in vivo** (not in vitro).
- The effect begins **with a delay** — 3–5 days (the existing factors must be broken down; t½ of factor VII is 6 hours, of factor II 60 hours). Therefore, **in acute thrombosis, warfarin is started together with heparin (LMWH)**, and the heparin is stopped when the INR has been in the target range on 2 consecutive days.
- In the first days of treatment, **protein C** (short t½) falls quickly — a temporary **hypercoagulable** state, rarely **warfarin skin necrosis** (in protein C deficiency).
- **Laboratory monitoring: prothrombin time and INR** (international normalized ratio). For most indications the target INR is **2.0–3.0**; with a mechanical mitral valve — 2.5–3.5. **Uses:** atrial fibrillation (especially with mitral stenosis and **mechanical heart valves** — in these cases direct oral anticoagulants are **contraindicated**), deep vein thrombosis and PE (treatment and secondary prevention), antiphospholipid syndrome.

CAUTION

Warfarin has very many interactions (narrow therapeutic range, metabolism via CYP2C9 and CYP3A4, 99% protein binding):

- **enhance its effect (bleeding risk)**: amiodarone, fluconazole and other azoles, metronidazole, many antibiotics (co-trimoxazole, fluoroquinolones, macrolides — they also reduce the intestinal flora), NSAIDs and aspirin (pharmacodynamic), SSRIs, acute alcohol intake, liver diseases;
- **reduce its effect**: rifampicin, carbamazepine, phenytoin, phenobarbital, St John's wort, foods rich in vitamin K (spinach, cabbage, broccoli, green tea — intake should be kept stable).

Whenever a new drug is prescribed, the INR should be checked frequently.

Adverse effects: bleeding (the most important), skin necrosis, “purple toe” syndrome, **teratogenicity** (in the 1st trimester warfarin embryopathy — hypoplasia of the nose and bones; in the 3rd trimester — fetal bleeding) — **contraindicated in pregnancy**.

Table 14.4. *Care in high INR and bleeding related to warfarin*

Situation	Management
INR 3.0–4.5, no bleeding	Omit 1–2 doses or reduce the dose, monitor the INR
INR 4.5–10, no bleeding	Stop warfarin temporarily; if the bleeding risk is high — vitamin K 1–2.5 mg orally
INR > 10, no bleeding	Stop; vitamin K 2.5–5 mg orally
Major (life-threatening) bleeding	Stop warfarin; prothrombin complex concentrate (or fresh frozen plasma) + vitamin K 5–10 mg intravenously slowly

3.4. Direct oral anticoagulants (DOACs)

Rivaroxaban, apixaban, edoxaban — direct inhibitors of **factor Xa**; **dabigatran** — a direct inhibitor of **thrombin** (the prodrug dabigatran etexilate).

Advantages (over warfarin): fixed doses, **no routine laboratory monitoring needed**, rapid onset and offset of action (hours), no food interactions, few drug interactions, **lower risk of intracerebral hemorrhage**.

Disadvantages and limitations:

- **contraindicated with mechanical heart valves and moderate-to-severe mitral stenosis** (warfarin is used);
- dependent on renal function (especially dabigatran — 80% renal): dose adjustment by GFR, not used in severe renal failure;
- contraindicated in pregnancy and breastfeeding;
- interactions with strong CYP3A4 and P-glycoprotein inhibitors/inducers;
- a missed dose means rapid loss of protection (adherence is important). **Antidotes:** dabigatran — **idarucizumab** (a monoclonal antibody fragment); Xa inhibitors — **andexanet alfa** (if unavailable — prothrombin complex concentrate).

Uses: stroke prevention in nonvalvular atrial fibrillation, deep vein thrombosis and PE (treatment and prevention), prevention after orthopedic surgery; low-dose rivaroxaban — with aspirin in IHD and peripheral artery disease.

Table 14.5. *Comparison of anticoagulants*

Feature	Unfractionated heparin	LMWH (enoxaparin)	Warfarin	DOACs (apixaban, rivaroxaban, dabigatran)
Route of administration	IV, SC	SC	Oral	Oral
Mechanism	AT III + IIa and Xa	AT III + mainly Xa	Vitamin K antagonist (II, VII, IX, X)	Direct Xa or IIa
Onset of action	Immediately	3–5 hours	3–5 days	2–4 hours
Monitoring	aPTT	Not needed	INR	Not needed
Antidote	Protamine sulfate	Protamine (partial)	Vitamin K, prothrombin complex	Idarucizumab (dabigatran), andexanet alfa
Pregnancy	Allowed	Drug of choice	Contraindicated	Contraindicated
Mechanical valve	Allowed	Allowed	Drug of choice	Contraindicated

CLINICAL SIGNIFICANCE

Anticoagulation in atrial fibrillation: according to the CHA_2DS_2-VASc score (HF — 1, AH — 1, age ≥ 75 — 2, diabetes — 1, stroke/TIA — 2, vascular disease — 1, age 65–74 — 1, female sex — 1), an oral anticoagulant **is recommended** at ≥ 2 points in men and ≥ 3 in women. In nonvalvular fibrillation **DOACs are preferred to warfarin**. The bleeding risk is assessed with the HAS-BLED score — a high score is not a reason to withhold the anticoagulant but a basis for correcting modifiable risk factors (AH, NSAIDs, alcohol).

4. Fibrinolytic (thrombolytic) drugs

Mechanism. They convert plasminogen into plasmin; plasmin breaks down the fibrin in the thrombus (dissolves the thrombus).

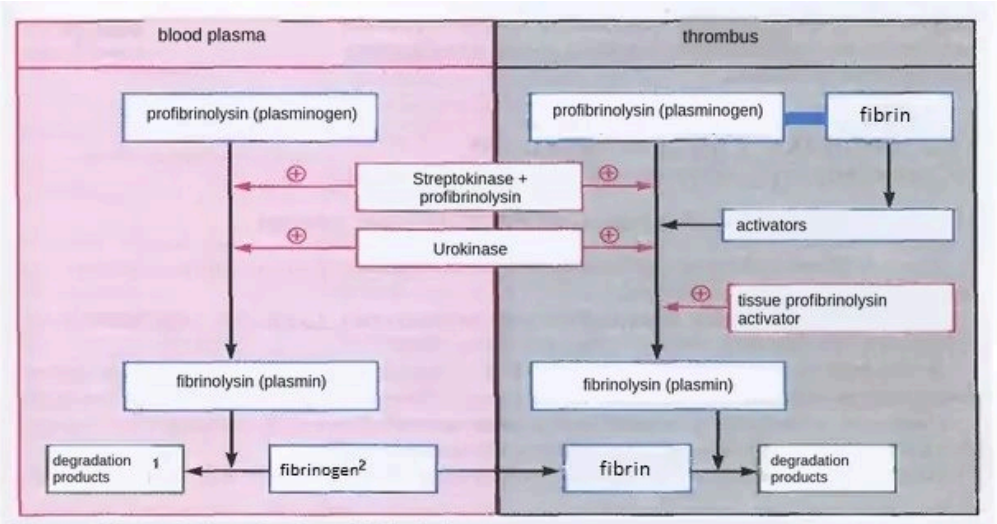


Figure 14.3. *Mechanism of action of fibrinolytic drugs: conversion of profibrinolysin (plasminogen) into fibrinolysin (plasmin) in the blood plasma and inside the thrombus (teaching material of the TMA Department of Pharmacology)*

Table 14.6. *Fibrinolytic drugs*

Preparation	Fibrin selectivity	Features
Streptokinase	None (systemic plasminemia, fibrinogen falls sharply)	A streptococcal protein — an antigen: allergic reactions, hypotension; not used again (antibodies for 6 months to 10 years); inexpensive
Urokinase	Low	Not antigenic
Alteplase (recombinant t-PA)	High (activates fibrin-bound plasminogen)	Ischemic stroke, infarction, PE; short t½ (5 minutes) — bolus + infusion
Tenecteplase	Very high	A single bolus (weight-based) — convenient in the prehospital setting in infarction
Reteplase	High	Two boluses
Prourokinase (Purolase)	High	In CIS countries

Uses:

- **ST-elevation myocardial infarction** — if percutaneous coronary intervention (stenting) is not possible within 120 minutes; up to **12 hours** from symptom onset (most effective in the first 3 hours — **"time is muscle"**);
- **ischemic stroke** — up to **4.5 hours** from symptom onset (alteplase, tenecteplase), after hemorrhage has been excluded on CT (**"time is brain"**);
- hemodynamically unstable (massive) **PE**;
- thrombosis of catheters and shunts.

CAUTION

Absolute contraindications to thrombolysis: a history of intracerebral hemorrhage at any time; ischemic stroke in the last 3–6 months (other than the acute stroke); a brain tumor or vascular malformation; active internal bleeding (except menstruation); suspected aortic dissection; severe trauma, head injury or major surgery in the last month; uncontrolled AH (> 185/110 mm Hg). The main complication is **bleeding, especially intracerebral hemorrhage**; antidote — tranexamic acid, cryoprecipitate (fibrinogen).

5. Drugs that stop bleeding (hemostatics)

5.1. Antifibrinolytics

Tranexamic acid and **aminocaproic acid** are lysine analogs; they block the binding of plasminogen to fibrin and inhibit fibrinolysis.

Uses of tranexamic acid: massive bleeding in trauma (within the first 3 hours — reduces mortality), **postpartum hemorrhage**, menorrhagia, dental procedures (in hemophilia), orthopedic and cardiac surgery, overdose of fibrinolytics. Contraindications: active thrombosis, DIC (hypercoagulable phase); seizures at high doses.

5.2. Coagulants

- **Vitamin K — phytomenadione** (K_1, natural, fat-soluble; oral and slow intravenous) and **menadione (Vikasol, K_3, synthetic)**. Uses: warfarin overdose, vitamin K deficiency (biliary obstruction, obstructive jaundice, prolonged antibiotic therapy, malabsorption), **prevention of hemorrhagic disease of the newborn** (phytomenadione 1 mg intramuscularly after birth). The effect begins in 6–24 hours (the factors must be synthesized). Rapid intravenous injection of

phytomenadione causes anaphylactoid reactions. Vikasol may cause hemolysis and jaundice in newborns — phytomenadione is preferred.

- **Prothrombin complex concentrate** (factors II, VII, IX, X) — rapid treatment of severe warfarin-related bleeding.
- **Clotting factor concentrates**: factor VIII (hemophilia A), factor IX (hemophilia B), von Willebrand factor; **desmopressin** increases the release of factor VIII and von Willebrand factor in mild hemophilia A and von Willebrand disease.
- **Local hemostatics**: hemostatic collagen sponge, thrombin, hydrogen peroxide, calcium alginate, epinephrine (topical).
- **Etamsylate** reduces capillary permeability and increases platelet adhesion; widely used in CIS countries, with a limited evidence base.

6. Drugs affecting hematopoiesis (brief)

6.1. Iron preparations

Iron deficiency anemia is the most common anemia (pregnant women, women with heavy periods, children, chronic blood loss).

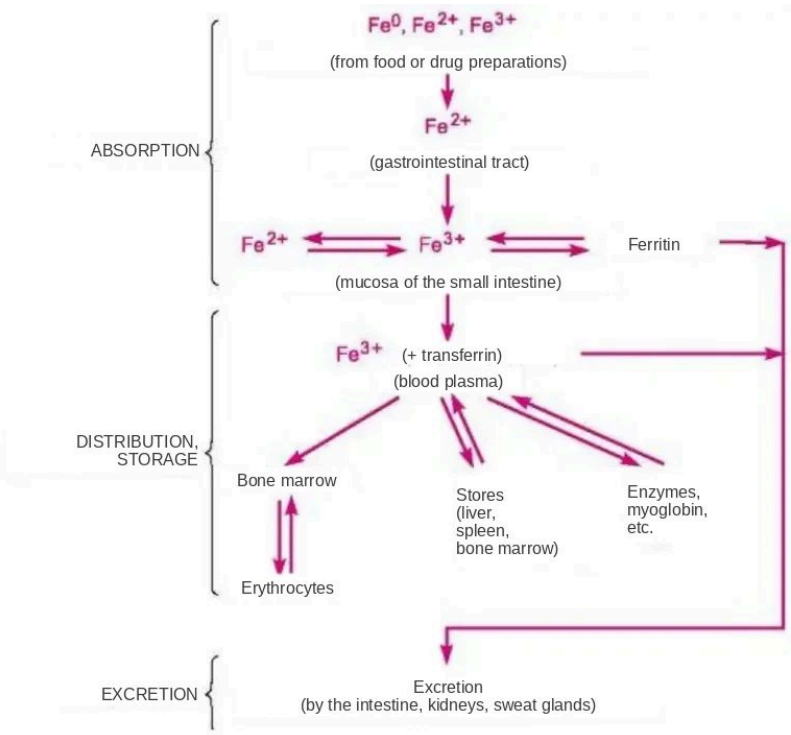


Figure 14.4. *Absorption, distribution, storage and excretion of iron (teaching material of the TMA Department of Pharmacology)*

Iron is absorbed mainly in **the duodenum and the upper jejunum**, in the divalent (**Fe^{2+}**) form; ascorbic acid improves absorption; antacids, PPIs, tetracyclines, calcium and tea (tannins) reduce it. In the blood it is carried by transferrin and stored as ferritin and hemosiderin.

- **Oral preparations** (first choice): iron(II) sulfate, fumarate, gluconate; iron(III) hydroxide polymaltose complex. The dose is based on **elemental iron**: 100–200 mg a day for adults (or 60–100 mg every other day — better absorption, fewer adverse effects). After hemoglobin normalizes, treatment continues for another **3 months** to replenish stores.

- **Adverse effects**: nausea, epigastric pain, constipation or diarrhea, **black discoloration of the stool** (must be distinguished from bleeding); liquid preparations stain tooth enamel.
- **Parenteral (intravenous) preparations** — ferric carboxymaltose, iron sucrose, ferric derisomaltose: when oral preparations are not tolerated or are ineffective, in malabsorption, chronic kidney disease, heart failure, the 2nd–3rd trimesters of pregnancy. Anaphylactic reactions are possible — give where resuscitation is available.

> **CAUTION**
> **Acute iron poisoning** is one of the common causes of fatal poisoning in children (sweet tablets!): vomiting (with blood), diarrhea, then a "false improvement", followed by shock, metabolic acidosis and liver failure. **Antidote — deferoxamine** (intravenously). Activated charcoal does not bind iron.

6.2. Vitamin B_{12} and folic acid

- **Cyanocobalamin, hydroxocobalamin (B_{12})** — in **megaloblastic (pernicious) anemia**, especially with intrinsic factor deficiency (atrophic gastritis, gastric resection), and with long-term metformin and PPI use; only parenterally or orally at high doses. B_{12} deficiency causes **neurological disorders** (subacute combined degeneration of the spinal cord).
- **Folic acid** — folate deficiency anemia; **when planning pregnancy** (400 mcg/day from 1–3 months before pregnancy and in the 1st trimester) — **prevention of neural tube defects**; during treatment with methotrexate.

> **CAUTION**
> In B_{12} deficiency, giving **folic acid** alone improves the anemia but **worsens the neurological disorders**. In megaloblastic anemia, the B_{12} level should be measured first, or both should be given together.

6.3. Stimulants of erythropoiesis and leukopoiesis

- **Erythropoietins** (epoetin alfa and beta, darbepoetin) — anemia of **chronic kidney disease**, chemotherapy-induced anemia. Hemoglobin is not raised above 120 g/l (risk of thrombosis, stroke, AH); iron stores must be adequate.
- **Granulocyte colony-stimulating factors** (filgrastim, pegfilgrastim) — neutropenia after chemotherapy.

7. Clinical part

Table 14.7. *Antithrombotic treatment in clinical situations*

Situation	Treatment
ST-elevation myocardial infarction	Aspirin 150–300 mg chewed + a $P2Y_{12}$ blocker (ticagrelor or clopidogrel) + heparin; primary coronary intervention or thrombolysis (alteplase, tenecteplase)
Stable IHD	Aspirin 75–100 mg (or clopidogrel)
Ischemic stroke (≤ 4.5 hours)	Thrombolysis (alteplase); then antiplatelet agents
Nonvalvular atrial fibrillation	DOACs (apixaban, rivaroxaban, dabigatran, edoxaban)
Mechanical heart valve	Warfarin (INR 2.5–3.5)
Deep vein thrombosis / PE	LMWH or a DOAC (rivaroxaban, apixaban); for at least 3 months
Postoperative prevention	LMWH (enoxaparin 40 mg SC once a day) or a DOAC

Situation	Treatment
Thrombosis in pregnancy	LMWH
Anticoagulation in severe renal failure	Unfractionated heparin, warfarin
HIT	Argatroban, bivalirudin, fondaparinux
Massive bleeding in trauma	Tranexamic acid (within 3 hours)

CLINICAL SIGNIFICANCE

Stopping antithrombotic drugs before surgery: aspirin — often continued (in patients with high cardiovascular risk); clopidogrel — 5 days before; warfarin — 5 days before (with LMWH "bridging" at high thrombotic risk); apixaban and rivaroxaban — 24–48 hours (depending on renal function); dabigatran — 48–96 hours. The decision is made jointly with the surgeon, anesthesiologist and cardiologist.

Practical skills

Skill 1. Starting and monitoring warfarin treatment

1. Determine the baseline INR, complete blood count, and liver and kidney function.
2. Start warfarin at 2.5–5 mg a day (2.5 mg in the elderly, in liver disease and with low body weight).
3. In acute thrombosis, give it together with LMWH for at least 5 days, until the INR is above 2.0 on 2 consecutive days.
4. Check the INR on days 3–5, then every 2–3 days, and once stable — every 4 weeks.
5. Explain to the patient: take the drug at the same time every day, eat foods rich in vitamin K in stable amounts, tell the doctor about new drugs, signs of bleeding (blood in the urine, black stool, bleeding gums, bruising).

Skill 2. Key prescriptions

PRESCRIPTION 14.1. ACETYLSALICYLIC ACID (ANTIPLATELET)

Rp.: Tabulettas Acidi acetylsalicylici 0.1 obductas N. 30
D.S. Take 1 tablet once a day after meals.

PRESCRIPTION 14.2. CLOPIDOGREL

Rp.: Clopidogreli 0.075
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet once a day.

PRESCRIPTION 14.3. TICAGRELOR

Rp.: Ticagreloris 0.09
D.t.d. N. 56 in tabulettis obductis
S. Take 1 tablet 2 times a day (together with aspirin 75–100 mg).

PRESCRIPTION 14.4. HEPARIN

Rp.: Heparini 5000 U/ml — 5 ml
D.t.d. N. 5
S. An intravenous bolus of 80 U/kg, then an infusion of 18 U/kg/hour; under aPTT monitoring.

PRESCRIPTION 14.5. ENOXAPARIN

Rp.: Solutionis Enoxaparini natrii 0.04 — 0.4 ml
D.t.d. N. 10 in syringis
S. Inject 0.4 ml under the skin of the abdomen once a day (thrombosis prevention).

PRESCRIPTION 14.6. WARFARIN

Rp.: Warfarini 0.0025
D.t.d. N. 100 in tabulettis
S. Take 2 tablets once a day in the evening at the same time; the next dose according to the INR.

PRESCRIPTION 14.7. APIXABAN

Rp.: Apixabani 0.005
D.t.d. N. 60 in tabulettis obductis
S. Take 1 tablet 2 times a day.

PRESCRIPTION 14.8. RIVAROXABAN

Rp.: Rivaroxabani 0.02
D.t.d. N. 28 in tabulettis obductis
S. Take 1 tablet once a day with the evening meal.

PRESCRIPTION 14.9. ALTEPLASE

Rp.: Alteplasi 0.05
D.t.d. N. 2
S. In ischemic stroke 0.9 mg/kg (maximum 90 mg): 10% as an intravenous bolus, the rest as an infusion over 60 minutes.

PRESCRIPTION 14.10. PROTAMINE SULFATE

Rp.: Solutionis Protamini sulfatis 1% — 5 ml
D.t.d. N. 5 in ampullis
S. Intravenously slowly (over 10 minutes); 1 mg of protamine neutralizes 100 U of heparin.

PRESCRIPTION 14.11. TRANEXAMIC ACID

Rp.: Solutionis Acidi tranexamici 5% — 10 ml
D.t.d. N. 5 in ampullis
S. 1 g intravenously over 10 minutes, then 1 g as an infusion over 8 hours (bleeding in trauma).

PRESCRIPTION 14.12. PHYTOMENADIONE (VITAMIN K_1)

Rp.: Solutionis Phytomenadioni 1% — 1 ml
D.t.d. N. 5 in ampullis
S. 0.1 ml (1 mg) intramuscularly to the newborn once (prevention of hemorrhagic disease).

PRESCRIPTION 14.13. IRON(II) SULFATE

Rp.: Tabulettas Ferrosi sulfatis 0.32 (100 mg elemental iron) retard N. 30
D.S. Take 1 tablet once a day 1 hour before meals, without tea or milk.

PRESCRIPTION 14.14. CYANOCOBALAMIN

Rp.: Solutionis Cyanocobalamini 0.05% — 1 ml
D.t.d. N. 10 in ampullis
S. 1 ml (500 mcg) intramuscularly every other day.

PRESCRIPTION 14.15. FOLIC ACID

Rp.: Acidi folici 0.0004
D.t.d. N. 90 in tabulettis
S. Take 1 tablet once a day (when planning pregnancy and in the 1st trimester).

Review questions

1. Stages of hemostasis: vascular-platelet hemostasis, the coagulation cascade, fibrinolysis.
2. Classification of drugs affecting blood coagulation.
3. Mechanism of the antiplatelet action of aspirin. Why is a low dose used?
4. Differences between $P2Y_{12}$ blockers. The interaction between clopidogrel and omeprazole.
5. Mechanism, monitoring, adverse effects and antidote of unfractionated heparin.
6. Heparin-induced thrombocytopenia: mechanism and treatment.
7. Advantages of LMWH over unfractionated heparin.
8. Mechanism of warfarin. Why is its effect delayed and why is it started together with heparin?
9. What is the INR? Target values.
10. Interactions of warfarin and care in excessive anticoagulation.
11. Direct oral anticoagulants: advantages, limitations and antidotes.
12. Fibrinolytics: mechanism, uses, contraindications.
13. Antifibrinolytics and vitamin K preparations.
14. Iron preparations: absorption, rules of use, adverse effects, poisoning.
15. Uses of vitamin B_{12} and folic acid.

Test questions

1. Which enzyme in platelets does aspirin irreversibly inhibit?
   A) Cyclooxygenase-1
   B) Lipoxygenase
   C) Adenylate cyclase
   D) Thrombin
   E) Phosphodiesterase

2. An antiplatelet agent that is a prodrug activated via CYP2C19:
   A) Clopidogrel
   B) Eptifibatide
   C) Cangrelor
   D) Aspirin
   E) Ticagrelor

3. Through which natural factor is the anticoagulant effect of heparin exerted?
   A) Antithrombin III
   B) Plasminogen
   C) Protein C
   D) Von Willebrand factor
   E) Vitamin K

4. The parameter used to monitor treatment with unfractionated heparin:
   A) INR
   B) Fibrinogen
   C) Bleeding time
   D) Platelet count
   E) aPTT

5. The antidote in heparin overdose:
   A) Vitamin K
   B) Idarucizumab
   C) Protamine sulfate
   D) Tranexamic acid
   E) Deferoxamine

6. The anticoagulant of choice for treating and preventing thrombosis in pregnancy:
   A) Dabigatran
   B) Enoxaparin
   C) Warfarin
   D) Apixaban
   E) Rivaroxaban

7. The synthesis of which clotting factors does warfarin disrupt?
   A) I, V, VIII
   B) VIII, IX
   C) XI, XII
   D) III, IV
   E) II, VII, IX, X

8. The parameter used to monitor treatment with warfarin:
   A) Platelet aggregation
   B) INR
   C) aPTT
   D) D-dimer
   E) Anti-Xa activity

9. A drug that enhances the effect of warfarin and increases the risk of bleeding:
   A) St John's wort
   B) Amiodarone
   C) Carbamazepine
   D) Phenobarbital
   E) Rifampicin

10. The anticoagulant of choice in a patient with a mechanical heart valve:
    A) Dabigatran
    B) Warfarin
    C) Aspirin
    D) Rivaroxaban
    E) Apixaban

11. The specific antidote to dabigatran:
    A) Vitamin K
    B) Idarucizumab
    C) Andexanet alfa
    D) Tranexamic acid
    E) Protamine sulfate

12. A fibrinolytic with antigenic properties that is not used again:
    A) Alteplase
    B) Reteplase
    C) Streptokinase
    D) Urokinase
    E) Tenecteplase

13. Within what time from symptom onset is thrombolysis performed in ischemic stroke?
    A) 48 hours
    B) 1 week
    C) 4.5 hours
    D) 12 hours
    E) 24 hours

14. A drug that reduces mortality from massive bleeding in trauma within the first 3 hours:
    A) Vikasol
    B) Etamsylate
    C) Calcium chloride
    D) Tranexamic acid
    E) Heparin

15. The antidote in acute poisoning with iron preparations in children:
    A) Protamine sulfate
    B) Unithiol
    C) Deferoxamine
    D) Activated charcoal
    E) EDTA

Clinical cases

Case 1

A 72-year-old woman takes warfarin for atrial fibrillation (stable at INR 2.4). She was prescribed co-trimoxazole for a urinary tract infection and fluconazole for foot candidiasis. After 6 days she developed nosebleeds and many bruises under the skin; INR 7.8.

Questions: 1) The cause of the INR rise? 2) Management? 3) What should be done in the future?

Answer: 1) Co-trimoxazole and fluconazole inhibited CYP2C9 and slowed warfarin metabolism (co-trimoxazole also displaces it from proteins and reduces the intestinal flora) — the anticoagulant effect increased sharply. 2) Stop warfarin temporarily; INR 4.5–10 with minor bleeding — vitamin K 1–2.5 mg orally; local control of the nosebleed; recheck the INR after 24 hours; if there is major bleeding — prothrombin complex concentrate + intravenous vitamin K. 3) Choose antibiotics with few interactions (nitrofurantoin or fosfomycin); monitor the INR frequently when a new drug is prescribed; consider switching to a DOAC (apixaban) in nonvalvular fibrillation.

Case 2

A 60-year-old man is receiving enoxaparin after knee replacement surgery. On day 8 his platelets fell from 250 to 90 × 10^9/l, and deep vein thrombosis of the left leg was found.

Questions: 1) Diagnosis? 2) Management.

Answer: 1) Heparin-induced thrombocytopenia (HIT) — an immune mechanism (antibodies to the heparin-PF4 complex), a platelet fall of more than 50%, on days 5–14 and with paradoxical thrombosis. 2) Stop all heparins immediately (including heparin solutions for flushing catheters); a non-heparin anticoagulant — intravenous argatroban or bivalirudin, or fondaparinux; once platelets recover, a DOAC or warfarin (warfarin only when platelets are > 150, otherwise there is a risk of skin necrosis and venous gangrene); platelet transfusion is not recommended; antibody testing; record heparin allergy in the medical records.

Case 3

A 28-year-old woman complains of weakness, dizziness, brittle nails and a craving to eat chalk. Her periods are heavy and prolonged. Hemoglobin 88 g/l, MCV 68 fl, ferritin 6 ng/ml. She is planning a pregnancy.

Questions: 1) Diagnosis? 2) Treatment plan? 3) Additional advice about planning pregnancy.

Answer: 1) Iron deficiency anemia (microcytic, low ferritin) caused by menorrhagia. 2) An oral iron(II) preparation — elemental iron 100 mg once a day or every other day, on an empty stomach, with ascorbic acid, without tea or milk; the effect is assessed by a rise in hemoglobin after 2–4 weeks; after hemoglobin normalizes, continue for another 3 months (to replenish stores); warn about dark stools; a gynecologist's consultation to find the cause of the menorrhagia (tranexamic acid if needed). If oral iron is not tolerated — intravenous ferric carboxymaltose. 3) Start folic acid 400 mcg/day 1–3 months before pregnancy (prevention of neural tube defects) and restore iron stores before pregnancy.

Quick review table

Table 14.8. *Drugs acting on the blood system — summary*

Group	Preparations	Mechanism	Monitoring / antidote	Key note
COX inhibitor	Aspirin 75–100 mg	Irreversible COX-1 blockade, TXA_2 ↓	—	Stomach, bleeding, Reye syndrome
$P2Y_{12}$ blockers	Clopidogrel, prasugrel, ticagrelor	ADP receptor blockade	—	Dual antiplatelet therapy with aspirin; clopidogrel — CYP2C19
GP IIb/IIIa blockers	Eptifibatide, tirofiban	The final pathway of aggregation	—	Intravenous only, during coronary intervention

Group	Preparations	Mechanism	Monitoring / antidote	Key note
Unfractionated heparin	Heparin	AT III + IIa and Xa	aPTT / protamine	HIT, osteoporosis
LMWH	Enoxaparin, nadroparin	AT III + Xa	Not needed / protamine (partial)	Drug of choice in pregnancy; kidneys
Vitamin K antagonists	Warfarin	VKORC1 — II, VII, IX, X	INR / vitamin K, prothrombin complex	Many interactions; contraindicated in pregnancy; drug of choice with a mechanical valve
DOACs	Apixaban, rivaroxaban, dabigatran	Direct Xa / IIa	Not needed / andexanet, idarucizumab	Contraindicated with a mechanical valve
Fibrinolytics	Alteplase, tenecteplase, streptokinase	Plasminogen → plasmin	— / tranexamic acid	Infarction (12 hours), stroke (4.5 hours), PE
Antifibrinolytics	Tranexamic acid	Blockade of plasminogen-fibrin binding	—	Trauma, postpartum hemorrhage
Vitamin K	Phytomenadione, Vikasol	Synthesis of II, VII, IX, X	—	Prevention in newborns, warfarin
Iron preparations	Iron sulfate, polymaltose, carboxymaltose	Hemoglobin synthesis	— / deferoxamine	On an empty stomach, 3 more months
B_{12}, folic acid	Cyanocobalamin, folic acid	DNA synthesis	—	Folate alone is dangerous in B_{12} deficiency

TOPIC 15

Hormonal preparations of protein and polypeptide structure. Anti-inflammatory drugs

Learning objectives

This topic brings together two large and practically very important sections. Among **protein and polypeptide hormones**, the most important is **insulin**: diabetes mellitus affects about 10% of the population, and insulin and other glucose-lowering drugs are part of a GP's daily practice. Hypothalamic and pituitary hormones, and thyroid and parathyroid hormones are used in endocrinology, obstetrics and pediatrics. **Anti-inflammatory drugs** — nonsteroidal (NSAIDs) and steroidal (glucocorticoids) — are among the most consumed drugs in the world, and their adverse effects (gastrointestinal bleeding, kidney injury, corticosteroid complications) are also very common.

The student should know:

- the classification of hormonal preparations and the preparations of hypothalamic-pituitary hormones;
- thyroid hormones and antithyroid drugs;
- the classification of insulin preparations, the principles of insulin therapy and the management of hypoglycemia;
- the mechanism and uses of oral and other glucose-lowering drugs (metformin, sulfonylureas, DPP-4 inhibitors, GLP-1 agonists, SGLT2 inhibitors);
- inflammatory mediators, and the classification, mechanism, uses and adverse effects of NSAIDs;
- the mechanism, effects, uses and adverse effects of glucocorticoids and the rules for withdrawing them.

The student should be able to:

- write prescriptions for drugs such as insulin, metformin, gliclazide, levothyroxine, thiamazole, oxytocin, ibuprofen, diclofenac, paracetamol, prednisolone and dexamethasone;
- provide emergency care in hypoglycemia;
- assess the patient's gastrointestinal and cardiovascular risk when choosing an NSAID.

1. Classification of hormonal preparations

Table 15.1. *Classification of hormonal preparations by chemical structure*

Group	Hormones and preparations
Protein and polypeptide structure	Hypothalamic hormones (gonadoliberin, somatostatin), pituitary hormones (growth hormone, ACTH, TSH, gonadotropins, prolactin, oxytocin, vasopressin), insulin, glucagon, calcitonin, parathyroid hormone, GLP-1
Amino acid derivatives	Thyroid hormones (thyroxine, triiodothyronine), catecholamines
Steroid structure	Glucocorticoids, mineralocorticoids, estrogens, progestins, androgens, anabolic steroids

REMEMBER

Protein and polypeptide hormones **are broken down in the gastrointestinal tract** — so they are given mainly **parenterally** (injection, intranasally). Exceptions: desmopressin (nasal spray, sublingual and oral tablets), the oral form of semaglutide (with a special absorption enhancer).

2. Preparations of hypothalamic and pituitary hormones

Table 15.2. *Preparations of hypothalamic and pituitary hormones*

Hormone	Preparations	Main use
Somatotropin (growth hormone)	Somatropin (recombinant)	Growth hormone deficiency in children (pituitary dwarfism), Turner syndrome
Somatostatin analogs	Octreotide, lanreotide	Acromegaly, neuroendocrine tumors, bleeding from esophageal varices
Gonadoliberin analogs	Goserelin, leuprorelin, triptorelin	With continuous use they “switch off” the pituitary: prostate cancer, endometriosis, precocious puberty; with pulsatile use — infertility
Gonadotropins	Menotropin, chorionic gonadotropin, follitropin	Infertility, ovulation induction, cryptorchidism
Corticotropin (ACTH)	Tetracosactide	Diagnosis of adrenal insufficiency
Inhibitors of prolactin secretion (dopamine agonists)	Bromocriptine, cabergoline	Hyperprolactinemia, prolactinoma, suppression of lactation
Oxytocin	Oxytocin, carbetocin	Augmentation (stimulation) of labor, prevention and treatment of postpartum hemorrhage, improving milk ejection during lactation
Vasopressin (antidiuretic hormone) and its analogs	Vasopressin, desmopressin, terlipressin	Central diabetes insipidus (desmopressin), nocturnal enuresis, hemophilia A and von Willebrand disease (desmopressin), variceal bleeding (terlipressin), septic shock (vasopressin)

CLINICAL SIGNIFICANCE

Oxytocin stimulates oxytocin receptors of the uterine myometrium and strengthens rhythmic contractions; the sensitivity of the myometrium rises sharply toward the end of pregnancy. Its use carries a risk of **uterine hyperstimulation** (fetal hypoxia, uterine rupture) and, at large doses, of an antidiuretic effect (**water intoxication**, hyponatremia); it is given only with an infusion pump under CTG monitoring. Other drugs acting on the myometrium are studied in 5th-semester independent study.

CAUTION

During treatment with **desmopressin** (especially in the elderly and in children with enuresis) there is a risk of **hyponatremia** and water intoxication: fluid intake in the evening must be restricted.

3. Thyroid hormones and antithyroid drugs

3.1. Thyroid hormones

The thyroid gland produces **thyroxine (T_4)** and **triiodothyronine (T_3)**. T_4 is converted in the periphery into the more active T_3. Through nuclear receptors they regulate basal metabolism, heat

production, growth and development (especially of the CNS), and cardiac function.

Levothyroxine (L-thyroxine) is the drug of choice for treating **hypothyroidism** (replacement therapy). T½ ≈ 7 days — once a day; the dose is titrated according to **TSH** (every 6–8 weeks). The full replacement dose in adults is ≈ 1.6 mcg/kg/day; **in the elderly and in IHD it is started at a small dose** (12.5–25 mcg) so as not to provoke myocardial ischemia. It is taken **on an empty stomach**, 30–60 minutes before breakfast, separately from other drugs (calcium, iron, PPIs, antacids). The dose requirement increases in pregnancy.

Other uses: congenital hypothyroidism (in newborns — early start is crucial for mental development), after thyroidectomy, TSH suppression in thyroid cancer.

Adverse effects (in overdose) — drug-induced thyrotoxicosis: tachycardia, atrial fibrillation, angina, weight loss, tremor, osteoporosis.

> **CAUTION**
> Using levothyroxine **for weight loss** is dangerous and prohibited.

3.2. Antithyroid drugs

Table 15.3. *Antithyroid drugs*

Group	Preparations	Mechanism	Use	Adverse effects
Thionamides	Thiamazole (Mercazolil), propylthiouracil	Inhibit thyroid peroxidase and stop hormone synthesis; PTU also reduces the peripheral conversion of $T_4 \rightarrow T_3$	Diffuse toxic goiter (Graves disease), preparation for surgery or radioiodine	Agranulocytosis (blood count immediately with fever or sore throat!), rash, hepatotoxicity (PTU — severe), teratogenicity (thiamazole — in the 1st trimester)
Iodine preparations (at large doses)	Potassium iodide, Lugol's solution	Rapidly stop hormone release (Wolff–Chaikoff effect)	Thyroid storm, before surgery (reduce the blood supply of the gland)	The effect is temporary ("escape" after 2–3 weeks)
Radioactive iodine	Sodium iodide ^{131}I	Destroys gland tissue by radiation	Graves disease, toxic nodule, cancer	Hypothyroidism; contraindicated in pregnancy
Beta-blockers	Propranolol	Reduce sympathetic signs, slightly inhibit $T_4 \rightarrow T_3$	Symptoms of thyrotoxicosis	—

> **REMEMBER**
> Treatment of thyrotoxicosis in pregnancy: **in the 1st trimester — propylthiouracil** (thiamazole is teratogenic), **in the 2nd–3rd trimesters — thiamazole** (PTU is hepatotoxic). During treatment with thionamides, the patient must be warned about **signs of agranulocytosis** (fever, sore throat, mouth ulcers), and in such cases the drug must be stopped immediately and the white cell count checked.

Prevention of iodine deficiency (endemic goiter) — iodized salt, potassium iodide 200–250 mcg/day in pregnant women. Uzbekistan is considered an iodine-deficient area.

3.3. Hormones regulating calcium metabolism

The **parathyroid hormone** analog **teriparatide** (in severe osteoporosis, stimulates bone formation, daily SC). **Calcitonin** inhibits osteoclasts (hypercalcemia, Paget disease; now rarely). Drugs for treating osteoporosis (bisphosphonates, denosumab, vitamin D) are studied in 5th-semester independent study.

4. Insulin and glucose-lowering drugs

4.1. Physiology of insulin

Insulin is the hormone of the pancreatic **beta cells**; it consists of A and B chains (51 amino acids). Glucose enters the beta cell and generates ATP → **ATP-dependent K^+ channels close** → depolarization → Ca^{2+} enters → insulin is released.

Insulin acts through a **tyrosine kinase receptor**:

- it increases glucose entry into muscle and fat cells (GLUT4 transporters move to the membrane);
- it enhances glycogen synthesis in the liver and inhibits **gluconeogenesis and glycogenolysis**;
- it inhibits lipolysis and enhances fat synthesis (stops the formation of **ketone bodies**);
- it increases protein synthesis;
- it shifts **potassium into cells** (activation of Na^+/K^+-ATPase).

4.2. Insulin preparations

Modern preparations are **recombinant human insulin** and its **analogs** (with a modified amino acid sequence and improved pharmacokinetics). The concentration is usually **100 U/ml** (U-100).

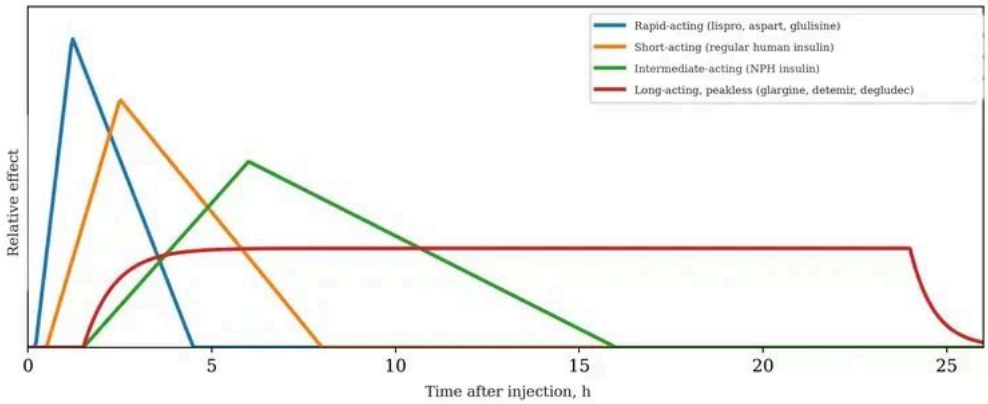


Figure 15.1. *Action profiles of insulin preparations (author's diagram)*

Table 15.4. *Classification of insulin preparations*

Group	Preparations	Onset of action	Peak	Duration	Use
Ultra-short-acting (rapid-acting analogs)	Lispro, aspart, glulisine	5–15 minutes	1–2 hours	3–5 hours	Before meals (bolus), insulin pumps
Short-acting (regular human insulin)	Actrapid, Humulin R	30 minutes	2–4 hours	6–8 hours	30 minutes before meals; intravenously — diabetic ketoacidosis, hyperkalemia
Intermediate-acting	NPH (isophane) insulin	1–2 hours	4–10 hours	12–18 hours	Basal (2 times a day)

Group	Preparations	Onset of action	Peak	Duration	Use
Long-acting analogs	Glargine, detemir	1–2 hours	Almost peakless	20–24 hours	Basal, once a day
Ultra-long-acting	Degludec, glargine U-300	1–2 hours	Peakless	36–42 hours	Basal; little nocturnal hypoglycemia
Premixed	NPH 70/30, aspart 70/30	—	Two peaks	12–18 hours	Simple regimen (2 times a day)

REMEMBER

Only short-acting (regular, soluble) insulin can be given intravenously; NPH and premixed insulins are suspensions — not for intravenous use. In diabetic ketoacidosis: **fluids (0.9% NaCl) → intravenous infusion of short-acting insulin** (0.1 U/kg/hour) → **potassium** (insulin shifts potassium into cells and causes hypokalemia) → glucose is added when blood glucose falls to 14 mmol/l.

Insulin therapy regimens:

- **basal-bolus (intensive)** — the standard in type 1 diabetes: a long-acting insulin (basal, ≈ 50% of the daily dose) + a short- or ultra-short-acting insulin before each meal (bolus); or an insulin pump;
- **in type 2 diabetes** — adding an evening basal insulin to oral agents (starting with 10 U or 0.1–0.2 U/kg, titrated by fasting morning glucose). **Uses:** type 1 diabetes mellitus (vital), type 2 diabetes when other agents are ineffective or in decompensation, **gestational diabetes** (the drug of choice in pregnancy), diabetic ketoacidosis and the hyperosmolar state, surgery and severe illness, **hyperkalemia** (with glucose).

Adverse effects: hypoglycemia (the most common and dangerous), weight gain, **lipodystrophy** at the injection site (sites must be rotated), hypokalemia, allergic reactions (rare).

4.3. Hypoglycemia

Hypoglycemia is glucose **below 3.9 mmol/l** (clinically significant — below 3.0). Causes: an excessive dose of insulin or a sulfonylurea, a missed meal, physical exertion, alcohol, renal failure.

Signs: adrenergic — sweating, tremor, palpitations, hunger, anxiety; neuroglycopenic — headache, confusion, unusual behavior, visual disturbances, seizures, coma. **Beta-blockers mask the adrenergic signs.**

REMEMBER

Care in hypoglycemia (the “rule of 15”): if conscious — **15–20 g of rapidly absorbed carbohydrate** (3–4 glucose tablets, 150–200 ml of sweet juice), recheck glucose after 15 minutes, repeat if needed, then a meal with complex carbohydrates. If **unconscious — 40% glucose 20–40 ml intravenously** (or 10% glucose) or **glucagon 1 mg** intramuscularly or subcutaneously (nasal powder 3 mg) — relatives can also give it at home. Fluids must not be poured into the mouth of an unconscious patient (aspiration).

4.4. Non-insulin glucose-lowering drugs

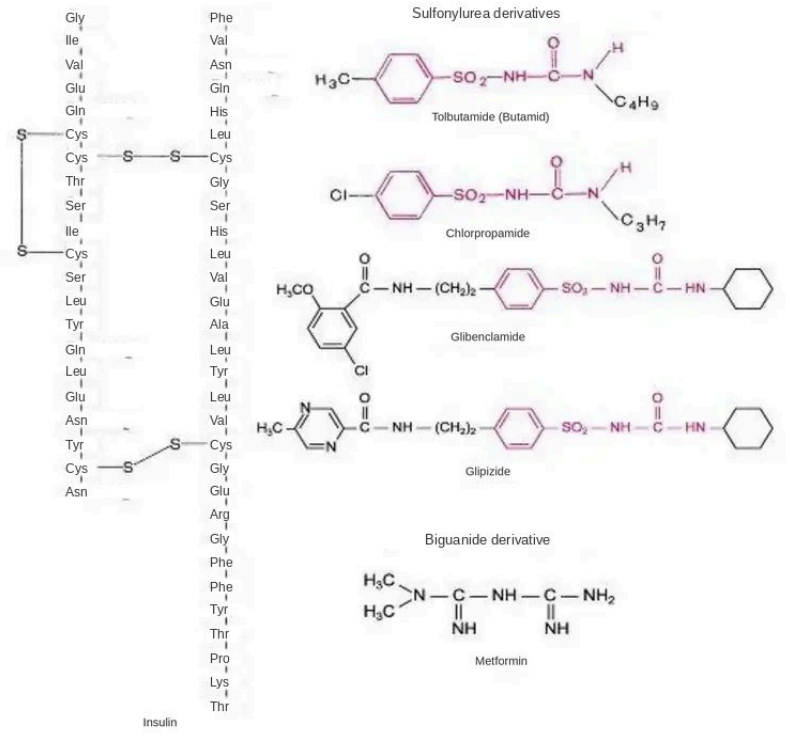


Figure 15.2. *The primary structure of insulin and the chemical structure of oral glucose-lowering drugs (sulfonylurea derivatives, metformin) (teaching material of the TMA Department of Pharmacology)*

Table 15.5. *Non-insulin glucose-lowering drugs*

Group	Preparations	Mechanism	Risk of hypoglycemia	Effect on weight	Main adverse effects and features
Biguanides	Metformin	Reduces hepatic gluconeogenesis (AMPK activation), increases peripheral insulin sensitivity	None (in monotherapy)	Neutral or decreases	Gastrointestinal disturbances (diarrhea, metallic taste — start with a small dose, with food), B_{12} deficiency, very rarely — lactic acidosis (contraindicated at GFR < 30; contrast agents, hypoxia, alcoholism)
Sulfonylurea derivatives	Gliclazide, glimepiride, glibenclamide	Close KATP channels in beta cells and increase insulin secretion	High (glibenclamide — the highest)	Increases	Hypoglycemia (dangerous in the elderly and in renal failure), weight gain
Meglitinides	Repaglinide, nateglinide	Like sulfonylureas, but short-acting	Intermediate	Increases	Before each meal

Group	Preparations	Mechanism	Risk of hypoglycemia	Effect on weight	Main adverse effects and features
Thiazolidinediones (glitazones)	Pioglitazone	A PPAR-gamma agonist — increases insulin sensitivity	Absent	Increases	Fluid retention, heart failure (contraindicated), bone fractures, risk of bladder cancer
DPP-4 inhibitors (gliptins)	Sitagliptin, vildagliptin, saxagliptin, linagliptin	Reduce the breakdown of incretins (GLP-1, GIP), increase glucose-dependent insulin secretion and reduce glucagon	Absent	Neutral	Well tolerated; rarely pancreatitis, joint pain
GLP-1 receptor agonists	Semaglutide, liraglutide, dulaglutide; tirzepatide (GIP/GLP-1)	Glucose-dependent insulin secretion ↑, glucagon ↓, gastric emptying slows, appetite decreases	Absent	Decreases markedly	Nausea, vomiting, pancreatitis, gallstones; reduce cardiovascular complications (semaglutide, liraglutide); treatment of obesity
SGLT2 inhibitors (gliflozins)	Empagliflozin, dapagliflozin, canagliflozin	Stop glucose reabsorption in the kidney — glucosuria	Absent	Decreases	Genital mycoses and urinary tract infections, hypovolemia, euglycemic ketoacidosis (stopped 3 days before surgery); reduce mortality and complications in heart failure and chronic kidney disease
Alpha-glucosidase inhibitors	Acarbose	Slows the breakdown of carbohydrates in the intestine	Absent	Neutral	Flatulence, diarrhea

CLINICAL SIGNIFICANCE

The modern principle of treating type 2 diabetes mellitus: first line — lifestyle changes + **metformin** (unless contraindicated). If the patient has **atherosclerotic cardiovascular disease, heart failure or chronic kidney disease**, an **SGLT2 inhibitor** and/or a **GLP-1 agonist** is added regardless of the glucose level (organ-protective effect). GLP-1 agonists are preferred in obesity. The target HbA_{1c} for most patients is **< 7%** (individualized in the elderly and frail, avoiding hypoglycemia).

CAUTION

Older textbooks presented sulfonylureas (especially glibenclamide) and biguanides such as buformin and phenformin as the main drugs. **Phenformin and buformin were withdrawn from the market because of lactic acidosis**; glibenclamide is not recommended in the elderly because it causes prolonged hypoglycemia — **gliclazide MR** or glimepiride is preferred.

5. Anti-inflammatory drugs

5.1. Inflammatory mediators and the arachidonic acid cascade

Arachidonic acid is released from cell membrane phospholipids by **phospholipase A_2**. It is metabolized by two pathways:

- **the cyclooxygenase (COX)** pathway → prostaglandin G_2/H_2 → **prostaglandins** (PGE_2 — pain, fever, inflammation, gastric protection, renal blood flow; PGI_2 — prostacyclin), **thromboxane A_2**;
- **the lipoxygenase** pathway → **leukotrienes** (LTB_4 — chemotaxis; LTC_4, D_4, E_4 — bronchospasm, vascular permeability).

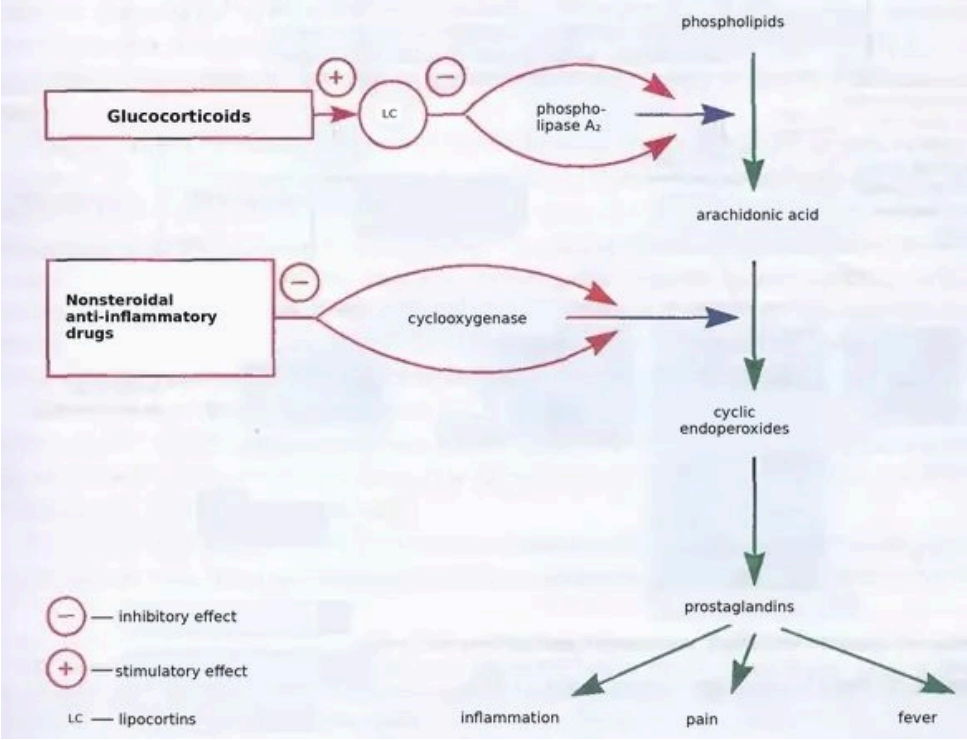


Figure 15.3. *Sites of action of anti-inflammatory drugs: glucocorticoids inhibit phospholipase A_2 (via lipocortin), NSAIDs inhibit cyclooxygenase (teaching material of the TMA Department of Pharmacology)*

Table 15.6. *Cyclooxygenase isoforms*

Isoform	Expression	Function
COX-1 (“constitutive”, “housekeeping”)	Constant: stomach, kidneys, platelets, endothelium	PGs protecting the gastric mucosa (mucus, bicarbonate, blood flow), renal blood flow, TXA_2 in platelets
COX-2 (“inducible”)	In the inflammatory focus under the influence of cytokines; also constitutive in the CNS, kidneys and endothelium	PGs for inflammation, pain and fever; prostacyclin in the endothelium

5.2. Nonsteroidal anti-inflammatory drugs (NSAIDs)

Mechanism. By inhibiting COX, they reduce prostaglandin synthesis. As a result:

1. **anti-inflammatory** effect (vascular permeability and edema decrease);
2. **analgesic** effect — peripheral (fewer PGs that sensitize nociceptors to bradykinin and histamine) and central;
3. **antipyretic** effect — PGE_2 synthesis in the hypothalamus decreases (normal temperature is not lowered);
4. **antiplatelet** effect — TXA_2 in platelets decreases (aspirin — irreversibly).

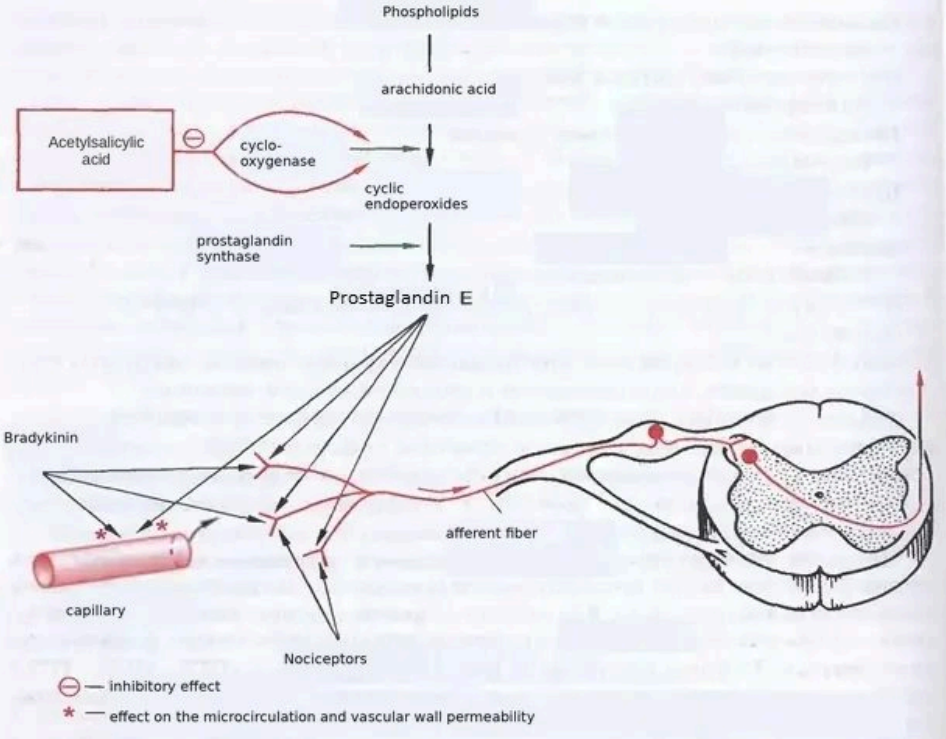


Figure 15.4. *Mechanism of the analgesic action of acetylsalicylic acid: by reducing prostaglandin E synthesis it lowers the sensitivity of nociceptors to bradykinin (teaching material of the TMA Department of Pharmacology)*

Table 15.7. *Classification of NSAIDs*

Group	Preparations	Features
Nonselective inhibitors of COX-1 and COX-2	Acetylsalicylic acid, ibuprofen, naproxen, ketoprofen, diclofenac, indomethacin, ketorolac, piroxicam, metamizole	Ibuprofen — relatively safe for the stomach (at low doses); diclofenac — potent, high cardiovascular risk; indomethacin — potent, the most adverse effects; ketorolac — a strong analgesic (no more than 5 days); naproxen — relatively low cardiovascular risk
Preferential COX-2 inhibitors	Meloxicam, nimesulide	Nimesulide — hepatotoxicity (restricted in many countries)
Selective COX-2 inhibitors (coxibs)	Celecoxib, etoricoxib	Low gastrointestinal risk, no antiplatelet effect, increased cardiovascular thrombotic risk
Analgesic-antipyretics (weak anti-inflammatory effect)	Paracetamol, metamizole (Analgin)	Paracetamol hardly inhibits COX in the periphery; no effect on the stomach or platelets

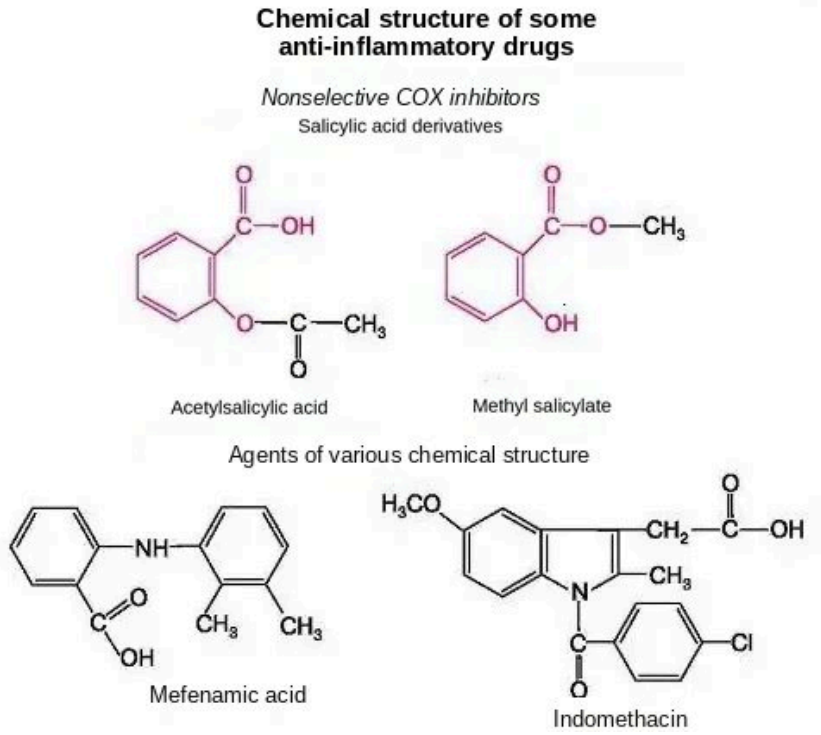


Figure 15.5. *Chemical structure of some anti-inflammatory drugs (teaching material of the TMA Department of Pharmacology)*

Uses: pain syndrome (headache, toothache, muscle and joint pain, dysmenorrhea, renal colic, postoperative pain), rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, **gout attack**, fever; aspirin — antiplatelet; indomethacin — closure of a patent ductus arteriosus in premature infants.

Table 15.8. *Adverse effects of NSAIDs*

System	Adverse effect	Mechanism / risk factors
Gastrointestinal	Dyspepsia, erosions and ulcers, bleeding, perforation (NSAID gastropathy)	Inhibition of COX-1 — fewer protective PGs; risk factors: age > 65, ulcer history, glucocorticoids, anticoagulants, SSRIs, high dose, H. pylori
Kidney	Acute kidney injury, sodium and water retention, hyperkalemia, interstitial nephritis, papillary necrosis	Reduced renal PGs; the “triple whammy” — NSAID + ACE inhibitor/ARB + diuretic — is especially dangerous
Cardiovascular	Rise in BP, worsening of heart failure, risk of myocardial infarction and stroke (coxibs, diclofenac)	Disturbed prostacyclin/TXA_2 balance, sodium retention
Respiration	Aspirin (NSAID)-induced asthma — bronchospasm, rhinosinusitis, nasal polyps	Arachidonic acid is “shunted” to the lipoxygenase pathway — leukotrienes increase
Liver	Raised transaminases (diclofenac, nimesulide)	—
Blood	Bleeding (antiplatelet effect); agranulocytosis (metamizole)	—
Pregnancy	In the 3rd trimester — premature closure of the ductus arteriosus, oligohydramnios, delayed labor	Avoid NSAIDs after 20 weeks
Children	Reye syndrome (aspirin)	Aspirin is contraindicated for fever under 16 years

CLINICAL SIGNIFICANCE

Prescribing NSAIDs **to patients at high gastrointestinal risk**: paracetamol or a topical NSAID (gel) where possible; if necessary — the lowest effective dose for the shortest time, **together with a PPI**, or a selective COX-2 inhibitor + a PPI. In patients at high cardiovascular risk — **naproxen** or low-dose ibuprofen; avoid coxibs and diclofenac. Ibuprofen may interfere with the cardioprotective antiplatelet effect of aspirin — aspirin is taken at least 30 minutes before ibuprofen.

5.3. Paracetamol

Paracetamol (acetaminophen) is an analgesic and antipyretic; its anti-inflammatory effect is very weak. Its mechanism is not fully clear: it inhibits COX mainly in the CNS (in the periphery, the high peroxide levels in the inflammatory focus inactivate it) and acts on the serotonergic and cannabinoid systems. It has almost no effect on the gastric mucosa, platelets or kidneys — it is the **first-choice** analgesic-antipyretic **in children, pregnant women and the elderly** and in gastric diseases.

Dose: 500–1000 mg for adults, **no more than 4 g a day** (2–3 g with liver disease, alcoholism, starvation, low body weight); **10–15 mg/kg** for children, every 4–6 hours, no more than 60 mg/kg a day.

CAUTION

Paracetamol poisoning — hepatotoxicity. At therapeutic doses, paracetamol is conjugated mainly with glucuronide and sulfate; a small part is converted by CYP2E1 into the toxic **NAPQI**, which is detoxified by **glutathione**. In overdose glutathione is depleted → NAPQI destroys hepatocytes → **liver necrosis** (on days 2–4; almost no signs on the first day!). **Antidote — N-acetylcysteine** (restores glutathione stores), most effective within 8 hours; measure the blood paracetamol concentration (using the nomogram). Risk factors: alcoholism, starvation, CYP inducers. Many combined “cold” remedies contain paracetamol — risk of **hidden overdose**.

5.4. Metamizole (Analgin)

A strong analgesic and antipyretic, antispasmodic. Because of **agranulocytosis** (rare but potentially fatal), it is banned or restricted in many countries; it is widely used in CIS countries. Rapid intravenous injection causes hypotension and anaphylaxis.

6. Glucocorticoids

6.1. Mechanism

Glucocorticoids (cortisol and its synthetic analogs), being lipophilic molecules, enter the cell and bind to the cytoplasmic **glucocorticoid receptor**; the complex moves into the nucleus and changes gene transcription:

- **transrepression** — suppresses the expression of inflammatory genes (via NF-kappaB, AP-1): cytokines (IL-1, IL-6, TNF-alpha), COX-2, iNOS and adhesion molecules decrease;
- **transactivation** — increases the synthesis of anti-inflammatory proteins (**lipocortin-1 (annexin-1)**, which inhibits phospholipase A_2 — as a result **both prostaglandins and leukotrienes** decrease); the metabolic adverse effects also arise mainly through this pathway.

6.2. Pharmacological effects

Table 15.9. *Effects of glucocorticoids*

Effect	Essence
Anti-inflammatory	All stages of inflammation (alteration, exudation, proliferation) are suppressed
Immunosuppressive	The number and activity of lymphocytes (especially T cells) decrease, antibody formation and cytokine synthesis decrease
Antiallergic	Release of mediators from mast cells and basophils decreases
Antishock	Vascular sensitivity to catecholamines increases, BP rises
Carbohydrate metabolism	Gluconeogenesis ↑ — hyperglycemia, “steroid diabetes”
Protein metabolism	Catabolism — muscle atrophy, skin thinning, slow wound healing, growth retardation in children
Fat metabolism	Redistribution of fat — cushingoid appearance (face, neck, abdomen)
Water-salt metabolism	Sodium and water retention, potassium loss (depends on mineralocorticoid activity)
Calcium metabolism	Intestinal absorption ↓, renal excretion ↑ — osteoporosis
Blood	Neutrophils ↑, lymphocytes, eosinophils and monocytes ↓

6.3. Preparations

Table 15.10. *Comparison of glucocorticoid preparations*

Preparation	Equivalent dose (mg)	Anti-inflammatory activity (cortisol = 1)	Mineralocorticoid activity	Duration of action
Hydrocortisone (cortisol)	20	1	1 (high)	Short (8–12 hours)
Prednisolone	5	4	0,8	Intermediate (12–36 hours)
Methylprednisolone	4	5	0,5	Intermediate
Triamcinolone	4	5	0	Intermediate
Dexamethasone	0,75	25–30	0	Long (36–72 hours)
Betamethasone	0,6	25–30	0	Long
Fludrocortisone	—	10	125 (very high)	— (as a mineralocorticoid)

MEMORY AID
Equivalent doses: **“20 – 5 – 4 – 0.75”** — hydrocortisone 20 mg = prednisolone 5 mg = methylprednisolone 4 mg = dexamethasone 0.75 mg. Dexamethasone is chosen for fetal lung maturation and in cerebral edema (no mineralocorticoid effect, potent, crosses the placenta).

Topical preparations: **inhaled** (budesonide, fluticasone — Topic 9), **nasal sprays** (mometasone), **for the skin** (hydrocortisone 1% — weak, betamethasone, clobetasol — very potent), **eye drops**, intra-articular (triamcinolone, depot betamethasone).

6.4. Uses

- **replacement therapy** — adrenal insufficiency (Addison disease — hydrocortisone + fludrocortisone), congenital adrenal hyperplasia;
- **rheumatic and autoimmune diseases** — rheumatoid arthritis, systemic lupus erythematosus, vasculitis, polymyalgia;
- **allergic conditions** — anaphylaxis (after epinephrine, as an add-on), angioedema, severe dermatitis, allergic rhinitis (topical);
- **bronchial asthma and COPD** (inhaled maintenance therapy; systemic in exacerbations);
- **kidney diseases** (nephrotic syndrome), **bowel** (Crohn disease, ulcerative colitis), **liver** (autoimmune hepatitis);
- **hematology and oncology** — leukemias, lymphomas (in chemotherapy regimens), chemotherapy-induced vomiting;
- **transplantation** — prevention of rejection of the transplanted organ;
- **cerebral edema** (peritumoral — dexamethasone), bacterial meningitis (dexamethasone), severe COVID-19 and pneumonia (dexamethasone);
- **fetal lung maturation** — at risk of preterm birth (dexamethasone, betamethasone);
- **shock** — septic shock (resistant to vasopressors), adrenal crisis.

6.5. Adverse effects

Table 15.11. *Adverse effects of glucocorticoids (with long-term systemic use)*

System	Adverse effects
Endocrine-metabolic	Cushingoid appearance (moon face, abdominal obesity), steroid diabetes, dyslipidemia, weight gain, growth retardation in children
Hypothalamic-pituitary-adrenal	Atrophy of the adrenal cortex — acute adrenal insufficiency on abrupt withdrawal
Musculoskeletal	Osteoporosis and fractures, avascular necrosis of the femoral head, steroid myopathy
Cardiovascular	Arterial hypertension, edema, hypokalemia
Gastrointestinal	Ulcer (mainly together with NSAIDs), pancreatitis
Immune	Infections (bacterial, fungal, tuberculosis reactivation, opportunistic), masking of signs of infection
Skin	Thinning, striae, ecchymoses, acne, slow wound healing
Eye	Cataract (posterior subcapsular), glaucoma
Psychiatric	Insomnia, euphoria, depression, steroid psychosis

REMEMBER

The rule for withdrawing glucocorticoids: if systemic glucocorticoids have been taken **for more than 3 weeks** (or > 20 mg/day of prednisolone equivalent for more than 1 week), they **must not be stopped abruptly** — the dose is reduced gradually (to allow the hypothalamic-pituitary-adrenal axis to recover). Abrupt withdrawal causes **acute adrenal insufficiency** (severe weakness, hypotension, shock, hypoglycemia, hyponatremia) and relapse of the disease. In such patients the dose must be increased during **stress** (surgery, severe infection, trauma) (“stress dose”, intravenous hydrocortisone).

CLINICAL SIGNIFICANCE

Prevention during long-term glucocorticoid treatment: take once in the morning (in line with the circadian rhythm); the lowest effective dose; **calcium and vitamin D**, and a bisphosphonate if the risk is high (osteoporosis); a PPI if combined with an NSAID; monitoring of glucose, BP and intraocular pressure; screening for latent tuberculosis and hepatitis B before treatment; live vaccines are contraindicated.

7. Clinical part

Table 15.12. *Choice of analgesic in pain and fever*

Situation	Choice	To be avoided
Fever in a child	Paracetamol or ibuprofen (weight-based)	Aspirin (Reye syndrome), metamizole
Pain/fever in pregnancy	Paracetamol	NSAIDs (especially after 20 weeks)
History of gastric ulcer	Paracetamol; if necessary a coxib + a PPI	A nonselective NSAID without a PPI
IHD, heart failure	Paracetamol; naproxen for a short time	Diclofenac, coxibs, high-dose NSAIDs
Chronic kidney disease	Paracetamol	All systemic NSAIDs
Aspirin-induced asthma	Paracetamol (≤ 1 g), coxibs (with caution)	Nonselective NSAIDs
Gout attack	NSAIDs (naproxen, indomethacin), colchicine or a glucocorticoid	Aspirin (alters uric acid excretion)
Renal colic	NSAIDs (diclofenac, ketorolac)	—
Osteoarthritis (single joint)	Topical NSAIDs (gel), paracetamol	—

Practical skills

Skill 1. Teaching insulin injection (pen injector)

1. Wash your hands; check the insulin name and expiry date; mix NPH and premixed insulins by tipping the pen 10–20 times (without shaking).
2. Attach a new needle and do a 2 U “test” (priming to remove air).
3. Dial the dose.
4. Pinch a skin fold on the abdomen (fastest absorption — for short-acting insulin), thigh, upper arm or buttock.
5. Insert the needle at 90° (short needle), press the button, wait **10 seconds** and withdraw the needle.
6. Change the injection site every time (to prevent lipodystrophy).
7. An opened pen is kept at room temperature for 4 weeks; spare pens in the refrigerator (not frozen).

Skill 2. Key prescriptions

PRESCRIPTION 15.1. OXYTOCIN
Rp.: Oxytocini 5 U — 1 ml
D.t.d. N. 10 in ampullis
S. 10 U intramuscularly to prevent postpartum hemorrhage (by an obstetrician).

PRESCRIPTION 15.2. DESMOPRESSIN
Rp.: Desmopressini 0.0001
D.t.d. N. 30 in tabulettis
S. 1 tablet 2–3 times a day (in central diabetes insipidus; under sodium monitoring).

PRESCRIPTION 15.3. LEVOTHYROXINE
Rp.: Levothyroxini natrii 0.00005
D.t.d. N. 100 in tabulettis
S. Take 1 tablet once a day in the morning 30–60 minutes before breakfast.

PRESCRIPTION 15.4. THIAMAZOLE
Rp.: Thiamazoli 0.005
D.t.d. N. 50 in tabulettis
S. Take 2 tablets 3 times a day; if fever or a sore throat appears, see a doctor immediately.

PRESCRIPTION 15.5. INSULIN GLARGINE
Rp.: Insulini glargini 100 U/ml — 3 ml
D.t.d. N. 5 in calamis
S. 10 U subcutaneously once a day at the same time; titrate the dose by morning glucose.

PRESCRIPTION 15.6. INSULIN ASPART
Rp.: Insulini asparti 100 U/ml — 3 ml
D.t.d. N. 5 in calamis
S. Subcutaneously before each main meal (0–10 minutes), according to the amount of carbohydrate.

PRESCRIPTION 15.7. METFORMIN
Rp.: Metformini hydrochloridi 0.5
D.t.d. N. 60 in tabulettis obductis
S. 1 tablet once a day with the evening meal, increasing by 1 tablet each week, up to 2 g a day.

PRESCRIPTION 15.8. GLICLAZIDE MR
Rp.: Tabulettas Gliclazidi 0.06 modificatas N. 30
D.S. Take 1/2–1 tablet once a day with breakfast.

PRESCRIPTION 15.9. EMPAGLIFLOZIN
Rp.: Empagliflozini 0.01
D.t.d. N. 30 in tabulettis obductis
S. Take 1 tablet once a day in the morning.

PRESCRIPTION 15.10. SEMAGLUTIDE

Rp.: Solutionis Semaglutidi 1.34 mg/ml — 1.5 ml
D.t.d. N. 1 in calamo
S. 0.25 mg subcutaneously once a week for 4 weeks, then 0.5 mg.

PRESCRIPTION 15.11. GLUCAGON

Rp.: Glucagoni 0.001
D.t.d. N. 1
S. In severe hypoglycemia, dissolve with the diluent and inject 1 mg intramuscularly or subcutaneously.

PRESCRIPTION 15.12. GLUCOSE 40%

Rp.: Solutionis Glucosi 40% — 20 ml
D.t.d. N. 10 in ampullis
S. 20–40 ml as an intravenous push (in hypoglycemic coma).

PRESCRIPTION 15.13. IBUPROFEN

Rp.: Ibuprofeni 0.4
D.t.d. N. 20 in tabulettis obductis
S. Take 1 tablet 3 times a day after meals, for no more than 5 days.

PRESCRIPTION 15.14. IBUPROFEN SUSPENSION (FOR CHILDREN)

Rp.: Suspensionis Ibuprofeni 100 mg/5 ml — 100 ml
D.S. 10 mg/kg when the temperature exceeds 38.5 °C, up to 3 times a day; shake before use.

PRESCRIPTION 15.15. DICLOFENAC (AMPOULES)

Rp.: Solutionis Diclofenaci natrii 2.5% — 3 ml
D.t.d. N. 3 in ampullis
S. 3 ml deep intramuscularly once a day for 2–3 days.

PRESCRIPTION 15.16. CELECOXIB

Rp.: Capsulas Celecoxibi 0.2 N. 20
D.S. Take 1 capsule 1–2 times a day.

PRESCRIPTION 15.17. PARACETAMOL

Rp.: Paracetamoli 0.5
D.t.d. N. 20 in tabulettis
S. 1–2 tablets every 6 hours, no more than 8 tablets (4 g) a day.

PRESCRIPTION 15.18. PREDNISOLONE

Rp.: Prednisoloni 0.005
D.t.d. N. 100 in tabulettis
S. The daily dose once in the morning (at 8 a.m.), after a meal; do not stop the drug abruptly.

PRESCRIPTION 15.19. DEXAMETHASONE (AMPOULES)

Rp.: Solutionis Dexamethasoni phosphatis 0.4% — 1 ml
D.t.d. N. 10 in ampullis
S. 1–2 ml intravenously or intramuscularly.

PRESCRIPTION 15.20. HYDROCORTISONE CREAM
Rp.: Cremoris Hydrocortisoni 1% — 15.0
D.S. Apply a thin layer to the affected skin 2 times a day, for no more than 7 days.

Review questions

1. Classification of hormonal preparations. Why are protein hormones not given orally?
2. Preparations of hypothalamic and pituitary hormones and their uses.
3. Uses and risks of oxytocin and desmopressin.
4. Principles of treatment with levothyroxine.
5. Antithyroid drugs: mechanism, uses, adverse effects. The choice in pregnancy.
6. Physiological effects of insulin and classification of insulin preparations.
7. Which insulin can be given intravenously? Principles of treating diabetic ketoacidosis.
8. Signs of hypoglycemia and emergency care.
9. Mechanism, advantages and contraindications of metformin.
10. Mechanism and risks of sulfonylureas.
11. Mechanism and advantages of DPP-4 inhibitors, GLP-1 agonists and SGLT2 inhibitors.
12. Modern principles of treating type 2 diabetes.
13. The arachidonic acid cascade. The difference between COX-1 and COX-2.
14. Mechanism and main effects of NSAIDs. Classification.
15. Adverse effects of NSAIDs and their prevention.
16. Features, doses and poisoning of paracetamol.
17. Mechanism and pharmacological effects of glucocorticoids.
18. Comparison of glucocorticoid preparations.
19. Uses and adverse effects of glucocorticoids.
20. Rules for withdrawing glucocorticoids. What is a “stress dose”?

Test questions

1. A hormone used to augment labor and prevent postpartum hemorrhage:
   A) Desmopressin
   B) Prolactin
   C) Oxytocin
   D) Vasopressin
   E) Somatostatin
2. The drug of choice for treating central diabetes insipidus:
   A) Insulin
   B) Hydrochlorothiazide
   C) Tolvaptan
   D) Desmopressin
   E) Oxytocin

3. The most dangerous adverse effect of thionamides:
   A) Bronchospasm
   B) Agranulocytosis
   C) Osteoporosis
   D) Hypoglycemia
   E) Hypertrichosis

4. The drug of choice for thyrotoxicosis in the 1st trimester of pregnancy:
   A) Thiamazole
   B) Propylthiouracil
   C) Levothyroxine
   D) Radioactive iodine
   E) Long-term potassium iodide

5. An insulin that can be given intravenously:
   A) Premixed insulin 70/30
   B) NPH insulin
   C) Short-acting (regular) human insulin
   D) Insulin degludec
   E) Insulin glargine

6. An agent that relatives can give at home to an unconscious hypoglycemic patient:
   A) Insulin
   B) Prednisolone
   C) Metformin
   D) Glucagon
   E) Epinephrine

7. The first-line drug for type 2 diabetes that reduces hepatic gluconeogenesis and does not cause hypoglycemia:
   A) Glibenclamide
   B) Pioglitazone
   C) Metformin
   D) Acarbose
   E) Repaglinide

8. A drug that increases insulin secretion by closing KATP channels in beta cells:
   A) Empagliflozin
   B) Sitagliptin
   C) Gliclazide
   D) Metformin
   E) Pioglitazone

9. Drugs that cause glucosuria and reduce mortality in heart failure and chronic kidney disease:
   A) Thiazolidinediones
   B) Meglitinides
   C) SGLT2 inhibitors
   D) Alpha-glucosidase inhibitors
   E) Sulfonylureas

10. A glucose-lowering drug contraindicated in heart failure:
    A) Linagliptin
    B) Metformin
    C) Semaglutide
    D) Empagliflozin
    E) Pioglitazone

11. The main reason for the harmful effect of NSAIDs on the gastric mucosa:
    A) Inhibition of COX-2
    B) Inhibition of COX-1 and reduction of protective prostaglandins
    C) Histamine release
    D) Reduced leukotriene synthesis
    E) Reduced acid secretion

12. The antidote in paracetamol poisoning:
    A) Naloxone
    B) Deferoxamine
    C) Flumazenil
    D) N-acetylcysteine
    E) Atropine

13. A drug contraindicated for lowering fever in a viral infection in children under 16:
    A) Ibuprofen
    B) Physical cooling
    C) Paracetamol
    D) Acetylsalicylic acid
    E) Plenty of fluids

14. The glucocorticoid with no mineralocorticoid activity and the highest anti-inflammatory activity:
    A) Fludrocortisone
    B) Cortisone
    C) Dexamethasone
    D) Hydrocortisone
    E) Prednisolone

15. The main risk of abruptly stopping glucocorticoids after long-term use:
    A) Osteoporosis
    B) Hyperglycemia
    C) Cushingoid appearance
    D) Cataract
    E) Acute adrenal insufficiency

Clinical cases

Case 1

A 58-year-old man with type 2 diabetes mellitus (HbA_{1c} 8.4%), a myocardial infarction 2 years ago, obesity (BMI 34), GFR 65 ml/min. He currently takes only glibenclamide; in the last month he has had two episodes of nocturnal hypoglycemia.

Questions: 1) Shortcomings of the current treatment? 2) Optimal treatment plan.

Answer: 1) Glibenclamide is the sulfonylurea with the highest risk of hypoglycemia, it leads to weight gain and does not reduce cardiovascular complications. 2) Stop glibenclamide; metformin (allowed at GFR 65); because of atherosclerotic cardiovascular disease and obesity, a GLP-1 agonist (semaglutide) and/or an SGLT2 inhibitor (empagliflozin) regardless of glucose — they reduce cardiovascular complications, lower weight and do not cause hypoglycemia; statin, aspirin, BP control; target HbA_{1c} < 7%.

Case 2

A 19-year-old female student has type 1 diabetes. On the day of an exam she skipped breakfast but injected her usual insulin dose. Two hours later she developed tremor, heavy sweating and confusion, and then lost consciousness. The glucometer shows 1.9 mmol/l.

Questions: 1) Condition? 2) Emergency care?

Answer: 1) Severe hypoglycemia (hypoglycemic coma). 2) Do not pour anything into her mouth (aspiration); glucagon 1 mg intramuscularly or subcutaneously (or 3 mg intranasally) — bystanders can give it; a health worker — 40% glucose 40–60 ml as an intravenous push, then an infusion of 5–10% glucose; once conscious — a carbohydrate meal; analyze the cause and teach the patient not to inject insulin without eating; she should always carry a glucagon kit.

Case 3

A 45-year-old woman has been taking prednisolone 15 mg/day for 6 months for rheumatoid arthritis. Because she felt better, she stopped the drug abruptly on her own. Three days later she developed severe weakness, nausea, vomiting, abdominal pain, BP 80/50 mm Hg, glucose 3.2 mmol/l, sodium 128 mmol/l.

Questions: 1) Diagnosis? 2) Pathogenesis? 3) Treatment and prevention.

Answer: 1) Acute adrenal insufficiency (glucocorticoid withdrawal syndrome, secondary). 2) Long-term exogenous glucocorticoid intake suppressed the hypothalamic-pituitary-adrenal axis (ACTH secretion decreased, the adrenal cortex atrophied); when the drug was stopped, her own cortisol production was insufficient. 3) Hydrocortisone 100 mg intravenously immediately, then 200 mg/day as an infusion or every 6 hours; infusion of 0.9% NaCl and glucose; once stable, switch to oral prednisolone and reduce the dose gradually (over several weeks to months). Explain to the patient that glucocorticoids must not be stopped abruptly and that the dose must be increased in stressful situations; give her a “steroid card”.

Quick review table

Table 15.13. *Polypeptide hormones and anti-inflammatory drugs — summary*

Group	Preparations	Mechanism	Main use	Main risk
Oxytocin	Oxytocin, carbetocin	Myometrial contraction	Labor stimulation, hemorrhage prevention	Hyperstimulation, water intoxication
Vasopressin analogs	Desmopressin, terlipressin	V_2 / V_1	Diabetes insipidus, enuresis / bleeding	Hyponatremia
Thyroid hormones	Levothyroxine	Nuclear receptors	Hypothyroidism	Thyrotoxicosis in overdose
Thionamides	Thiamazole, PTU	Thyroid peroxidase ↓	Graves disease	Agranulocytosis

Group	Preparations	Mechanism	Main use	Main risk
Insulins	Aspart, lispro, regular, NPH, glargine, degludec	Tyrosine kinase receptor	Type 1, some type 2 diabetes, DKA	Hypoglycemia
Biguanides	Metformin	Gluconeogenesis ↓	Type 2 diabetes (first line)	Gastrointestinal, lactic acidosis (GFR < 30)
Sulfonylureas	Gliclazide, glimepiride	K_{ATP} closure	Type 2 diabetes	Hypoglycemia, weight
Incretins	Sitagliptin; semaglutide, liraglutide	DPP-4 ↓; GLP-1 agonist	Type 2 diabetes, obesity	GLP-1: nausea, pancreatitis
SGLT2 inhibitors	Empagliflozin, dapagliflozin	Glucosuria	Diabetes, HF, chronic kidney disease	Genital infections, ketoacidosis
Nonselective NSAIDs	Ibuprofen, diclofenac, naproxen, ketorolac	COX-1 and COX-2 ↓	Pain, inflammation, fever	Gastrointestinal bleeding, kidneys, cardiovascular
Coxibs	Celecoxib, etoricoxib	COX-2 ↓	Patients at high gastric risk	Cardiovascular thrombotic risk
Paracetamol	Paracetamol	Central COX ↓	Pain, fever (children, pregnant women)	Hepatotoxicity (> 4 g); antidote — N-acetylcysteine
Glucocorticoids	Hydrocortisone, prednisolone, methylprednisolone, dexamethasone	Gene transcription, phospholipase A_2 ↓	Inflammation, autoimmune, allergy, shock, replacement	Cushingoid appearance, diabetes, osteoporosis, infections; adrenal insufficiency on withdrawal

TOPIC 16

Antiseptic and disinfectant agents. Synthetic antibacterial agents

Learning objectives

The fight against infections is waged in two directions: destroying microorganisms in the environment and on the skin surface (**antisepsis and disinfection**) and treating infection inside the body (**chemotherapy**). Proper hand disinfection is considered the most effective measure for preventing hospital-acquired infections. Synthetic antibacterial agents (fluoroquinolones, sulfonamides, nitrofurans, nitroimidazoles) are widely used to treat urinary tract, intestinal and anaerobic infections. Antibiotic **resistance** is a global health problem of the 21st century; that is why knowing the principles of rational use of antimicrobials is mandatory for every doctor.

The student should know:

- the concepts of antisepsis, disinfection and sterilization and the classification of antiseptics;
- the mechanism, spectrum of activity and uses of the main groups of antiseptics;
- the general principles of chemotherapy, the targets of antimicrobial drugs and the mechanisms of resistance;
- the mechanism, uses and adverse effects of sulfonamides, co-trimoxazole, quinolones and fluoroquinolones, nitrofurans, nitroimidazoles, oxazolidinones and 8-hydroxyquinolines.

The student should be able to:

- write prescriptions for drugs such as ethyl alcohol, chlorhexidine, povidone-iodine, hydrogen peroxide, potassium permanganate, brilliant green, ciprofloxacin, levofloxacin, co-trimoxazole, nitrofurantoin and metronidazole;
- perform the technique of hygienic hand disinfection;
- choose a synthetic antibacterial agent for a urinary tract infection.

1. Basic concepts

- **Antisepsis** — destroying microorganisms on the skin, mucous membranes and wound surfaces or stopping their multiplication.
- **Disinfection** — destroying pathogenic microorganisms on objects in the environment (instruments, rooms, waste, clothing).
- **Sterilization** — complete destruction of all microorganisms, including spores (autoclave, dry heat, ethylene oxide, glutaraldehyde).
- **Chemotherapeutic agents** — agents that act selectively inside the body on microorganisms, parasites or tumor cells.

> **REMEMBER**
> The main difference between antiseptic and chemotherapeutic agents is **selective action (selectivity)**. Antiseptics are protoplasmic poisons: they act on microbial and human cells **nonselectively**, so they are used only on surfaces. Chemotherapeutic agents act on targets specific to microbes and cause little harm to the macroorganism — they are used systemically.

Types of antimicrobial action: **bacteriostatic** (stops multiplication) and **bactericidal** (kills). The type of action depends on concentration, the type of microbe and the conditions of the environment.

2. Antiseptic and disinfectant agents

2.1. Requirements for antiseptics and factors affecting their activity

An ideal antiseptic should be broad-spectrum, fast-acting, not lose its activity in the presence of organic matter (blood, pus), not damage tissue, not be absorbed, not cause allergy, and be inexpensive and stable. No such substance exists — each group has its own advantages and disadvantages.

Factors affecting activity: **concentration** and contact time, temperature, the pH of the environment, **organic matter** (blood, pus and proteins inactivate most antiseptics), the type of microbe (spores, mycobacteria and non-enveloped viruses are the most resistant).

2.2. Classification of antiseptics

Table 16.1. *Classification of antiseptic and disinfectant agents*

Group	Preparations	Mechanism	Main use
Alcohols	Ethyl alcohol 70%, isopropyl alcohol	Protein denaturation, membrane dissolution	Antisepsis of hands and injection sites
Halogens and halogen-containing agents	Povidone-iodine, alcoholic iodine solution 5%, Lugol's solution; chloramine B, sodium hypochlorite, chlorinated lime	Oxidation, halogenation of proteins	Surgical field, skin, wounds (iodophors); disinfection (chlorine-containing agents)
Oxidizing agents	Hydrogen peroxide 3%, potassium permanganate	Release of active oxygen	Wound cleaning (mechanical action — foam), irrigation, gargling; gastric lavage in poisoning ($KMnO_4$)
Biguanides	Chlorhexidine	Damage to the cell membrane	Hands, surgical field, oral cavity, catheters
Detergents (cationic surfactants)	Benzalkonium chloride, cetylpyridinium, Miramistin, decamethoxine	Disruption of membrane permeability	Mucous membranes, preservative in eye drops, disinfection
Aldehydes	Formaldehyde (formalin), glutaraldehyde	Alkylation of the amino groups of proteins	Disinfection and sterilization of instruments, fixation of tissues
Phenols	Phenol (carbolic acid), resorcinol, cresol, triclosan	Protein denaturation	Disinfection; resorcinol — skin diseases
Dyes	Brilliant green, methylene blue, ethacridine lactate (Rivanol)	Binding to the bacterial cell	Skin, minor wounds, purulent skin diseases
Salts of heavy metals	Silver nitrate, silver sulfadiazine, Protargol, mercury compounds (historically), zinc sulfate	Blocking the SH groups of proteins and enzymes	Burns (silver sulfadiazine), conjunctivitis, astringent/caustic
Acids and alkalis	Boric acid, salicylic acid, ammonia solution	Change in pH	Skin diseases (rarely used; boric acid is contraindicated in children)
Nitrofuran derivatives	Furacilin (nitrofural)	DNA damage	Wound irrigation, gargling

2.3. Features of the main antiseptics

Ethyl alcohol — the **70%** concentration is the most effective: 96% alcohol rapidly coagulates the proteins on the cell surface, forms a protective layer and does not penetrate inside; at 70%, the presence of water improves denaturation. It acts on bacteria, fungi and enveloped viruses; **it does not act on spores**. It is not used on open wounds (pain, tissue necrosis). Alcohol-based antiseptics with added glycerin do not dry the skin of the hands.

Iodine preparations. Alcoholic iodine solution — on the skin and around wounds; irritating, allergenic, and with long-term use — **iodism** (rhinitis, rash, increased salivation) and thyroid dysfunction. **Povidone-iodine** (an iodophor — a complex of iodine with polyvinylpyrrolidone) releases iodine slowly, irritates little, has a broad spectrum (including spores) and is widely used for surgical field antisepsis. It is not applied to large areas in newborns and pregnant women (iodine absorption — hypothyroidism).

Chlorhexidine — a potent antiseptic with a long (residual) effect; it “sticks” well to the skin. The **chlorhexidine + alcohol** combination is **the most effective** for antisepsis of the surgical field and the central venous catheter site. Its activity decreases with organic matter and soap. It must not come into contact with the eye, the middle ear or the meninges (it damages them); with prolonged use as a mouthwash it stains the teeth.

Hydrogen peroxide 3% — under the action of tissue catalase it releases oxygen, forms **foam** and cleans the wound mechanically; its antiseptic effect is weak and short. It helps stop capillary bleeding. Injecting it into deep closed cavities is dangerous (gas embolism).

Potassium permanganate — a strong oxidizing agent; depending on the concentration it is antiseptic (0.01–0.1%), astringent and caustic (2–5%). In some poisonings (alkaloids, phosphorus) a 0.02–0.1% solution is used for gastric lavage; the **crystals** must be completely dissolved — they burn the mucosa.

Brilliant green 1–2% — active against gram-positive bacteria (staphylococci); skin wounds, pyoderma, the umbilical wound. **Furacilin 0.02%** (1:5000) — irrigation of wounds and cavities; weak activity, many microorganisms are resistant.

> **CAUTION**
> Common incorrect practices: 1) applying iodine and brilliant green **into an open wound** — this damages tissue and slows healing (only around the wound); 2) using boric acid in children — it is absorbed through the skin and may cause severe poisoning; 3) pouring alcohol onto an open wound. The most important thing in cleaning minor wounds is **irrigation with plenty of clean water or saline**.

2.4. Hand disinfection

> **CLINICAL SIGNIFICANCE**
> According to the WHO, a large share of hospital-acquired infections is transmitted through the hands of health workers. **The 5 moments for hand hygiene** (WHO): 1) before touching a patient; 2) before an aseptic procedure; 3) after contact with body fluids; 4) after touching a patient; 5) after touching the patient's surroundings. If the hands are not visibly soiled, an **alcohol-based hand rub** (3–5 ml, rubbed for 20–30 seconds) is more effective than washing with soap; if they are visibly soiled, or after working with C. difficile or norovirus — soap and water (alcohol does not act on spores).

2.5. Laboratory class: studying the mechanism of action of antiseptics

This section corresponds to the 6th-semester class “Laboratory study of the mechanism of action of antiseptic agents”.

Experiment 1 (protein denaturation): solutions of ethyl alcohol (70% and 96%), phenol, formaldehyde, silver nitrate and copper sulfate are added to egg white solution in test tubes — **precipitation (coagulation)** of the protein is observed. Conclusion: most antiseptics denature the proteins of the microbial cell.

Experiment 2 (effect of oxidizing agents): when hydrogen peroxide is added to blood or a piece of meat, **foam** (oxygen) forms under the action of catalase; potassium permanganate solution becomes colorless on contact with organic matter (manganese(IV) oxide precipitates). Conclusion: the action of oxidizing agents is short and ends quickly in an organic environment.

Experiment 3 (effect of organic matter and concentration): antiseptics of different concentrations are applied to a microbial culture (paper disk method) and the zones of growth inhibition are measured; when serum is added, the zones shrink.

> **REMEMBER**
> Laboratory conclusions: the effect of antiseptics depends on **concentration and contact time**; **organic matter** (blood, pus, protein) reduces the activity of most antiseptics — so a wound must first be cleaned mechanically.

3. General principles of chemotherapy

3.1. Targets of antimicrobial drugs

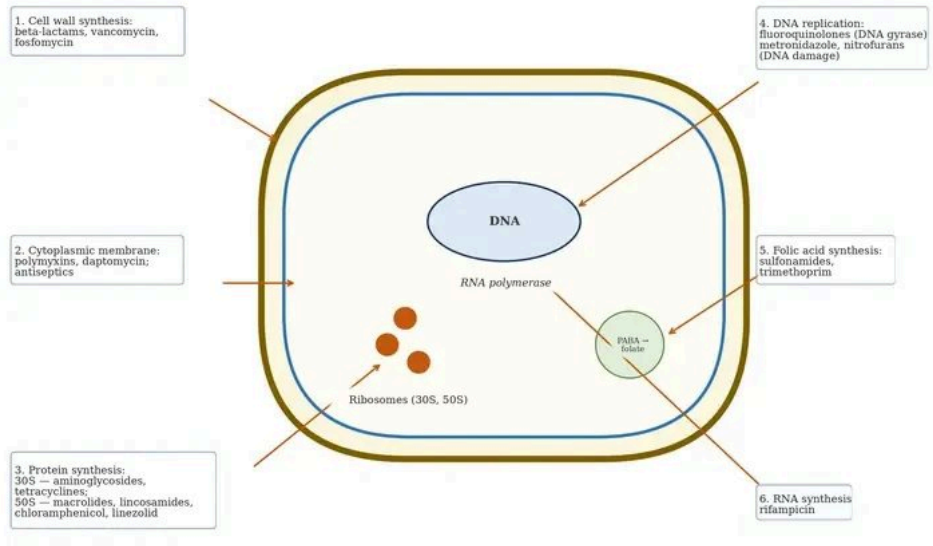


Figure 16.1. *The main targets of antimicrobial drugs in the bacterial cell (author's diagram)*

3.2. Mechanisms of resistance

Table 16.2. *Mechanisms of bacterial resistance to antimicrobial drugs*

Mechanism	Examples
Drug-inactivating enzymes	Beta-lactamases (penicillinases, extended-spectrum beta-lactamases — ESBL, carbapenemases); aminoglycoside-modifying enzymes

Mechanism	Examples
Alteration of the target	Penicillin-binding proteins (MRSA — PBP2a), DNA gyrase mutation (fluoroquinolones), ribosomal methylation (macrolides), alteration of the vancomycin target (VRE)
Reduced entry into the cell	Loss of porin channels (gram-negative bacteria)
Active efflux of the drug (efflux pumps)	Tetracyclines, fluoroquinolones, macrolides
Alternative metabolic pathway	Resistance to sulfonamides and trimethoprim
Biofilm formation	Infections of catheters and prostheses

CAUTION

Antimicrobial resistance is a global threat that causes millions of deaths a year. Factors that worsen it: unjustified use of antibiotics in viral infections (ARVI, influenza), over-the-counter sales, wrong dose and duration, antibiotics in animal farming. The WHO **AWaRe** classification divides antibiotics into “Access” (can be widely used), “Watch” (use with caution) and “Reserve” groups.

3.3. Basic principles of antimicrobial therapy

1. **Establishing the indication** — when a bacterial infection is likely (antibiotics do not act on viral infections).
2. **Identifying the pathogen** — where possible, take cultures before the antibiotic; first **empirical** therapy (according to the likely pathogens and local resistance), then switching to **targeted** therapy according to the results (de-escalation).
3. **Penetration of the drug into the site of infection** (urine, CNS, bone, prostate).
4. **Patient factors**: age, pregnancy, renal and hepatic function, allergy history, interactions.
5. **Optimal dose, route and duration** — the shortest effective course possible.
6. **Assessing effectiveness at 48–72 hours**.
7. Combinations only when indicated (tuberculosis, severe sepsis, H. pylori, prevention of resistance).

4. Synthetic antibacterial agents

Table 16.3. *Classification of synthetic antibacterial agents*

Group	Preparations	Mechanism
Sulfonamides	Sulfamethoxazole, sulfadiazine, sulfasalazine, sulfacetamide (Sulfacyl sodium), silver sulfadiazine	Competitively inhibit dihydropteroate synthetase (PABA antagonists) — bacteriostatic
Diaminopyrimidines	Trimethoprim	Inhibits bacterial dihydrofolate reductase
Combination	Co-trimoxazole (sulfamethoxazole + trimethoprim 5:1)	Two steps of folate synthesis — bactericidal
Quinolones (1st generation)	Nalidixic acid, pipemidic acid	DNA gyrase inhibitors (urinary tract only)

Group	Preparations	Mechanism
Fluoroquinolones	2nd generation: ciprofloxacin, ofloxacin, norfloxacin; 3rd–4th generations (“respiratory”): levofloxacin, moxifloxacin, gemifloxacin	Inhibitors of DNA gyrase (topoisomerase II) and topoisomerase IV — bactericidal
Nitrofurans	Nitrofurantoin, furazolidone, nifuroxazide, furazidin (Furagin), nitrofural (Furacilin)	Reactive metabolites that damage DNA and proteins
Nitroimidazoles	Metronidazole, tinidazole, ornidazole, secnidazole	Reduced in the anaerobic cell to form radicals that damage DNA — bactericidal
Oxazolidinones	Linezolid, tedizolid	Block the initiation of protein synthesis on the 50S ribosomal subunit
8-Hydroxyquinolines	Nitroxoline, clioquinol	Chelate metal ions
Fosfomycin (a synthetic antibiotic)	Fosfomycin trometamol	Blocks the first step of cell wall synthesis

4.1. Sulfonamides and co-trimoxazole

Mechanism. Bacteria synthesize folic acid themselves from **PABA** (para-aminobenzoic acid) (humans obtain it from food). Sulfonamides are structurally similar to PABA — they **competitively inhibit dihydropteroate synthetase**. **Trimethoprim** inhibits **dihydrofolate reductase** at the next step. Together (**co-trimoxazole**) they produce sequential blockade, synergy and a bactericidal effect.

> **REMEMBER**
> Sources of PABA reduce the effect of sulfonamides: **pus and necrotic tissue** (much PABA) and drugs that form PABA — **procaine (Novocaine), benzocaine (Anesthesin)**. That is why sulfonamides are ineffective when used together with procaine or in a purulent wound.

Uses of co-trimoxazole: urinary tract infections (if local resistance is low), **Pneumocystis jirovecii pneumonia** (treatment and prevention — the drug of choice in HIV infection), nocardiosis, toxoplasmosis, skin infections caused by MRSA, shigellosis. **Sulfasalazine** — ulcerative colitis and rheumatoid arthritis (5-aminosalicylic acid is released in the intestine). **Sulfacetamide** eye drops (conjunctivitis), **silver sulfadiazine** — burns.

Table 16.4. *Adverse effects of sulfonamides and co-trimoxazole*

Adverse effect	Comment
Allergic reactions, including Stevens–Johnson syndrome and toxic epidermal necrolysis	The most dangerous; stop immediately
Crystalluria	Precipitates in acidic urine — drink plenty of fluids
Hematologic: agranulocytosis, megaloblastic anemia (trimethoprim), hemolysis in G6PD deficiency	—
Hyperkalemia	Trimethoprim acts like amiloride (especially with ACE inhibitors)
Kernicterus in newborns	Displaces bilirubin from albumin — contraindicated in late pregnancy and in infants under 2 months
Teratogenicity (1st trimester — folate antagonism)	—
Enhances the effect of warfarin	CYP2C9 inhibition

4.2. Fluoroquinolones

Mechanism. By inhibiting bacterial **DNA gyrase** (topoisomerase II; mainly in gram-negatives) and **topoisomerase IV** (mainly in gram-positives), they stop DNA replication — a **concentration-dependent bactericidal** effect.

Table 16.5. *Generations and spectrum of fluoroquinolones*

Generation	Preparations	Spectrum	Main use
1st (non-fluorinated quinolones)	Nalidixic acid	Gram-negatives (concentration only in urine)	Historically — urinary tract
II	Ciprofloxacin, ofloxacin, norfloxacin	Gram-negatives (including Pseudomonas aeruginosa — ciprofloxacin), partly gram-positives, atypicals	Complicated urinary tract infections, pyelonephritis, intestinal infections (salmonellosis, shigellosis), prostatitis, Pseudomonas infections, anthrax
3rd–4th (“respiratory”)	Levofloxacin, moxifloxacin	+ pneumococci (including penicillin-resistant ones), atypicals (mycoplasma, chlamydia, legionella); moxifloxacin — anaerobes	Community-acquired pneumonia (alternative), severe sinusitis; drug-resistant tuberculosis (reserve)

CAUTION

Because of **serious adverse effects of fluoroquinolones** (FDA and EMA warnings), they are **not recommended as first choice** in uncomplicated cystitis, acute bronchitis and mild sinusitis:

- **tendinitis and tendon (Achilles) rupture** — especially in people over 60 and together with **glucocorticoids**;
- peripheral neuropathy (may be irreversible), CNS disorders (dizziness, insomnia, psychosis, **seizures** — especially with NSAIDs);
- **QT prolongation** and arrhythmias;
- aortic aneurysm and dissection;
- dysglycemia (hypo- and hyperglycemia), photosensitivity, C. difficile colitis;
- contraindicated **in children and adolescents** (damage to growing joint cartilage) and **in pregnancy and breastfeeding** (except for special indications).

Interactions: antacids, calcium, iron, zinc and dairy products form chelates with fluoroquinolones and sharply reduce their absorption (take 2 hours before or 6 hours after); ciprofloxacin inhibits CYP1A2 and increases the concentrations of **theophylline** and caffeine; it enhances the effect of warfarin.

4.3. Nitrofurans

Nitrofurantoin reaches therapeutic concentrations **only in urine** (low in blood). The **first-line** drug for **uncomplicated cystitis** (in women) (5 days), prevention of recurrent urinary tract infection. Resistance develops rarely. Ineffective and toxic at GFR < 30–45 ml/min. Adverse effects: nausea, brown urine; with long-term use — **pulmonary reactions** (acute pneumonitis, chronic fibrosis), peripheral neuropathy, hepatitis; hemolysis in G6PD deficiency and in newborns.

Nifuroxazide — not absorbed from the intestine, used in bacterial diarrhea (in CIS countries). **Furazolidone** — intestinal infections, giardiasis; inhibits MAO, **disulfiram-like reaction with alcohol**. **Furazidin (Furagin)** — urinary tract.

4.4. Nitroimidazoles

In the cells of anaerobic microorganisms and protozoa, the nitro group of **metronidazole** is **reduced**, forming free radicals that damage DNA (this process does not occur in aerobic cells — the basis of its selective action).

Spectrum: anaerobic bacteria (Bacteroides fragilis, Clostridioides difficile, Fusobacterium), **protozoa** (Trichomonas vaginalis, Giardia lamblia, Entamoeba histolytica), Gardnerella (bacterial vaginosis), H. pylori.

Uses: anaerobic infections (abdominal, pelvic, abscesses — often with other antibiotics), **trichomoniasis** (both partners are treated), bacterial vaginosis, amebiasis, giardiasis, mild C. difficile colitis (vancomycin/fidaxomicin are now preferred), H. pylori eradication, prophylaxis in colorectal surgery.

Adverse effects: metallic taste in the mouth, nausea, dark urine, **disulfiram-like reaction with alcohol** (facial flushing, tachycardia, vomiting — no alcohol during treatment and for 48–72 hours afterward), peripheral neuropathy and encephalopathy (with long-term and high-dose use), enhances the effect of warfarin; caution in the 1st trimester of pregnancy.

> **MEMORY AID**
> The spectrum of metronidazole — **“GET GAP on the Metro”**: **G**iardia, **E**ntamoeba, **T**richomonas, **G**ardnerella, **A**naerobes, **P** — H. pylori. In short: **“Anaerobes and protozoa — metronidazole”**.

4.5. Oxazolidinones and others

Linezolid — active against gram-positive bacteria, including **MRSA** and **vancomycin-resistant enterococci (VRE)**; oral bioavailability ≈ 100%. A reserve antibiotic. Adverse effects: **myelosuppression** (thrombocytopenia — with use for more than 2 weeks), peripheral and optic neuropathy, lactic acidosis; inhibits MAO — **serotonin syndrome with SSRIs**, hypertension with tyramine.

Nitroxoline (5-NOK) — urinary tract infections; colors the urine bright yellow.

Fosfomycin trometamol — a single dose in uncomplicated cystitis (3 g), first line; also allowed in pregnant women.

5. Clinical part

Table 16.6. *Choice of synthetic antibacterial agent in urinary tract infections (adults)*

Clinical situation	First choice	Alternative	Comment
Uncomplicated acute cystitis in women	Nitrofurantoin for 5 days or fosfomycin 3 g once	Pivmecillinam; co-trimoxazole (resistance < 20%)	Fluoroquinolones are not recommended
Acute pyelonephritis (mild-moderate)	Ciprofloxacin or levofloxacin for 7 days (resistance < 10%), cephalosporins	Co-trimoxazole (if susceptible)	Nitrofurantoin and fosfomycin are ineffective (low tissue concentration)
Cystitis / asymptomatic bacteriuria in pregnancy	Fosfomycin, amoxicillin/clavulanate, cephalexin	Nitrofurantoin (not in the weeks before delivery)	Fluoroquinolones, co-trimoxazole (1st and 3rd trimesters) — contraindicated
Acute prostatitis in men	Fluoroquinolones (2–4 weeks)	Co-trimoxazole	Penetrate the prostate well

Table 16.7. *Synthetic agents in other infections*

Infection	Choice
Trichomoniasis, bacterial vaginosis	Metronidazole (also the partner — in trichomoniasis)
Anaerobic abdominal infection	Metronidazole + a cephalosporin or a fluoroquinolone
Pneumocystis pneumonia	Co-trimoxazole (high dose)
MRSA (severe)	Linezolid, vancomycin
Giardiasis, amebiasis	Metronidazole, tinidazole
Conjunctivitis (bacterial)	Ofloxacin or sulfacetamide drops

Practical skills

Skill 1. Hygienic hand disinfection with an alcohol-based hand rub (according to the WHO, 20–30 seconds)

1. Put 3–5 ml of antiseptic into the palm (an amount that wets the hands completely).
2. Rub palm to palm.
3. Right palm over the back of the left hand (including between the fingers), and vice versa.
4. Rub palm to palm with the fingers interlaced.
5. Rub the backs of the bent fingers against the opposite palm.
6. Rub the thumbs with a rotational movement.
7. Rub the fingertips against the opposite palm with a rotational movement.
8. Continue until the hands are dry (without wiping).

Skill 2. Key prescriptions

PRESCRIPTION 16.1. ETHYL ALCOHOL 70%
Rp.: Spiritus aethylici 70% — 100 ml
D.S. For wiping the hands and the injection site.

PRESCRIPTION 16.2. CHLORHEXIDINE
Rp.: Solutionis Chlorhexidini bigluconatis 0.05% — 100 ml
D.S. For irrigating wounds and rinsing the mouth.

PRESCRIPTION 16.3. POVIDONE-IODINE
Rp.: Solutionis Povidoni-iodi 10% — 120 ml
D.S. For treating the surgical field and the skin around wounds.

PRESCRIPTION 16.4. ALCOHOLIC IODINE SOLUTION
Rp.: Solutionis Iodi spirituosae 5% — 10 ml
D.S. For applying to the skin around a wound.

PRESCRIPTION 16.5. HYDROGEN PEROXIDE
Rp.: Solutionis Hydrogenii peroxydi 3% — 100 ml
D.S. For cleaning wounds.

PRESCRIPTION 16.6. POTASSIUM PERMANGANATE

Rp.: Kalii permanganatis 5.0

D.S. Dissolve a few crystals completely in water to prepare a 0.05% (pale pink) solution; for irrigating a wound.

PRESCRIPTION 16.7. BRILLIANT GREEN

Rp.: Solutionis Viridis nitentis spirituosae 1% — 10 ml

D.S. Apply to minor skin wounds and pustules.

PRESCRIPTION 16.8. CIPROFLOXACIN

Rp.: Ciprofloxacini 0.5

D.t.d. N. 14 in tabulettis obductis

S. Take 1 tablet 2 times a day for 7 days, 2 hours before dairy products and antacids.

PRESCRIPTION 16.9. LEVOFLOXACIN

Rp.: Levofloxacini 0.5

D.t.d. N. 7 in tabulettis obductis

S. Take 1 tablet once a day.

PRESCRIPTION 16.10. CO-TRIMOXAZOLE

Rp.: Tabulettas “Co-trimoxazolum 960” N. 6

D.S. Take 1 tablet 2 times a day for 3 days, with plenty of fluids.

PRESCRIPTION 16.11. NITROFURANTOIN

Rp.: Capsulas Nitrofurantoini 0.1 N. 10

D.S. Take 1 capsule 2 times a day with food for 5 days.

PRESCRIPTION 16.12. FOSFOMYCIN

Rp.: Granulorum Fosfomycini trometamoli 3.0

D.t.d. N. 1

S. Dissolve the contents of the sachet in 1/2 glass of water and take once in the evening after emptying the bladder.

PRESCRIPTION 16.13. METRONIDAZOLE (TABLETS)

Rp.: Metronidazoli 0.5

D.t.d. N. 14 in tabulettis

S. Take 1 tablet 2 times a day for 7 days; no alcohol during treatment and for 3 days afterward.

PRESCRIPTION 16.14. METRONIDAZOLE (INFUSION)

Rp.: Solutionis Metronidazoli 0.5% — 100 ml

D.t.d. N. 10

S. 100 ml as an intravenous drip 3 times a day.

PRESCRIPTION 16.15. SULFACETAMIDE EYE DROPS

Rp.: Solutionis Sulfacetamidi natrii 20% — 10 ml

D.S. Instill 1–2 drops into each eye 4–6 times a day.

Review questions

1. Concepts of antisepsis, disinfection and sterilization. The difference between antiseptic and chemotherapeutic agents.
2. Requirements for antiseptics and factors affecting their activity.
3. Classification and mechanisms of antiseptics.
4. Why is 70% alcohol more effective than 96% alcohol?
5. Features of iodine preparations, chlorhexidine, hydrogen peroxide and potassium permanganate.
6. The 5 moments for hand hygiene and the disinfection technique.
7. How is the mechanism of action of antiseptics studied in a laboratory experiment?
8. Targets of antimicrobial drugs in the bacterial cell.
9. Mechanisms of resistance. The AWaRe classification.
10. Basic principles of antimicrobial therapy.
11. Mechanism of sulfonamides and trimethoprim. Uses and adverse effects of co-trimoxazole.
12. Why are sulfonamides ineffective in a purulent environment and with procaine?
13. Mechanism, generations and spectrum of fluoroquinolones.
14. Adverse effects, contraindications and interactions of fluoroquinolones.
15. Uses and limitations of nitrofurantoin.
16. Mechanism, spectrum, uses and adverse effects of metronidazole.
17. Features of linezolid.

Test questions

1. The most effective concentration of ethyl alcohol for hand antisepsis:
   A) 96%
   B) 100%
   C) 20%
   D) 70%
   E) 40%

2. Which microorganisms does ethyl alcohol not act on?
   A) Fungi
   B) Escherichia coli
   C) Enveloped viruses
   D) Staphylococci
   E) Bacterial spores

3. An antiseptic that cleans a wound mechanically by forming foam:
   A) Chlorhexidine
   B) Ethyl alcohol
   C) Hydrogen peroxide
   D) Brilliant green
   E) Furacilin

4. The combination considered most effective for surgical field antisepsis:
   A) Chlorhexidine + alcohol
   B) Brilliant green + iodine
   C) Boric acid + phenol
   D) Potassium permanganate + alcohol
   E) Furacilin + hydrogen peroxide

5. An aldehyde used to sterilize instruments:
   A) Ethacridine
   B) Glutaraldehyde
   C) Povidone-iodine
   D) Chlorhexidine
   E) Phenol

6. Mechanism of action of sulfonamides:
   A) Disruption of cell wall synthesis
   B) Competitive inhibition of dihydropteroate synthetase (PABA antagonism)
   C) Inhibition of DNA gyrase
   D) Inhibition of RNA polymerase
   E) Blockade of the 30S ribosome

7. A local anesthetic that reduces the antibacterial effect of sulfonamides:
   A) Ropivacaine
   B) Procaine (Novocain)
   C) Lidocaine
   D) Articaine
   E) Bupivacaine

8. The drug of choice for treating and preventing Pneumocystis pneumonia in HIV infection:
   A) Co-trimoxazole
   B) Nitrofurantoin
   C) Linezolid
   D) Ciprofloxacin
   E) Metronidazole

9. Mechanism of action of fluoroquinolones:
   A) Inhibition of beta-lactamase
   B) Disruption of folic acid synthesis
   C) Damage to the cell membrane
   D) Blockade of the 50S ribosome
   E) Inhibition of DNA gyrase and topoisomerase IV

10. An adverse effect of fluoroquinolones whose risk increases when they are used together with glucocorticoids:
    A) Crystalluria
    B) Disulfiram-like reaction
    C) Tendon rupture
    D) Agranulocytosis
    E) Hypokalemia

11. A group that must not be used in children and pregnant women (joint cartilage damage):
    A) Penicillins
    B) Cephalosporins
    C) Fosfomycin
    D) Fluoroquinolones
    E) Nitrofurans

12. A drug used as a single dose in uncomplicated cystitis:
    A) Linezolid
    B) Fosfomycin trometamol
    C) Metronidazole
    D) Sulfasalazine
    E) Ciprofloxacin

13. A drug that reaches therapeutic concentrations only in urine and is ineffective in pyelonephritis:
    A) Levofloxacin
    B) Ceftriaxone
    C) Nitrofurantoin
    D) Ciprofloxacin
    E) Co-trimoxazole

14. A drug that acts on anaerobes and protozoa and causes a disulfiram-like reaction with alcohol:
    A) Metronidazole
    B) Co-trimoxazole
    C) Linezolid
    D) Nitroxoline
    E) Ciprofloxacin

15. A synthetic antibacterial agent active against MRSA and VRE that may cause serotonin syndrome with SSRIs:
    A) Nitrofurantoin
    B) Linezolid
    C) Co-trimoxazole
    D) Levofloxacin
    E) Metronidazole

Clinical cases

Case 1

A 26-year-old woman (not pregnant) has complained of frequent painful urination and suprapubic discomfort for 1 day. No fever or back pain. No comorbidities. A friend advised her to take ciprofloxacin.

Questions: 1) Diagnosis? 2) Optimal treatment? 3) Why is ciprofloxacin not the first choice?

Answer: 1) Uncomplicated acute cystitis. 2) Nitrofurantoin 100 mg 2 times a day for 5 days or fosfomycin trometamol 3 g once; plenty of fluids. 3) Fluoroquinolones are not recommended in uncomplicated cystitis because of the risk of serious adverse effects (tendons, nerves, aorta, QT) and their contribution to resistance — they are reserved for pyelonephritis and complicated infections.

Case 2

A 70-year-old man is taking levofloxacin for community-acquired pneumonia; he also takes prednisolone 30 mg/day for a COPD exacerbation. On day 5, while walking, he felt a “snap” and sharp pain above the heel and cannot push off with his foot.

Questions: 1) What complication? 2) Risk factors? 3) Management.

Answer: 1) Achilles tendon rupture — a serious adverse effect of fluoroquinolones. 2) Age > 60, concomitant use of glucocorticoids (especially dangerous); renal failure and transplantation are also risk factors. 3) Stop levofloxacin immediately and switch to an antibiotic of another group (beta-lactam + macrolide), immobilize the leg, an orthopedic consultation (conservative or surgical treatment); record the adverse effect in the medical records; avoid fluoroquinolones in the future.

Case 3

A 32-year-old woman has a frothy, yellow-green, malodorous vaginal discharge and itching. Microscopy shows motile trichomonads. She says she drinks a glass of wine with dinner every day.

Questions: 1) Treatment plan? 2) What special warning is given to the patient?

Answer: 1) Metronidazole (500 mg 2 times a day for 7 days or 2 g once) or tinidazole; the sexual partner is treated at the same time, abstinence from sex until treatment is completed; testing for other sexually transmitted infections. 2) No alcohol during treatment and for at least 48–72 hours after it ends — metronidazole inhibits aldehyde dehydrogenase and causes a disulfiram-like reaction (facial flushing, headache, palpitations, nausea, vomiting); explain that the metallic taste in the mouth and dark urine are harmless.

Quick review table

Table 16.8. *Antiseptics and synthetic antibacterial agents — summary*

Group	Preparations	Mechanism	Main use	Important note
Alcohols	Ethyl alcohol 70%	Protein denaturation	Hands, skin	Does not act on spores
Halogens	Povidone-iodine, iodine, chlorine-containing agents	Oxidation, halogenation	Surgical field; disinfection	Iodism, hypothyroidism
Biguanides	Chlorhexidine	Membrane damage	Hands, surgical field, catheters	Keep away from the eyes and ears
Oxidizing agents	Hydrogen peroxide, $KMnO_4$	Active oxygen	Wound cleaning, irrigation	Short action
Dyes	Brilliant green, methylene blue	—	Skin wounds	Not into the wound
Aldehydes	Glutaraldehyde, formaldehyde	Alkylation	Sterilization of instruments	Toxic, not on the skin
Sulfonamides + trimethoprim	Co-trimoxazole	Sequential blockade of folate synthesis	Pneumocystosis, urinary tract infections, MRSA skin infections	Stevens–Johnson, hyperkalemia, kernicterus

Group	Preparations	Mechanism	Main use	Important note
Fluoroquinolones	Ciprofloxacin, levofloxacin, moxifloxacin	DNA gyrase, topoisomerase IV	Pyelonephritis, prostatitis, intestine, pneumonia, Pseudomonas	Tendons, neuropathy, QT, aorta; contraindicated in children and pregnant women; not with antacids
Nitrofurans	Nitrofurantoin, furazidin, nifuroxazide	DNA damage	Uncomplicated cystitis	Contraindicated at low GFR; pulmonary reactions
Nitroimidazoles	Metronidazole, tinidazole	DNA damage in the anaerobic cell	Anaerobes, trichomoniasis, giardiasis, amebiasis, bacterial vaginosis	Disulfiram-like reaction with alcohol
Oxazolidinones	Linezolid	50S, initiation of protein synthesis	MRSA, VRE	Myelosuppression, serotonin syndrome
Fosfomycin	Fosfomycin trometamol	Cell wall	Cystitis (3 g once)	Allowed in pregnancy

TOPIC 17

Antibiotics

Learning objectives

In 1928 A. Fleming discovered penicillin, in the 1940s it entered clinical practice, and mortality from bacterial infections fell sharply. Today antibiotics are the most prescribed group of drugs, but about half of them are prescribed without justification (mainly for viral infections). This leads to complications such as resistance, allergic reactions and C. difficile colitis. The most important skills for a GP are correctly determining whether an antibiotic is needed, choosing the first-line drug for common infections, and knowing the restrictions related to allergy and special groups (children, pregnant women, the elderly).

The student should know:

- the classification of antibiotics (by chemical structure, mechanism, spectrum and type of action);
- beta-lactam antibiotics (penicillins, cephalosporins, carbapenems, monobactams) and beta-lactamase inhibitors;
- glycopeptides, aminoglycosides, macrolides, tetracyclines, lincosamides, amphenicols, polymyxins and other antibiotics;
- the main adverse effects of antibiotics and their prevention;
- the principles of choosing antibiotics for common infections.

The student should be able to:

- write prescriptions for drugs such as benzylpenicillin, benzathine benzylpenicillin, amoxicillin, amoxicillin/clavulanate, ceftriaxone, cefuroxime, azithromycin, clarithromycin, doxycycline, gentamicin, vancomycin and clindamycin;
- choose an antibiotic in a patient with a history of penicillin allergy;
- draw up an empirical treatment plan for acute tonsillitis, otitis and community-acquired pneumonia.

1. Classification of antibiotics

Antibiotics are substances produced by microorganisms (fungi, actinomycetes, bacteria), or their semisynthetic and synthetic analogs, that selectively inhibit the growth of other microorganisms or kill them.

Table 17.1. *Classification of antibiotics by mechanism of action*

Mechanism	Groups	Type of action
Disruption of cell wall synthesis	Beta-lactams (penicillins, cephalosporins, carbapenems, monobactams), glycopeptides (vancomycin, teicoplanin), fosfomycin	Bactericidal
Damage to the cytoplasmic membrane	Polymyxins (polymyxin B, colistin), lipopeptides (daptomycin)	Bactericidal
Disruption of protein synthesis (30S subunit)	Aminoglycosides	Bactericidal
	Tetracyclines, glycylcyclines (tigecycline)	Bacteriostatic

Mechanism	Groups	Type of action
Disruption of protein synthesis (50S subunit)	Macrolides, lincosamides, amphenicols (chloramphenicol), oxazolidinones, fusidic acid	Mainly bacteriostatic
Disruption of nucleic acid synthesis	Rifamycins (RNA polymerase)	Bactericidal

MEMORY AID

Drugs acting on **30S** — "**AT 30**" — **A**minoglycosides, **T**etracyclines. Drugs acting on **50S** — "**CELL at 50**" — **C**hloramphenicol, **E**rythromycin (macrolides), **L**incosamides (clindamycin), **L**inezolid.

REMEMBER

Giving bactericidal and bacteriostatic antibiotics together (for example, penicillin + tetracycline) is usually inappropriate: beta-lactams act only on the cell wall of **growing** bacteria, while bacteriostatic drugs stop growth — antagonism is possible. Exception: special combinations (for example, penicillin + clindamycin in severe streptococcal infections — to stop toxin synthesis).

Pharmacodynamic characteristics: time-dependent antibiotics (beta-lactams, vancomycin, macrolides) — effectiveness depends on the time the concentration stays above the MIC: frequent dosing or prolonged infusion; **concentration-dependent** antibiotics (aminoglycosides, fluoroquinolones) — a high peak concentration is important: a large dose once a day; they have a **postantibiotic effect**.

2. Beta-lactam antibiotics

2.1. General mechanism

Beta-lactams have a **beta-lactam ring** in their molecule. They bind to bacterial **penicillin-binding proteins (PBPs)** — the **transpeptidases** that form the cross-links of peptidoglycan — and block the last step of cell wall synthesis. As a result, the growing bacterium cannot withstand the osmotic pressure and undergoes **lysis** (bactericidal). Human cells have no cell wall — that is why beta-lactams are **very little toxic**.

Resistance: beta-lactamases (break the beta-lactam ring), altered PBPs (**MRSA** — PBP2a; penicillin-resistant pneumococci), reduced porins and efflux in gram-negatives.

2.2. Penicillins

Table 17.2. *Classification of penicillins*

Group	Preparations	Spectrum	Main use
Natural (biosynthetic)	Benzylpenicillin (sodium and potassium salts, intravenous and intramuscular), benzathine benzylpenicillin (depot, intramuscular — 2–4 weeks), phenoxymethylpenicillin (oral)	Gram-positive cocci (streptococci, pneumococci — susceptible strains), meningococcus, gonococcus (now often resistant), treponema, clostridia, anthrax, listeria	Syphilis (all stages), prevention of rheumatic fever, streptococcal tonsillitis, scarlet fever, erysipelas, meningococcal infection, gas gangrene, leptospirosis
Antistaphylococcal (penicillinase-resistant)	Oxacillin, flucloxacillin, dicloxacillin	Staphylococci (penicillinase producers) — except MRSA	Staphylococcal infections (skin, bone, endocarditis)

Group	Preparations	Spectrum	Main use
Aminopenicillins (broad-spectrum)	Amoxicillin, ampicillin	+ E. coli, H. influenzae, enterococci, salmonella, listeria (non-beta-lactamase producers)	Acute otitis, sinusitis, tonsillitis, community-acquired pneumonia, urinary tract, H. pylori, endocarditis prophylaxis, listeriosis
Inhibitor-protected	Amoxicillin/clavulanate, ampicillin/sulbactam	+ beta-lactamase-producing staphylococci, H. influenzae, Moraxella, anaerobes, many enterobacteria	Complicated otitis, sinusitis, bites (dog, human), aspiration pneumonia, abdominal infections
Antipseudomonal	Piperacillin/tazobactam, ticarcillin/clavulanate	+ Pseudomonas aeruginosa, many gram-negatives, anaerobes	Severe hospital infections, neutropenic fever

CLINICAL SIGNIFICANCE

Amoxicillin is well absorbed orally (≈ 90%, independent of food), while ampicillin is poorly absorbed (≈ 40%) — amoxicillin is preferred for oral use. **Benzathine benzylpenicillin** (Bicillin), given once intramuscularly, maintains a low concentration for 2–4 weeks — ideal for treating syphilis and preventing recurrences of rheumatic fever.

CAUTION

Aminopenicillins and infectious mononucleosis: when ampicillin or amoxicillin is given to patients with Epstein–Barr virus infection (often misdiagnosed as “tonsillitis”), a **maculopapular rash** appears in 70–100% of cases; this is not a true allergy, but it is the reason to exclude mononucleosis when treating “tonsillitis” in a young patient.

2.3. Cephalosporins

Cephalosporins are derived from **cephalosporin C** (7-aminocephalosporanic acid); they are more resistant to beta-lactamases than penicillins. No cephalosporin acts on **enterococci or listeria**; only the 5th generation (ceftaroline) acts **on MRSA**.

Table 17.3. *Generations of cephalosporins*

Generation	Parenteral	Oral	Features of the spectrum	Main use
I	Cefazolin	Cephalexin, cefadroxil	Gram-positive cocci (staphylococci — MSSA, streptococci); weak against gram-negatives	Surgical prophylaxis (cefazolin), skin and soft tissue infections
II	Cefuroxime	Cefuroxime axetil, cefaclor	+ H. influenzae, Moraxella, some enterobacteria	Otitis, sinusitis, pneumonia, urinary tract

Generation	Parenteral	Oral	Features of the spectrum	Main use
III	Ceftriaxone, cefotaxime, ceftazidime (antipseudomonal), cefoperazone (antipseudomonal)	Cefixime, ceftibuten	Strong against gram-negatives; ceftriaxone and cefotaxime — pneumococci; cross the BBB	Bacterial meningitis, severe pneumonia, pyelonephritis, gonorrhea (ceftriaxone), Lyme disease, enteric fever; ceftazidime — Pseudomonas
IV	Cefepime	—	Broad: gram-positives + gram-negatives, Pseudomonas	Severe hospital infections, neutropenic fever
V	Ceftaroline, ceftobiprole	—	Also active against MRSA	Complicated skin infections with MRSA, pneumonia
New combinations	Ceftazidime/avibactam, ceftolozane/tazobactam, cefiderocol	—	Multidrug-resistant gram-negatives (ESBL, some carbapenemases)	Reserve

MEMORY AID

With each generation: **gram-negative activity increases**, **gram-positive activity** (up to the 3rd generation) **decreases**, and resistance to beta-lactamases and penetration of the BBB increase. Cephalosporins do not act on **"LAME"**: **L**isteria, **A**typicals (mycoplasma, chlamydia), **M**RSA (except the 5th generation), **E**nterococci.

CAUTION

Ceftriaxone: 1) in newborns (especially premature infants) it is not given intravenously together with **calcium-containing solutions** — ceftriaxone-calcium precipitates in the lungs and kidneys; 2) in newborns it displaces bilirubin from albumin (contraindicated in hyperbilirubinemia); 3) it may cause **pseudolithiasis** (sludge) in the gallbladder. **Cefoperazone** — a disulfiram-like reaction with alcohol and hypoprothrombinemia.

2.4. Carbapenems and monobactams

Carbapenems (meropenem, imipenem/cilastatin, ertapenem, doripenem) are **the broadest-spectrum** beta-lactams: gram-positives, gram-negatives (including **ESBL-producing** enterobacteria), anaerobes; meropenem and imipenem — Pseudomonas (not ertapenem). They do not act on MRSA, Stenotrophomonas or atypicals. **Reserve antibiotics** — severe hospital infections, sepsis, ESBL infections. **Imipenem** is broken down in the kidney by dehydropeptidase — it is given together with **cilastatin**; risk of seizures (especially with imipenem). Carbapenems sharply reduce valproate concentrations.

Monobactams — **aztreonam**: only against gram-negatives (including Pseudomonas); **almost no cross-allergy with penicillins** (except for the side chain it shares with ceftazidime) — an alternative for gram-negative infections in severe penicillin allergy.

2.5. Adverse effects of beta-lactams and allergy

Table 17.4. *Adverse effects of beta-lactams*

Adverse effect	Comment
Allergic reactions	Urticaria, rash, angioedema, anaphylactic shock (0.01–0.05%), Stevens–Johnson syndrome, interstitial nephritis, hemolytic anemia, serum sickness
Gastrointestinal disturbances, C. difficile colitis	Especially amoxicillin/clavulanate (diarrhea), 3rd-generation cephalosporins, carbapenems
Neurotoxicity (seizures)	Very high doses (benzylpenicillin, imipenem), renal failure; intrathecal administration is prohibited
Jarisch–Herxheimer reaction	In the first hours of treating syphilis (and other spirochetoses) — fever, chills, worsening of the rash (mass death of spirochetes)
Electrolyte disturbances	Benzylpenicillin potassium salt at large doses — hyperkalemia
Hematologic	Neutropenia, thrombocytopenia (with prolonged high-dose treatment)
Cholestatic hepatitis	Amoxicillin/clavulanate (in the elderly), oxacillin

CLINICAL SIGNIFICANCE

“Penicillin allergy” — about 10% of patients report it, but **more than 90%** of them are not actually allergic (often a viral rash in childhood or an adverse effect). A wrong label “condemns” the patient to less effective and more expensive antibiotics. The history must be clarified in detail: the type of reaction (urticaria, angioedema, anaphylaxis — true IgE-mediated; a mild rash — probably not) and its timing. In true severe allergy penicillins are contraindicated; **cross-reactivity with cephalosporins** depends mainly on **similar side chains** and is **less than 2%** (even less with the 2nd–3rd generations); in non-anaphylactic allergy, 3rd-generation cephalosporins are usually safe. Skin tests and a supervised drug challenge with an allergist make it possible to remove the “allergy” label.

CAUTION

The tradition of **intradermal penicillin testing** (before every injection in CIS countries) does not reliably predict allergy in patients without a history and may itself cause sensitization. Whenever any antibiotic is given for the first time, an **anaphylaxis kit** (epinephrine) must be ready and the patient must be observed for 30 minutes.

3. Glycopeptides

Vancomycin (and teicoplanin) binds to the D-alanyl-D-alanine part of peptidoglycan precursors and blocks cell wall synthesis. Only against **gram-positive** bacteria (it does not cross the outer membrane of gram-negatives): **MRSA**, coagulase-negative staphylococci, enterococci, **C. difficile**, penicillin-resistant pneumococci.

- Intravenously — MRSA infections (bacteremia, endocarditis, pneumonia, osteomyelitis), penicillin-resistant pneumococcal meningitis (with ceftriaxone).
- **Orally** (not absorbed) — **C. difficile colitis** (severe and recurrent).
- The dose is monitored by the minimum (**trough**) blood concentration or the AUC. **Adverse effects: “red man syndrome”** (histamine release with rapid infusion — flushing of the face, neck and chest, itching, hypotension; prevention — infusion over at least 60 minutes), **nephrotoxicity** (especially with aminoglycosides and piperacillin/tazobactam), ototoxicity, phlebitis.

4. Aminoglycosides

Table 17.5. *Generations of aminoglycosides*

Generation	Preparations	Property
I	Streptomycin, kanamycin, neomycin	Streptomycin — tuberculosis (reserve), plague, tularemia; neomycin — only topical and intestinal (very toxic)
II	Gentamicin, tobramycin, netilmicin	Gram-negatives, Pseudomonas (especially tobramycin)
III	Amikacin	Gentamicin-resistant strains; multidrug-resistant tuberculosis

Mechanism. They bind irreversibly to the **30S** ribosomal subunit and cause misreading of mRNA and synthesis of nonfunctional proteins — **bactericidal**, concentration-dependent, with a strong postantibiotic effect. Their entry into the bacterium is **oxygen-dependent — they do not act on anaerobes**; their activity decreases in an acidic environment (abscess).

Pharmacokinetics: not absorbed from the intestine (parenteral or topical); cross the BBB poorly; excreted almost unchanged by the **kidneys**.

Uses: severe gram-negative infections (sepsis, pyelonephritis, hospital-acquired pneumonia) — usually in combination with a beta-lactam; **enterococcal and streptococcal endocarditis** (synergy — the beta-lactam damages the cell wall and makes it easier for the aminoglycoside to enter); plague, tularemia; tuberculosis (amikacin); topical — eye drops (tobramycin), skin ointments.

> **REMEMBER**
>
> **Adverse effects of aminoglycosides — "NOTN"**: **N**ephrotoxicity (proximal tubules, usually reversible), **O**totoxicity (cochlear — irreversible hearing loss; vestibular — dizziness, ataxia; even one dose in people with a mitochondrial DNA mutation), **T**eratogenicity (damage to the fetal auditory nerve — contraindicated in pregnancy), **N**euromuscular blockade (enhances the effect of muscle relaxants, dangerous in myasthenia; treatment — calcium preparations). **Once-daily dosing** (the same total dose) reduces toxicity while preserving effectiveness; monitoring of renal function and blood concentration; the course is as short as possible (5–7 days); caution together with furosemide, vancomycin and other nephrotoxic drugs.

5. Macrolides

Preparations: 14-membered — **erythromycin**, **clarithromycin**, roxithromycin; 15-membered (azalides) — **azithromycin**; 16-membered — josamycin, spiramycin, midecamycin.

Mechanism. Binding to the 50S subunit blocks translocation of the peptide chain — bacteriostatic (bactericidal at high concentrations).

Spectrum: gram-positive cocci (streptococci, pneumococci — resistance is growing), **atypical bacteria** (**Mycoplasma pneumoniae, Chlamydia, Legionella**), Bordetella pertussis, Campylobacter, H. pylori (clarithromycin), H. influenzae (azithromycin, clarithromycin), toxoplasma (spiramycin), mycobacteria (MAC — clarithromycin, azithromycin).

Uses: community-acquired pneumonia (atypical pathogens; in severe forms in combination with a beta-lactam), streptococcal tonsillitis and other infections **in penicillin allergy**, **whooping cough**, chlamydial infections (azithromycin 1 g once), H. pylori eradication (clarithromycin), campylobacteriosis, **trachoma**, travelers' diarrhea; azithromycin — long-term low-dose prevention of exacerbations in COPD and bronchiectasis (anti-inflammatory effect).

Pharmacokinetics: azithromycin reaches very high concentrations in tissues and inside cells, t½ — 68 hours; **short courses** (3–5 days).

Adverse effects: gastrointestinal disturbances (erythromycin is a **motilin agonist**, strongly so; it is also used as a prokinetic), **QT prolongation** and arrhythmias (especially with other QT-prolonging drugs), cholestatic hepatitis, ototoxicity (at high doses); in newborns, erythromycin and azithromycin — **pyloric stenosis**.

> **CAUTION**
> **Erythromycin and clarithromycin are strong CYP3A4 inhibitors** (azithromycin — very weak): they increase the concentrations of **statins** (simvastatin, atorvastatin — rhabdomyolysis), **warfarin**, carbamazepine, cyclosporine, tacrolimus, colchicine, calcium antagonists, ergot alkaloids, some antiarrhythmics and DOACs. When there is a risk of interaction, **azithromycin** is preferred.

6. Tetracyclines

Preparations: tetracycline, **doxycycline** (preferred — well absorbed, 1–2 times a day, no dose change in renal failure), **minocycline**; glycylcyclines — **tigecycline** (reserve, multidrug-resistant bacteria), omadacycline, eravacycline.

Mechanism. Reversible binding to the 30S subunit stops the binding of aminoacyl-tRNA to the ribosome — bacteriostatic.

Spectrum: broad — gram-positives and gram-negatives, **intracellular and atypical pathogens**: rickettsiae (**drugs of choice**), chlamydiae, mycoplasmas, **borreliae (Lyme disease)**, brucellae (with streptomycin or rifampicin), Vibrio cholerae, anthrax, plague, tularemia, leptospirae, some protozoa (malaria prophylaxis — doxycycline), MRSA (doxycycline, minocycline), Propionibacterium acnes.

Uses: rickettsioses (epidemic typhus, Q fever), Lyme disease, brucellosis, chlamydial infections, atypical pneumonia, acne and rosacea, cholera, malaria prophylaxis, H. pylori (in a quadruple regimen).

> **REMEMBER**
> **Tetracyclines are contraindicated in children under 8, pregnant women and breastfeeding women**: they bind calcium and accumulate in growing **bones and teeth** — **yellow-brown discoloration of tooth enamel**, enamel hypoplasia, slowed bone growth. Other adverse effects: **photosensitivity** (sun rash — doxycycline), esophageal ulcer (take the capsule with plenty of water, upright), gastrointestinal disturbances, candidiasis and superinfections, hepatotoxicity (in pregnant women), raised intracranial pressure; expired tetracycline — Fanconi syndrome. Absorption is reduced by chelation with **calcium, magnesium, aluminum, iron and milk**.

> **CAUTION**
> According to modern data, **a short course of doxycycline** (≤ 21 days) hardly causes tooth discoloration in children under 8; therefore in life-threatening **rickettsioses** (for example, Rocky Mountain spotted fever) doxycycline is the drug of choice at any age.

7. Other antibiotics

7.1. Lincosamides

Clindamycin (and lincomycin) — act on the **50S** subunit; bacteriostatic. **Gram-positive cocci** (staphylococci, including some MRSA, streptococci) and **anaerobes** (Bacteroides, clostridia —

except C. difficile). It penetrates bone tissue well and inhibits toxin synthesis.

Uses: skin and soft tissue infections, **osteomyelitis**, lung abscess and aspiration pneumonia, pelvic and abdominal anaerobic infections, odontogenic infections, **toxic shock syndrome** and necrotizing fasciitis (with penicillin — to stop toxin synthesis), streptococcal infections **in penicillin allergy**, acne (topical), bacterial vaginosis (vaginal cream), toxoplasmosis and pneumocystosis (with other drugs).

Adverse effects: C. difficile colitis (pseudomembranous colitis) — one of the antibiotics with the highest risk; diarrhea, rash.

7.2. Amphenicols

Chloramphenicol (levomycetin) — acts on the **50S** subunit; broad-spectrum, bacteriostatic; crosses the BBB well. Because of severe adverse effects it is now a **reserve** drug (bacterial meningitis — in severe penicillin allergy, enteric fever, rickettsioses — when there is no other option); topically — eye drops, ointments (Levomekol).

> **CAUTION**
> Dangerous adverse effects of chloramphenicol (levomycetin): 1) **aplastic anemia** — dose-independent, idiosyncratic, often irreversible and fatal (one in 40,000 cases), described even after eye drops; 2) dose-dependent **reversible myelosuppression**; 3) **"gray baby syndrome"** in newborns — accumulation due to immature hepatic glucuronidation, with gray-bluish skin color, hypothermia, vomiting, abdominal distension and collapse. In CIS countries chloramphenicol tablets were widely used for diarrhea — this **is not recommended**.

7.3. Polymyxins and others

- **Polymyxin B** and **colistin (polymyxin E)** bind to the lipopolysaccharides and phospholipids of the gram-negative bacterial membrane and damage the membrane like a detergent — bactericidal. **The last reserve** in infections with gram-negative bacteria resistant to many drugs, including **carbapenems** (Acinetobacter, Pseudomonas, Klebsiella). Nephrotoxic and neurotoxic. Topically — eye and ear drops.
- **Daptomycin** — against gram-positives (MRSA, VRE); membrane depolarization; inactivated by pulmonary surfactant — **ineffective in pneumonia**; myopathy (CK monitoring).
- **Fusidic acid** — staphylococci; a topical cream for skin infections.
- **Mupirocin** — topical (nasal ointment) — eradication of staphylococcal carriage (including MRSA), impetigo.
- **Rifamycins** (rifampicin, rifabutin, rifaximin) — in Topic 18; rifaximin is not absorbed from the intestine (travelers' diarrhea, hepatic encephalopathy).

8. Clinical use of antibiotics

8.1. Empirical choice in common infections

Table 17.6. *Choice of antibiotics in common infections in outpatient practice (adults)*

Infection	Is an antibiotic needed?	First choice	In penicillin allergy
Acute respiratory viral infection, acute bronchitis	Absent	Symptomatic treatment	—

Infection	Is an antibiotic needed?	First choice	In penicillin allergy
Acute streptococcal tonsillitis (high Centor/McIsaac score or positive rapid test)	Yes	Phenoxymethylpenicillin or amoxicillin for 10 days; benzathine benzylpenicillin once	Azithromycin for 5 days, clarithromycin, clindamycin; cephalexin (if no anaphylaxis)
Acute otitis media (children)	Often observation for 48–72 hours; under 6 months, severe or bilateral — yes	Amoxicillin (high dose)	A macrolide; cefuroxime
Acute bacterial sinusitis (symptoms > 10 days or “double-sickening”)	Sometimes	Amoxicillin or amoxicillin/clavulanate	Doxycycline; levofloxacin (reserve)
Community-acquired pneumonia (outpatient, without comorbidities)	Yes	Amoxicillin (high dose)	Doxycycline or a macrolide
Community-acquired pneumonia (with comorbidities)	Yes	Amoxicillin/clavulanate ± a macrolide	A “respiratory” fluoroquinolone
Community-acquired pneumonia (severe, in hospital)	Yes	Ceftriaxone + azithromycin (or clarithromycin)	Levofloxacin, moxifloxacin
Cellulitis, erysipelas	Yes	Phenoxymethylpenicillin/benzylpenicillin (erysipelas), cephalexin or flucloxacillin	Clindamycin
Animal bite	Yes (prophylaxis)	Amoxicillin/clavulanate	Doxycycline + metronidazole
Syphilis (primary, secondary)	Yes	Benzathine benzylpenicillin 2.4 million U once intramuscularly	Doxycycline for 14 days (in pregnant women — desensitization and penicillin)
Gonorrhea	Yes	Ceftriaxone 500 mg–1 g once intramuscularly (+ doxycycline for chlamydia)	With a specialist
Chlamydial infection	Yes	Doxycycline for 7 days	Azithromycin 1 g once (in pregnant women)
Bacterial meningitis (adults)	Yes, urgently	Ceftriaxone (or cefotaxime) + vancomycin; > 50 years and in immunosuppression + ampicillin (listeria); dexamethasone	With a specialist

CLINICAL SIGNIFICANCE

If meningococcal infection is suspected before hospital (fever, a hemorrhagic rash that does not fade on pressure, meningeal signs), the doctor must give **benzylpenicillin** or **ceftriaxone** intravenously or intramuscularly immediately, without waiting for transport: every hour of delay increases the risk of death.

8.2. Antibiotics in special groups of patients

Table 17.7. *Safety of antibiotics in pregnancy and childhood*

Relatively safe (in pregnancy)	With caution	Contraindicated
Penicillins, cephalosporins, azithromycin, erythromycin (except estolate), clindamycin, nitrofurantoin (not before delivery), fosfomycin, metronidazole (2nd–3rd trimesters)	Co-trimoxazole (except the 1st and 3rd trimesters), clarithromycin	Tetracyclines, fluoroquinolones, aminoglycosides (ototoxicity; except in life-threatening situations), chloramphenicol (gray baby), tigecycline

CAUTION

Antibiotics contraindicated in children: **tetracyclines** (under 8; a short course of doxycycline is an exception), **fluoroquinolones** (under 18, except for special indications — cartilage), **chloramphenicol** (newborns — gray baby syndrome), **ceftriaxone** (in newborns with hyperbilirubinemia and with calcium-containing solutions), **co-trimoxazole** and sulfonamides (under 2 months — kernicterus).

8.3. Antibiotic-associated diarrhea and C. difficile

Almost any antibiotic (especially **clindamycin, cephalosporins, fluoroquinolones, amoxicillin/clavulanate**) can disrupt the intestinal flora and lead to overgrowth and toxin production by **Clostridioides difficile** — from watery diarrhea to **pseudomembranous colitis** and toxic megacolon. Treatment: stop the causative antibiotic; **oral vancomycin** or **fidaxomicin** (metronidazole only in mild cases when nothing else is available); in recurrent cases — fecal microbiota transplantation. Loperamide and other antiperistaltic agents are contraindicated. Prevention — no unjustified antibiotics, hand hygiene (with soap; alcohol does not act on spores).

Practical skills

Skill 1. Checklist before prescribing an antibiotic

1. Is a bacterial infection likely? (No antibiotic is needed for a viral infection.)
2. The likely type of pathogen and local resistance?
3. Were specimens taken before the antibiotic (blood, urine, sputum culture)?
4. Allergy history (what reaction, when)?
5. Pregnancy, breastfeeding, age?
6. Renal and hepatic function — dose adjustment?
7. Drug interactions (statins, warfarin, QT-prolonging drugs)?
8. Are the dose, route, interval and **duration** clearly defined?
9. Is there a plan to assess the effect after 48–72 hours?

Skill 2. Key prescriptions

PRESCRIPTION 17.1. BENZYLPENICILLIN SODIUM SALT AND DILUENT

Rp.: Benzylpenicillini-natrii 1 000 000 U
D.t.d. N. 30
S. Dissolve the contents of the vial in 2 ml of water for injection and inject 1 million U intramuscularly every 4 hours.
#
Rp.: Aquae pro injectionibus 2 ml
D.t.d. N. 30 in ampullis
S. For dissolving benzylpenicillin.

PRESCRIPTION 17.2. BENZATHINE BENZYLPENICILLIN

Rp.: Benzathini benzylpenicillini 2 400 000 U
D.t.d. N. 1
S. Suspend the contents of the vial in the diluent and inject once deep intramuscularly, divided between both buttocks (in syphilis).

PRESCRIPTION 17.3. PHENOXYMETHYLPENICILLIN

Rp.: Phenoxymethylpenicillini 0.5
D.t.d. N. 40 in tabulettis
S. Take 1 tablet 4 times a day 1 hour before meals for 10 days.

PRESCRIPTION 17.4. AMOXICILLIN

Rp.: Capsulas Amoxicillini 0.5 N. 21
D.S. Take 1 capsule 3 times a day for 7 days.

PRESCRIPTION 17.5. AMOXICILLIN/CLAVULANATE

Rp.: Tabulettas “Amoxicillinum + Acidum clavulanicum” 875 mg/125 mg N. 14
D.S. Take 1 tablet 2 times a day at the start of a meal for 7 days.

PRESCRIPTION 17.6. CEFTRIAXONE

Rp.: Ceftriaxoni 1.0
D.t.d. N. 7
S. Dissolve the contents of the vial in 10 ml of water for injection and inject intravenously slowly (over 2–4 minutes) once a day.

PRESCRIPTION 17.7. CEFUROXIME AXETIL

Rp.: Cefuroximi axetili 0.5
D.t.d. N. 14 in tabulettis obductis
S. Take 1 tablet 2 times a day after meals.

PRESCRIPTION 17.8. AZITHROMYCIN

Rp.: Azithromycini 0.5
D.t.d. N. 3 in tabulettis obductis
S. Take 1 tablet once a day for 3 days.

PRESCRIPTION 17.9. CLARITHROMYCIN
Rp.: Clarithromycini 0.5
D.t.d. N. 14 in tabulettis obductis
S. Take 1 tablet 2 times a day.

PRESCRIPTION 17.10. DOXYCYCLINE
Rp.: Capsulas Doxycyclini 0.1 N. 14
D.S. Take 1 capsule 2 times a day with plenty of water, in an upright position; avoid the sun.

PRESCRIPTION 17.11. GENTAMICIN
Rp.: Solutionis Gentamicini sulfatis 4% — 2 ml
D.t.d. N. 10 in ampullis
S. 5 mg/kg once a day intravenously or intramuscularly (under renal function monitoring).

PRESCRIPTION 17.12. VANCOMYCIN
Rp.: Vancomycini 1.0
D.t.d. N. 10
S. Dissolve the contents of the vial in 200 ml of 0.9% NaCl and give as an intravenous drip every 12 hours over at least 60–120 minutes.

PRESCRIPTION 17.13. CLINDAMYCIN
Rp.: Capsulas Clindamycini 0.3 N. 21
D.S. Take 1 capsule 3 times a day for 7 days.

PRESCRIPTION 17.14. MUPIROCIN
Rp.: Unguenti Mupirocini 2% — 15.0
D.S. Apply to the nasal passages 2–3 times a day for 5 days.

Review questions

1. Classification of antibiotics (by mechanism, spectrum and type of action).
2. Time-dependent and concentration-dependent antibiotics. Clinical significance.
3. Mechanism of action of beta-lactams and mechanisms of resistance.
4. Classification, spectrum and uses of penicillins.
5. Beta-lactamase inhibitors and protected penicillins.
6. Generations, spectrum and uses of cephalosporins. Which microbes do they not act on?
7. Features of carbapenems and aztreonam.
8. Adverse effects of beta-lactams. The approach to a history of “penicillin allergy”.
9. Uses and adverse effects of vancomycin.
10. Mechanism, uses, adverse effects and dosing features of aminoglycosides.
11. Spectrum, uses and interactions of macrolides.
12. Spectrum, uses and contraindications of tetracyclines.
13. Features and risks of clindamycin and chloramphenicol.
14. The place of polymyxins and daptomycin.
15. Empirical choice of antibiotics in common infections.
16. Antibiotics contraindicated in pregnancy and childhood.

17. C. difficile infection: risk factors and treatment.

Test questions

1. The target of beta-lactam antibiotics:
   A) DNA gyrase
   B) Penicillin-binding proteins (transpeptidases)
   C) RNA polymerase
   D) Dihydrofolate reductase
   E) 30S ribosome

2. The drug of choice for treating syphilis:
   A) Ceftriaxone
   B) Azithromycin
   C) Ciprofloxacin
   D) Benzathine benzylpenicillin
   E) Gentamicin

3. An antibiotic that often causes a rash when given in infectious mononucleosis:
   A) Azithromycin
   B) Doxycycline
   C) Ceftriaxone
   D) Clindamycin
   E) Amoxicillin

4. A microorganism on which no cephalosporin acts:
   A) Escherichia coli
   B) Enterococci
   C) Haemophilus influenzae
   D) Streptococcus pneumoniae
   E) Neisseria meningitidis

5. The main cephalosporin for empirical treatment of bacterial meningitis:
   A) Cefuroxime axetil
   B) Cefixime
   C) Cephalexin
   D) Cefazolin
   E) Ceftriaxone

6. A beta-lactam with almost no cross-allergy with penicillins:
   A) Meropenem
   B) Aztreonam
   C) Ampicillin
   D) Cephalexin
   E) Oxacillin

7. An antibiotic that causes “red man syndrome” with rapid infusion:
   A) Vancomycin
   B) Ceftriaxone
   C) Doxycycline
   D) Gentamicin
   E) Azithromycin

8. An irreversible adverse effect of aminoglycosides:
   A) Tooth discoloration
   B) Photosensitivity
   C) Ototoxicity
   D) Gastrointestinal disturbance
   E) Nephrotoxicity

9. Which microorganisms do aminoglycosides not act on?
   A) Escherichia coli
   B) Anaerobes
   C) Staphylococci
   D) Pseudomonas aeruginosa
   E) Gram-negative aerobes

10. A macrolide that increases the risk of rhabdomyolysis when given together with statins:
    A) Spiramycin
    B) Josamycin
    C) Azithromycin
    D) Midecamycin
    E) Clarithromycin

11. A group that does not act on the pathogens of atypical pneumonia (mycoplasma, chlamydia, legionella):
    A) Beta-lactams
    B) “Respiratory” fluoroquinolones
    C) Macrolides
    D) Doxycycline
    E) Tetracyclines

12. The drug of choice in rickettsioses and Lyme disease:
    A) Vancomycin
    B) Amoxicillin
    C) Ceftazidime
    D) Doxycycline
    E) Gentamicin

13. Antibiotics contraindicated in children under 8 because of tooth enamel discoloration:
    A) Penicillins
    B) Cephalosporins
    C) Macrolides
    D) Tetracyclines
    E) Carbapenems

14. An antibiotic that causes “gray baby syndrome” in newborns and aplastic anemia:
    A) Ampicillin
    B) Erythromycin
    C) Gentamicin
    D) Chloramphenicol
    E) Clindamycin

15. One of the antibiotics with the highest risk of pseudomembranous colitis (C. difficile), and its treatment:
    A) Doxycycline; treatment — penicillin
    B) Vancomycin; treatment — ceftriaxone
    C) Gentamicin; treatment — loperamide
    D) Clindamycin; treatment — oral vancomycin or fidaxomicin
    E) Azithromycin; treatment — clindamycin

Clinical cases

Case 1

A 22-year-old student presents with fever (38.8 °C) for 2 days, severe sore throat and difficulty swallowing. No cough. The tonsils are enlarged with purulent exudate, and the anterior cervical lymph nodes are enlarged and tender. The rapid strep test is positive. At age 10 she had a “rash” after amoxicillin (no urticaria or swelling).

Questions: 1) Diagnosis? 2) Is an antibiotic needed, and which one? 3) How do you assess the “allergy” history?

Answer: 1) Acute streptococcal (group A) tonsillitis (Centor score 4, positive test). 2) Yes — to prevent complications (rheumatic fever, peritonsillar abscess) and to shorten the duration of symptoms. A mild childhood rash (without urticaria or anaphylaxis) is low-risk — probably a viral rash; it can be clarified with an allergist. Cephalexin (a 1st-generation cephalosporin) with caution, or phenoxymethylpenicillin for 10 days after allergy has been excluded; if the patient or doctor is worried — azithromycin for 5 days or clindamycin. 3) A detailed history: the type of reaction, its timing, its treatment; for low-risk rashes, allergy testing (a drug challenge) to remove the “allergy” label.

Case 2

A 65-year-old patient received clarithromycin for community-acquired pneumonia. He takes simvastatin 40 mg, amlodipine and warfarin. After 5 days he developed severe muscle pain and weakness, dark urine and nosebleeds. CK is markedly raised, INR 5.8.

Questions: 1) The cause of these conditions? 2) Management, and which antibiotic would have been preferable?

Answer: 1) Clarithromycin is a strong CYP3A4 inhibitor: the simvastatin concentration rose and rhabdomyolysis developed (muscle pain, myoglobinuria, raised CK); warfarin metabolism slowed and the INR rose — bleeding. The amlodipine concentration may also rise. 2) Stop clarithromycin and simvastatin, stop warfarin temporarily and give vitamin K (INR 4.5–10 with minor bleeding); in rhabdomyolysis — plenty of intravenous fluids, monitoring of creatinine and potassium. Replace the antibiotic with amoxicillin (± azithromycin — it hardly affects CYP3A4). Whenever an antibiotic is prescribed, interactions with the patient's other drugs must be checked.

Case 3

A 70-year-old woman received ceftriaxone and levofloxacin in hospital for pneumonia. On day 8 of treatment she developed watery diarrhea 8–10 times a day, abdominal pain, fever of 38.5 °C and leukocytosis of 18×10^9/l. Stool C. difficile toxin is positive.

Questions: 1) Diagnosis? 2) Treatment? 3) What must not be done?

Answer: 1) Clostridioides difficile infection (antibiotic-associated colitis), severe form (white cells > 15 × 10^9/l is a severity criterion; creatinine is also monitored). 2) Stop other antibiotics where possible; oral vancomycin 125 mg 4 times a day for 10 days or fidaxomicin; fluid and electrolyte replacement; contact isolation, handwashing with soap (alcohol does not act on spores); stop the PPI if it is not needed. 3) Loperamide and other antiperistaltic agents — risk of toxic megacolon; intravenous vancomycin (it does not reach the intestine).

Quick review table

Table 17.8. *Antibiotics — summary*

Group	Preparations	Mechanism	Main use	Main risk / note
Natural penicillins	Benzylpenicillin, benzathine benzylpenicillin, phenoxymethylpenicillin	PBPs, cell wall	Syphilis, streptococcal infections, rheumatic fever prevention, meningococcus	Allergy, Jarisch–Herxheimer
Aminopenicillins	Amoxicillin, ampicillin	—	Otitis, sinusitis, pneumonia, tonsillitis	Rash in mononucleosis
Protected	Amoxicillin/clavulanate, piperacillin/tazobactam	+ beta-lactamase inhibitor	Complicated infections, bites; Pseudomonas	Diarrhea, hepatitis
Cephalosporins	Cefazolin, cefuroxime, ceftriaxone, ceftazidime, cefepime, ceftaroline	—	Prophylaxis, meningitis, pneumonia, gonorrhea	No action on enterococci, listeria, atypicals
Carbapenems	Meropenem, imipenem, ertapenem	—	ESBL, severe hospital infections	Reserve; seizures (imipenem)
Monobactam	Aztreonam	—	Gram-negatives, penicillin allergy	—
Glycopeptides	Vancomycin	D-Ala-D-Ala	MRSA; oral — C. difficile	Red man, nephrotoxicity
Aminoglycosides	Gentamicin, amikacin, streptomycin	30S, bactericidal	Gram-negative sepsis, endocarditis (combination)	Nephro- and ototoxicity, pregnancy; none against anaerobes
Macrolides	Azithromycin, clarithromycin, erythromycin	50S	Atypical pneumonia, streptococci in allergy, whooping cough, chlamydia, H. pylori	QT; CYP3A4 (clarithromycin, erythromycin)
Tetracyclines	Doxycycline, minocycline, tigecycline	30S	Rickettsioses, Lyme, chlamydia, acne, malaria prophylaxis	Children (under 8), pregnancy; photosensitivity
Lincosamides	Clindamycin	50S	Anaerobes, osteomyelitis, toxin syndromes	C. difficile
Amphenicols	Chloramphenicol	50S	Reserve; topical	Aplastic anemia, gray baby

Group	Preparations	Mechanism	Main use	Main risk / note
Polymyxins	Colistin, polymyxin B	Membrane	Carbapenem-resistant gram-negatives	Nephro- and neurotoxicity
Others	Daptomycin, mupirocin, fusidic acid, rifaximin	Various	MRSA, topical, intestine	Daptomycin — ineffective in pneumonia

TOPIC 18

Antituberculosis drugs. Antiviral drugs

Learning objectives

Tuberculosis ranks among the leading causes of death from infectious diseases; **drug-resistant tuberculosis** is a serious problem in Uzbekistan and the Central Asian region. Treating tuberculosis requires a long-term combination of several drugs and strict adherence by the patient. **Antiviral drugs** have changed radically in recent decades: HIV infection has become a chronic, manageable disease, and hepatitis C is cured completely with 8–12 weeks of tablets. At the same time, many "immunomodulators" and "antivirals" are widely sold without proven effectiveness — a GP must be able to tell evidence-based treatment apart.

The student should know:

- the classification of antituberculosis drugs, and the mechanism, adverse effects and interactions of the main (first-line) drugs;
- the principles of treating tuberculosis, standard regimens and approaches to treating drug-resistant tuberculosis;
- the stages of viral replication and the targets of antiviral drugs;
- drugs against influenza, herpes, cytomegalovirus, hepatitis B and C, HIV and COVID-19;
- the place of interferons and interferon inducers.

The student should be able to:

- write prescriptions for drugs such as isoniazid, rifampicin, pyrazinamide, ethambutol, acyclovir, valacyclovir and oseltamivir;
- recognize and manage the adverse effects of antituberculosis drugs;
- determine the indications for antiviral treatment in influenza and herpes infections.

1. Antituberculosis drugs

1.1. Features of mycobacteria and principles of treatment

Mycobacterium tuberculosis is a slowly multiplying bacterium (division time 18–24 hours) whose cell wall is rich in **mycolic acids** (acid-fast, impermeable to many drugs) and which can live inside macrophages. The disease focus contains different populations: rapidly multiplying (in cavities), slowly multiplying (in an acidic environment, inside macrophages) and "dormant" (persistent) forms.

> **REMEMBER**
> **The basic principles of tuberculosis treatment**: 1. **Combination therapy** — at least 3–4 drugs: with monotherapy, **resistance** develops quickly. 2. **Long-term treatment** — to eliminate persistent forms (in drug-susceptible tuberculosis — at least 4–6 months). 3. **Two phases**: **intensive** (2 months — rapid reduction of the number of mycobacteria) and **continuation** (4 months — elimination of the rest). 4. **Directly observed therapy (DOT)** — a health worker watches every dose being taken (or video observation). 5. **Drug susceptibility testing** before treatment (rapid molecular tests — Xpert MTB/RIF detects rifampicin resistance in 2 hours).

1.2. Classification of antituberculosis drugs

Table 18.1. *Classification of antituberculosis drugs (based on the WHO approach)*

Group	Preparations
First-line (main) — in drug-susceptible tuberculosis	Isoniazid (H), rifampicin (R), pyrazinamide (Z), ethambutol (E); rifapentine, rifabutin
In drug-resistant tuberculosis — group A	Bedaquiline, levofloxacin / moxifloxacin, linezolid
Group B	Clofazimine, cycloserine (terizidone)
Group C	Delamanid, pyrazinamide, ethambutol, carbapenems (with amoxicillin/clavulanate), amikacin, streptomycin, ethionamide / prothionamide, para-aminosalicylic acid (PAS)

CAUTION

Older textbooks divided antituberculosis drugs into “first-line” and “second-line (reserve)”, and streptomycin and kanamycin were widely used. Current WHO recommendations do not recommend injectable drugs (kanamycin, capreomycin) for drug-resistant tuberculosis because of **ototoxicity** and low effectiveness; **oral regimens** (bedaquiline, pretomanid, linezolid, moxifloxacin) are used instead.

1.3. First-line drugs

Table 18.2. *First-line antituberculosis drugs*

Preparation	Mechanism	Effect	Main adverse effects	Special notes
Isoniazid (INH)	Inhibits mycolic acid synthesis (a prodrug — activated by the KatG enzyme)	Bactericidal against rapidly multiplying bacteria	Hepatotoxicity (increases with age and alcohol), peripheral neuropathy (B_6 antagonist), seizures, lupus-like syndrome	Pyridoxine (B_6) prophylaxis; slow and fast acetylators; seizures in overdose — intravenous pyridoxine
Rifampicin	Inhibits DNA-dependent RNA polymerase	Bactericidal (all populations, especially persisters)	Hepatotoxicity, orange-red discoloration of body fluids (urine, tears — contact lenses!), flu-like syndrome (with intermittent dosing), thrombocytopenia	The strongest CYP450 inducer — many interactions
Pyrazinamide	Converted to pyrazinoic acid, it disrupts membrane energetics in an acidic environment	Bactericidal against mycobacteria in an acidic environment (inside macrophages)	Hepatotoxicity (the most hepatotoxic), hyperuricemia and gout, arthralgia, photosensitivity	Only in the intensive phase (2 months)
Ethambutol	Inhibits arabinosyltransferase and disrupts the arabinogalactan of the cell wall	Bacteriostatic	Optic neuritis — reduced visual acuity and red-green color discrimination (dose-dependent, usually reversible)	Monitoring of vision; caution in young children (vision is hard to assess); the dose is reduced in renal failure

Preparation	Mechanism	Effect	Main adverse effects	Special notes
Streptomycin	30S ribosome (an aminoglycoside)	Bactericidal (extracellular)	Oto- and nephrotoxicity	Now rarely used; contraindicated in pregnancy

MEMORY AID

The first-line drugs — **"RIPE"**: **R**ifampicin, **I**soniazid, **P**yrazinamide, **E**thambutol. Their adverse effects: **R** — **R**ed urine and **R**aised (induced) enzymes; **I** — **I**nflamed nerves (neuropathy) and liver; **P** — **P**odagra (gout) and liver; **E** — **E**yes. Three drugs (H, R, Z) are hepatotoxic, while ethambutol is harmless to the liver.

CAUTION

Rifampicin has very many interactions: it sharply **reduces** the concentrations of **oral contraceptives** (unexpected pregnancy — nonhormonal contraception is needed), warfarin, DOACs, glucocorticoids, sulfonylureas, antiretroviral drugs (protease inhibitors, most integrase inhibitors), azole antifungals, methadone, cyclosporine, tacrolimus, digoxin, beta-blockers and calcium antagonists. In tuberculosis with HIV, **rifabutin** (a weak inducer) is sometimes used instead of rifampicin.

1.4. Drugs for treating drug-resistant tuberculosis

- **Bedaquiline** inhibits mycobacterial **ATP synthase** (a new mechanism); bactericidal. Adverse effects: **QT prolongation**, hepatotoxicity.
- **Pretomanid** and **delamanid** — nitroimidazooxazines; they disrupt mycolic acid synthesis and cellular respiration. QT prolongation, neuropathy.
- **Linezolid** — 50S ribosome (Topic 16); **myelosuppression**, peripheral and **optic neuropathy** (with long-term use).
- **Levofloxacin, moxifloxacin** — DNA gyrase (Topic 16).
- **Clofazimine** — red-brown skin pigmentation, QT prolongation.
- **Cycloserine** — cell wall synthesis; **neuropsychiatric** adverse effects (depression, psychosis, seizures) — given with pyridoxine.

Table 18.3. *The main tuberculosis treatment regimens (based on WHO recommendations)*

Situation	Regimen
Drug-susceptible tuberculosis (standard)	2 months of HRZE (intensive phase) + 4 months of HR (continuation phase) — 6 months in total
Drug-susceptible tuberculosis (over 12 years, selected patients)	4-month regimen: rifapentine + isoniazid + pyrazinamide + moxifloxacin
Non-severe tuberculosis in children	4-month regimen (2HRZ(E) + 2HR)
Tuberculous meningitis, bone and joint tuberculosis	The continuation phase is extended (9–12 months); in meningitis + a glucocorticoid
Multidrug-/rifampicin-resistant tuberculosis (MDR/RR-TB)	BPaLM — bedaquiline + pretomanid + linezolid + moxifloxacin for 6 months (over 14 years) or individualized longer regimens
Latent tuberculosis infection (preventive treatment)	3 months of weekly rifapentine + isoniazid (3HP), 4 months of rifampicin (4R), 3 months of isoniazid + rifampicin (3HR) or 6–9 months of isoniazid

CLINICAL SIGNIFICANCE

Hepatotoxicity of antituberculosis drugs is the most common reason for stopping treatment. Liver enzymes are measured before treatment; in risk groups (alcoholism, hepatitis B and C, HIV, old age, pregnancy) they are monitored regularly. If symptoms appear (nausea, vomiting, loss of appetite, jaundice, right upper quadrant pain) or transaminases rise more than 3-fold with symptoms (more than 5-fold without symptoms), **all hepatotoxic drugs are stopped**, and once the enzymes normalize they are restarted one by one.

CAUTION

Stopping treatment on one's own or taking drugs incorrectly is the main cause of **drug-resistant tuberculosis**. A patient infected with a resistant strain transmits it to others. That is why explaining to the patient the need for the full course of treatment and ensuring DOT are of crucial importance.

2. Antiviral drugs

2.1. Stages of viral replication and drug targets

Viruses are obligate intracellular parasites — they use the mechanisms of the host cell. That is why it is difficult to create drugs that act selectively on viruses; drugs target **virus-specific enzymes and processes**.

Table 18.4. *Stages of viral replication and the drugs acting on them*

Step	Drugs
Attachment to and entry into the cell	Maraviroc (HIV — a CCR5 antagonist), enfuvirtide (HIV — a fusion inhibitor), palivizumab and nirsevimab (RSV — monoclonal antibodies)
Uncoating of the virus	Amantadine, rimantadine (influenza A — M_2 ion channel; now not recommended because of resistance)
Nucleic acid synthesis (polymerases)	Acyclovir and its analogs (herpes DNA polymerase), ganciclovir, foscarnet, sofosbuvir (HCV RNA polymerase), remdesivir, molnupiravir (SARS-CoV-2), baloxavir (influenza endonuclease)
Reverse transcription	Nucleoside/nucleotide (NRTIs — tenofovir, lamivudine, abacavir, zidovudine) and non-nucleoside (NNRTIs — efavirenz, nevirapine, rilpivirine) reverse transcriptase inhibitors (HIV; tenofovir, entecavir — HBV)
Integration	Integrase inhibitors (HIV — dolutegravir, raltegravir, bictegravir)
Maturation of viral proteins (proteolysis)	Protease inhibitors (HIV — darunavir, lopinavir; HCV — glecaprevir; SARS-CoV-2 — nirmatrelvir)
Release of virions from the cell	Neuraminidase inhibitors (influenza A and B — oseltamivir, zanamivir)
The viral replication complex	NS5A inhibitors (HCV — velpatasvir, ledipasvir, pibrentasvir)

2.2. Anti-influenza drugs

- **Neuraminidase inhibitors** — **oseltamivir** (oral), zanamivir (inhaled): by inhibiting neuraminidase they stop the release of new virions from the infected cell and their spread. They act on influenza A and B. Effective when started **within 48 hours** of symptom onset — they shorten the illness by ≈ 1 day and reduce complications and hospitalizations **in risk groups** (the elderly, pregnant women, children under 2, chronic diseases, immunosuppression); in severe and

hospitalized patients at any time. Prophylaxis — after contact. Adverse effects: nausea, vomiting; rarely neuropsychiatric disorders in children. Zanamivir — bronchospasm in asthma and COPD.

- **Baloxavir marboxil** inhibits the cap-dependent endonuclease of the viral polymerase; a single dose.
- **Amantadine, rimantadine** — influenza A only; almost all modern strains are **resistant** — not recommended (amantadine is used in Parkinson's disease).

CLINICAL SIGNIFICANCE

The most effective prevention of influenza is **annual vaccination**, especially for risk groups and health workers. Antiviral drugs do not replace vaccination.

CAUTION

The effectiveness of **umifenovir (Arbidol), Ingavirin, Kagocel, oxolinic ointment, tilorone** and similar "antiviral" and "immunomodulatory" preparations widely sold in CIS countries **has not been proven in quality randomized trials** for influenza and ARVI; they are not included in WHO or international clinical recommendations. For most ARVIs (rhinovirus, adenovirus) there is no specific antiviral treatment — symptomatic treatment is enough.

2.3. Drugs against herpesviruses

Acyclovir is a guanosine analog and a **prodrug**: viral **thymidine kinase** (only in the infected cell) converts it into the monophosphate, and then cellular kinases phosphorylate it to the triphosphate. Acyclovir triphosphate inhibits the viral **DNA polymerase** and, by being incorporated into the DNA chain, stops its elongation. It has almost no effect on uninfected cells — **high selectivity**, low toxicity.

Table 18.5. *Drugs against herpesviruses*

Preparation	Features	Main use
Acyclovir	Low oral bioavailability (15–30%) — 5 times a day; intravenous, topical (cream, eye ointment)	Herpes simplex (labial, genital), herpes encephalitis (intravenously — urgently), chickenpox and herpes zoster (severe cases, immunosuppression), neonatal herpes
Valacyclovir	A prodrug of acyclovir; bioavailability ≈ 55% — 2–3 times a day	Genital herpes (treatment and suppressive therapy), herpes zoster (within 72 hours — reduces the risk of postherpetic neuralgia)
Famciclovir	A prodrug of penciclovir	Herpes, shingles
Ganciclovir, valganciclovir	Active against cytomegalovirus (viral phosphotransferase UL97); myelosuppression (neutropenia), teratogenic	CMV infection — in immunosuppression, after transplantation, CMV retinitis in HIV
Foscarnet	Independent of thymidine kinase — against acyclovir-resistant strains	Resistant herpes, CMV; nephrotoxic, electrolyte disturbances
Cidofovir	—	Resistant CMV; nephrotoxic

REMEMBER

The main adverse effect of acyclovir is **crystalluria and acute kidney injury** with large intravenous doses (prevention — adequate hydration, slow infusion, dosing according to renal function); rarely neurotoxicity (in the elderly and in renal failure). Herpesviruses **are never completely eliminated** from the body — the drugs only stop replication and do not act on latent infection.

2.4. Drugs against viral hepatitis

- **Hepatitis B** (chronic): **tenofovir** (tenofovir disoproxil fumarate or alafenamide) and **entecavir** — nucleoside/nucleotide analogs; highly effective, with very little resistance; usually taken **long term** (often for life); they suppress the virus and reduce the risk of cirrhosis and hepatocellular carcinoma. Tenofovir is also used **in pregnancy** (to prevent mother-to-child transmission). Pegylated interferon alfa — a finite course in selected patients. Abrupt withdrawal — a severe flare of hepatitis.
- **Hepatitis C**: **direct-acting antivirals (DAAs)** — **sofosbuvir** (an NS5B polymerase inhibitor) + **velpatasvir** or daclatasvir (NS5A inhibitors), glecaprevir/pibrentasvir; an **8–12-week** course eliminates the virus (cure) in **more than 95%** of patients. Pangenotypic regimens do not require genotyping. Screening for hepatitis B before treatment is mandatory (risk of HBV reactivation); sofosbuvir with amiodarone — severe bradycardia.

CAUTION
Hepatitis C used to be treated with pegylated interferon and ribavirin for 24–48 weeks — low effectiveness and severe adverse effects (flu-like syndrome, depression, cytopenias, hemolytic anemia). Now **interferon-free DAA regimens** are the standard; the WHO aims to eliminate hepatitis C by 2030.

2.5. Drugs against HIV infection (antiretroviral therapy)

HIV is a retrovirus; its RNA is converted into DNA by **reverse transcriptase** and integrated into the host genome by **integrase**; new viral proteins mature with the help of **protease**.

Table 18.6. *Groups of antiretroviral drugs*

Group	Preparations	Main adverse effects
Nucleoside/nucleotide reverse transcriptase inhibitors (NRTIs)	Tenofovir (TDF, TAF), lamivudine, emtricitabine, abacavir, zidovudine	Tenofovir TDF — kidneys and bone; abacavir — hypersensitivity (HLA-B*5701 test); zidovudine — anemia
Non-nucleoside reverse transcriptase inhibitors (NNRTIs)	Efavirenz, nevirapine, rilpivirine, doravirine	Efavirenz — CNS (strange dreams, depression); nevirapine — hepatotoxicity, rash
Integrase inhibitors (INSTIs)	Dolutegravir, bictegravir, raltegravir, cabotegravir (long-acting injection)	Well tolerated; weight gain; insomnia
Protease inhibitors (PIs)	Darunavir, lopinavir, atazanavir (“boosted” with ritonavir or cobicistat)	Metabolic disturbances, gastrointestinal; many interactions via CYP3A4
Entry inhibitors	Maraviroc, enfuvirtide	Special situations

REMEMBER
Principles of antiretroviral therapy (ART): ART is started **as soon as possible in every patient diagnosed with HIV, regardless of the CD4 count**. The standard first-line regimen (WHO) is **tenofovir + lamivudine (or emtricitabine) + dolutegravir** (TLD, in a single tablet, once a day). The goal is to suppress the viral load to an undetectable level: **“Undetectable = Untransmittable”** (U = U). ART is taken for life; adherence prevents resistance.

Prevention: post-exposure prophylaxis (PEP) — a 28-day course of ART **within 72 hours** (ideally within 2 hours) after a risky contact (including a needlestick injury in a health worker); **pre-exposure**

prophylaxis (PrEP) — tenofovir/emtricitabine in high-risk groups; prevention of mother-to-child transmission — ART in pregnancy.

2.6. Drugs against COVID-19

- **Nirmatrelvir/ritonavir** — an inhibitor of the main protease (Mpro) of SARS-CoV-2; ritonavir slows its metabolism. In patients **at high risk** of severe disease, **within 5 days** of symptom onset. **Ritonavir is a strong CYP3A4 inhibitor**: dangerous interactions with statins, antiarrhythmics, DOACs and some anticonvulsants.
- **Remdesivir** — an RNA polymerase inhibitor; intravenous; in hospitalized patients and in high-risk outpatients.
- **Molnupiravir** — introduces mutations into the viral RNA; when there are no other options; contraindicated in pregnancy.
- Anti-inflammatory treatment in severe COVID-19 — **dexamethasone** (reduces mortality in patients requiring oxygen), tocilizumab, baricitinib; anticoagulants (thrombosis prevention).

2.7. Interferons

Interferons are proteins produced by cells in response to a virus; they induce an antiviral state in cells (enzymes that break down viral RNA, inhibition of protein synthesis) and enhance the immune response.

- **Interferon alfa** (pegylated) — chronic hepatitis B (selected patients), some tumors (hairy cell leukemia, melanoma); **interferon beta** — multiple sclerosis; **interferon gamma** — chronic granulomatous disease.
- **Adverse effects** (parenteral): flu-like syndrome (fever, myalgia), depression and suicidal thoughts, cytopenias, autoimmune thyroid disease, fatigue.

> **CAUTION**
> **Intranasal and rectal interferon preparations** and **interferon inducers** (tilorone, Kagocel, Cycloferon) are widely used in CIS countries to treat and prevent ARVI and influenza, but their effectiveness has not been confirmed in studies that meet the standards of modern evidence-based medicine.

3. Clinical part

Table 18.7. *Indications for antiviral treatment in viral infections*

Infection	Treatment	Comment
Influenza (in a risk group or severe)	Oseltamivir 75 mg 2 times a day for 5 days	Most effective within 48 hours; always in severe cases
Influenza (healthy young person, mild)	Symptomatic; oseltamivir optional	—
Cold sores (mild)	Topical acyclovir or no treatment; if frequently recurrent — a short course of valacyclovir	—
Genital herpes (first episode)	Valacyclovir or acyclovir for 7–10 days	Suppressive therapy — with more than 6 recurrences a year

Infection	Treatment	Comment
Herpes zoster (> 50 years, ophthalmic branch, immunosuppression)	Valacyclovir 1 g 3 times a day for 7 days (within 72 hours)	Prevention — vaccination (recombinant vaccine)
Herpes encephalitis	Intravenous acyclovir 10 mg/kg every 8 hours for 14–21 days	Started as soon as it is suspected
Chronic hepatitis B	Tenofovir or entecavir	Often long-term
Chronic hepatitis C	Sofosbuvir/velpatasvir for 12 weeks (or glecaprevir/pibrentasvir for 8 weeks)	Cure > 95%
HIV	Tenofovir + lamivudine + dolutegravir	As soon as possible after diagnosis
COVID-19 (in a risk group)	Nirmatrelvir/ritonavir for 5 days (within 5 days)	Check interactions

Practical skills

Skill 1. Monitoring during antituberculosis treatment

1. Before treatment: liver enzymes, bilirubin, creatinine, uric acid, HIV test, hepatitis B and C, visual acuity and color vision (ethambutol), pregnancy test.
2. Every month: body weight, adverse effects (liver, neuropathy, vision, joints), sputum test (at the end of month 2 — conversion).
3. Pyridoxine (B_6) 25–50 mg/day — with isoniazid (mandatory in diabetes, alcoholism, HIV, pregnancy, malnutrition).
4. Women taking rifampicin — nonhormonal contraception.
5. DOT — observation of every dose.

Skill 2. Key prescriptions

PRESCRIPTION 18.1. ISONIAZID
Rp.: Isoniazidi 0.3
D.t.d. N. 30 in tabulettis
S. Take 1 tablet once a day in the morning on an empty stomach, together with pyridoxine.

PRESCRIPTION 18.2. RIFAMPICIN
Rp.: Capsulas Rifampicini 0.15 N. 60
D.S. Take 4 capsules (600 mg) once a day 1 hour before meals; the urine may turn red.

PRESCRIPTION 18.3. PYRAZINAMIDE
Rp.: Pyrazinamidi 0.5
D.t.d. N. 90 in tabulettis
S. Take 3 tablets once a day (intensive phase, 2 months).

PRESCRIPTION 18.4. ETHAMBUTOL
Rp.: Ethambutoli 0.4
D.t.d. N. 90 in tabulettis
S. Take 3 tablets once a day; seek help immediately if vision changes.

PRESCRIPTION 18.5. FIXED-DOSE COMBINATION TABLET
Rp.: Tabulettas “Rifampicinum + Isoniazidum + Pyrazinamidum + Ethambutolum” 150/75/400/275 mg N. 84
D.S. Take 4 tablets once a day (weight 55–70 kg) in the intensive phase.

PRESCRIPTION 18.6. PYRIDOXINE
Rp.: Pyridoxini hydrochloridi 0.025
D.t.d. N. 50 in tabulettis
S. Take 1 tablet once a day together with isoniazid.

PRESCRIPTION 18.7. OSELTAMIVIR
Rp.: Capsulas Oseltamiviri 0.075 N. 10
D.S. Take 1 capsule 2 times a day for 5 days.

PRESCRIPTION 18.8. ACYCLOVIR (TABLETS)
Rp.: Acicloviri 0.4
D.t.d. N. 35 in tabulettis
S. Take 1 tablet 5 times a day (every 4 hours, with a break at night) for 7 days.

PRESCRIPTION 18.9. ACYCLOVIR (CREAM)
Rp.: Cremoris Aciclovіri 5% — 5.0
D.S. Apply to the blisters on the lips 5 times a day, starting from the first signs.

PRESCRIPTION 18.10. ACYCLOVIR (INTRAVENOUS)
Rp.: Aciclovіri 0.25
D.t.d. N. 30
S. 10 mg/kg every 8 hours as an intravenous infusion over 1 hour (in herpes encephalitis), with adequate hydration.

PRESCRIPTION 18.11. VALACYCLOVIR
Rp.: Valaciclovіri 1.0
D.t.d. N. 21 in tabulettis obductis
S. Take 1 tablet 3 times a day for 7 days (in herpes zoster).

PRESCRIPTION 18.12. TENOFOVIR
Rp.: Tenofoviri disoproxili fumaratis 0.3
D.t.d. N. 30 in tabulettis obductis
S. Take 1 tablet once a day with food (chronic hepatitis B).

Review questions

1. Features of mycobacteria and their effect on treatment.
2. Basic principles of tuberculosis treatment. Why are combined and long-term treatment needed?
3. Classification of antituberculosis drugs.
4. Mechanism and adverse effects of isoniazid, and the role of pyridoxine.
5. Mechanism, adverse effects and interactions of rifampicin.
6. Features and adverse effects of pyrazinamide and ethambutol.
7. Treatment regimen for drug-susceptible tuberculosis. Prevention of latent tuberculosis.

8. Modern approaches to treating drug-resistant tuberculosis.
9. Management of hepatotoxicity of antituberculosis drugs.
10. Stages of viral replication and targets of antiviral drugs.
11. Anti-influenza drugs: mechanism, indications, timing.
12. Mechanism of acyclovir and the basis of its selectivity. Uses and adverse effects.
13. Drugs against cytomegalovirus.
14. Drugs for treating hepatitis B and C.
15. Groups of antiretroviral drugs and the principles of ART. PEP and PrEP.
16. Drugs against COVID-19.
17. Interferons and their uses. The problem of unproven “antiviral” preparations.

Test questions

1. An antituberculosis drug that inhibits mycolic acid synthesis and is prescribed together with pyridoxine:
   A) Streptomycin
   B) Isoniazid
   C) Ethambutol
   D) Rifampicin
   E) Bedaquiline
2. A drug that colors urine and tears orange-red:
   A) Isoniazid
   B) Ethambutol
   C) Pyrazinamide
   D) Linezolid
   E) Rifampicin
3. An antituberculosis drug that reduces the effectiveness of oral contraceptives:
   A) Pyrazinamide
   B) Streptomycin
   C) Ethambutol
   D) Isoniazid
   E) Rifampicin
4. A drug that leads to reduced red-green color discrimination:
   A) Bedaquiline
   B) Ethambutol
   C) Pyrazinamide
   D) Isoniazid
   E) Rifampicin
5. An antituberculosis drug that causes hyperuricemia and gout:
   A) Ethambutol
   B) Rifampicin
   C) Pyrazinamide
   D) Bedaquiline
   E) Moxifloxacin

6. The standard regimen for treating drug-susceptible tuberculosis:
   A) 2 months of HRZE + 4 months of HR
   B) 1 month of streptomycin
   C) 2 months of HR + 4 months of HRZE
   D) 6 months of isoniazid monotherapy
   E) 2 weeks of HRZE

7. A drug that inhibits mycobacterial ATP synthase and is used in drug-resistant tuberculosis:
   A) Cycloserine
   B) Pretomanid
   C) Clofazimine
   D) Bedaquiline
   E) Linezolid

8. The antidote that stops seizures in isoniazid overdose:
   A) Naloxone
   B) Atropine
   C) Flumazenil
   D) Pyridoxine
   E) Acetylcysteine

9. A drug that inhibits neuraminidase in influenza:
   A) Oseltamivir
   B) Umifenovir
   C) Remdesivir
   D) Rimantadine
   E) Acyclovir

10. Within what time from symptom onset should oseltamivir be started to be most effective?
    A) 7 days
    B) 48 hours
    C) 10 days
    D) 2 weeks
    E) Timing does not matter

11. An antiviral drug activated by viral thymidine kinase:
    A) Foscarnet
    B) Dolutegravir
    C) Oseltamivir
    D) Tenofovir
    E) Acyclovir

12. The drug of choice in cytomegalovirus infection:
    A) Rimantadine
    B) Oseltamivir
    C) Lamivudine
    D) Ganciclovir
    E) Acyclovir cream

13. The modern regimen that cures hepatitis C in more than 95% of cases:
    A) Interferon alfa monotherapy
    B) Ribavirin monotherapy
    C) Sofosbuvir + velpatasvir
    D) Tenofovir
    E) Acyclovir

14. An integrase inhibitor included in the standard first-line regimen for HIV infection:
    A) Zidovudine
    B) Dolutegravir
    C) Efavirenz
    D) Darunavir
    E) Maraviroc

15. A protease inhibitor used in COVID-19 in combination with a strong CYP3A4 inhibitor:
    A) Dexamethasone
    B) Molnupiravir
    C) Remdesivir
    D) Nirmatrelvir (with ritonavir)
    E) Oseltamivir

Clinical cases

Case 1

A 34-year-old man with pulmonary tuberculosis (mycobacteria in sputum, Xpert MTB/RIF — rifampicin-susceptible) is being treated with the HRZE regimen. In week 5 he developed nausea, vomiting, loss of appetite and yellowing of the sclerae. ALT is 6 times the upper limit of normal, bilirubin is raised. The patient drinks alcohol every day.

Questions: 1) Condition? 2) Which drugs may be the cause? 3) Management.

Answer: 1) Drug-induced hepatitis (during antituberculosis treatment). 2) Isoniazid, rifampicin and pyrazinamide are hepatotoxic; alcohol increases the risk; other causes (viral hepatitis) must be excluded. 3) Stop all hepatotoxic drugs; if the patient's condition and the severity of the disease require it — a temporary liver-sparing regimen (ethambutol + levofloxacin ± linezolid); when the enzymes and bilirubin approach normal, restart the drugs one by one (rifampicin first, then isoniazid; do not restart pyrazinamide and extend the continuation phase); abstinence from alcohol; DOT.

Case 2

A 28-year-old woman is being treated for tuberculosis with a rifampicin-containing regimen. She takes a combined oral contraceptive and does not want to become pregnant.

Questions: What advice would you give?

Answer: Rifampicin is a strong CYP3A4 inducer; it speeds up the metabolism of estrogens and progestins and sharply reduces contraceptive effectiveness. During treatment and for at least 4 weeks after rifampicin is stopped, nonhormonal contraception (a copper intrauterine device, condoms) or a method not affected by inducers (depot medroxyprogesterone injection, a levonorgestrel intrauterine system) is recommended. She should also be warned that rifampicin colors urine and other fluids red and stains soft contact lenses.

Case 3

A 68-year-old man has had severe burning pain along the intercostal spaces of the right half of the chest for 2 days, followed by clusters of blisters on reddened skin in the same area.

Questions: 1) Diagnosis? 2) Treatment? 3) Prevention.

Answer: 1) Herpes zoster (shingles) — reactivation of the varicella-zoster virus. 2) Valacyclovir 1 g 3 times a day for 7 days (or acyclovir 800 mg 5 times a day) — within 72 hours of the onset of the rash (over 50 years — to reduce the risk of postherpetic neuralgia); analgesics for pain (paracetamol, NSAIDs; for severe pain — gabapentinoids or a low-dose TCA); dose adjustment according to renal function. If the ophthalmic branch is involved — an urgent ophthalmologist. 3) Vaccination with the recombinant zoster vaccine for people over 50.

Quick review table

Table 18.8. *Antituberculosis and antiviral drugs — summary*

Group	Preparations	Mechanism	Main use	Important adverse effect / note
Isoniazid	H	Mycolic acid synthesis	Tuberculosis (all regimens), latent tuberculosis	Hepatitis, neuropathy — pyridoxine
Rifampicin	R	RNA polymerase	Tuberculosis, latent tuberculosis	Red fluids, strong CYP inducer
Pyrazinamide	Z	Membrane energetics in an acidic environment	Tuberculosis (intensive phase)	Hepatitis, gout
Ethambutol	E	Arabinogalactan synthesis	Tuberculosis	Optic neuritis
MDR-TB drugs	Bedaquiline, pretomanid, linezolid, moxifloxacin	ATP synthase, various	BPaLM regimen	QT, myelosuppression, neuropathy
Neuraminidase inhibitors	Oseltamivir, zanamivir	Release of virions	Influenza A and B (within 48 hours)	Nausea
Antiherpetic	Acyclovir, valacyclovir, famciclovir	Viral DNA polymerase (activated by thymidine kinase)	Herpes, shingles, encephalitis	Crystalluria (IV)
Anti-CMV	Ganciclovir, valganciclovir, foscarnet	DNA polymerase	CMV infection	Myelosuppression; foscarnet — nephrotoxic
Anti-HBV	Tenofovir, entecavir	Reverse transcriptase (polymerase)	Chronic hepatitis B	Long-term; flare on withdrawal
Anti-HCV (DAAs)	Sofosbuvir, velpatasvir, glecaprevir/pibrentasvir	NS5B, NS5A, NS3/4A	Hepatitis C (8–12 weeks)	HBV reactivation; bradycardia with amiodarone
ART	Tenofovir, lamivudine, dolutegravir, efavirenz, darunavir	Reverse transcriptase, integrase, protease	HIV, PEP, PrEP	Adherence, interactions

Group	Preparations	Mechanism	Main use	Important adverse effect / note
COVID-19	Nirmatrelvir/ritonavir, remdesivir, molnupiravir	Protease, RNA polymerase	Early treatment in risk groups	Ritonavir — CYP3A4 inhibitor
Interferons	Interferon alfa, beta	Antiviral state, immunomodulation	Hepatitis B (selected), multiple sclerosis	Flu-like syndrome, depression

Answers to the test questions

Topic 1: 1-A; 2-B; 3-D; 4-D; 5-A; 6-D; 7-D; 8-A; 9-E; 10-A; 11-B; 12-B; 13-C; 14-B; 15-D.

Topic 2: 1-B; 2-B; 3-C; 4-C; 5-E; 6-B; 7-E; 8-E; 9-B; 10-B; 11-A; 12-B; 13-C; 14-C; 15-C.

Topic 3: 1-A; 2-C; 3-A; 4-E; 5-A; 6-B; 7-E; 8-D; 9-D; 10-C; 11-C; 12-A; 13-D; 14-B; 15-B.

Topic 4: 1-A; 2-C; 3-C; 4-C; 5-D; 6-B; 7-B; 8-B; 9-E; 10-C; 11-E; 12-E; 13-E; 14-C; 15-A.

Topic 5: 1-C; 2-A; 3-B; 4-C; 5-D; 6-A; 7-C; 8-E; 9-B; 10-E; 11-A; 12-A; 13-C; 14-B; 15-A.

Topic 6: 1-C; 2-C; 3-E; 4-A; 5-D; 6-C; 7-C; 8-B; 9-B; 10-A; 11-C; 12-E; 13-D; 14-D; 15-D.

Topic 7: 1-B; 2-D; 3-D; 4-B; 5-A; 6-E; 7-D; 8-A; 9-C; 10-C; 11-D; 12-D; 13-D; 14-A; 15-A.

Topic 8: 1-E; 2-A; 3-D; 4-B; 5-C; 6-C; 7-A; 8-A; 9-D; 10-D; 11-D; 12-D; 13-A; 14-C; 15-D.

Topic 9: 1-A; 2-A; 3-A; 4-C; 5-A; 6-D; 7-C; 8-D; 9-E; 10-D; 11-C; 12-E; 13-B; 14-B; 15-C.

Topic 10: 1-C; 2-A; 3-D; 4-B; 5-C; 6-E; 7-E; 8-B; 9-A; 10-C; 11-A; 12-B; 13-B; 14-B; 15-A.

Topic 11: 1-B; 2-C; 3-B; 4-B; 5-A; 6-C; 7-D; 8-A; 9-D; 10-A; 11-C; 12-C; 13-D; 14-D; 15-A.

Topic 12: 1-E; 2-D; 3-E; 4-C; 5-D; 6-A; 7-C; 8-C; 9-B; 10-B; 11-A; 12-A; 13-B; 14-B; 15-D.

Topic 13: 1-D; 2-D; 3-B; 4-D; 5-C; 6-A; 7-C; 8-B; 9-C; 10-D; 11-B; 12-B; 13-C; 14-C; 15-A.

Topic 14: 1-A; 2-A; 3-A; 4-E; 5-C; 6-B; 7-E; 8-B; 9-B; 10-B; 11-B; 12-C; 13-C; 14-D; 15-C.

Topic 15: 1-C; 2-D; 3-B; 4-B; 5-C; 6-D; 7-C; 8-C; 9-C; 10-E; 11-B; 12-D; 13-D; 14-C; 15-E.

Topic 16: 1-D; 2-E; 3-C; 4-A; 5-B; 6-B; 7-B; 8-A; 9-E; 10-C; 11-D; 12-B; 13-C; 14-A; 15-B.

Topic 17: 1-B; 2-D; 3-E; 4-B; 5-E; 6-B; 7-A; 8-C; 9-B; 10-E; 11-A; 12-D; 13-D; 14-D; 15-D.

Topic 18: 1-B; 2-E; 3-E; 4-B; 5-C; 6-A; 7-D; 8-D; 9-A; 10-B; 11-E; 12-D; 13-C; 14-B; 15-D.

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